

SEACAR SAV Analysis

Last compiled on 03 August, 2026

Contents

Important Notes	1
Alligator Harbor Aquatic Preserve	2
Apalachicola Bay Aquatic Preserve	6
Apalachicola National Estuarine Research Reserve	10
Banana River Aquatic Preserve	14
Big Bend Seagrasses Aquatic Preserve	18
Biscayne Bay Aquatic Preserve	22
Boca Ciega Bay Aquatic Preserve	26
Cape Haze Aquatic Preserve	30
Cockroach Bay Aquatic Preserve	34
Estero Bay Aquatic Preserve	38
Florida Keys National Marine Sanctuary	42
Fort Pickens State Park Aquatic Preserve	46
Gasparilla Sound-Charlotte Harbor Aquatic Preserve	50
Indian River-Malabar to Vero Beach Aquatic Preserve	54
Indian River-Vero Beach to Ft. Pierce Aquatic Preserve	58
Jensen Beach to Jupiter Inlet Aquatic Preserve	62
Lemon Bay Aquatic Preserve	66
Lignumvitae Key Aquatic Preserve	69
Loxahatchee River-Lake Worth Creek Aquatic Preserve	71
Matlacha Pass Aquatic Preserve	75
Mosquito Lagoon Aquatic Preserve	79
Nature Coast Aquatic Preserve	83

Pine Island Sound Aquatic Preserve	87
Pinellas County Aquatic Preserve	91
Rookery Bay Aquatic Preserve	95
Rookery Bay National Estuarine Research Reserve	99
St. Andrews Aquatic Preserve	103
St. Joseph Bay Aquatic Preserve	107
St. Martins Marsh Aquatic Preserve	111
Terra Ceia Aquatic Preserve	115
Summary of SEACAR_SAV_BB_script_website.R	118

Important Notes

The purpose of this script is to provide a report summary of SAV analysis. The script used for analysis is SEACAR_SAV_BB_script_website.R.

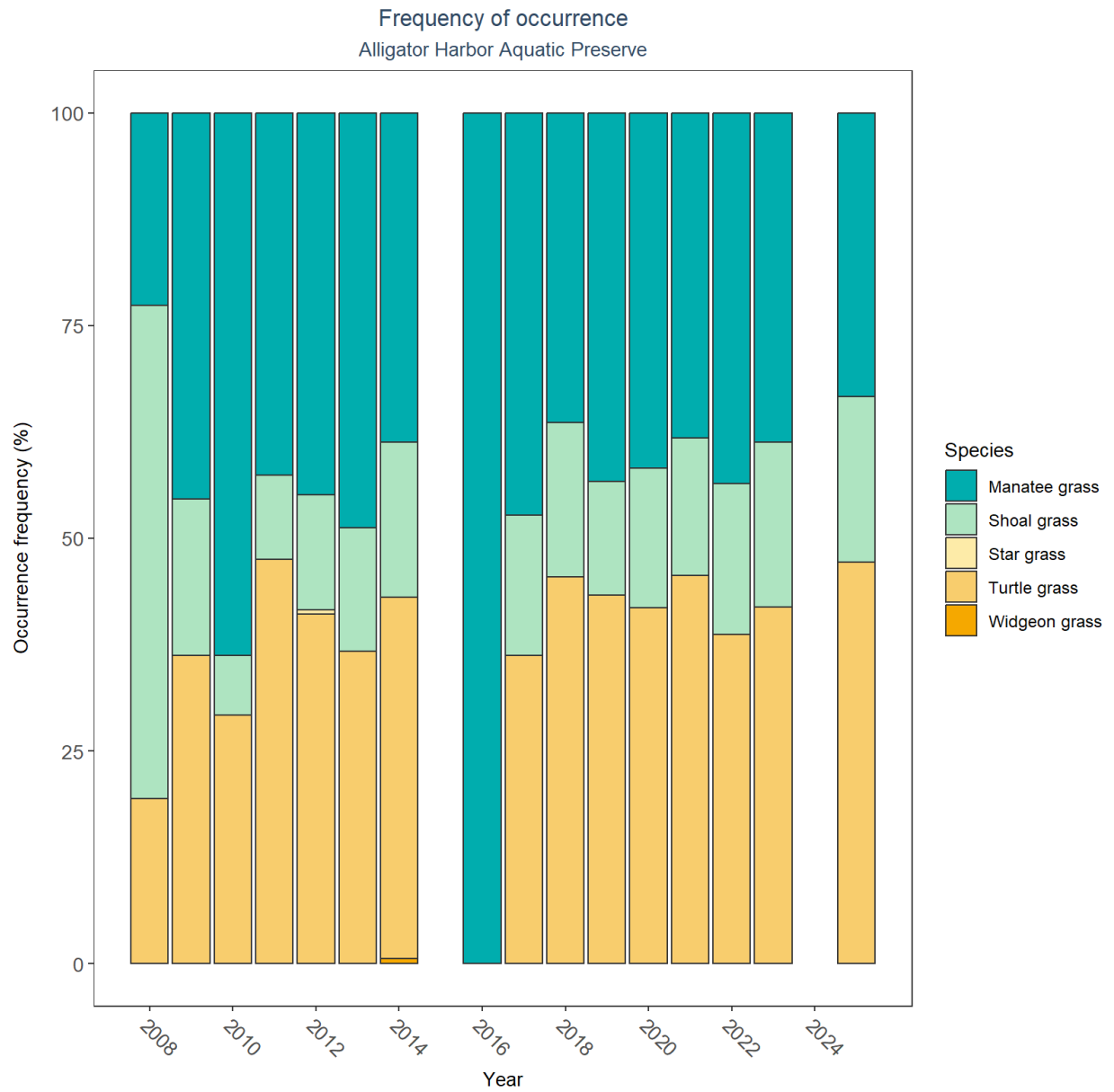
All scripts and outputs can be found on the SEACAR GitHub repository:

https://github.com/FloridaSEACAR/SEACAR_Trend_Analyses

This script is based off of code originally written by Stephen Durham with comments by Marcus W. Beck. Modified by Tyler Hill.

The file being used for the analysis is: **All_SAV_Parameters-2026-Jun-04.txt**

Alligator Harbor Aquatic Preserve



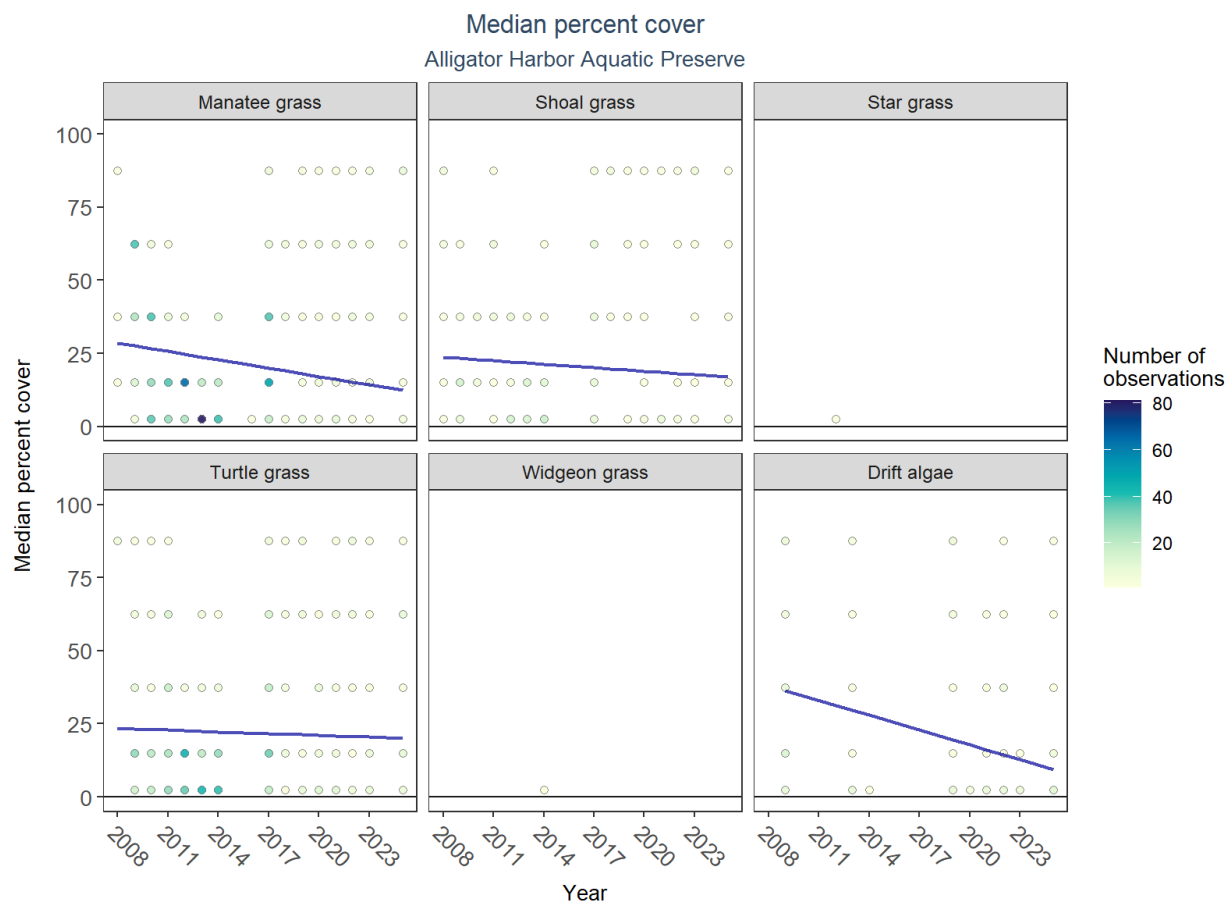
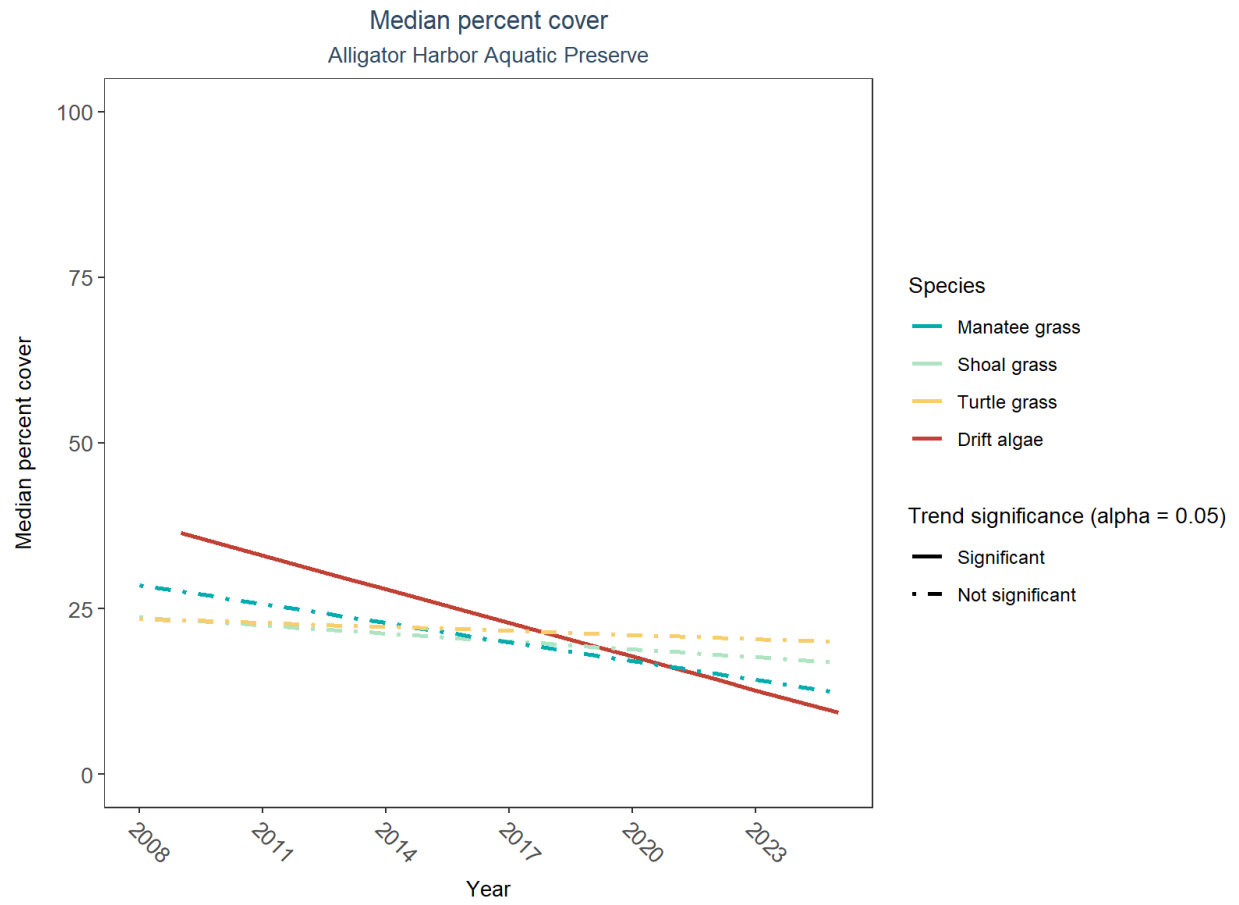


Table 1: SAV LME Results for Alligator Harbor Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Drift algae	Decreasing trend	2009 - 2025	61.87	-1.70	0.02
Shoal grass	No detectable trend	2008 - 2025	29.30	-0.40	0.49
Star grass	Insufficient data	2012 - 2012	-	-	-
Widgeon grass	Insufficient data	2014 - 2014	-	-	-
Manatee grass	No detectable trend	2008 - 2025	41.95	-0.96	0.08
Turtle grass	No detectable trend	2008 - 2025	26.30	-0.20	0.72

An annual decrease in percent cover was observed for drift algae (-1.70%). Manatee grass, shoal grass, and turtle grass showed no detectable change in percent cover. Trends in percent cover could not be evaluated for star grass and widgeon grass due to insufficient data.



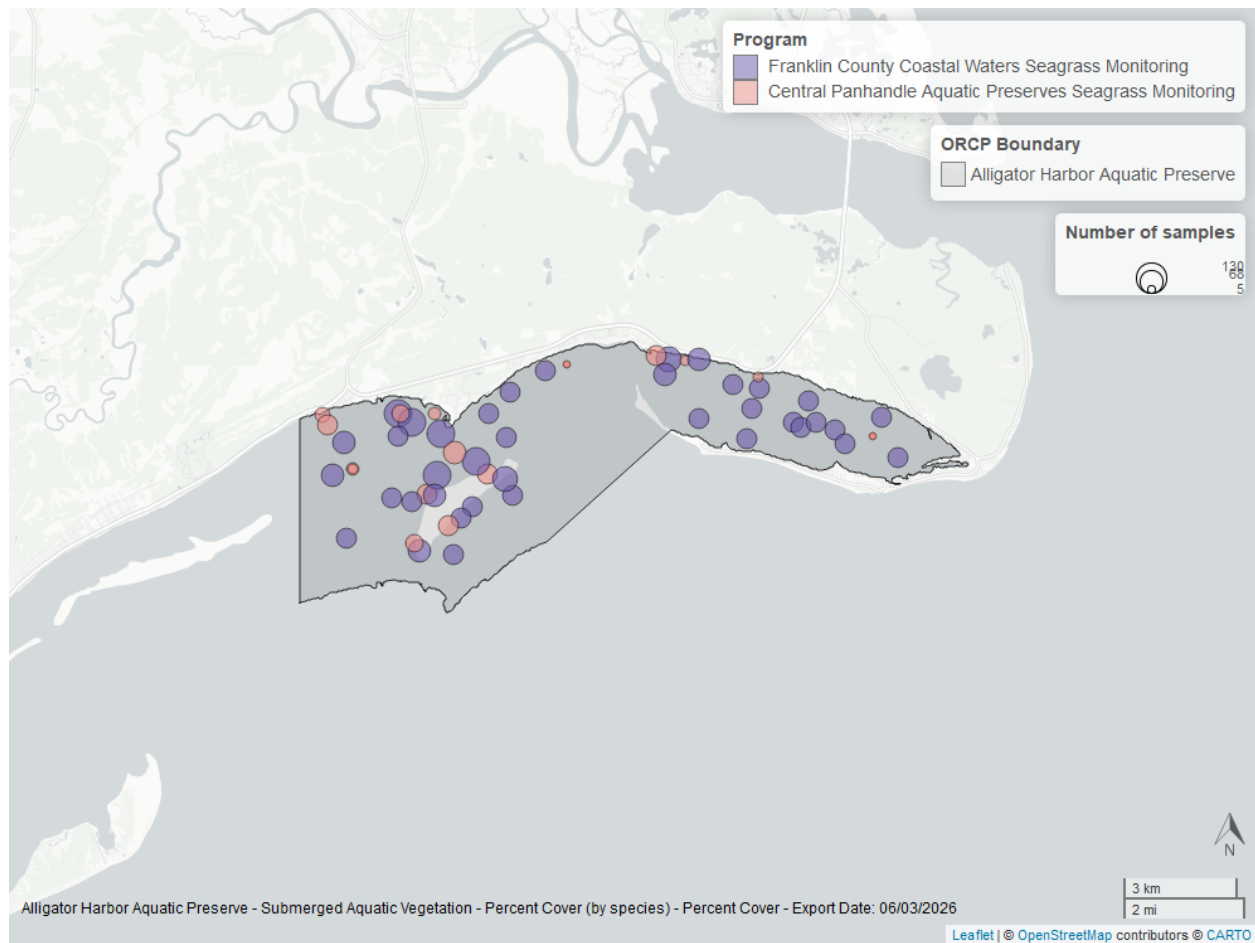
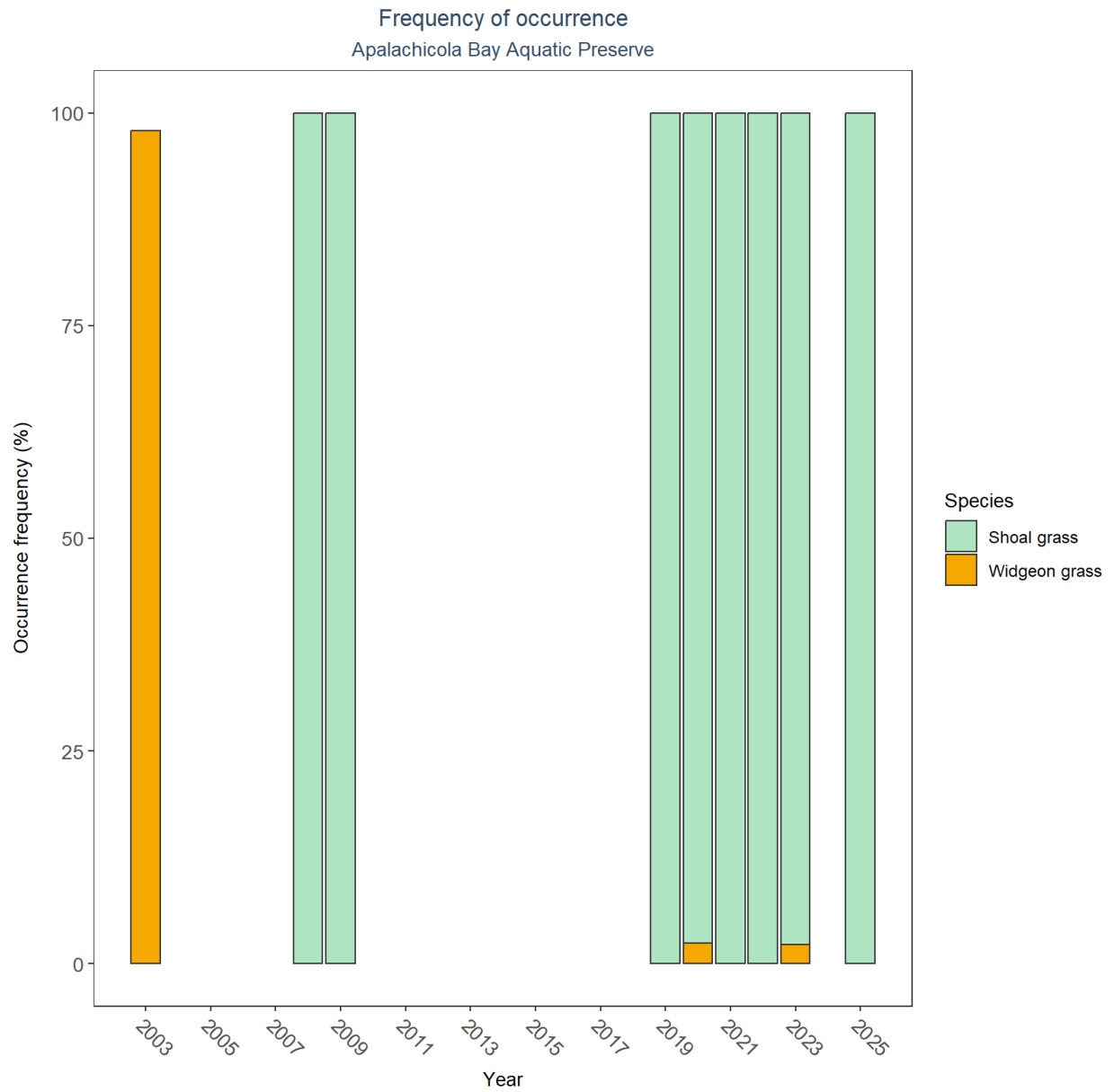


Figure 1: Map showing SAV sampling sites within the boundaries of *Alligator Harbor Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Apalachicola Bay Aquatic Preserve



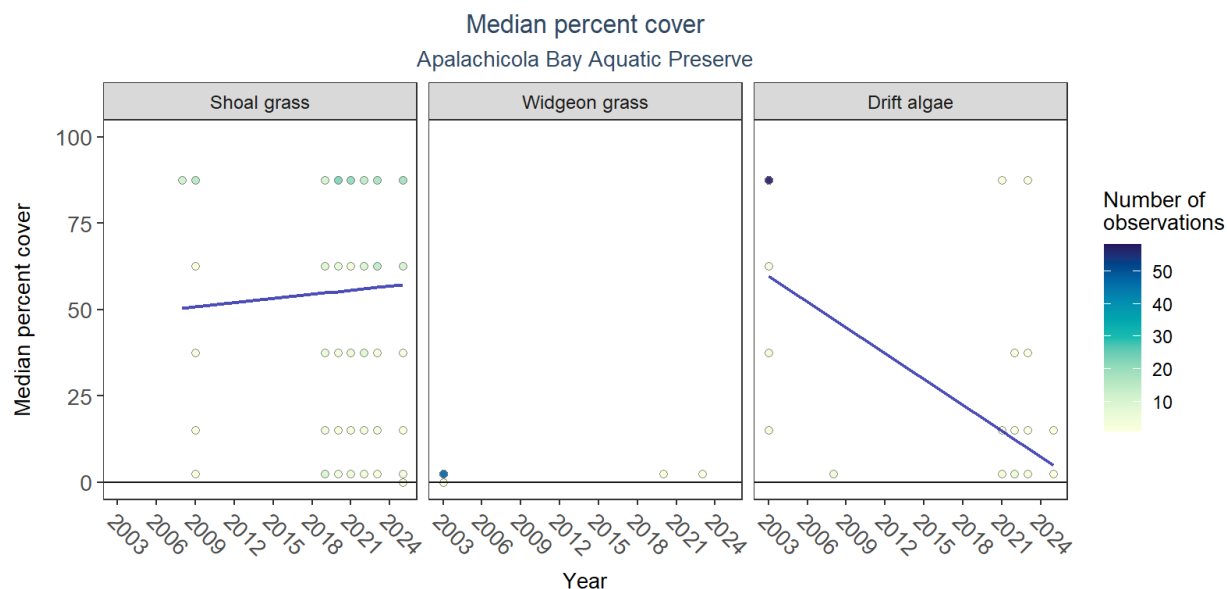
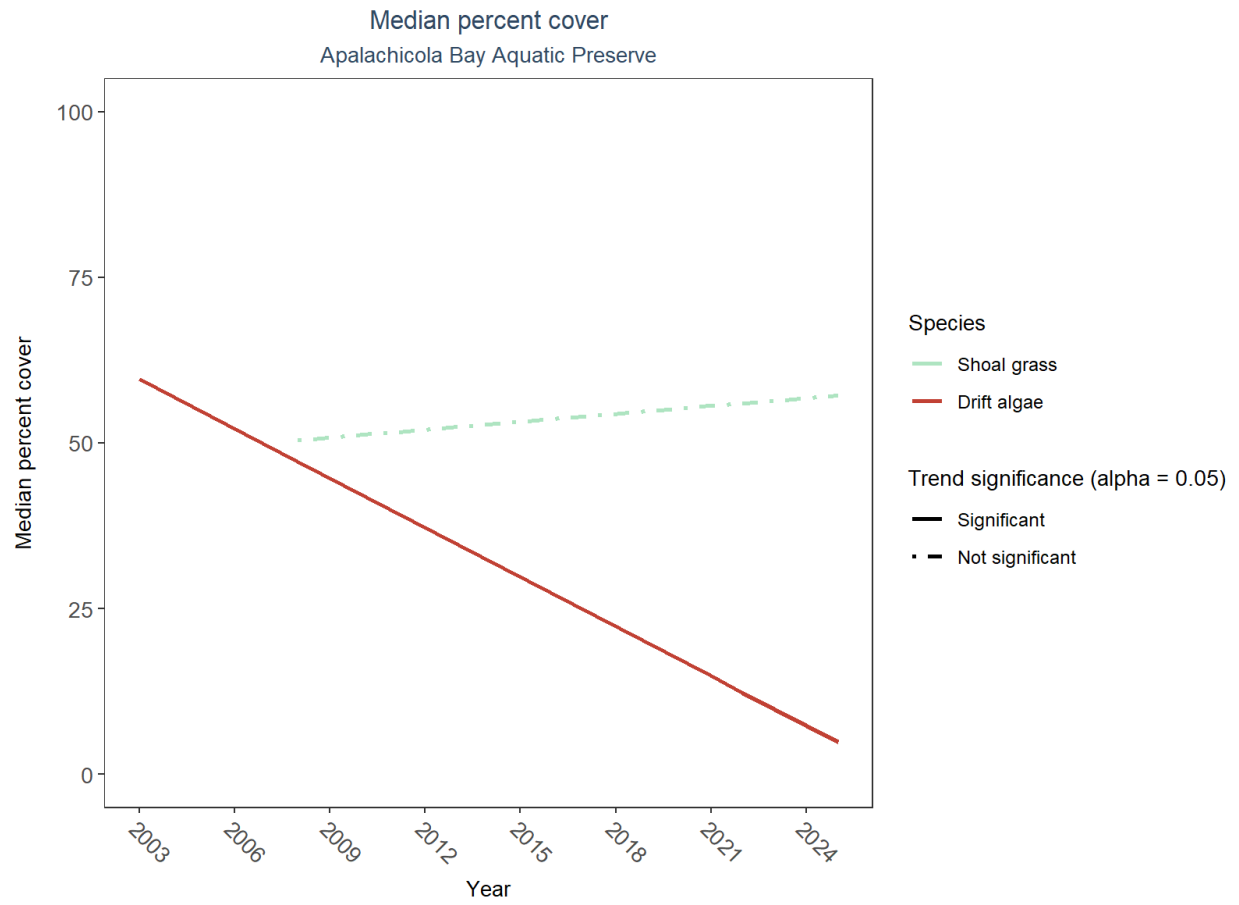


Table 2: SAV LME Results for Apalachicola Bay Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Drift algae	Decreasing trend	2003 - 2025	82.06	-2.49	0.0
Shoal grass	No detectable trend	2008 - 2025	44.92	0.40	0.7
Widgeon grass	Insufficient data	2003 - 2023	-	-	-

An annual decrease in percent cover was observed for drift algae (-2.49%). No detectable change in percent cover was observed for shoal grass. Trends in percent cover could not be evaluated for widgeon grass due to insufficient data.



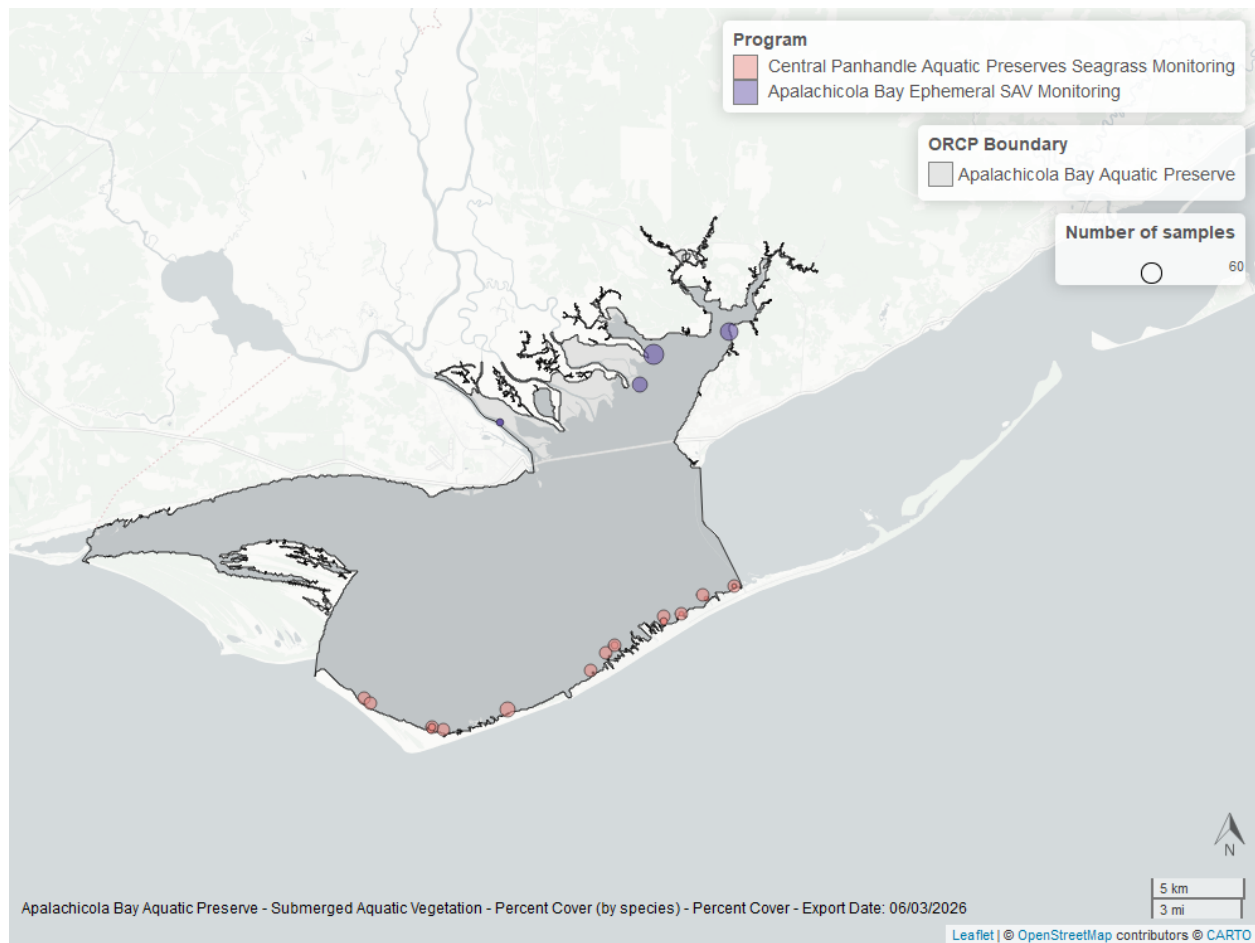
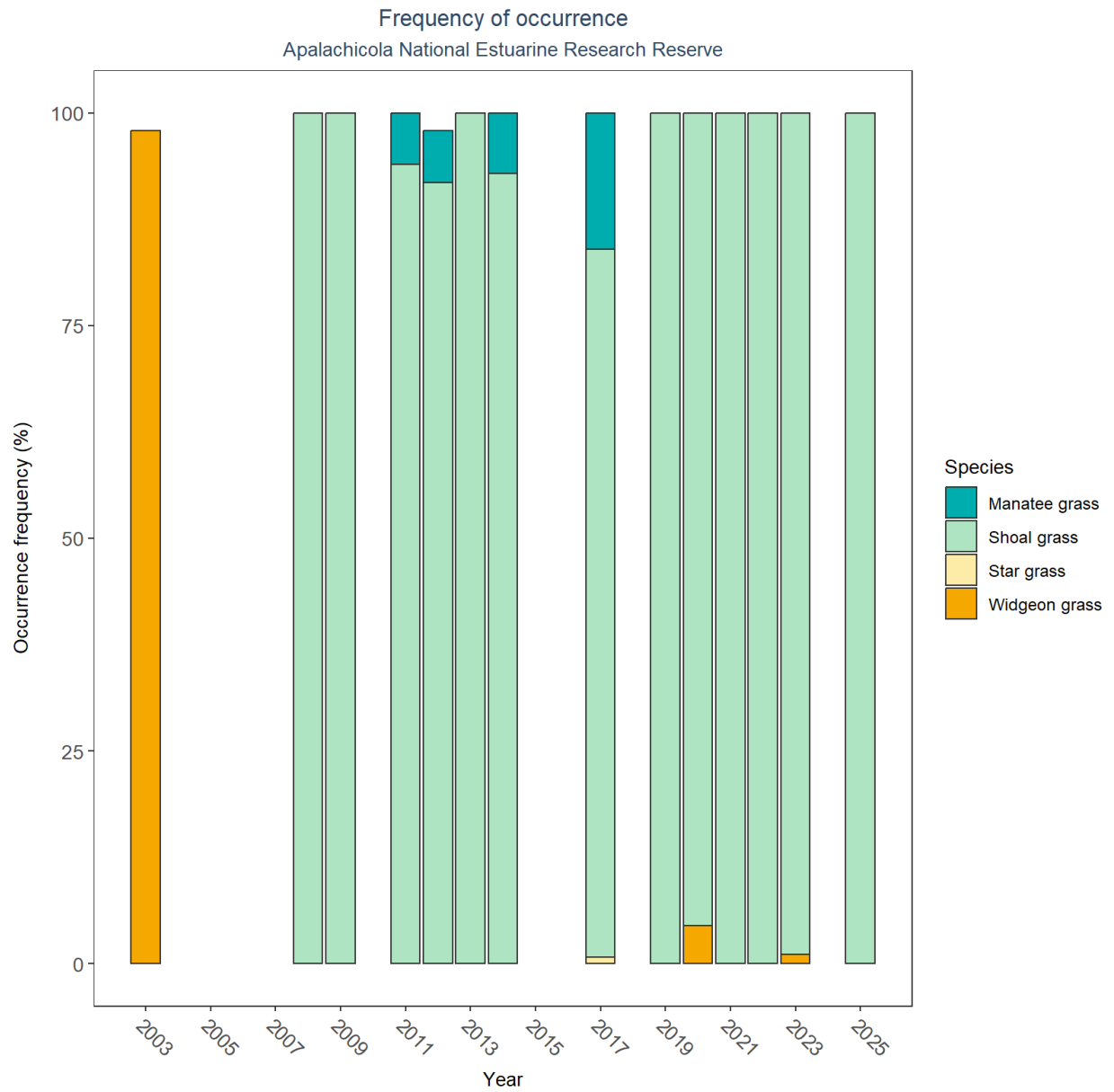


Figure 2: Map showing SAV sampling sites within the boundaries of *Apalachicola Bay Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Apalachicola National Estuarine Research Reserve



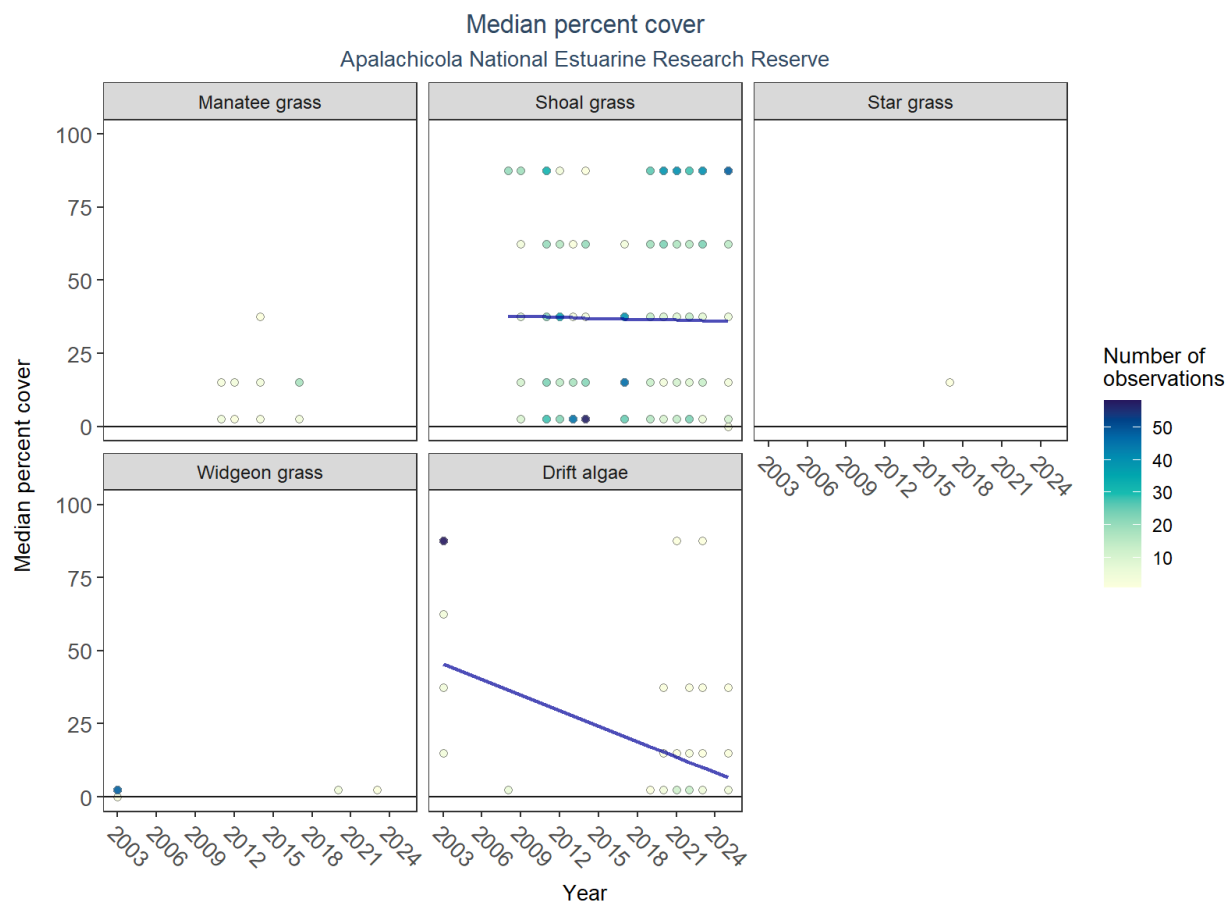
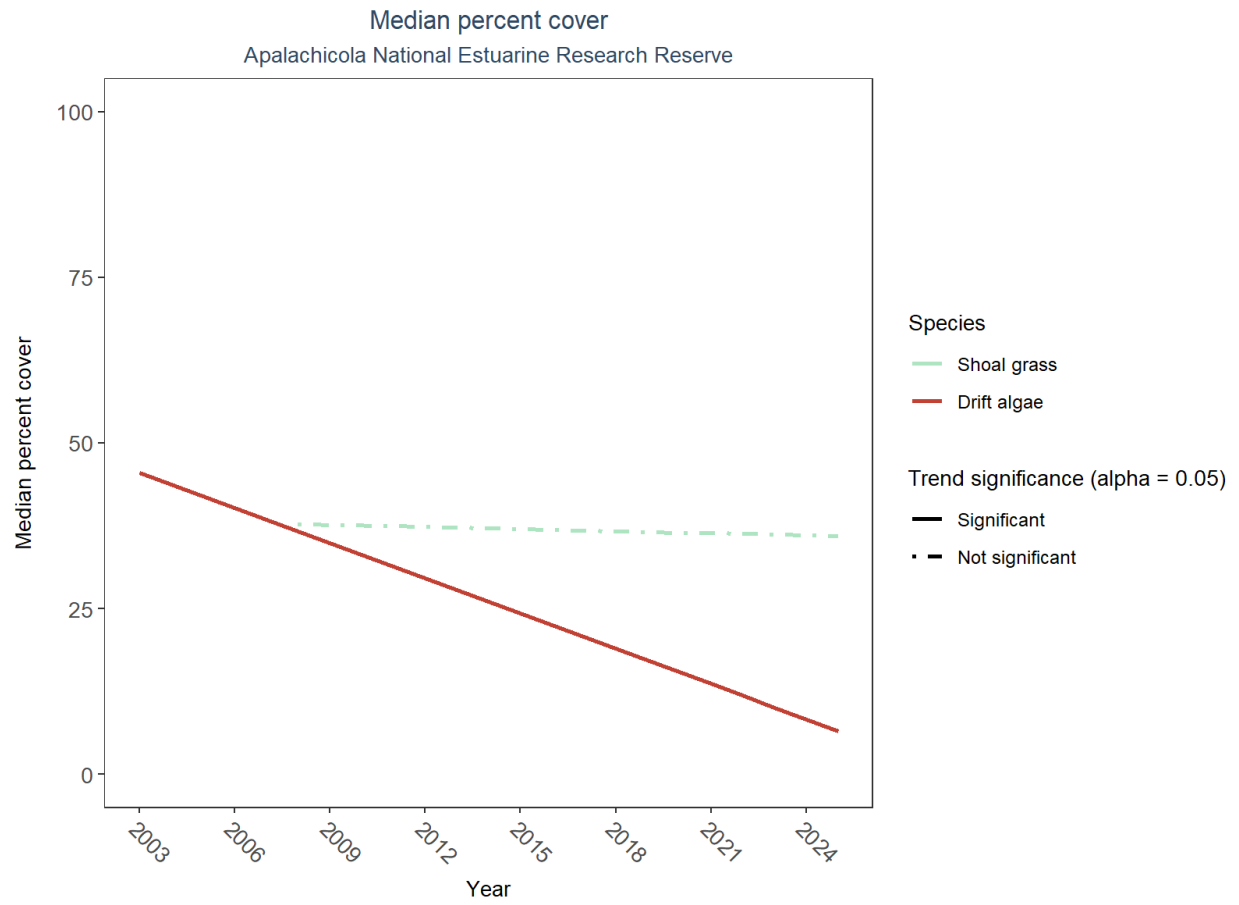


Table 3: SAV LME Results for Apalachicola National Estuarine Research Reserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Drift algae	Decreasing trend	2003 - 2025	61.41	-1.77	0.02
Shoal grass	No detectable trend	2008 - 2025	39.25	-0.11	0.86
Star grass	Insufficient data	2017 - 2017	-	-	-
Widgeon grass	Insufficient data	2003 - 2023	-	-	-
Manatee grass	Insufficient data	2011 - 2017	-	-	-

An annual decrease in percent cover was observed for drift algae (-1.77%). No detectable change in percent cover was observed for shoal grass. Trends in percent cover could not be evaluated for manatee grass, star grass, and widgeon grass due to insufficient data.



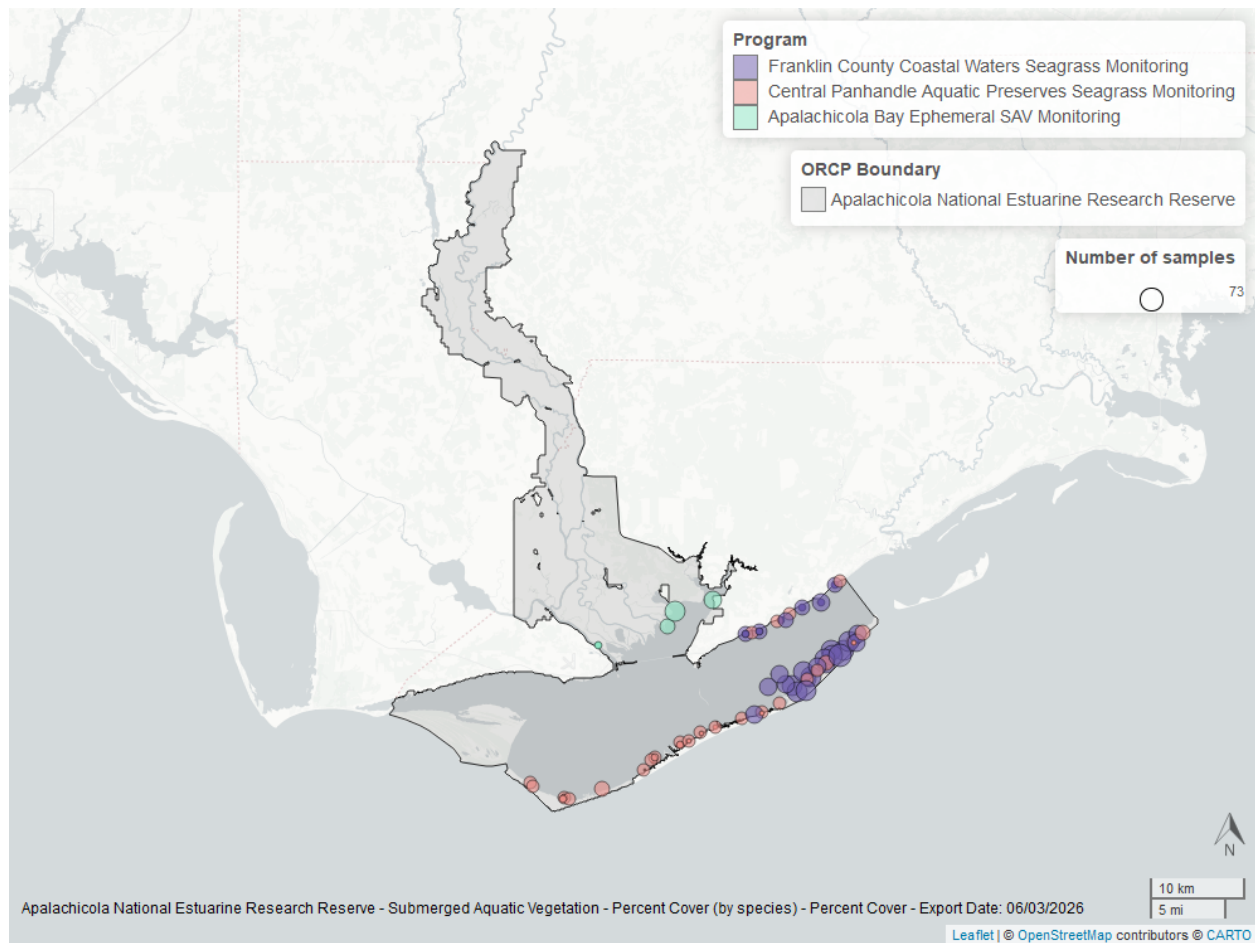
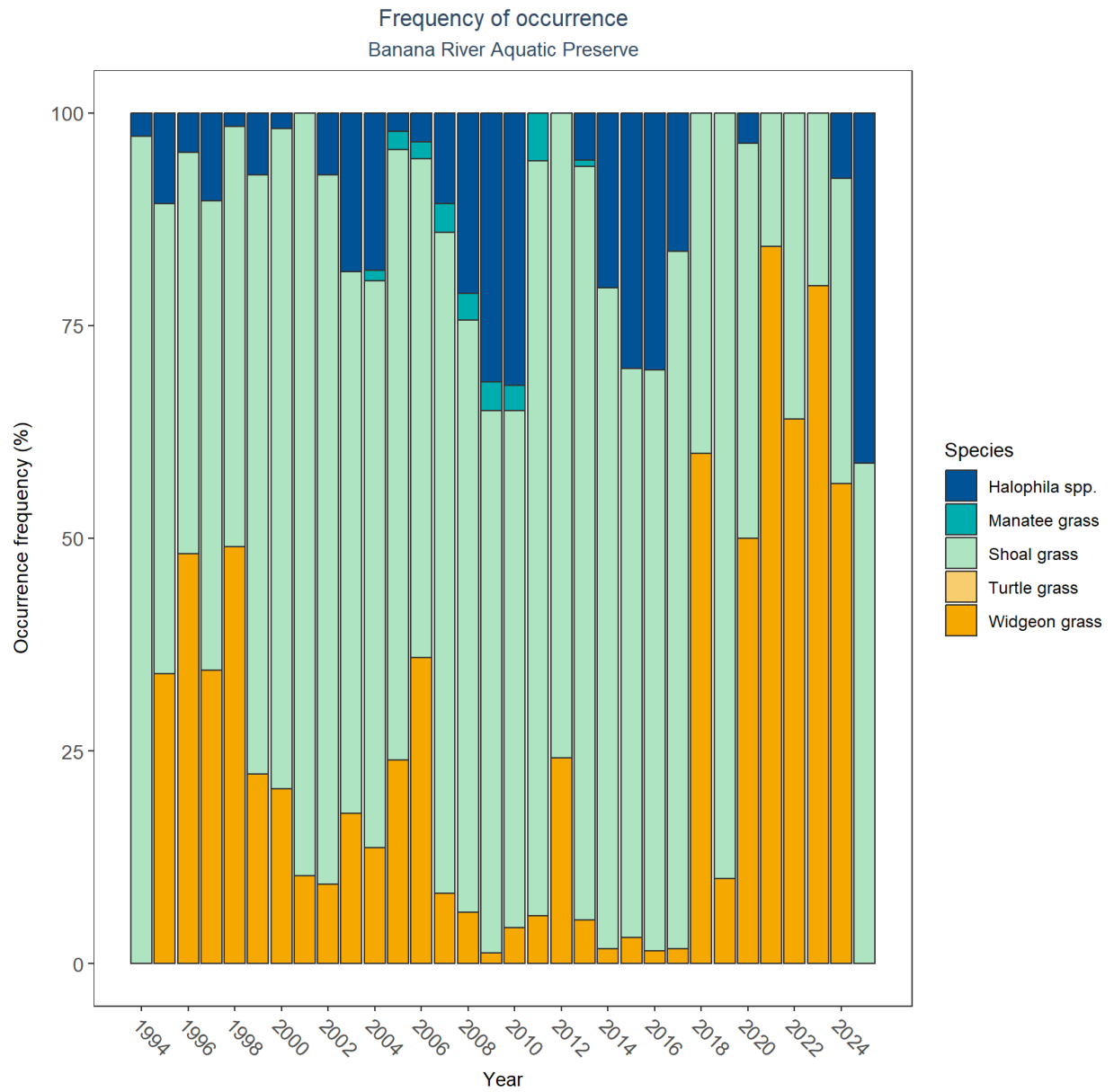


Figure 3: Map showing SAV sampling sites within the boundaries of *Apalachicola National Estuarine Research Reserve*. The point size reflects the number of samples at a given sampling site.

Banana River Aquatic Preserve



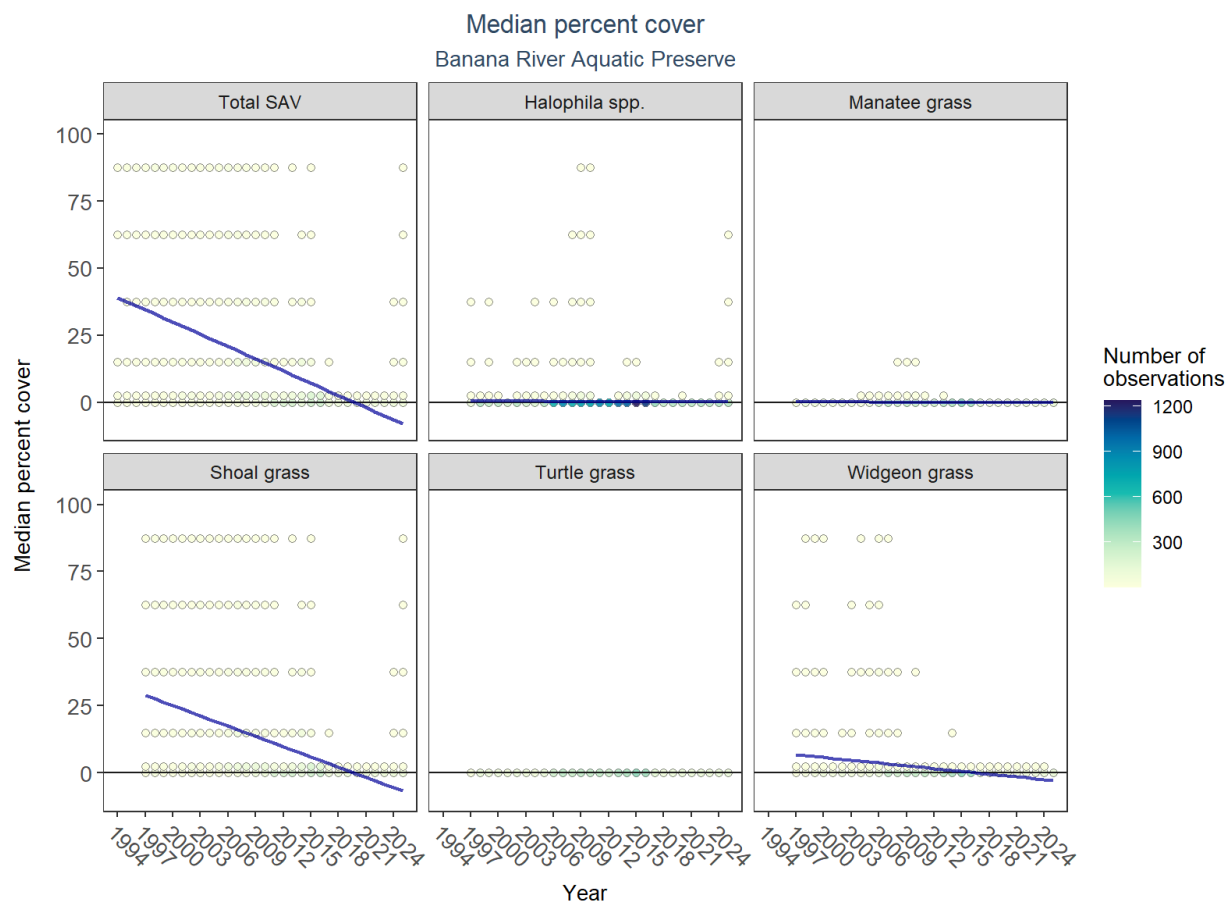
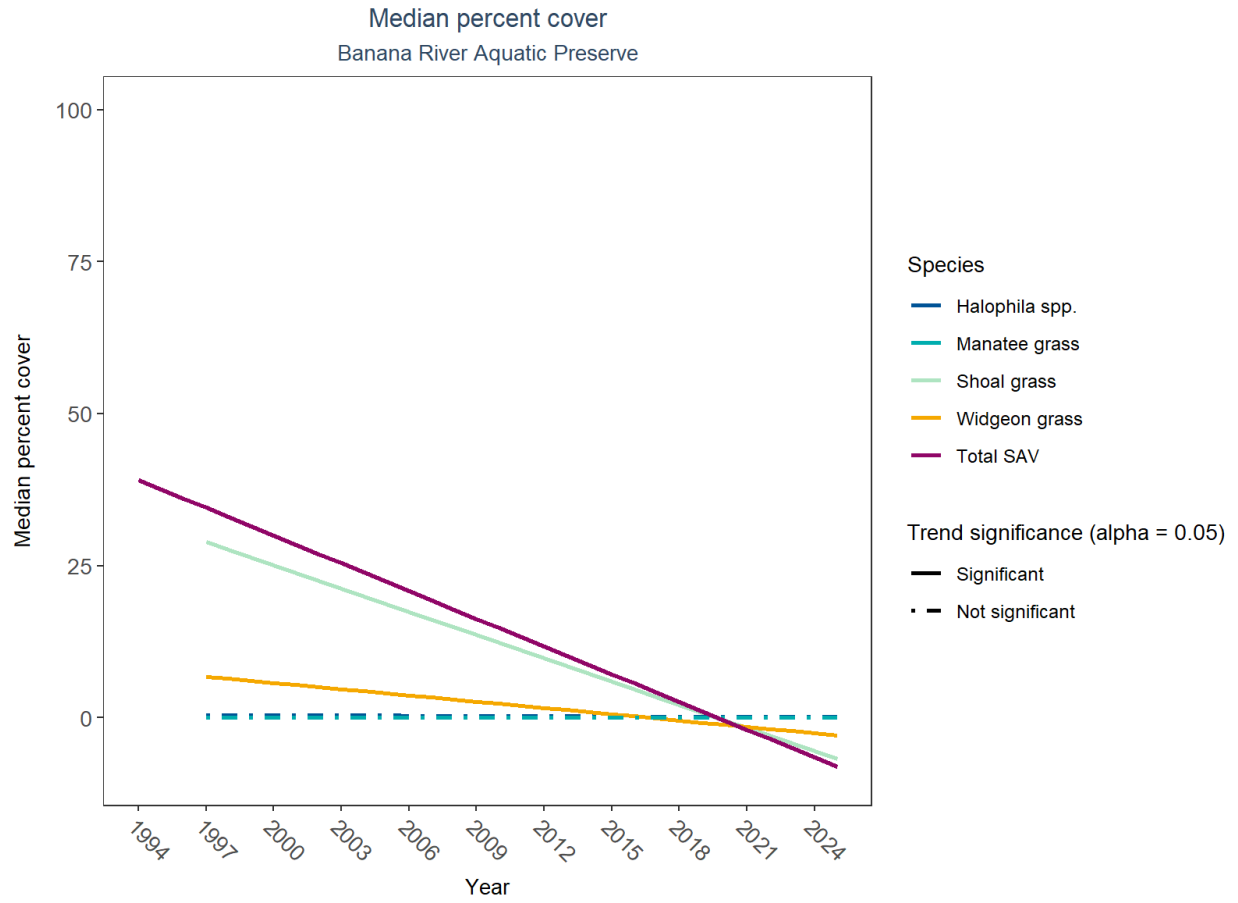


Table 4: SAV LME Results for Banana River Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Shoal grass	Decreasing trend	1997 - 2025	32.71	-1.27	0.00
Halophila spp.	No detectable trend	1997 - 2025	0.50	-0.01	0.14
Widgeon grass	Decreasing trend	1997 - 2025	7.80	-0.34	0.00
Manatee grass	No detectable trend	1997 - 2025	0.09	0.00	0.07
Turtle grass	Model did not fit the available data	1997 - 2025	-	-	-
Total SAV	Decreasing trend	1994 - 2025	39.07	-1.52	0.00

Annual decreases in percent cover were observed for total SAV (-1.52%), shoal grass (-1.27%), and widgeon grass (-0.34%). No detectable change in percent cover was observed for *Halophila* spp. and manatee grass. The model could not be fitted for turtle grass.



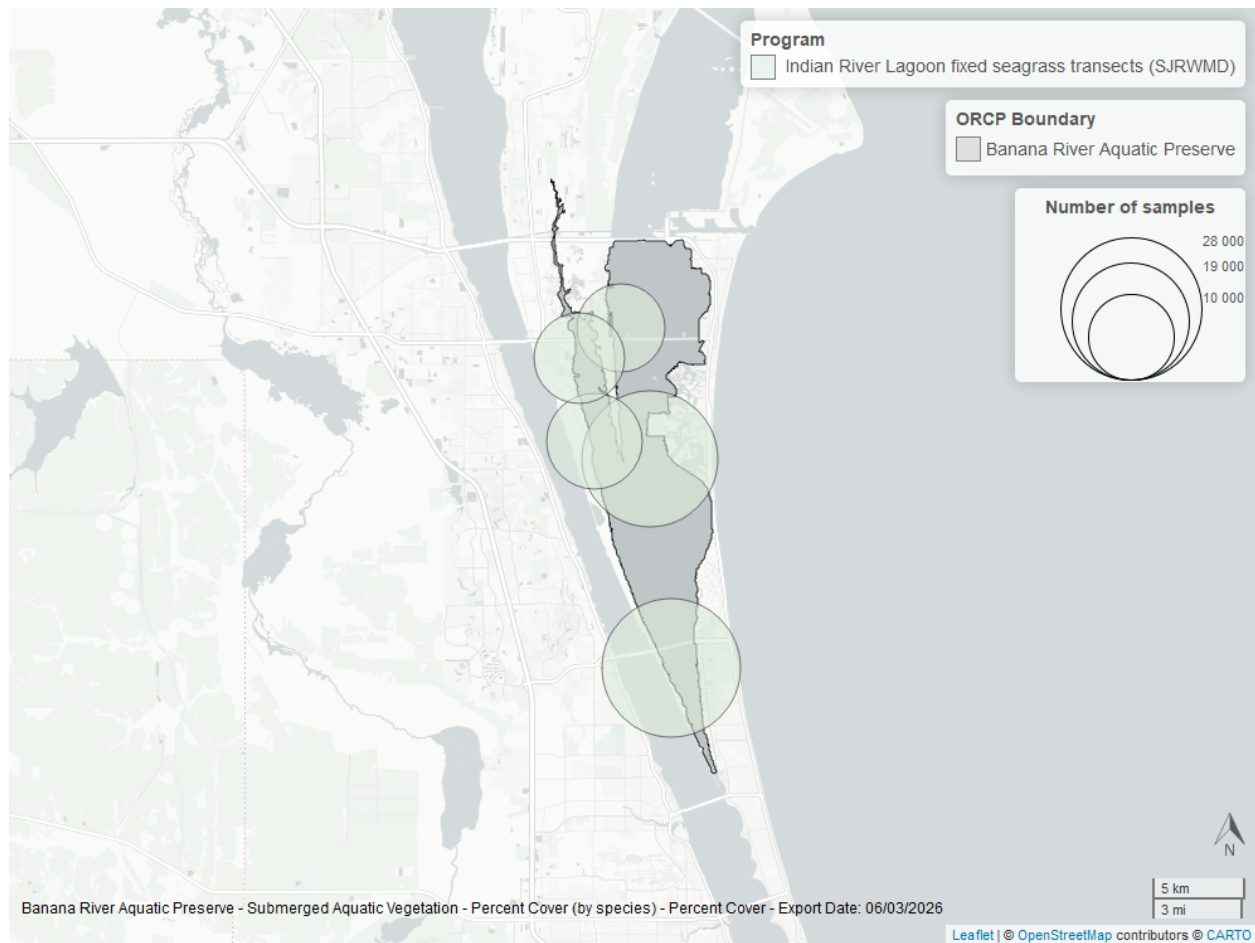
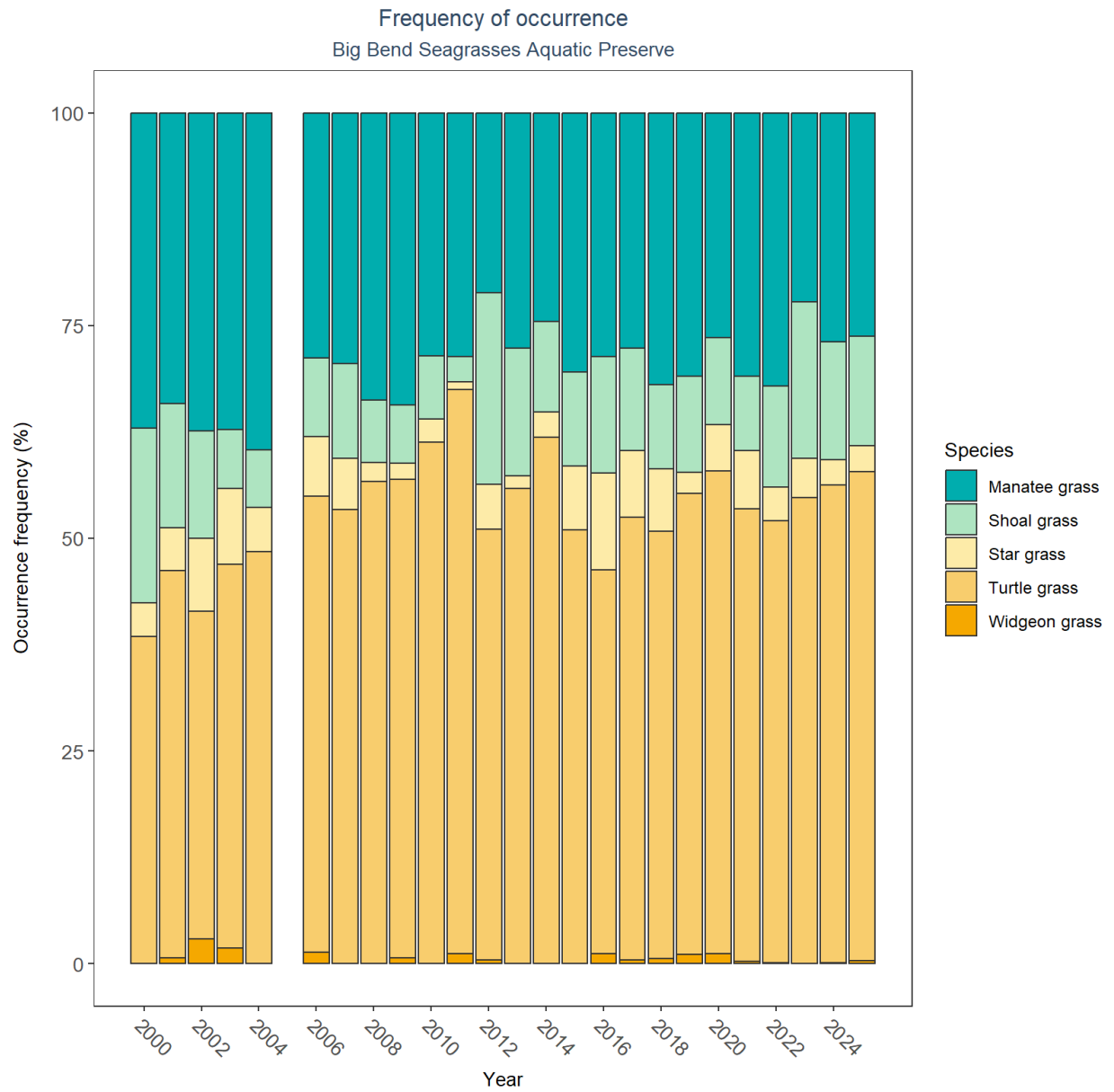


Figure 4: Map showing SAV sampling sites within the boundaries of *Banana River Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Big Bend Seagrasses Aquatic Preserve



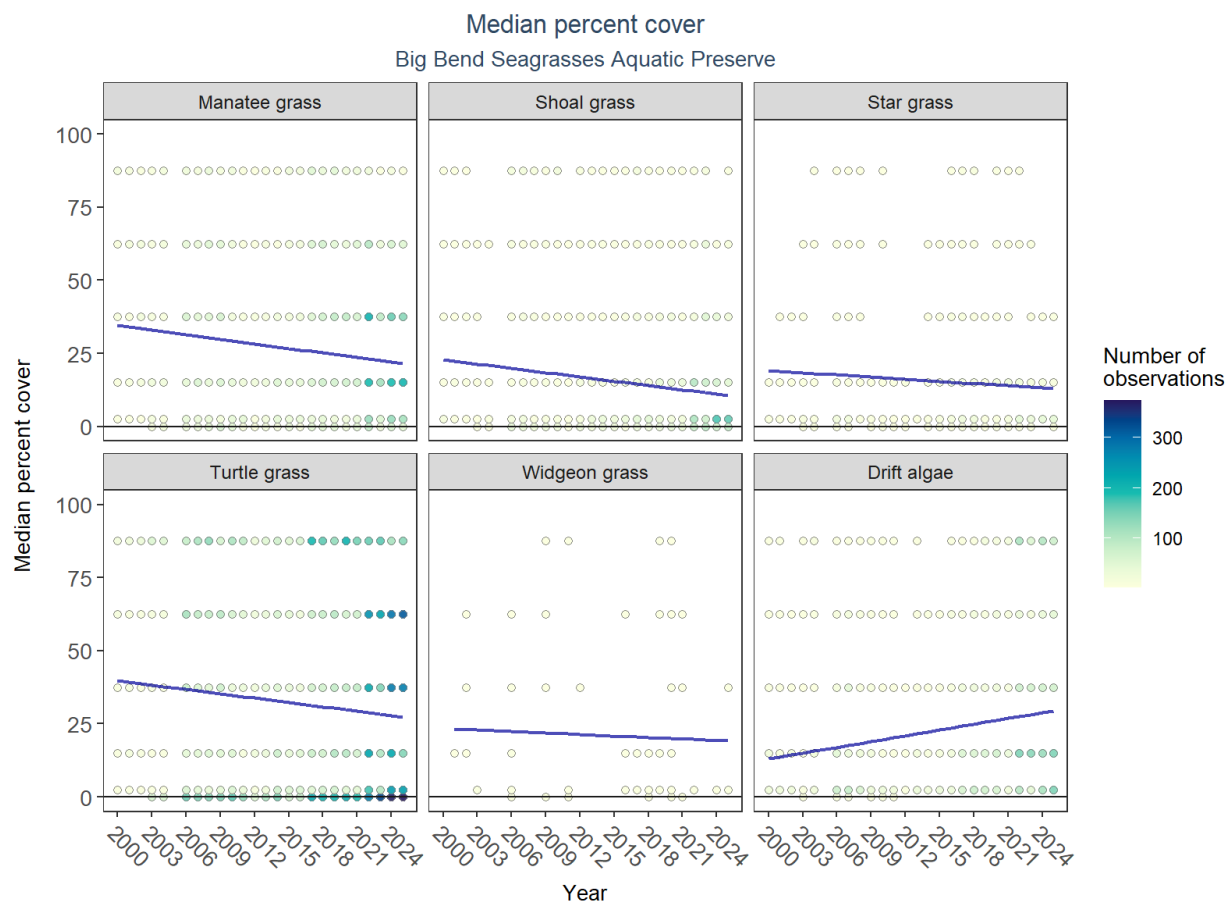
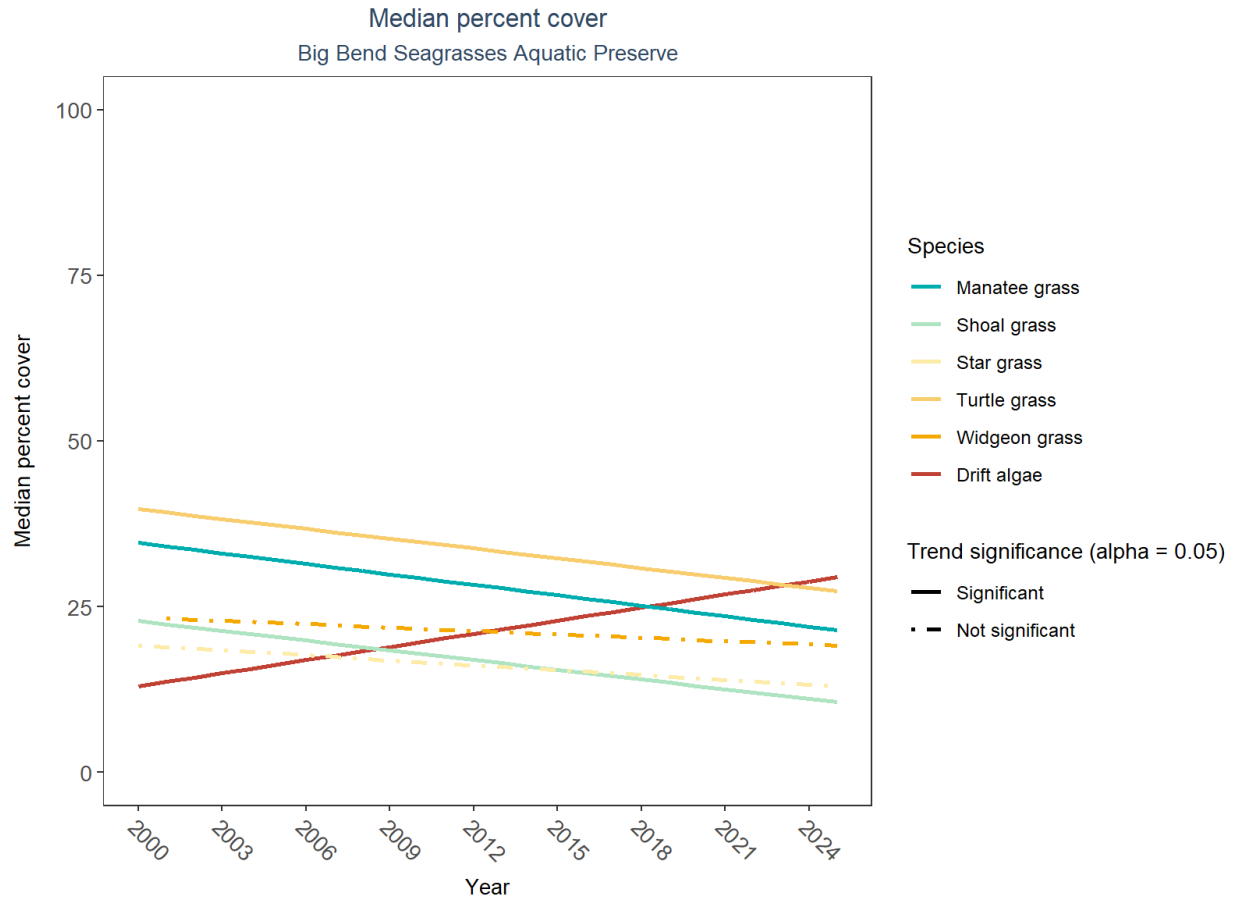


Table 5: SAV LME Results for Big Bend Seagrasses Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Drift algae	Increasing trend	2000 - 2025	9.03	0.66	0.00
Shoal grass	Decreasing trend	2000 - 2025	25.78	-0.49	0.00
Star grass	No detectable trend	2000 - 2025	20.62	-0.25	0.05
Widgeon grass	No detectable trend	2001 - 2025	24.41	-0.17	0.71
Manatee grass	Decreasing trend	2000 - 2025	37.79	-0.53	0.00
Turtle grass	Decreasing trend	2000 - 2025	42.68	-0.49	0.00

An annual increase in percent cover was observed for drift algae (0.66%). Annual decreases in percent cover were observed for manatee grass (-0.53%), shoal grass (-0.49%), and turtle grass (-0.49%). No detectable change in percent cover was observed for star grass and widgeon grass.



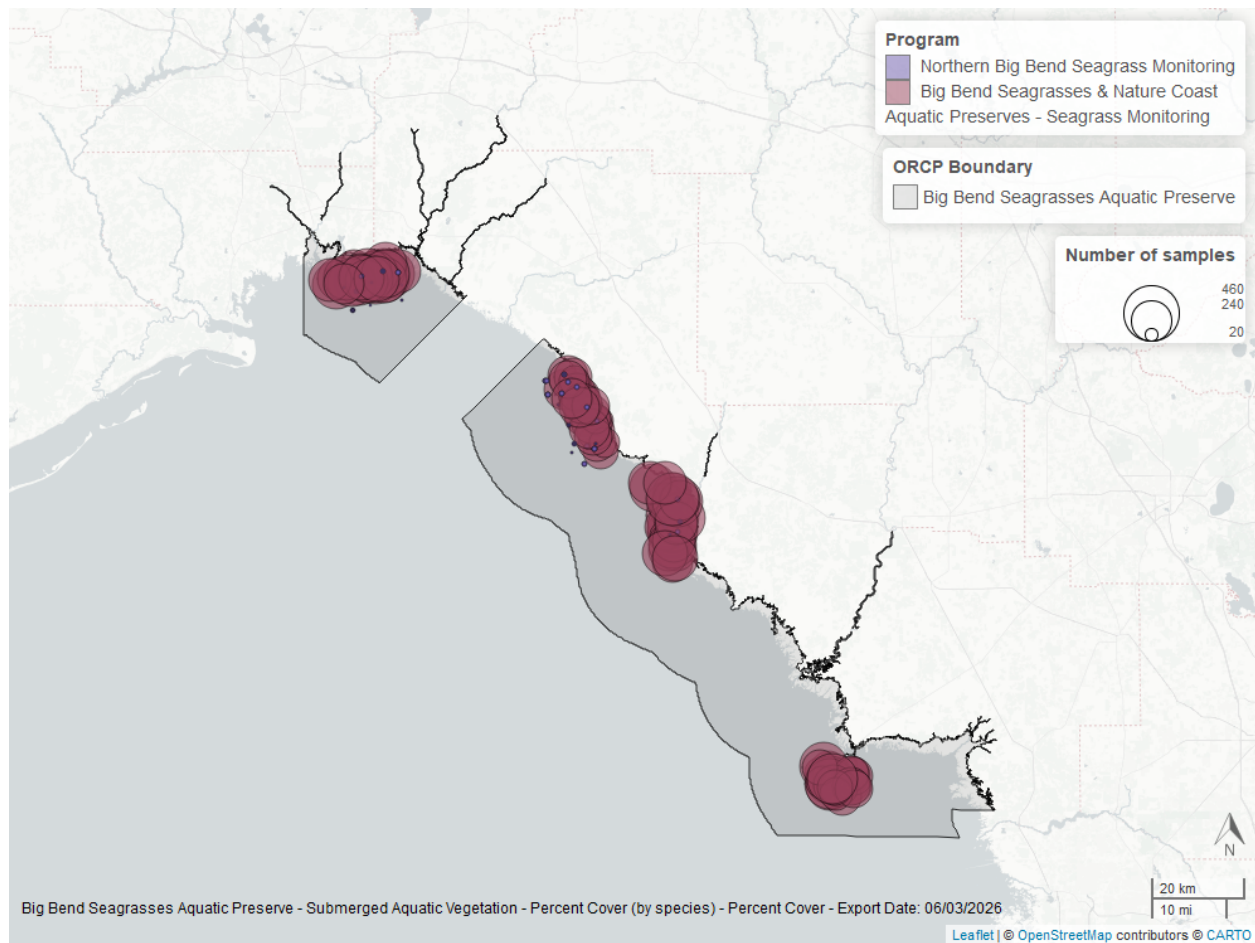
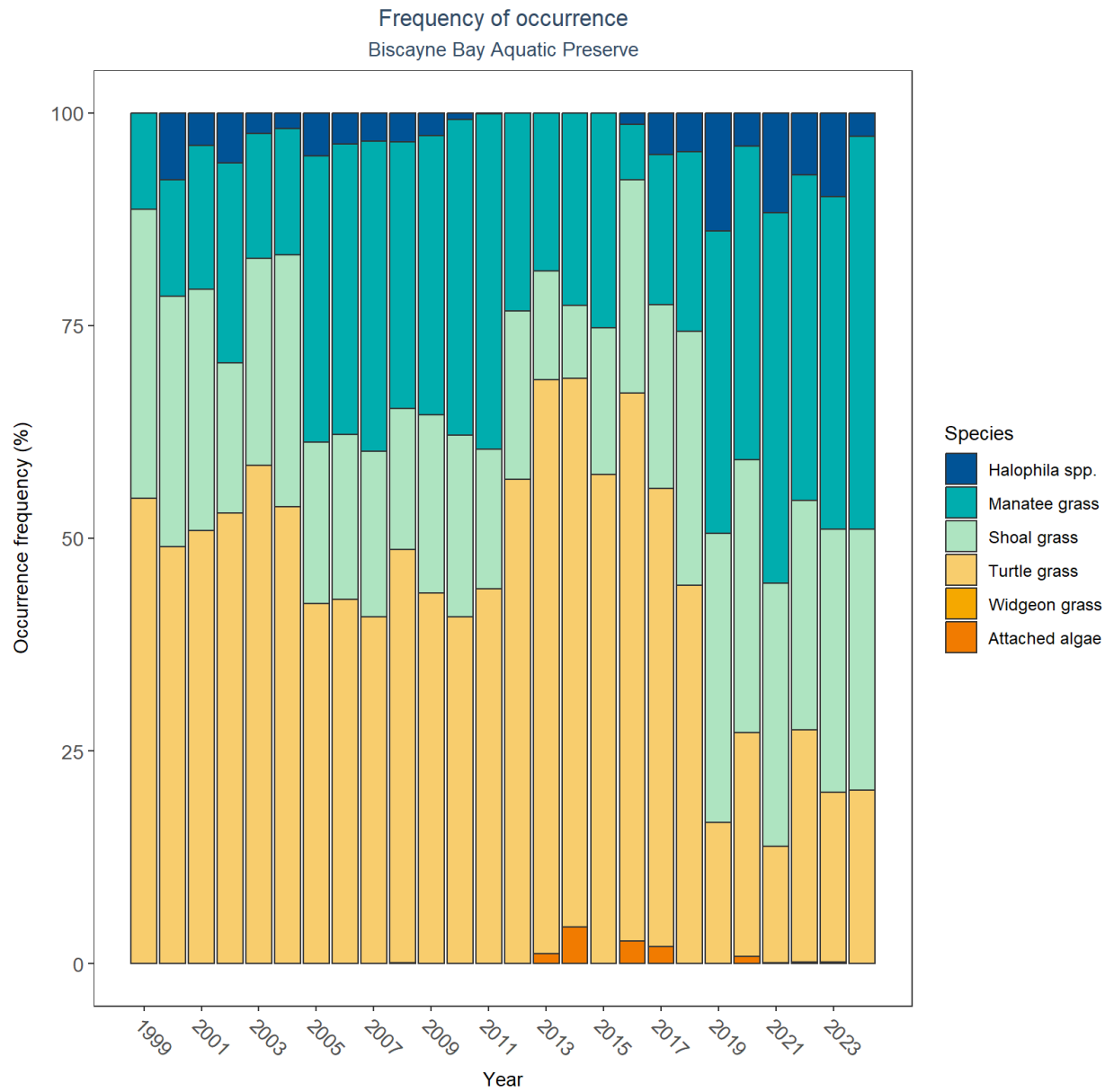


Figure 5: Map showing SAV sampling sites within the boundaries of *Big Bend Seagrasses Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Biscayne Bay Aquatic Preserve



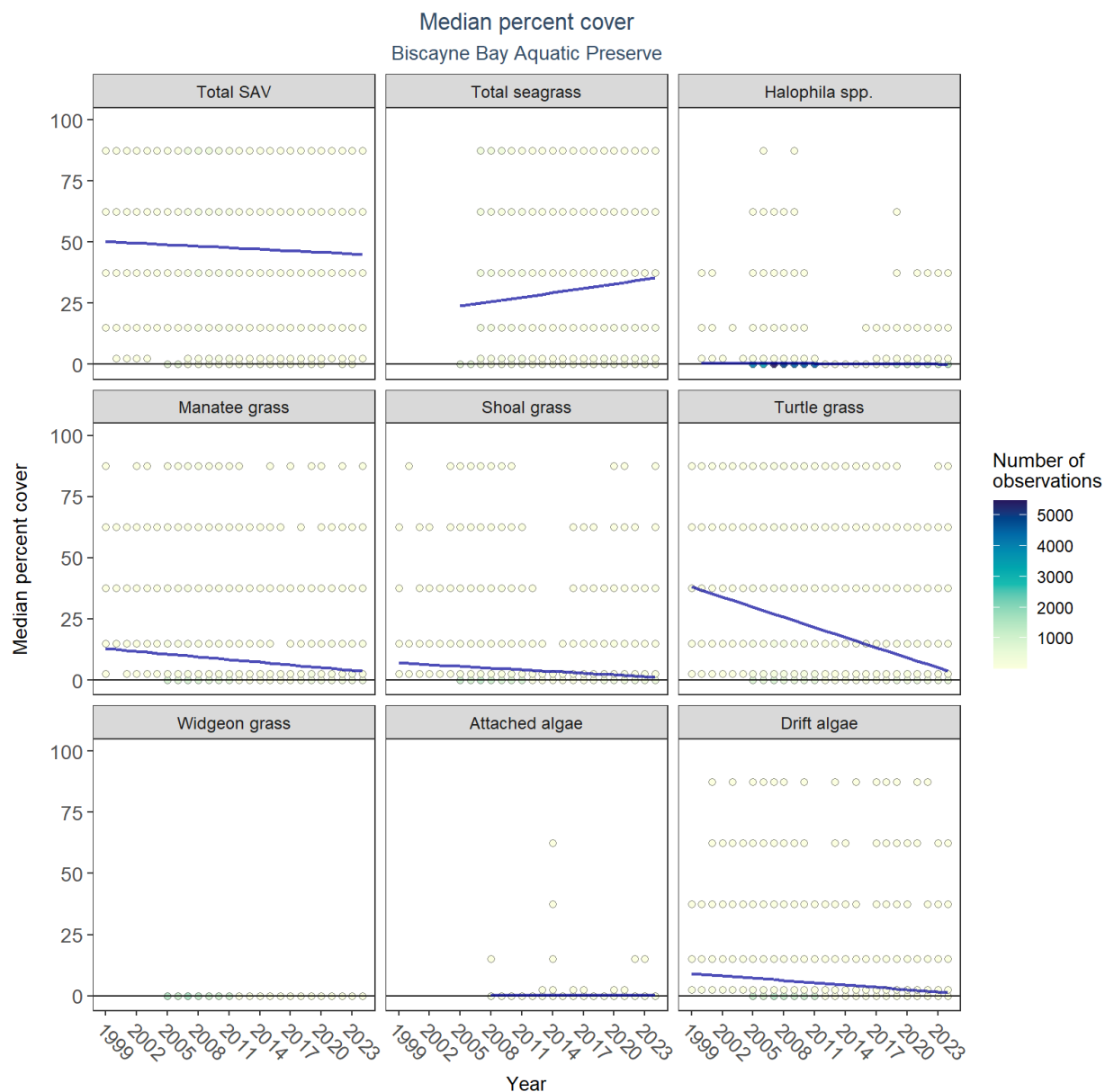
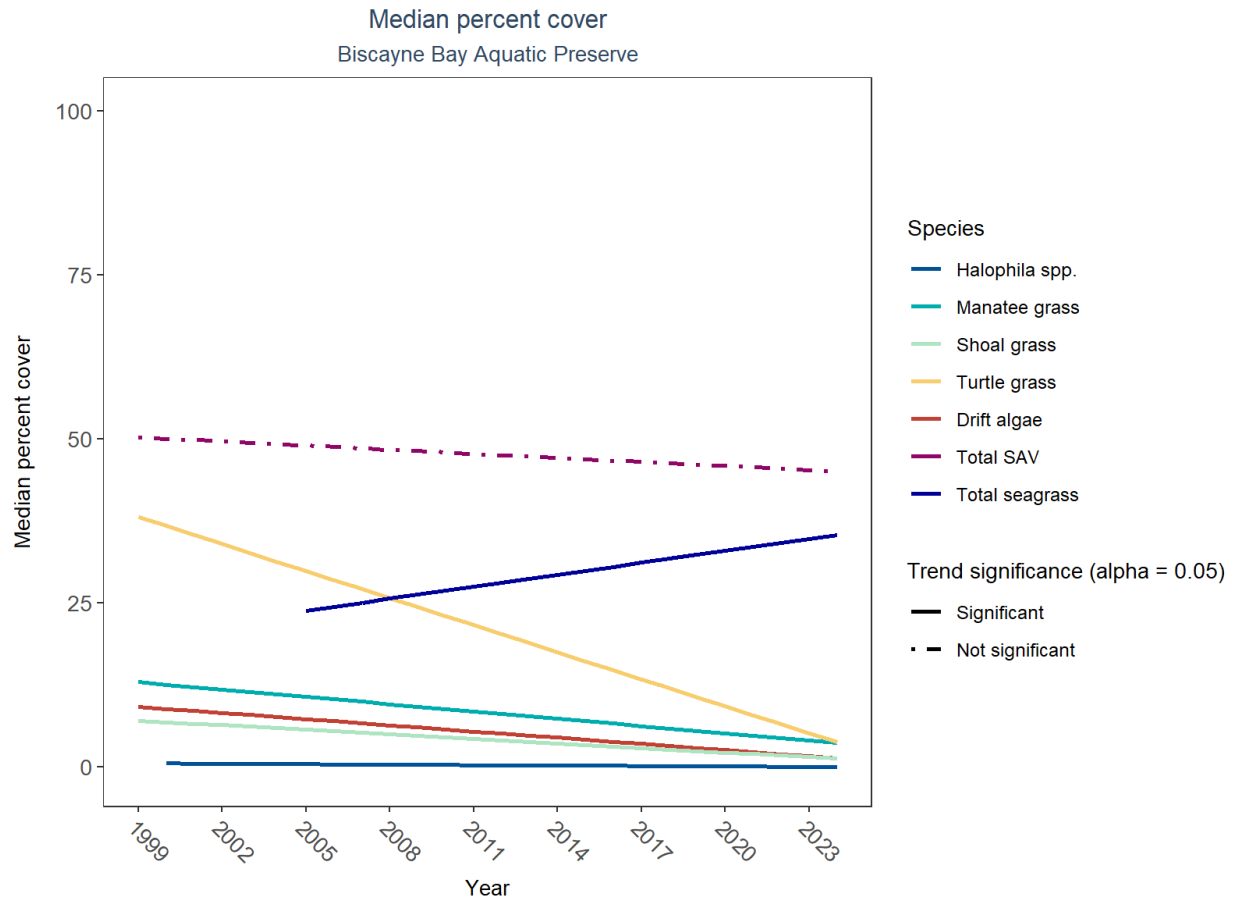


Table 6: SAV LME Results for Biscayne Bay Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Attached algae	No detectable trend	2008 - 2024	-0.03	0.01	0.10
Drift algae	Decreasing trend	1999 - 2024	10.73	-0.31	0.00
Shoal grass	Decreasing trend	1999 - 2024	8.21	-0.23	0.00
Halophila spp.	Decreasing trend	2000 - 2024	0.67	-0.02	0.01
Widgeon grass	Model did not fit the available data	2005 - 2024	-	-	-
Manatee grass	Decreasing trend	1999 - 2024	14.75	-0.37	0.00
Turtle grass	Decreasing trend	1999 - 2024	44.97	-1.37	0.00
Total SAV	No detectable trend	1999 - 2024	51.24	-0.21	0.26
Total seagrass	Increasing trend	2005 - 2024	17.14	0.61	0.00

An annual increase in percent cover was observed for total seagrass (0.61%). Annual decreases in percent cover were observed for *Halophila* spp. (-0.02%), manatee grass (-0.37%), shoal grass (-0.23%), turtle grass (-1.37%), and drift algae (-0.31%). No detectable change in percent cover was observed for total SAV and attached algae. The model could not be fitted for widgeon grass.



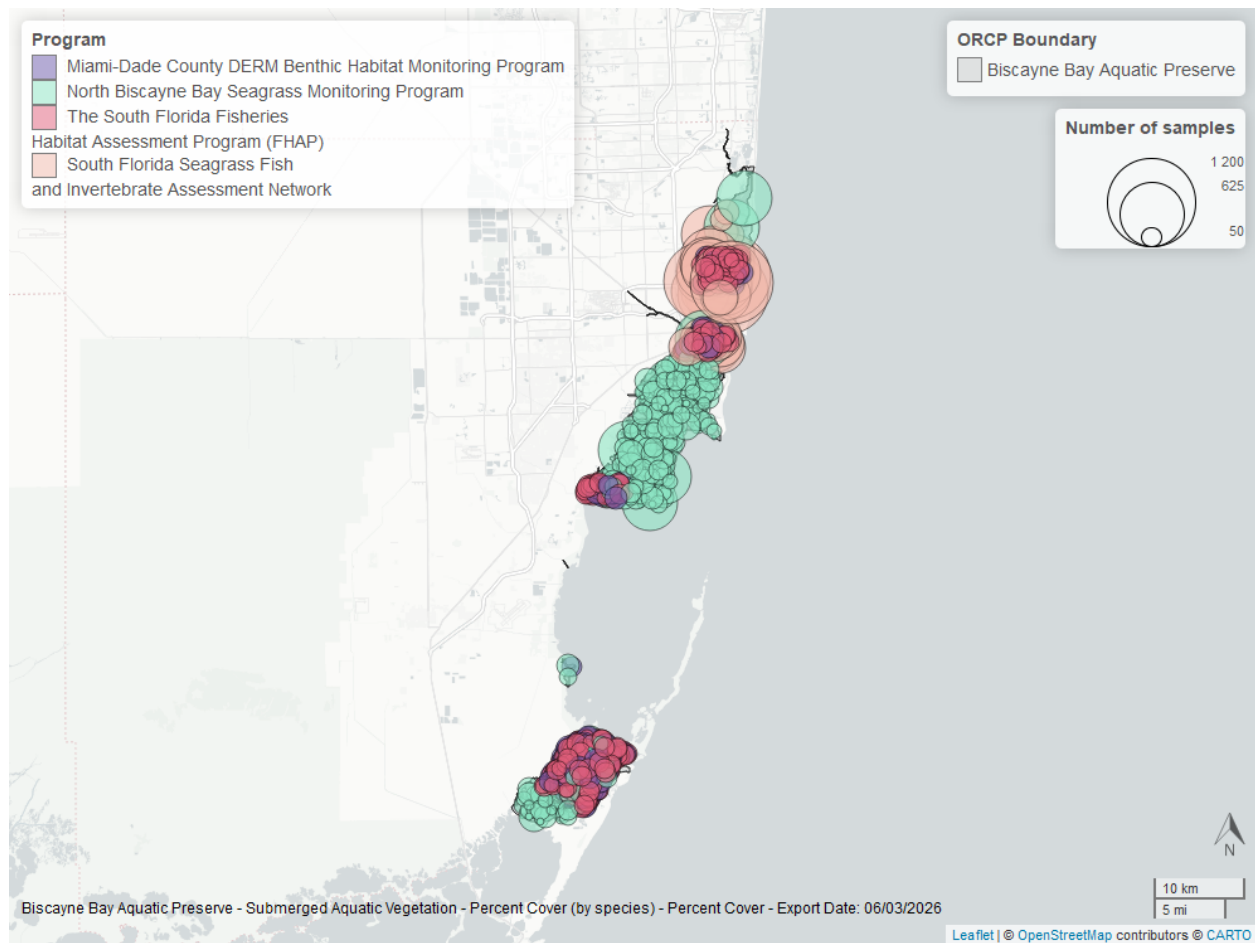
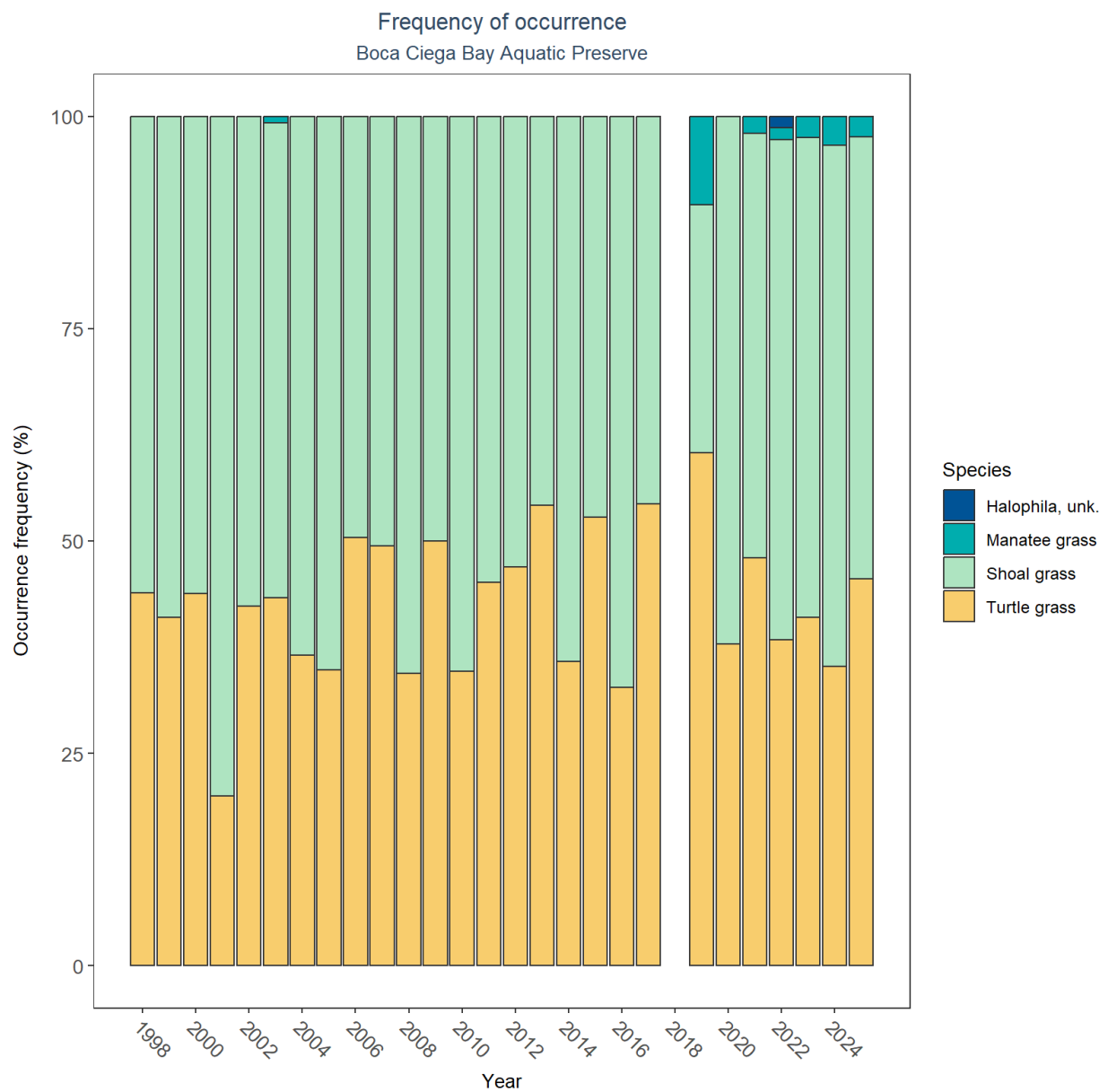


Figure 6: Map showing SAV sampling sites within the boundaries of *Biscayne Bay Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Boca Ciega Bay Aquatic Preserve



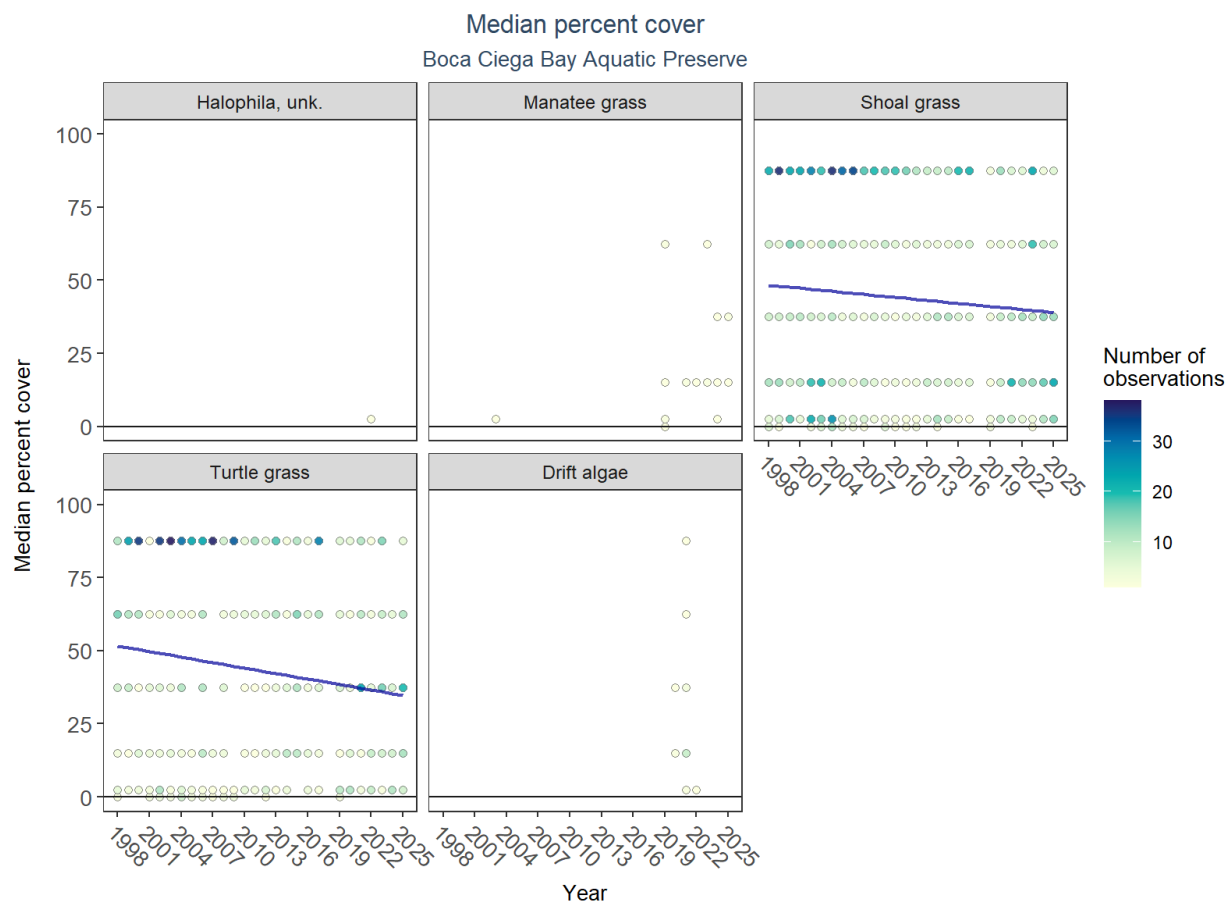
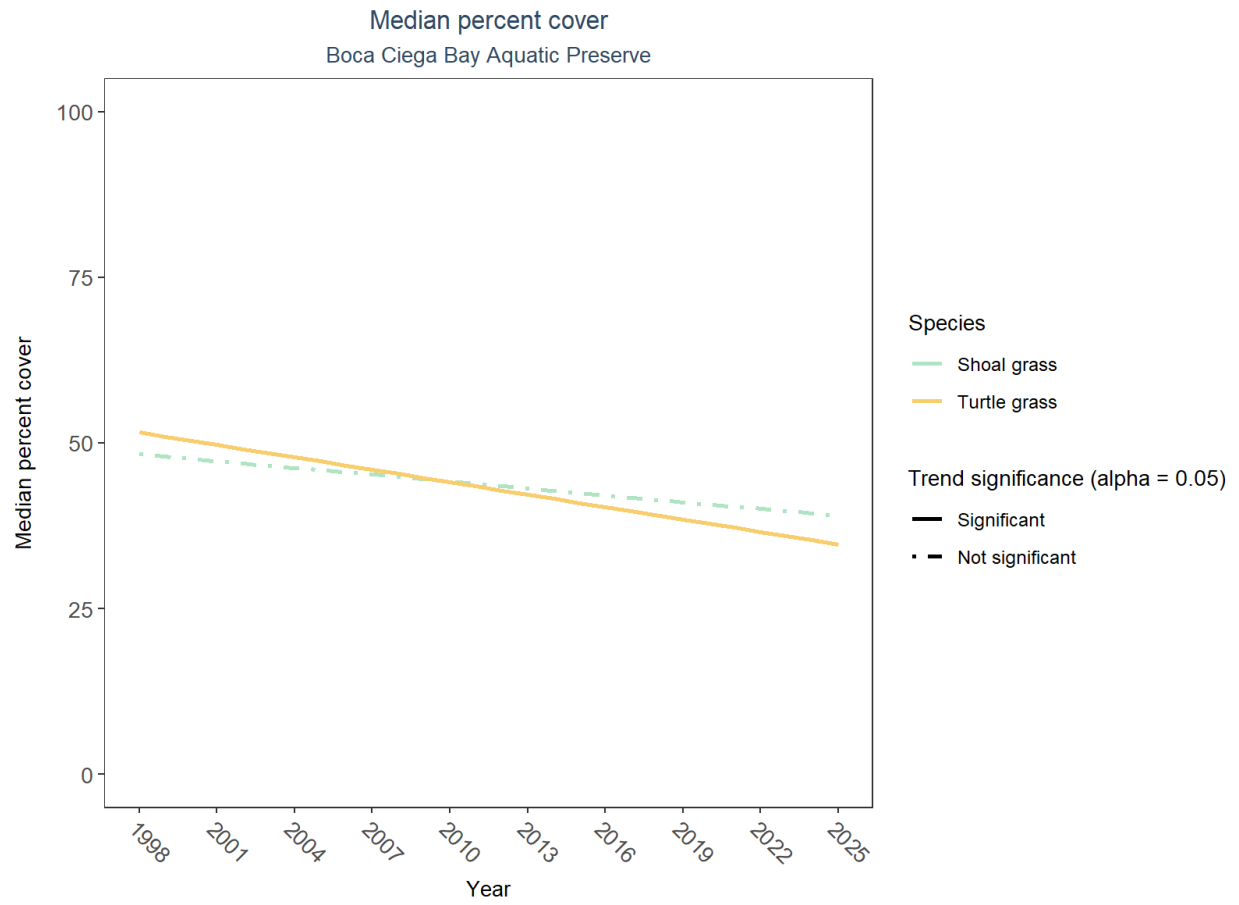


Table 7: SAV LME Results for Boca Ciega Bay Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Drift algae	Insufficient data	2020 - 2022	-	-	-
Shoal grass	No detectable trend	1998 - 2025	49.73	-0.35	0.05
Manatee grass	Model did not fit the available data	2003 - 2025	-	-	-
Turtle grass	Decreasing trend	1998 - 2025	54.11	-0.63	0.02
Halophila, unk.	Insufficient data	2022 - 2022	-	-	-

An annual decrease in percent cover was observed for turtle grass (-0.63%). No detectable change in percent cover was observed for shoal grass. Trends in percent cover could not be evaluated for unknown *Halophila* and drift algae due to insufficient data, and the model could not be fitted for manatee grass.



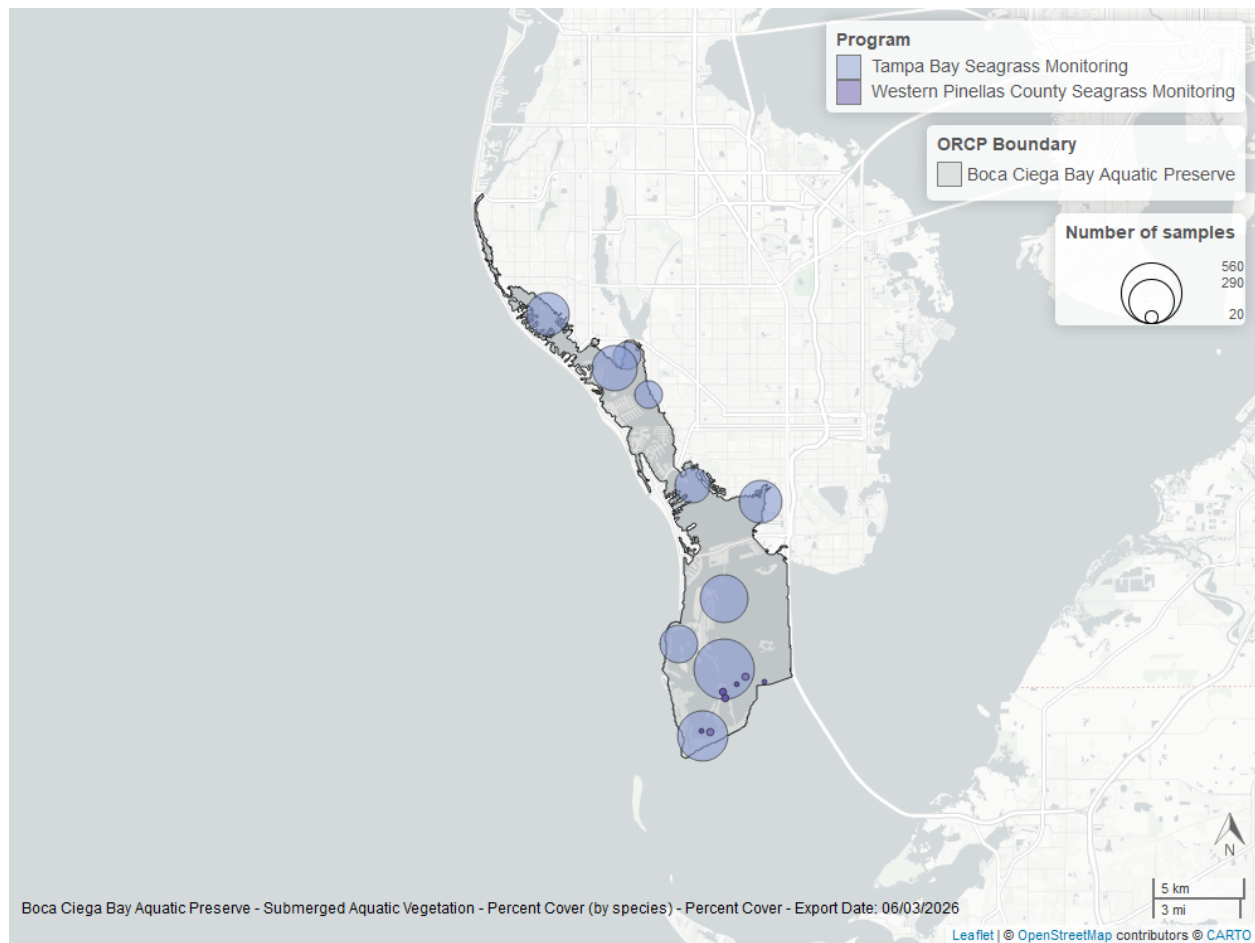
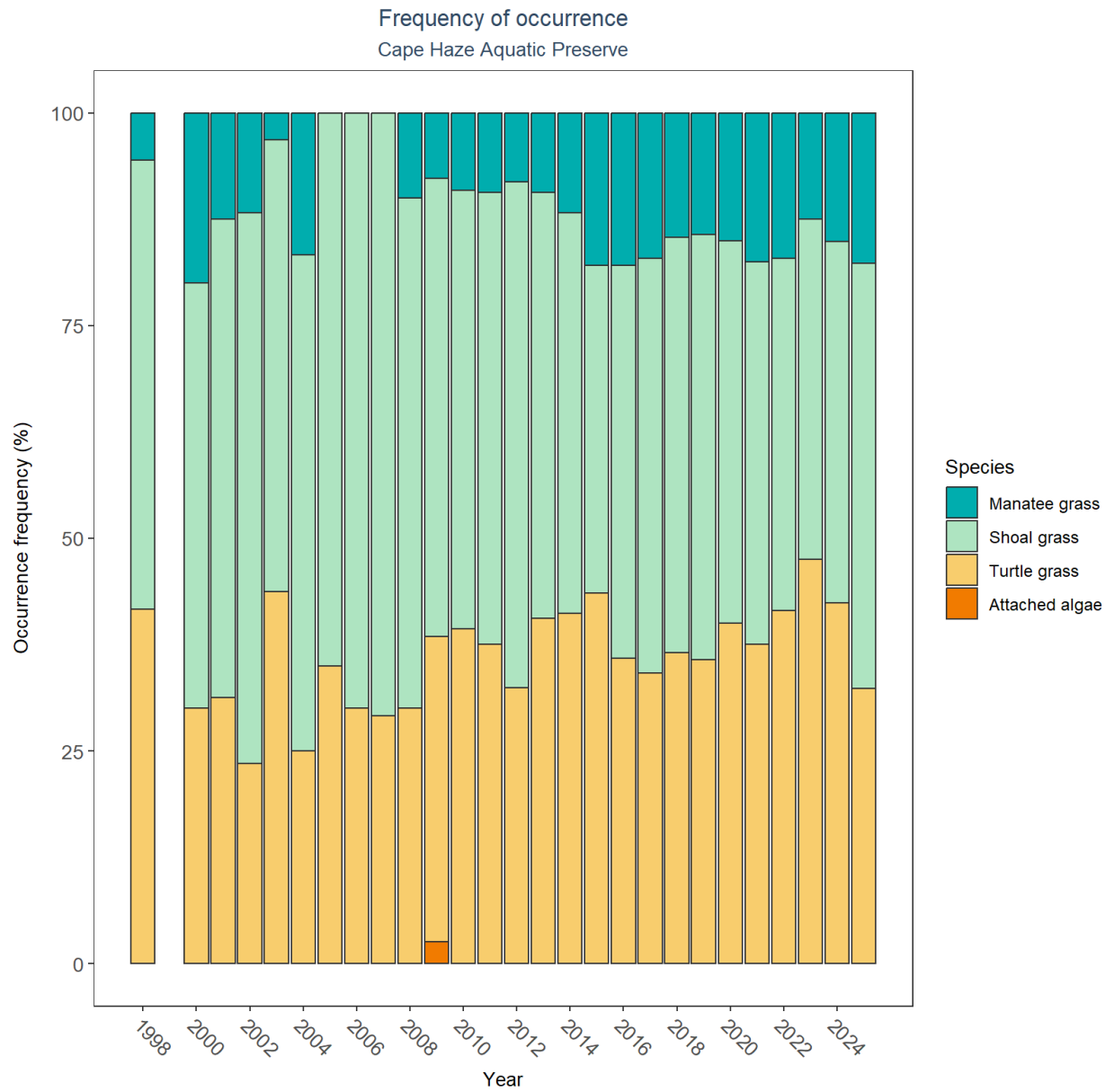


Figure 7: Map showing SAV sampling sites within the boundaries of *Boca Ciega Bay Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Cape Haze Aquatic Preserve



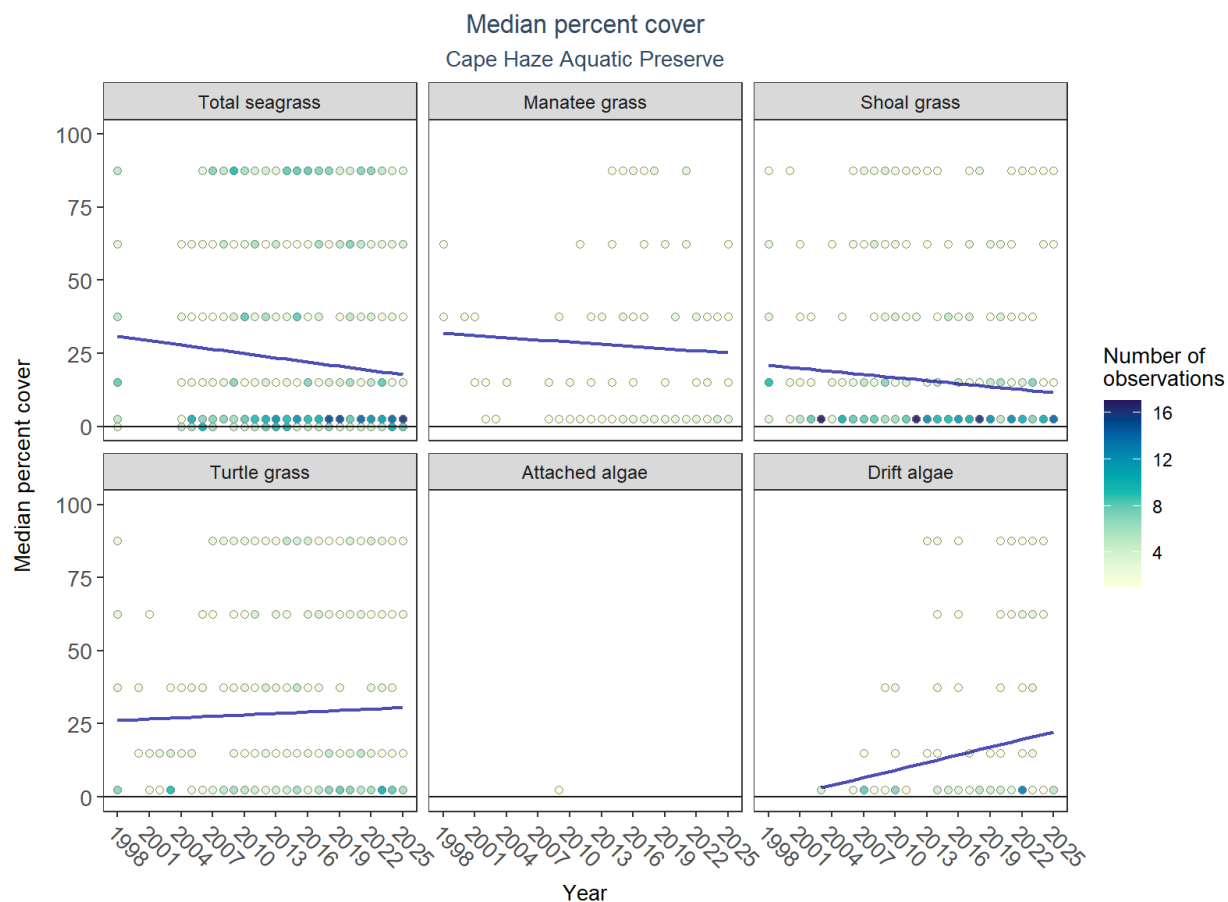
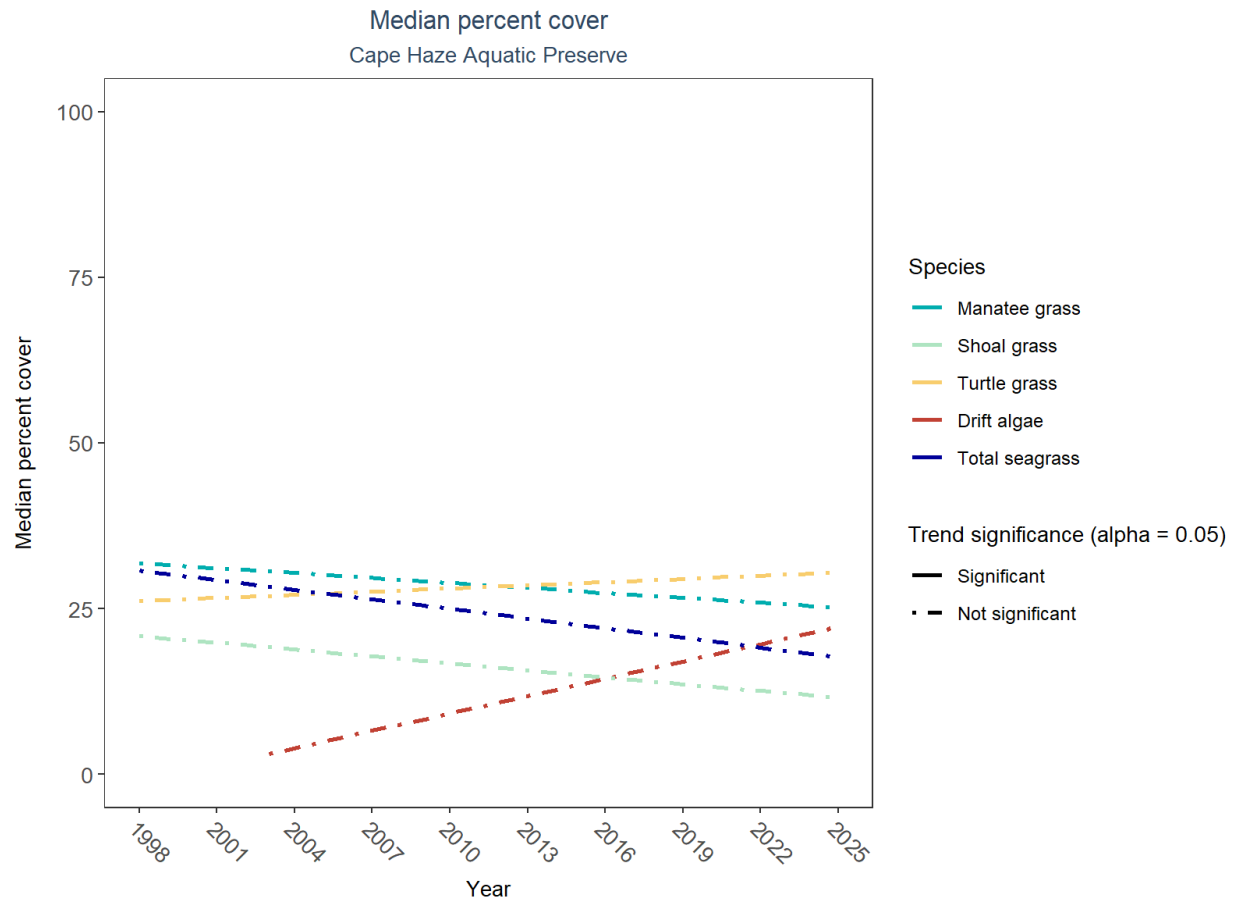


Table 8: SAV LME Results for Cape Haze Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Attached algae	Insufficient data	2009 - 2009	-	-	-
Drift algae	No detectable trend	2003 - 2025	-4.73	0.87	0.12
Shoal grass	No detectable trend	1998 - 2025	22.33	-0.35	0.28
Manatee grass	No detectable trend	1998 - 2025	32.87	-0.25	0.74
Turtle grass	No detectable trend	1998 - 2025	25.49	0.16	0.62
Total seagrass	No detectable trend	1998 - 2025	32.72	-0.48	0.21

Total seagrass, manatee grass, shoal grass, turtle grass, and drift algae showed no detectable change in percent cover. Trends in percent cover could not be evaluated for attached algae due to insufficient data.



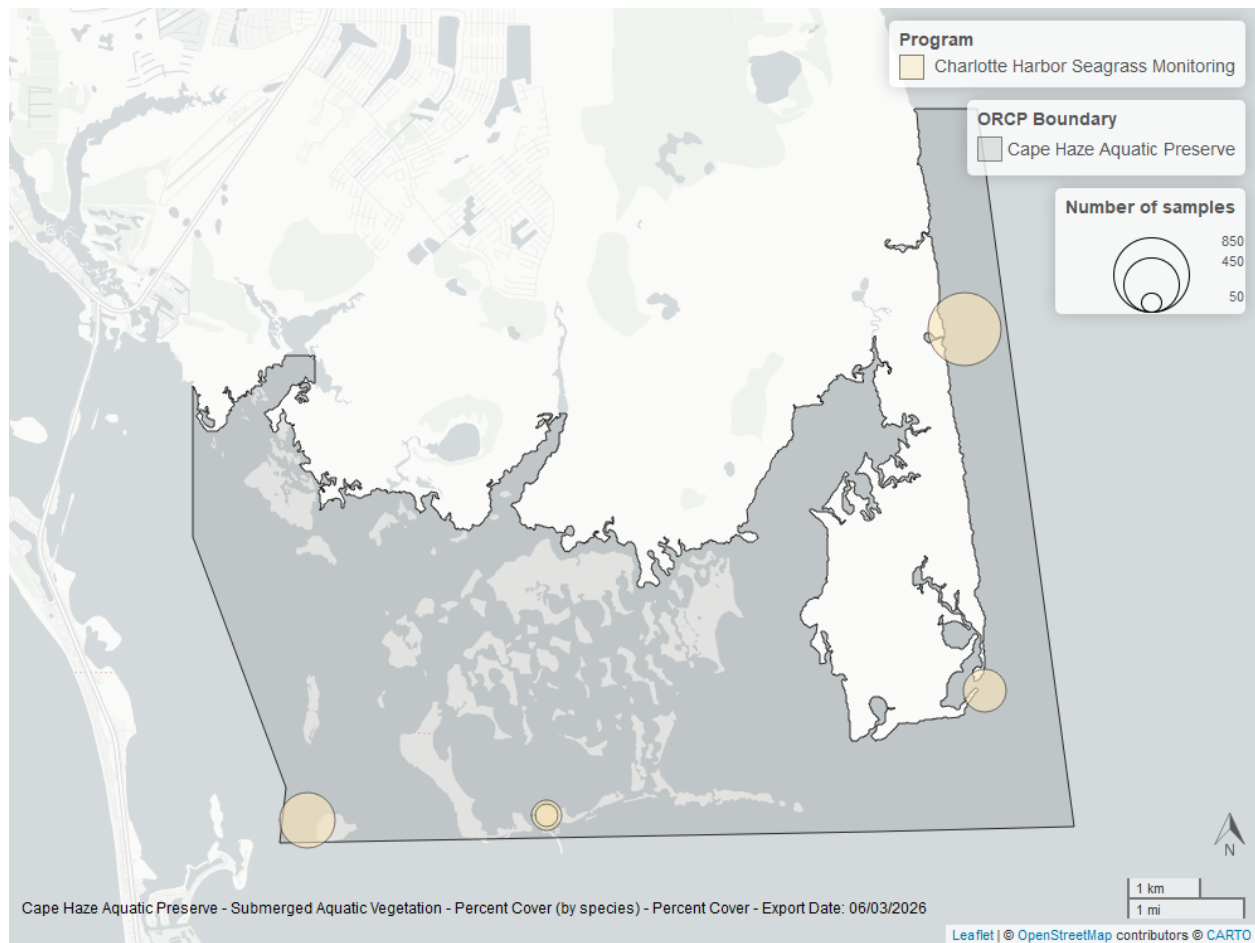
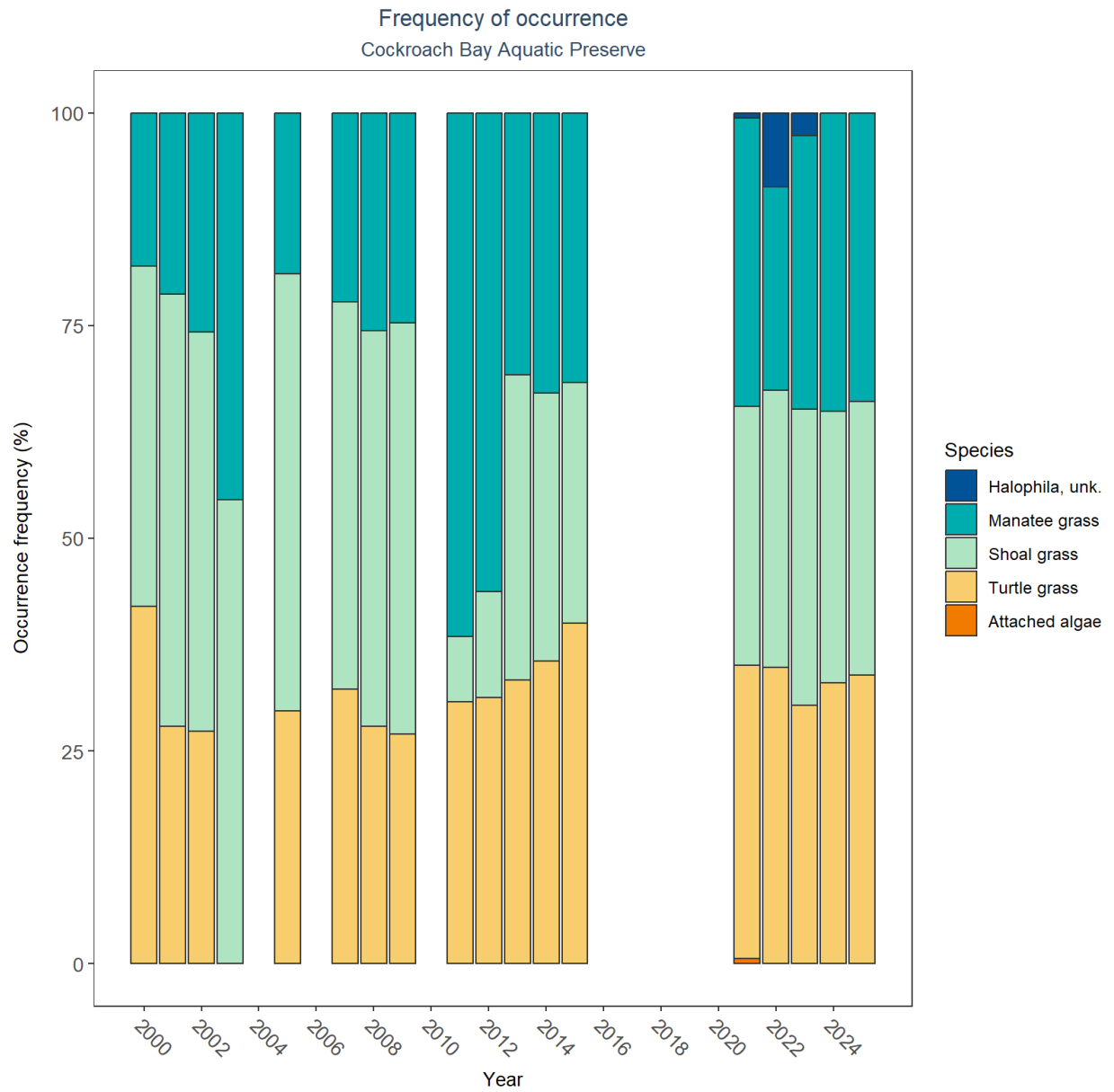


Figure 8: Map showing SAV sampling sites within the boundaries of *Cape Haze Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Cockroach Bay Aquatic Preserve



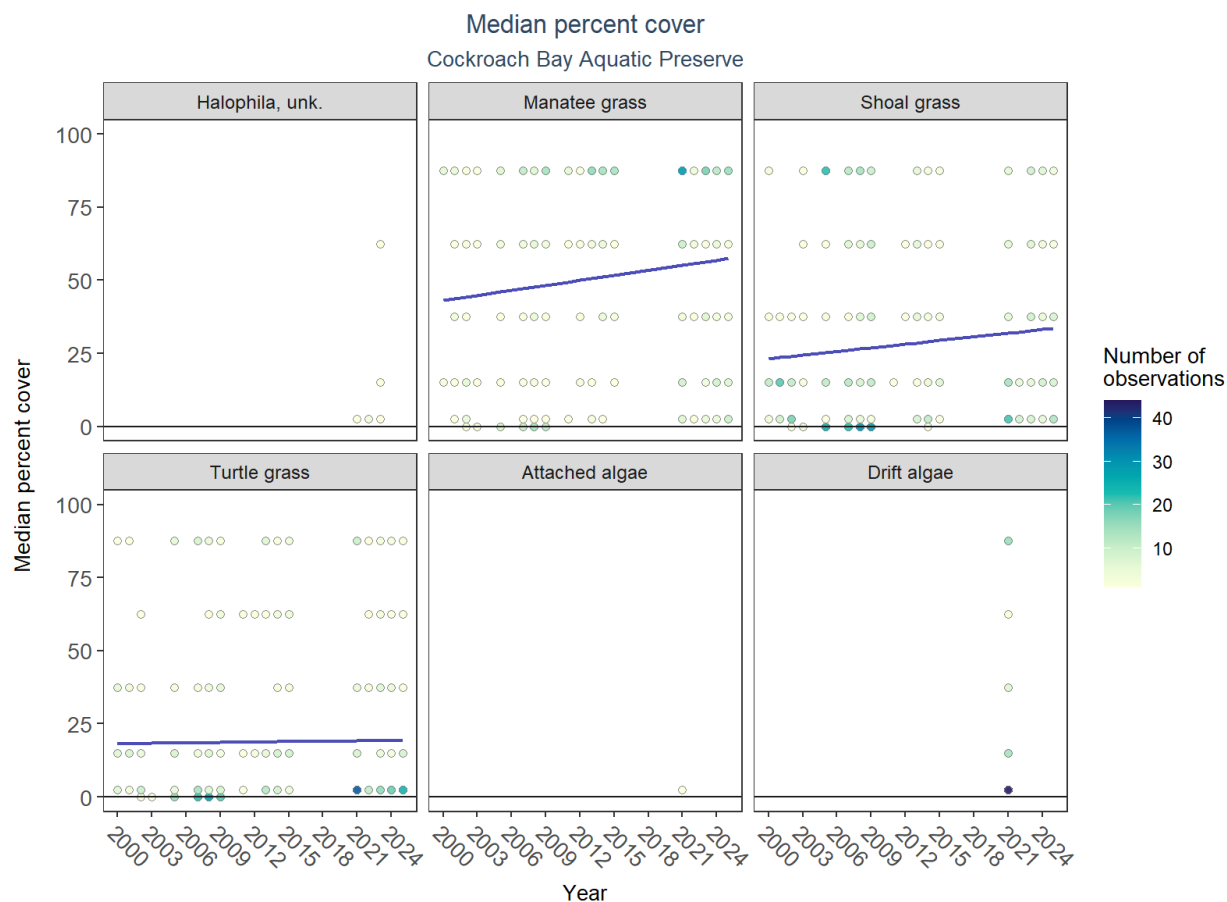
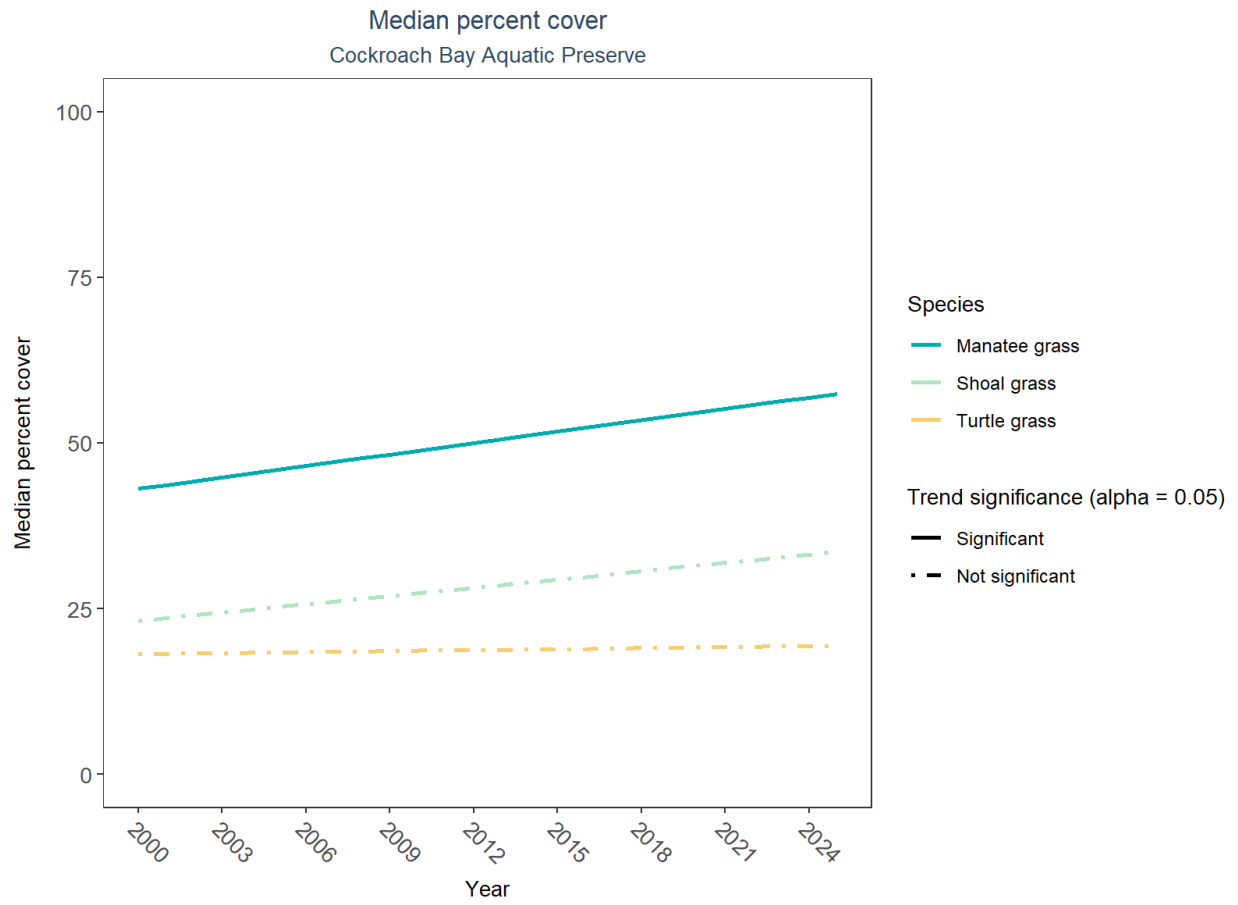


Table 9: SAV LME Results for Cockroach Bay Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Attached algae	Insufficient data	2021 - 2021	-	-	-
Drift algae	Insufficient data	2021 - 2021	-	-	-
Shoal grass	No detectable trend	2000 - 2025	20.69	0.42	0.05
Manatee grass	Increasing trend	2000 - 2025	39.66	0.57	0.02
Turtle grass	No detectable trend	2000 - 2025	17.87	0.05	0.80
Halophila, unk.	Insufficient data	2021 - 2023	-	-	-

An annual increase in percent cover was observed for manatee grass (0.57%). No detectable change in percent cover was observed for shoal grass and turtle grass. Trends in percent cover could not be evaluated for unknown *Halophila*, attached algae, and drift algae due to insufficient data.



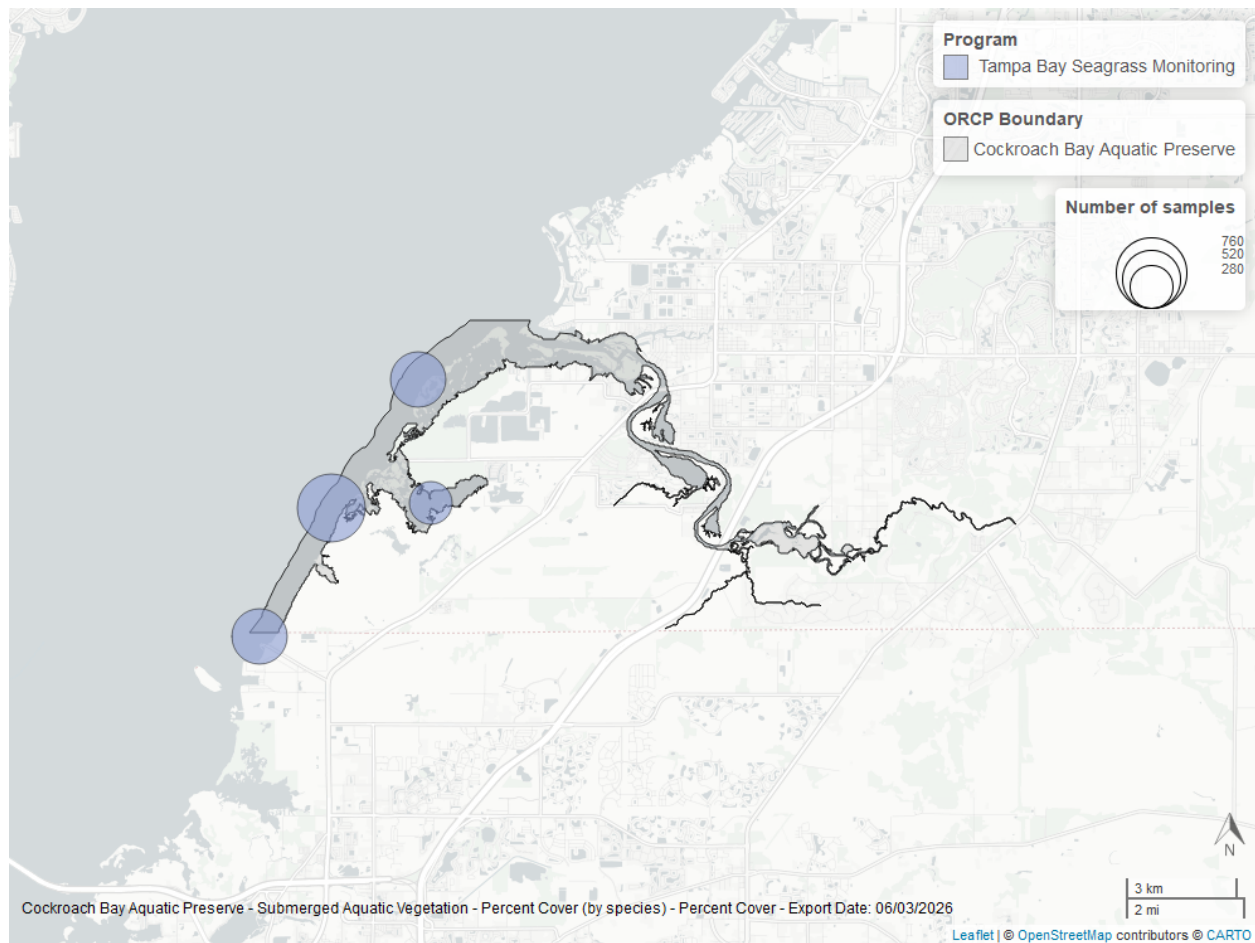
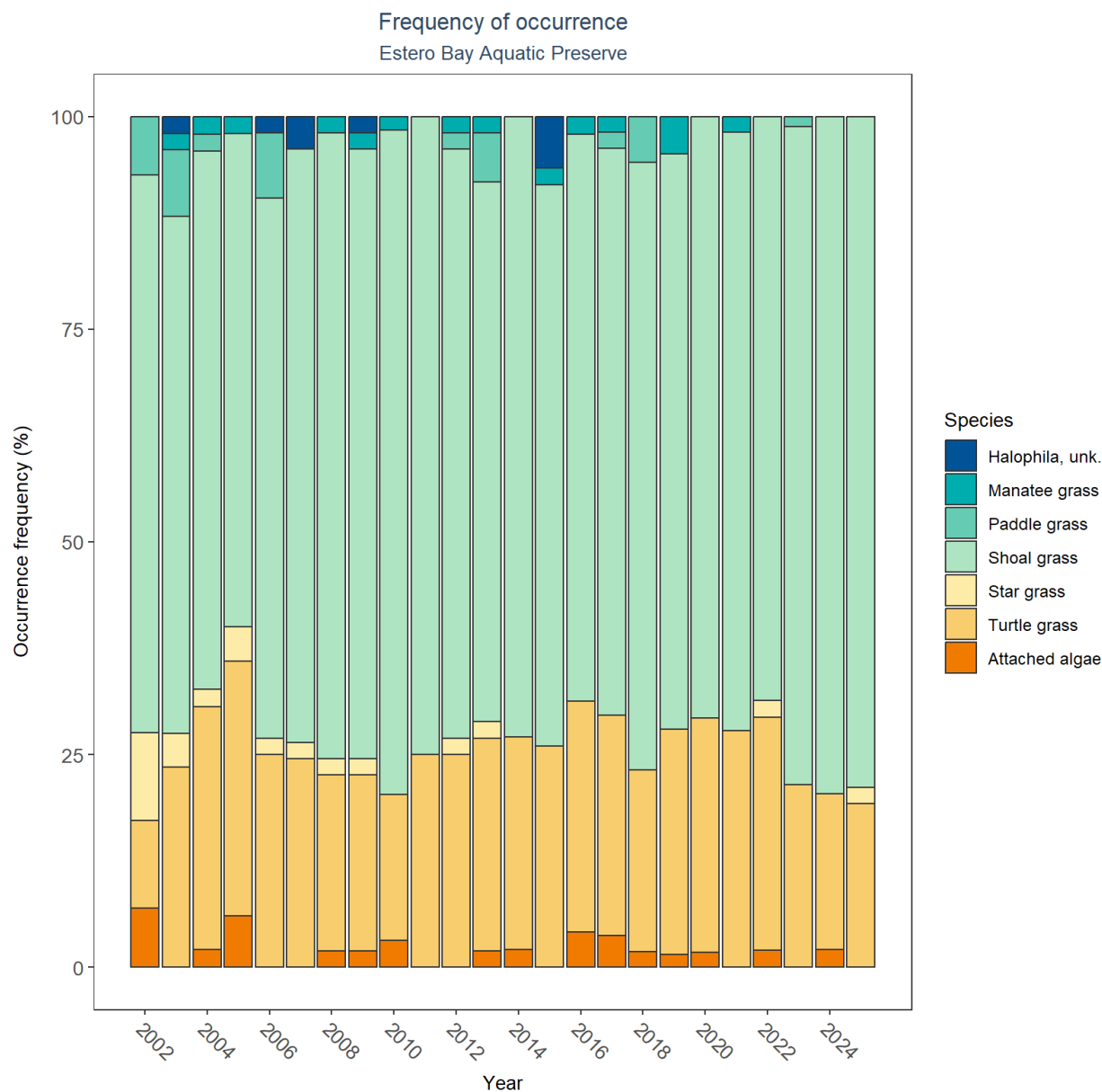


Figure 9: Map showing SAV sampling sites within the boundaries of *Cockroach Bay Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Estero Bay Aquatic Preserve



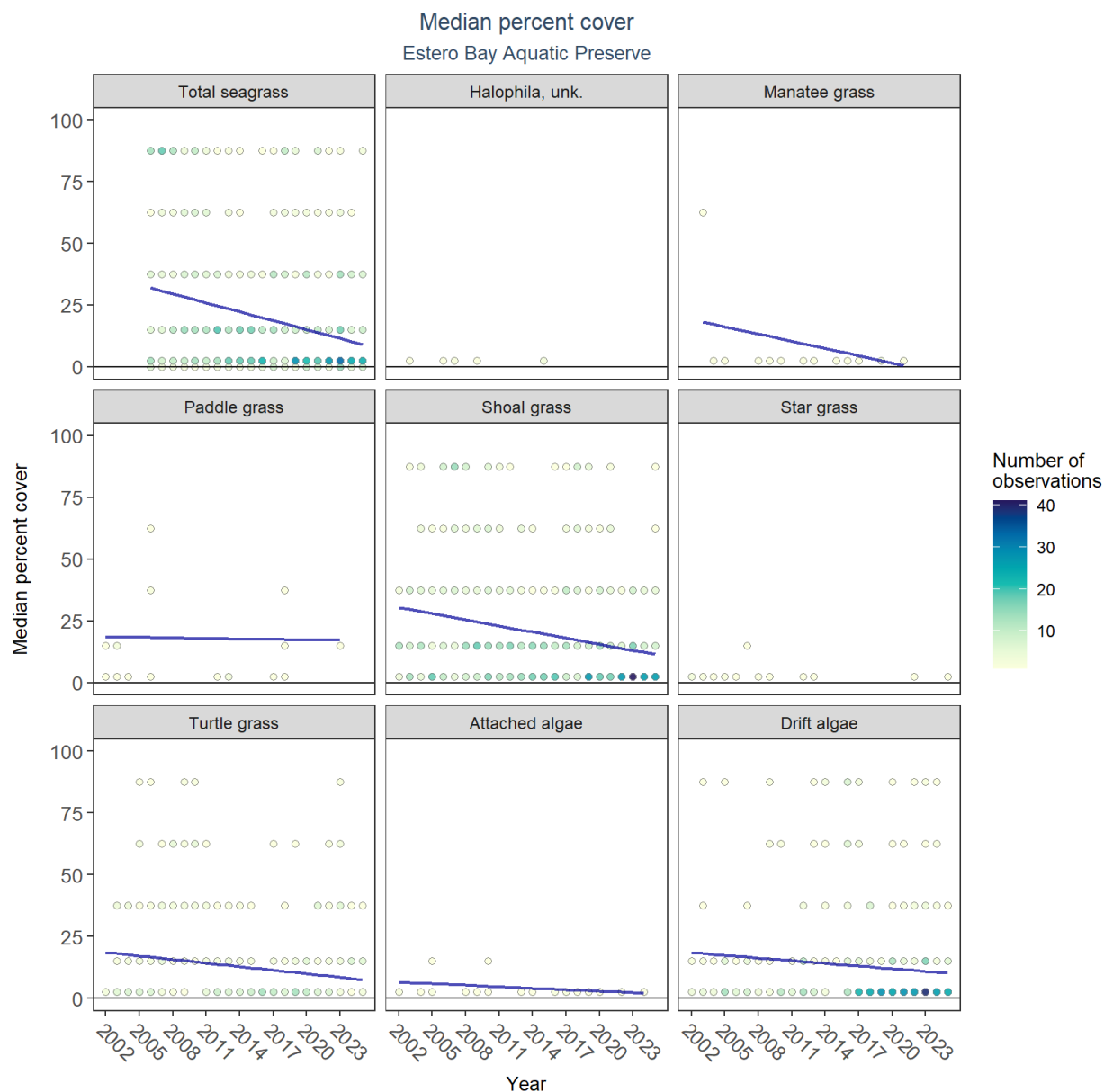
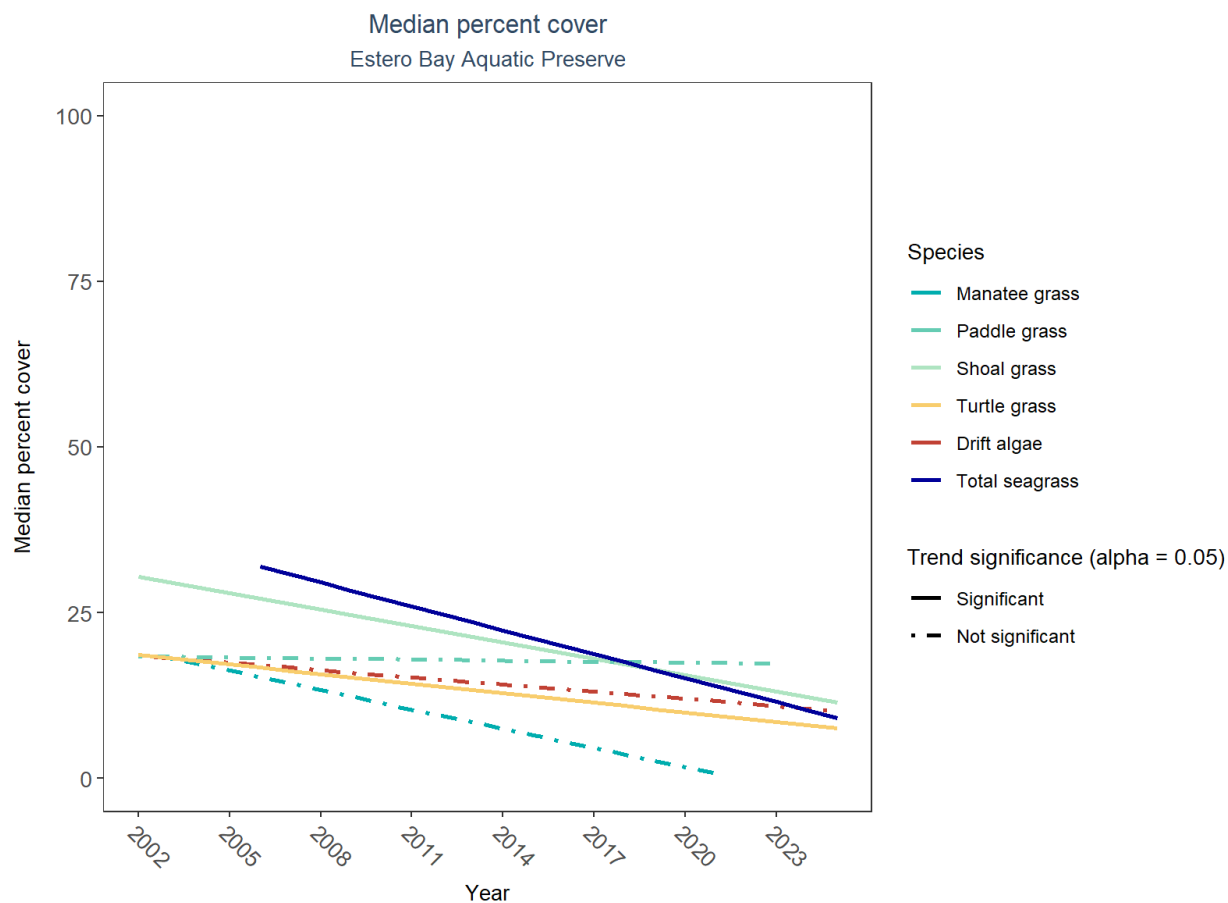


Table 10: SAV LME Results for Estero Bay Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Attached algae	No detectable trend	2002 - 2024	8.13	-0.20	0.39
Drift algae	No detectable trend	2002 - 2025	21.42	-0.36	0.23
Shoal grass	Decreasing trend	2002 - 2025	37.00	-0.82	0.00
Paddle grass	No detectable trend	2002 - 2023	18.90	-0.06	0.92
Star grass	Model did not fit the available data	2002 - 2025	-	-	-
Manatee grass	No detectable trend	2003 - 2021	26.94	-0.97	0.35
Turtle grass	Decreasing trend	2002 - 2025	22.48	-0.48	0.01
Total seagrass	Decreasing trend	2006 - 2025	46.37	-1.20	0.03
Halophila, unk.	Model did not fit the available data	2003 - 2015	-	-	-

Annual decreases in percent cover were observed for total seagrass (-1.20%), shoal grass (-0.82%), and turtle grass (-0.48%). Manatee grass, paddle grass, attached algae, and drift algae showed no detectable change in percent cover. A model could not be fitted for unknown *Halophila* and star grass.



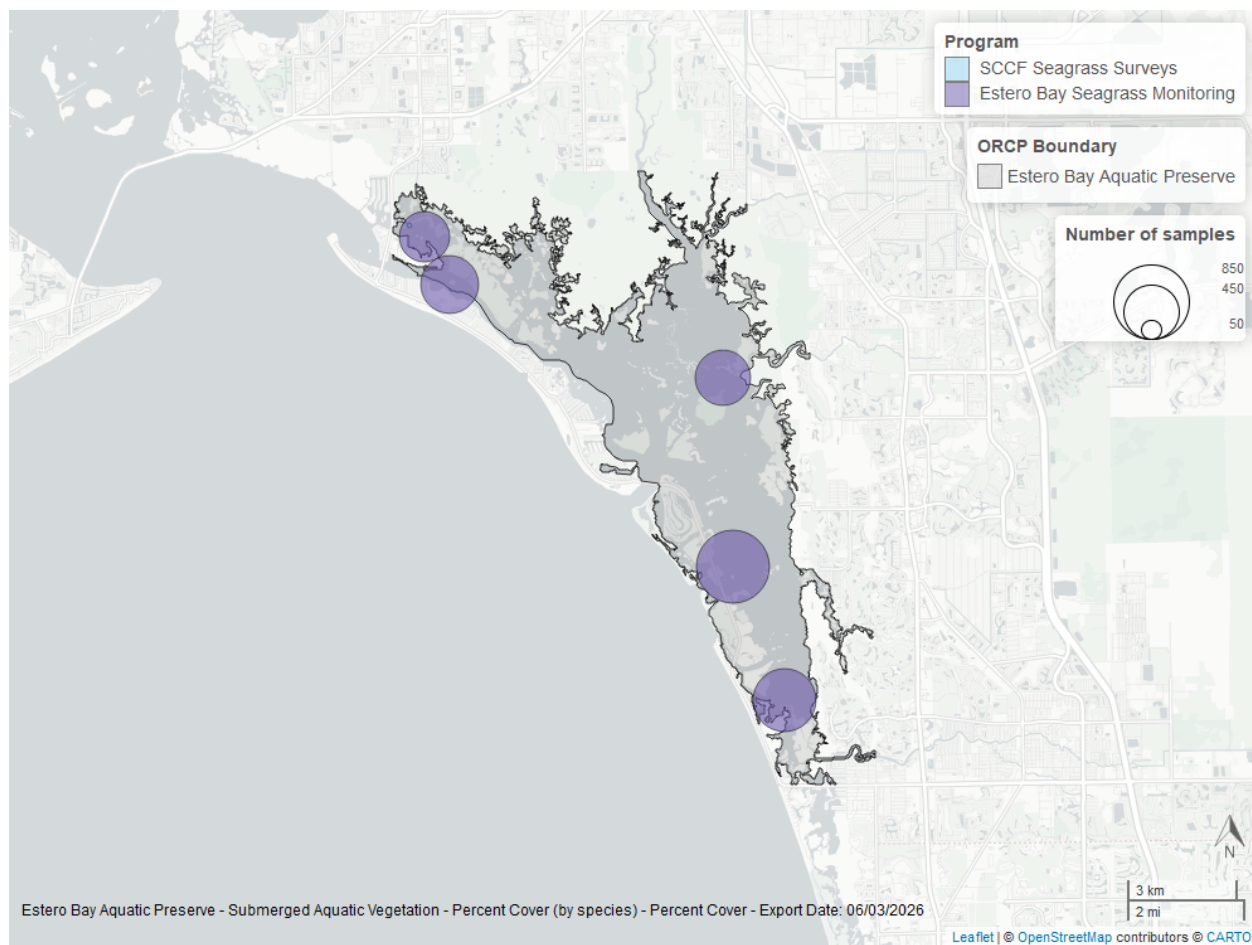
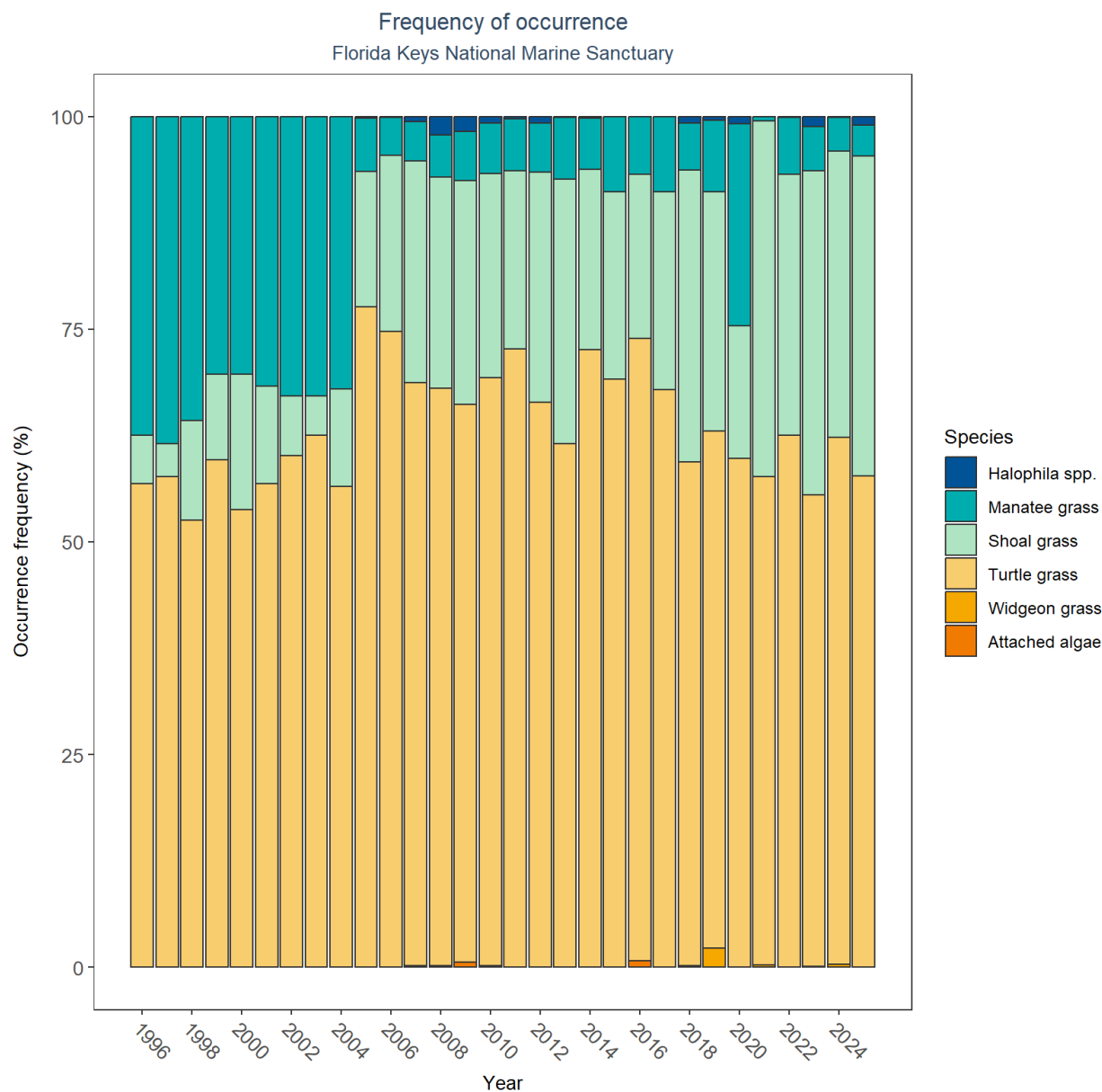


Figure 10: Map showing SAV sampling sites within the boundaries of *Estero Bay Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Florida Keys National Marine Sanctuary



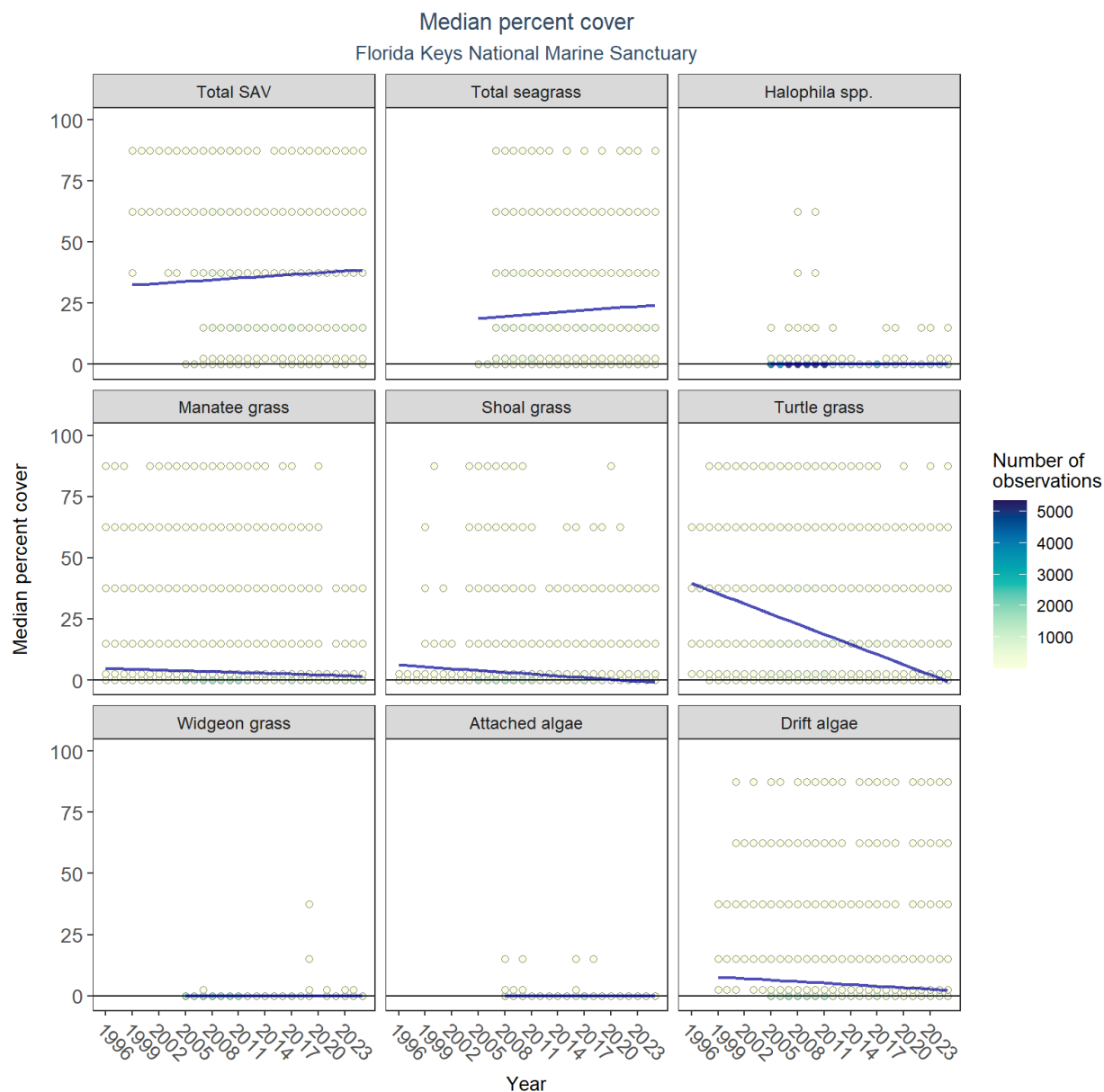
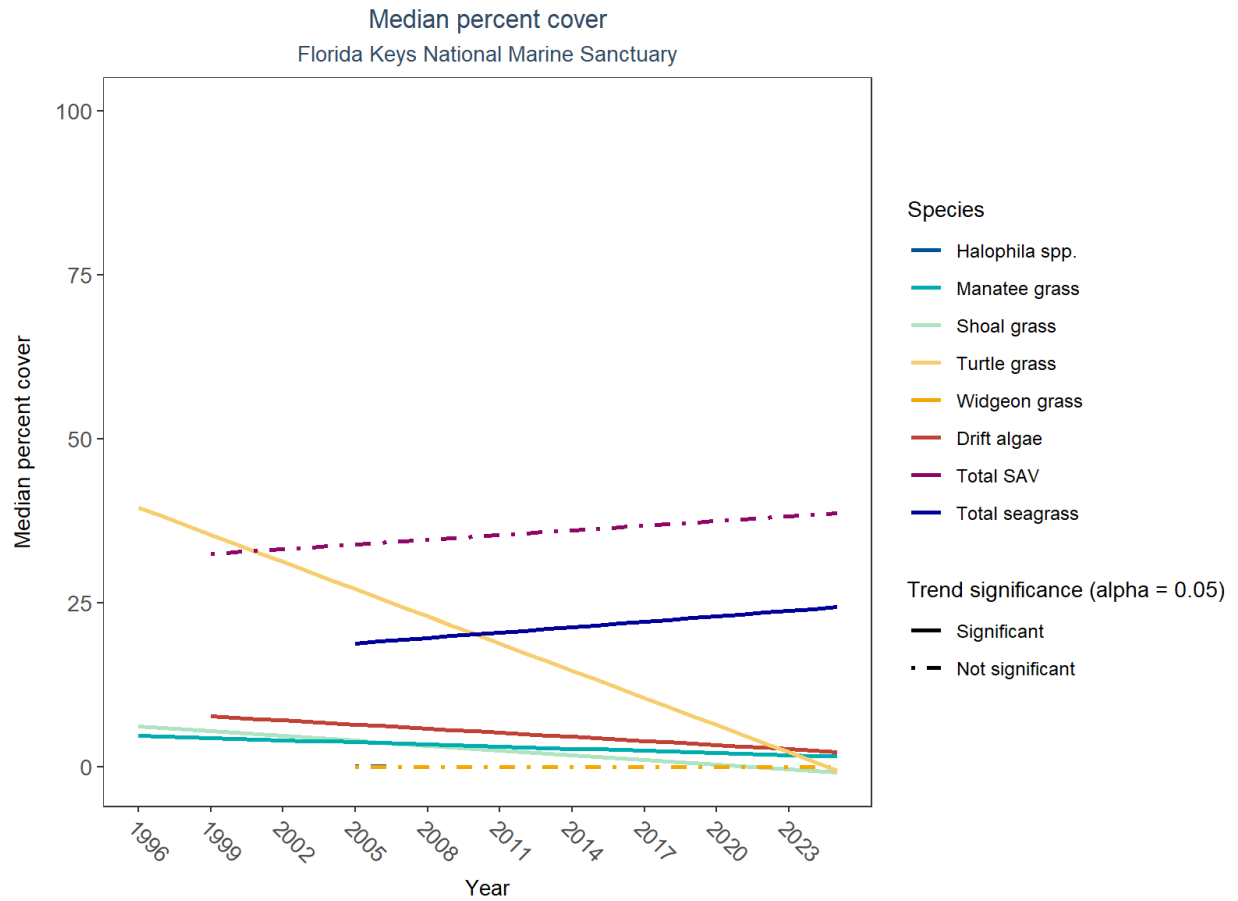


Table 11: SAV LME Results for Florida Keys National Marine Sanctuary

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Attached algae	No detectable trend	2008 - 2025	0.07	0.00	0.20
Drift algae	Decreasing trend	1999 - 2025	8.74	-0.21	0.00
Shoal grass	Decreasing trend	1996 - 2025	6.66	-0.24	0.00
Halophila spp.	No detectable trend	2005 - 2025	0.07	0.00	0.41
Widgeon grass	No detectable trend	2005 - 2025	-0.03	0.00	0.13
Manatee grass	Decreasing trend	1996 - 2025	4.94	-0.11	0.02
Turtle grass	Decreasing trend	1996 - 2025	42.29	-1.38	0.00
Total SAV	No detectable trend	1999 - 2025	31.28	0.24	0.12
Total seagrass	Increasing trend	2005 - 2025	15.77	0.28	0.00

An annual increase in percent cover was observed for total seagrass (0.28%). Annual decreases in percent cover were observed for manatee grass (-0.11%), shoal grass (-0.24%), turtle grass (-1.38%), and drift algae (-0.21%). Total SAV, *Halophila* spp., widgeon grass, and attached algae showed no detectable change in percent cover.



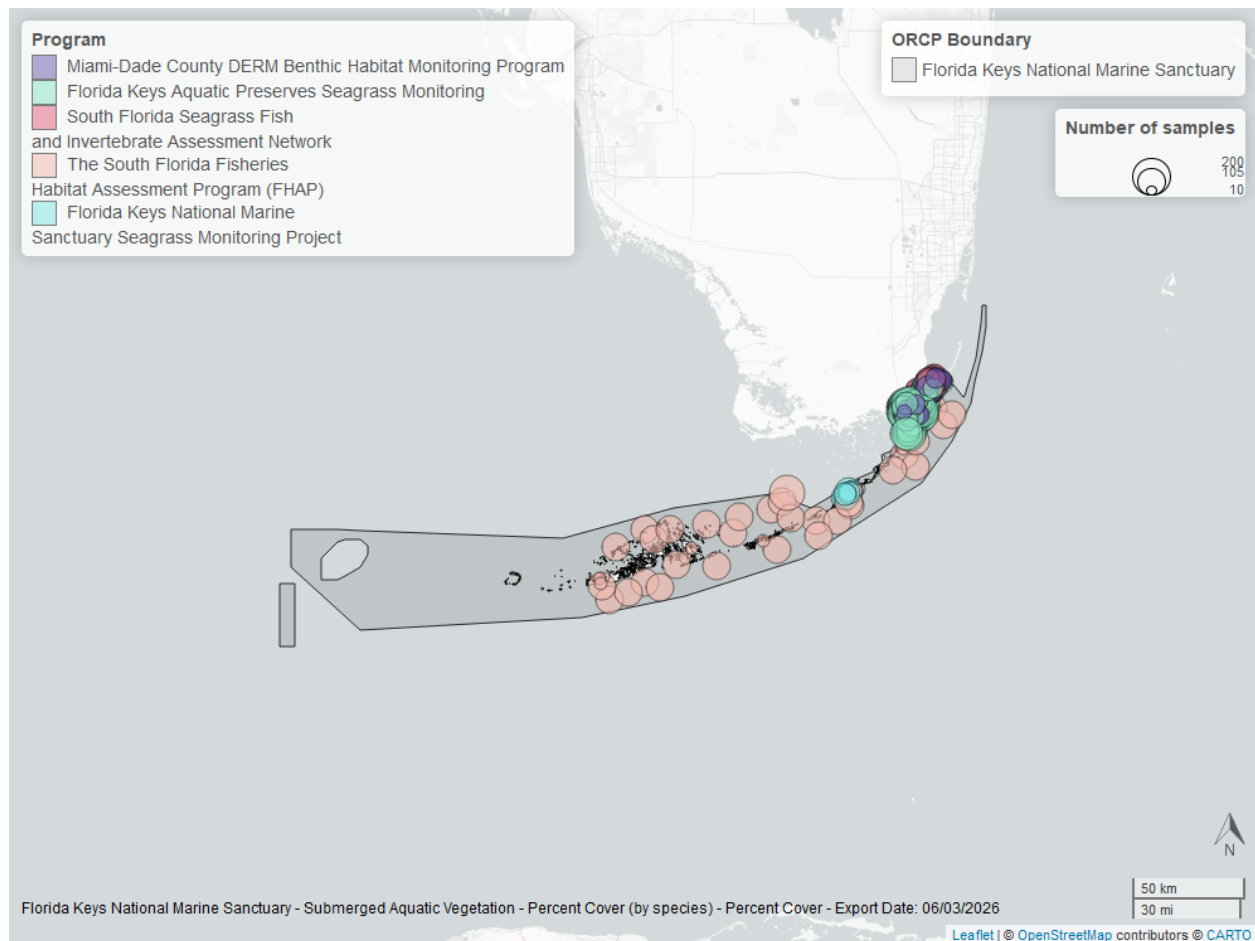
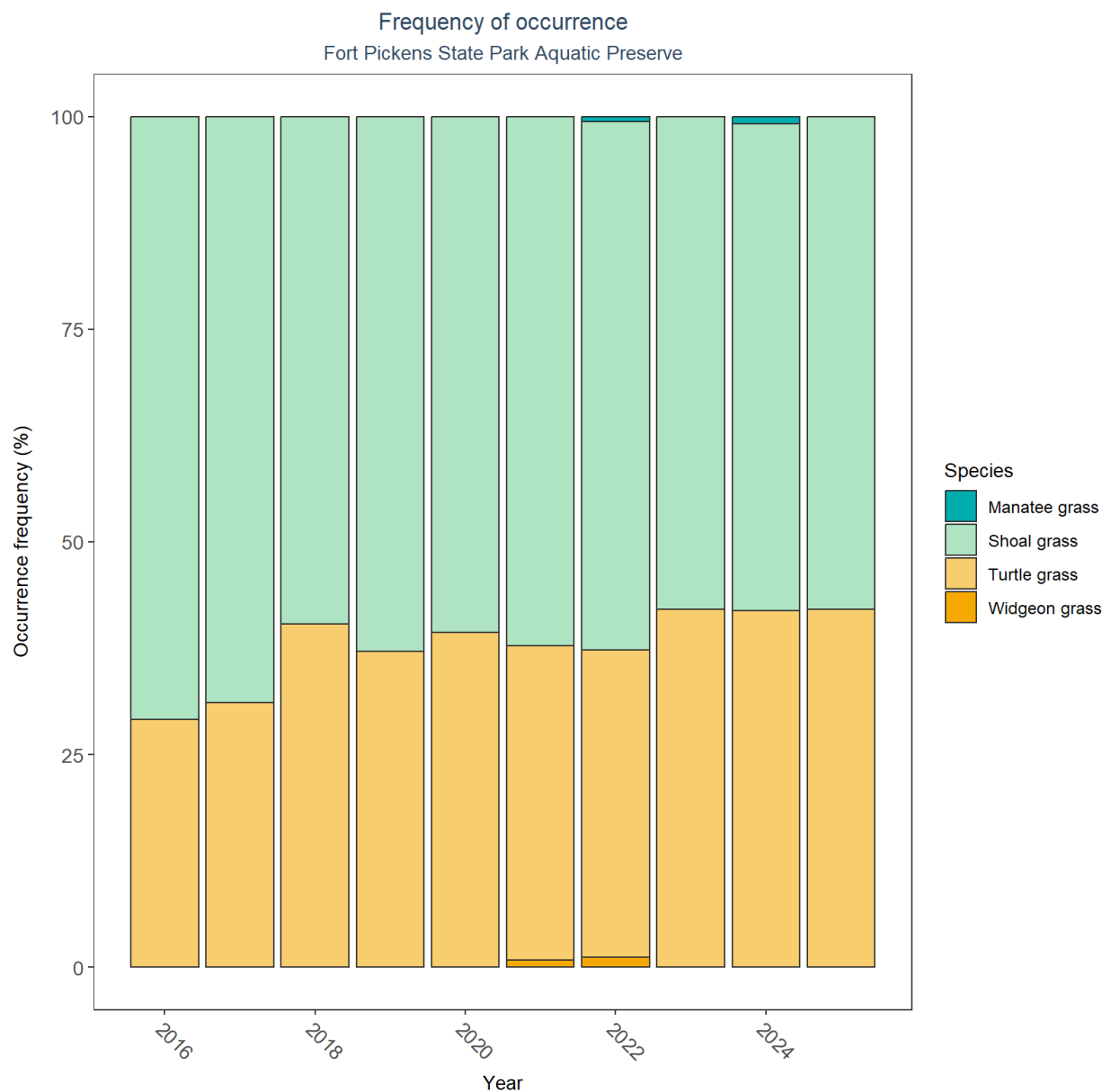


Figure 11: Map showing SAV sampling sites within the boundaries of *Florida Keys National Marine Sanctuary*. The point size reflects the number of samples at a given sampling site.

Fort Pickens State Park Aquatic Preserve



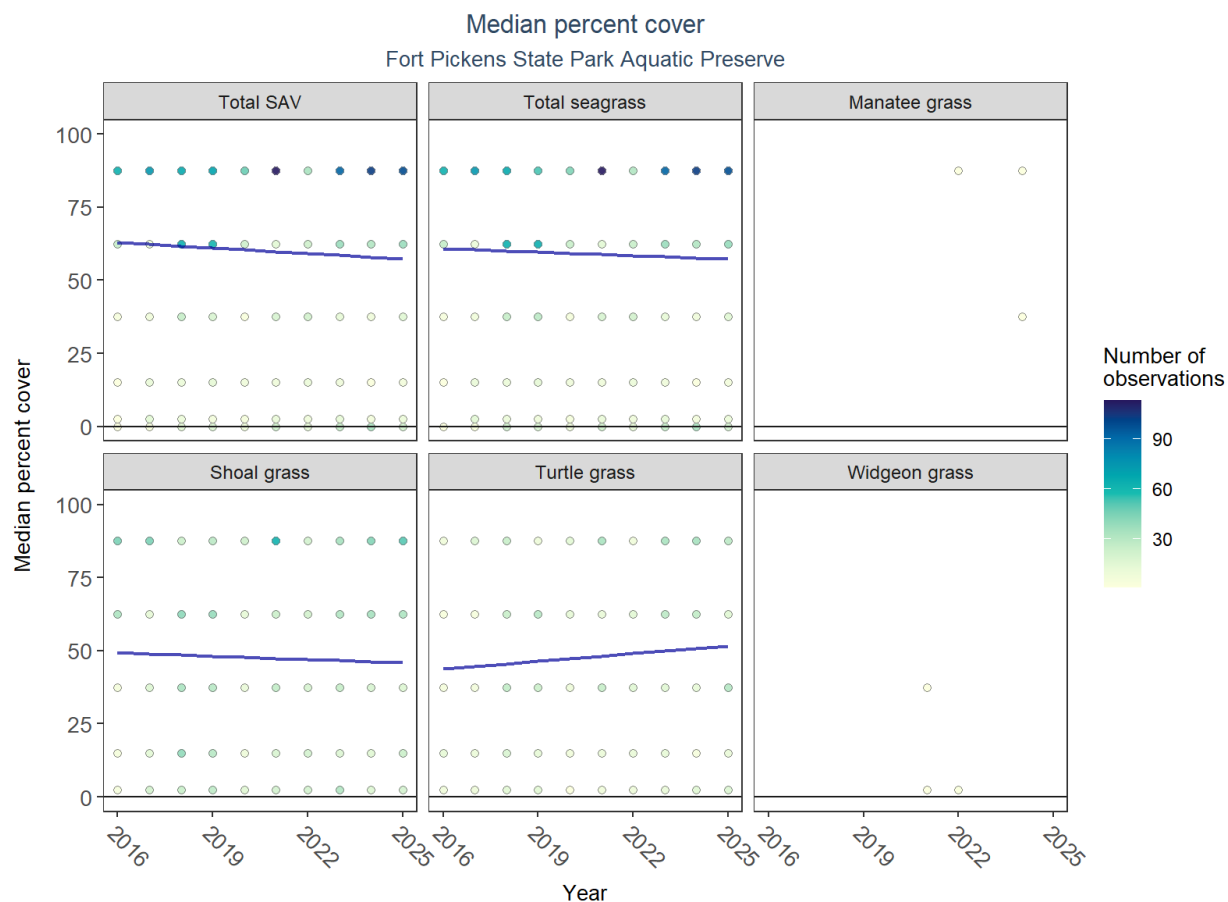
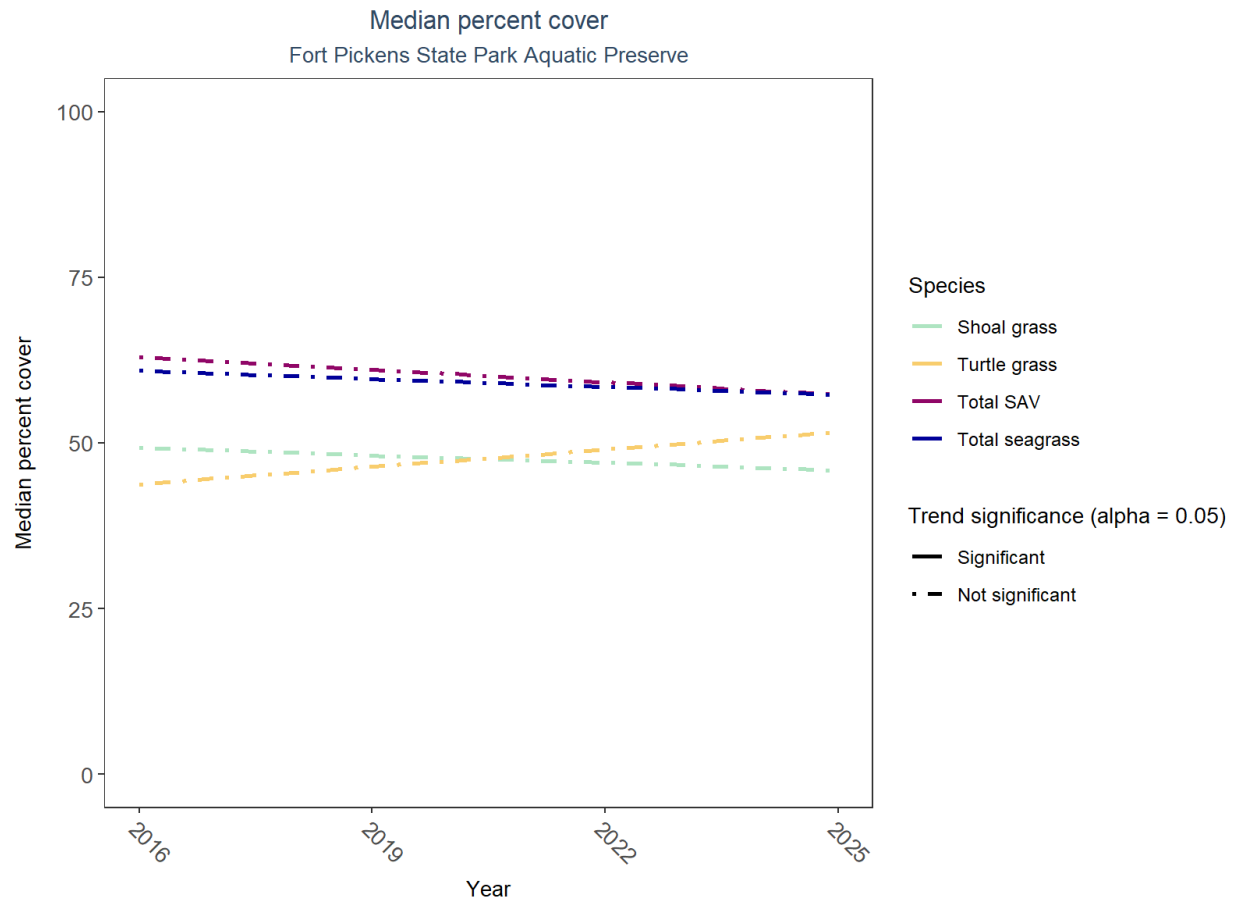


Table 12: SAV LME Results for Fort Pickens State Park Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Shoal grass	No detectable trend	2016 - 2025	57.58	-0.38	0.30
Widgeon grass	Insufficient data	2021 - 2022	-	-	-
Manatee grass	Insufficient data	2022 - 2024	-	-	-
Turtle grass	No detectable trend	2016 - 2025	24.56	0.87	0.14
Total SAV	No detectable trend	2016 - 2025	76.79	-0.63	0.22
Total seagrass	No detectable trend	2016 - 2025	69.86	-0.41	0.45

Total SAV, total seagrass, shoal grass, and turtle grass showed no detectable change in percent cover. Trends in percent cover could not be evaluated for manatee grass and widgeon grass due to insufficient data.



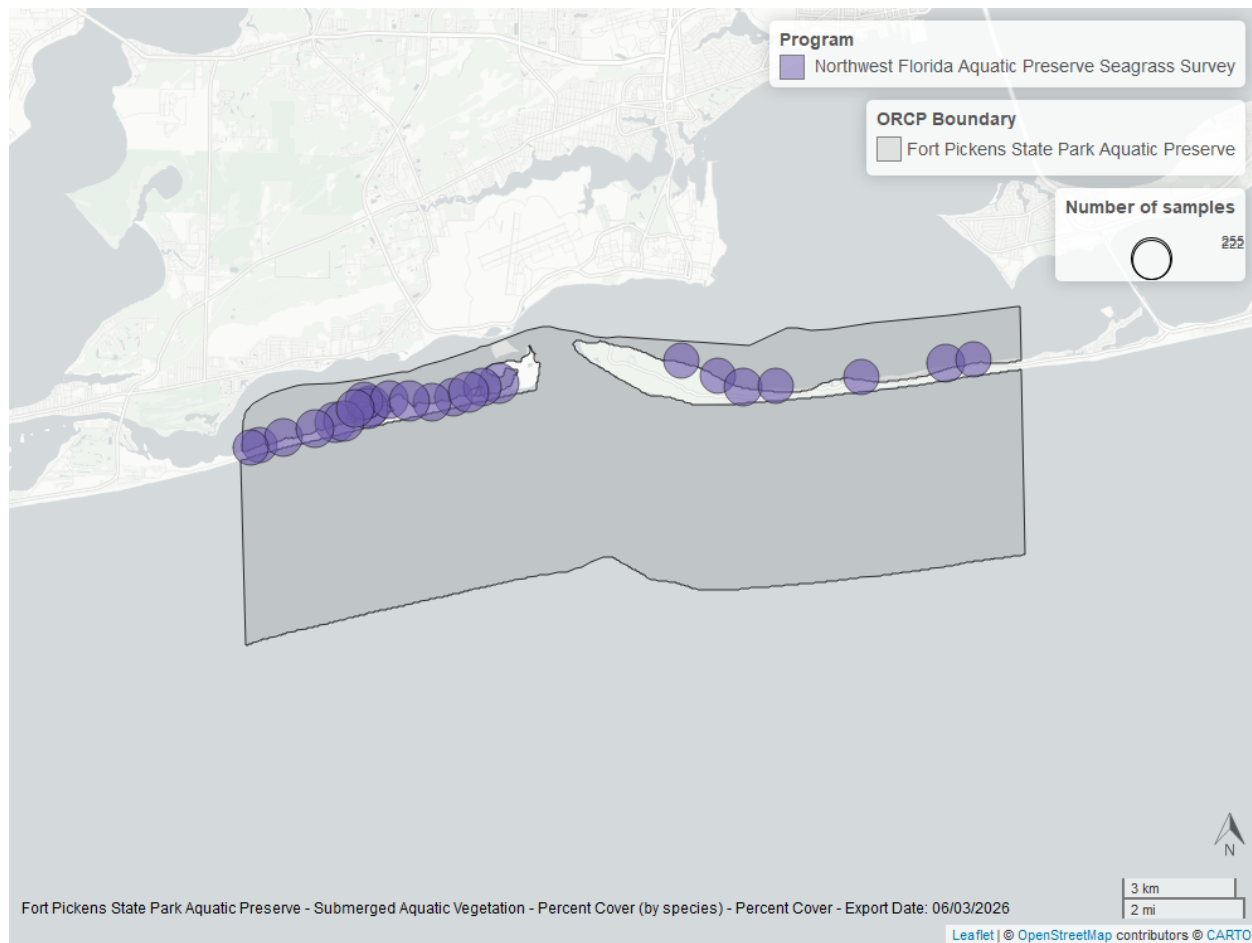
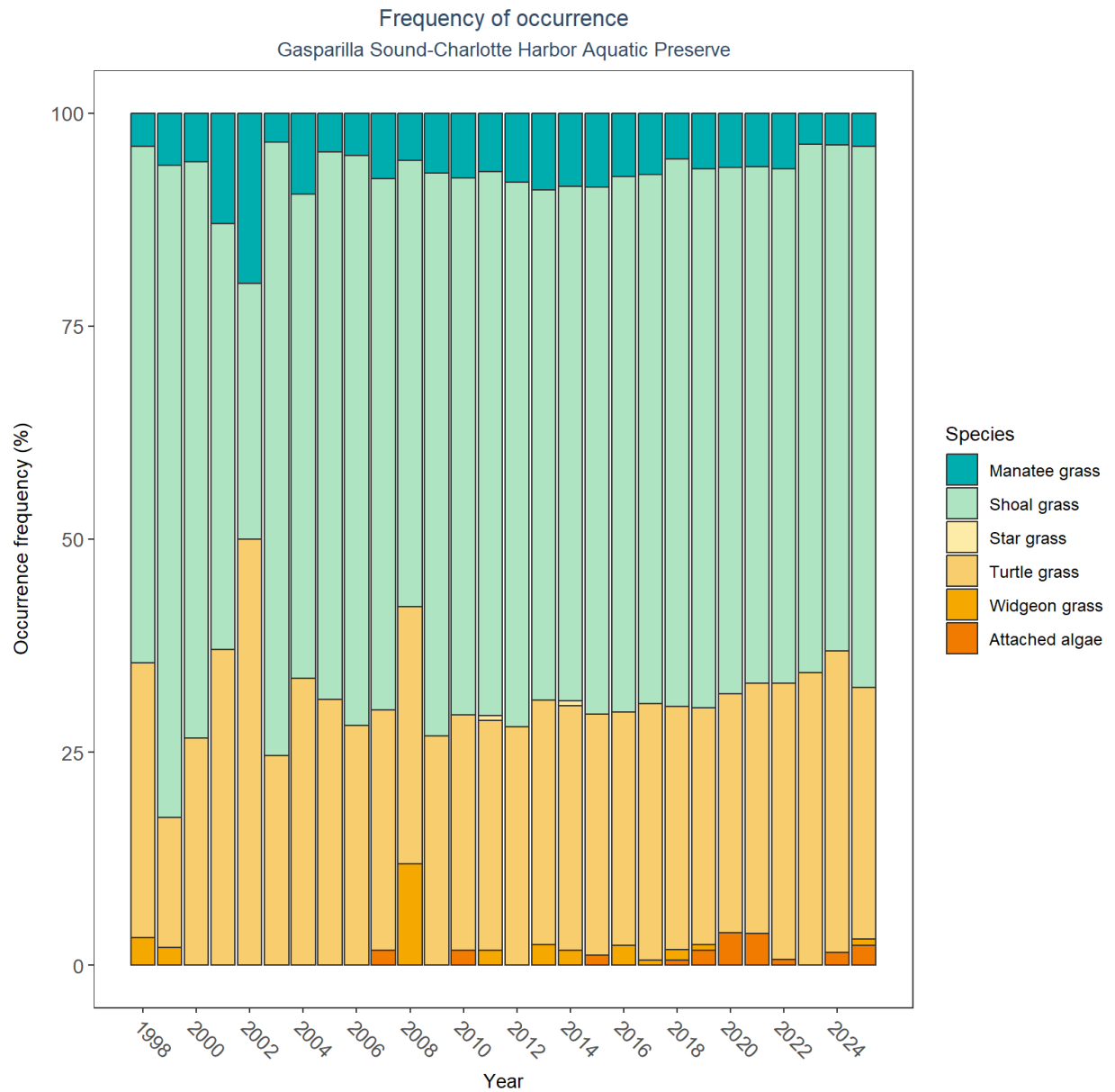


Figure 12: Map showing SAV sampling sites within the boundaries of *Fort Pickens State Park Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Gasparilla Sound-Charlotte Harbor Aquatic Preserve



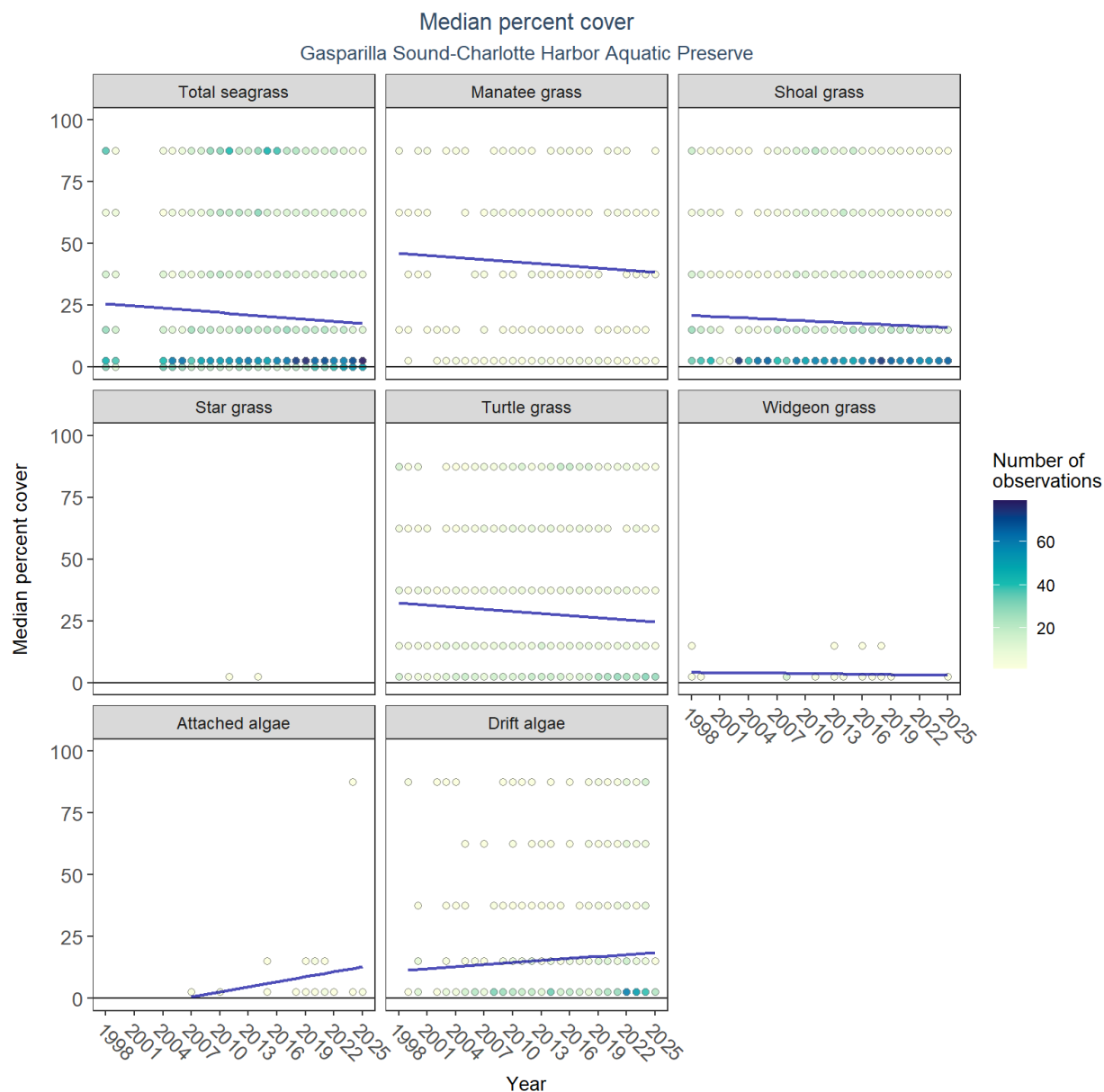
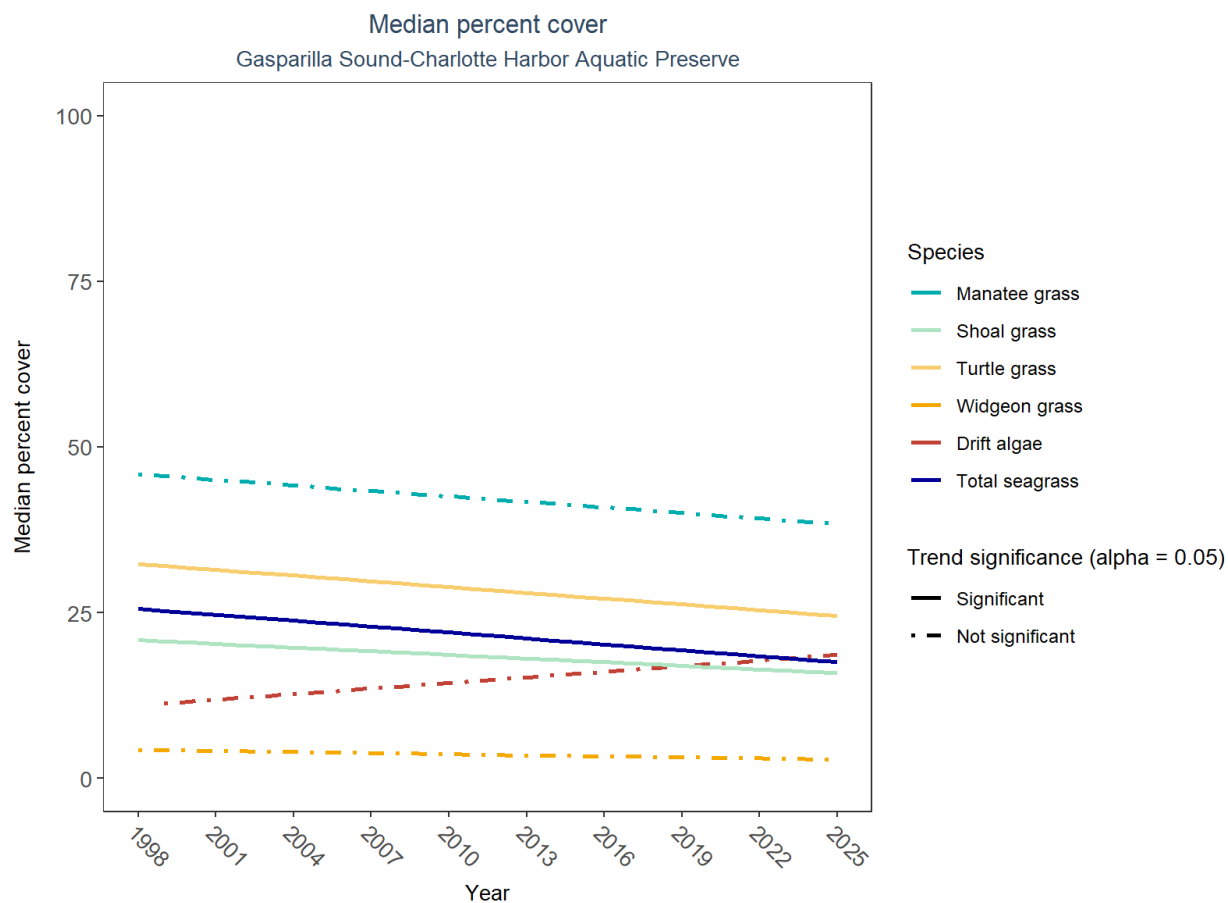


Table 13: SAV LME Results for Gasparilla Sound-Charlotte Harbor Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Attached algae	No detectable trend	2007 - 2025	-8.32	0.68	0.26
Drift algae	No detectable trend	1999 - 2025	9.88	0.28	0.28
Shoal grass	Decreasing trend	1998 - 2025	21.59	-0.18	0.01
Star grass	Insufficient data	2011 - 2014	-	-	-
Widgeon grass	No detectable trend	1998 - 2025	4.54	-0.05	0.70
Manatee grass	No detectable trend	1998 - 2025	47.03	-0.28	0.56
Turtle grass	Decreasing trend	1998 - 2025	33.51	-0.29	0.02
Total seagrass	Decreasing trend	1998 - 2025	26.78	-0.30	0.00

Annual decreases in percent cover were observed for total seagrass (-0.30%), shoal grass (-0.18%), and turtle grass (-0.29%). Manatee grass, widgeon grass, attached algae, and drift algae showed no detectable change in percent cover. Trends in percent cover could not be evaluated for star grass due to insufficient data.



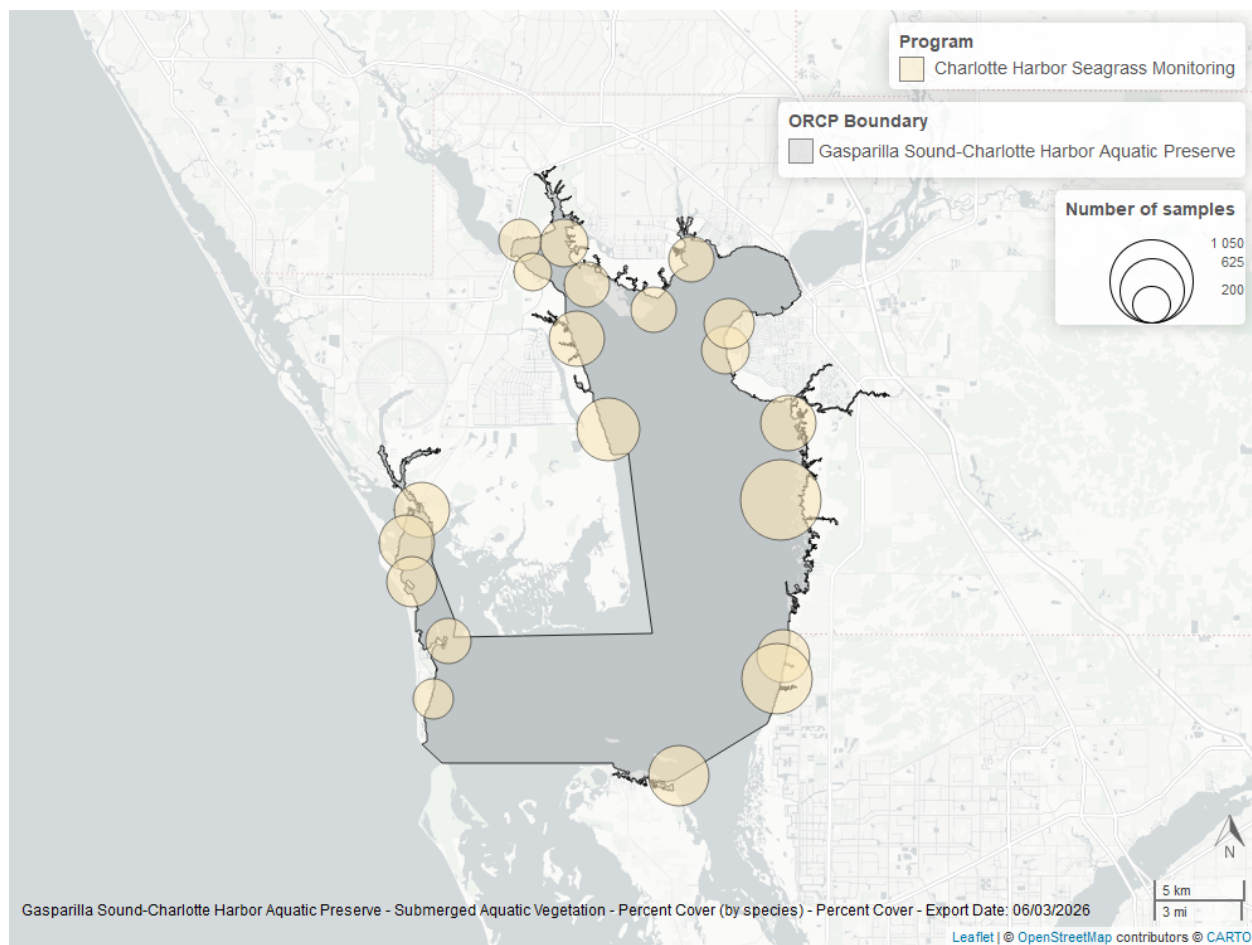
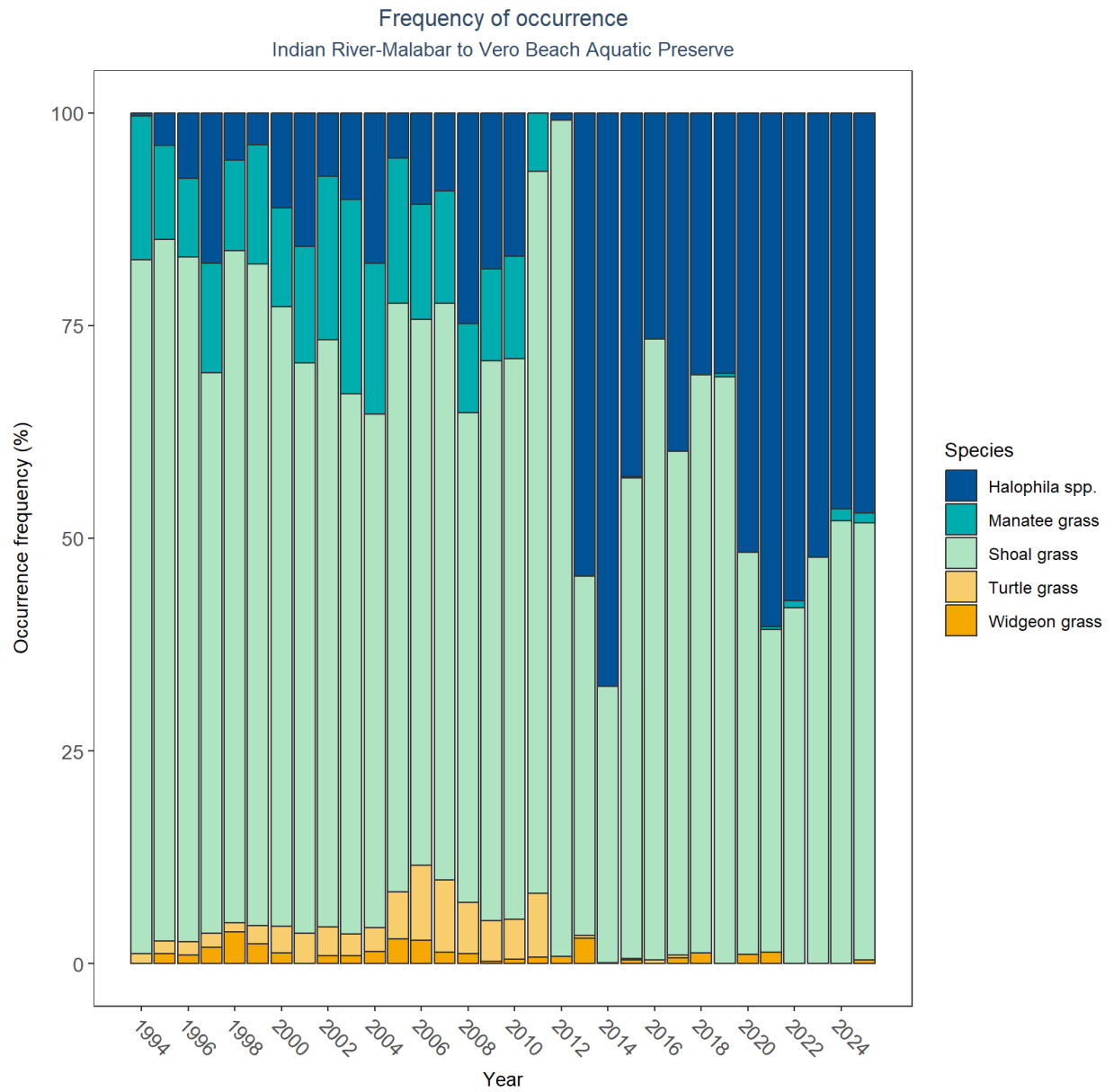


Figure 13: Map showing SAV sampling sites within the boundaries of *Gasparilla Sound-Charlotte Harbor Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Indian River-Malabar to Vero Beach Aquatic Preserve



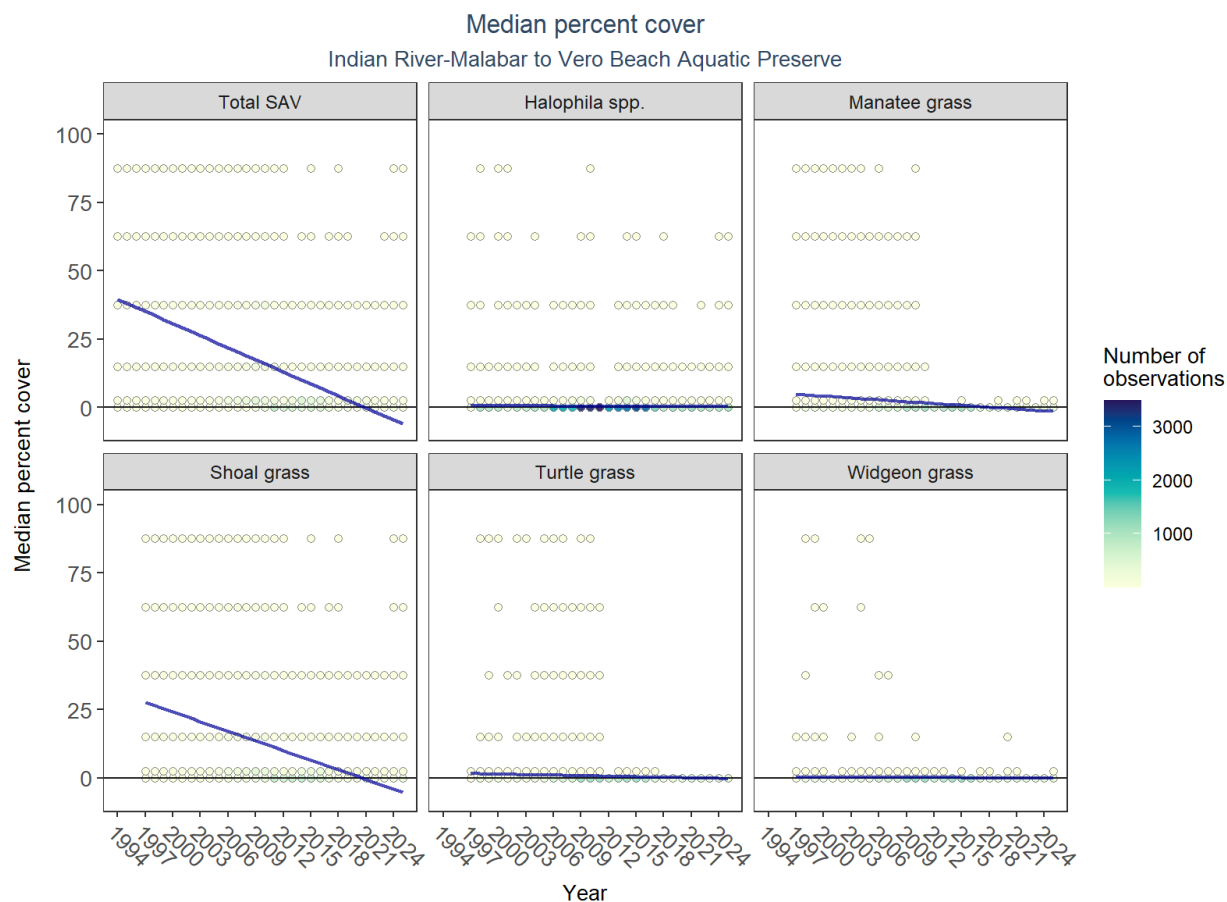
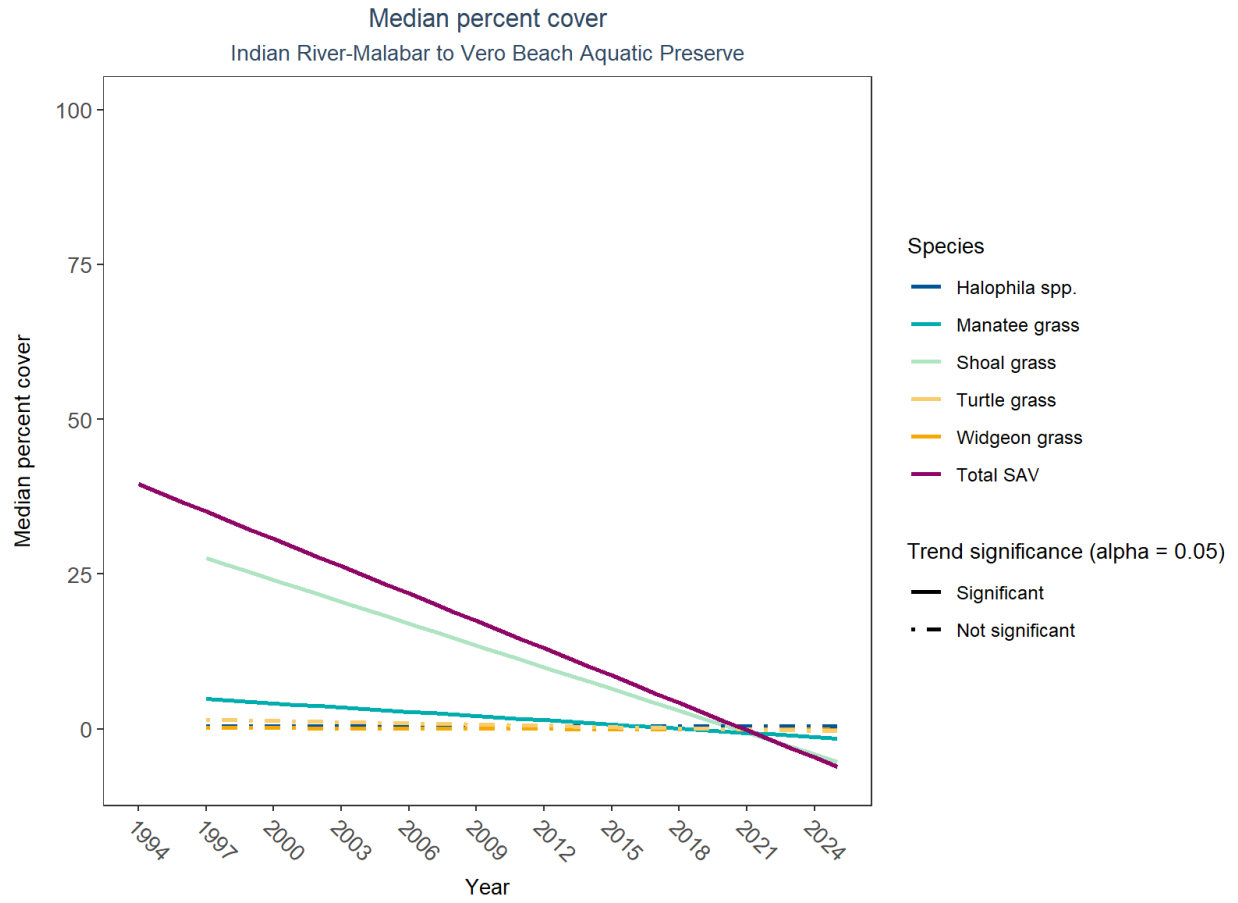


Table 14: SAV LME Results for Indian River-Malabar to Vero Beach Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Shoal grass	Decreasing trend	1997 - 2025	31.12	-1.17	0.00
Halophila spp.	No detectable trend	1997 - 2025	0.49	0.00	0.79
Widgeon grass	No detectable trend	1997 - 2025	0.24	-0.01	0.10
Manatee grass	Decreasing trend	1997 - 2025	5.52	-0.23	0.00
Turtle grass	No detectable trend	1997 - 2025	1.73	-0.07	0.10
Total SAV	Decreasing trend	1994 - 2025	39.51	-1.47	0.00

Annual decreases in percent cover were observed for total SAV (-1.47%), manatee grass (-0.23%), and shoal grass (-1.17%). *Halophila* spp., turtle grass, and widgeon grass showed no detectable change in percent cover.



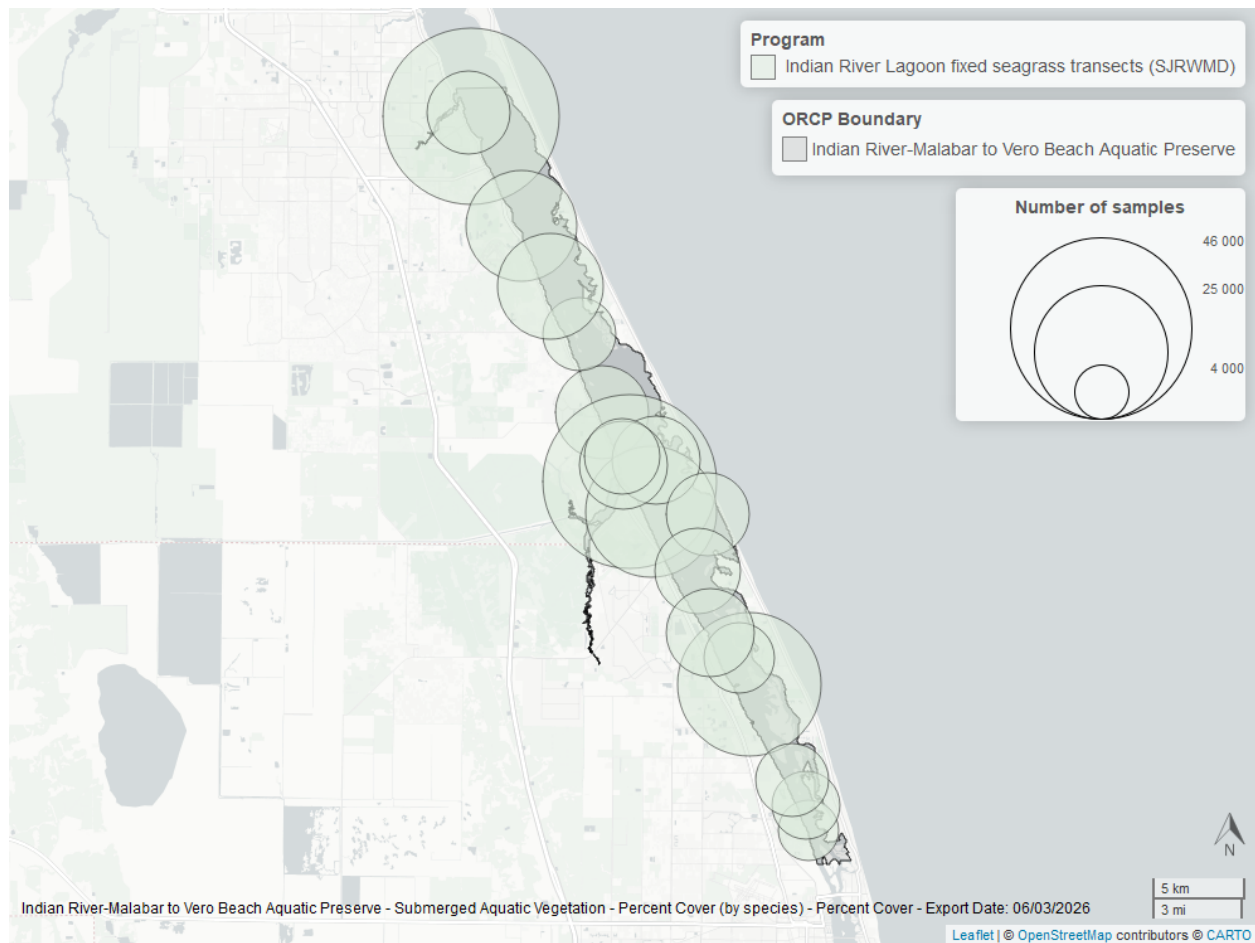
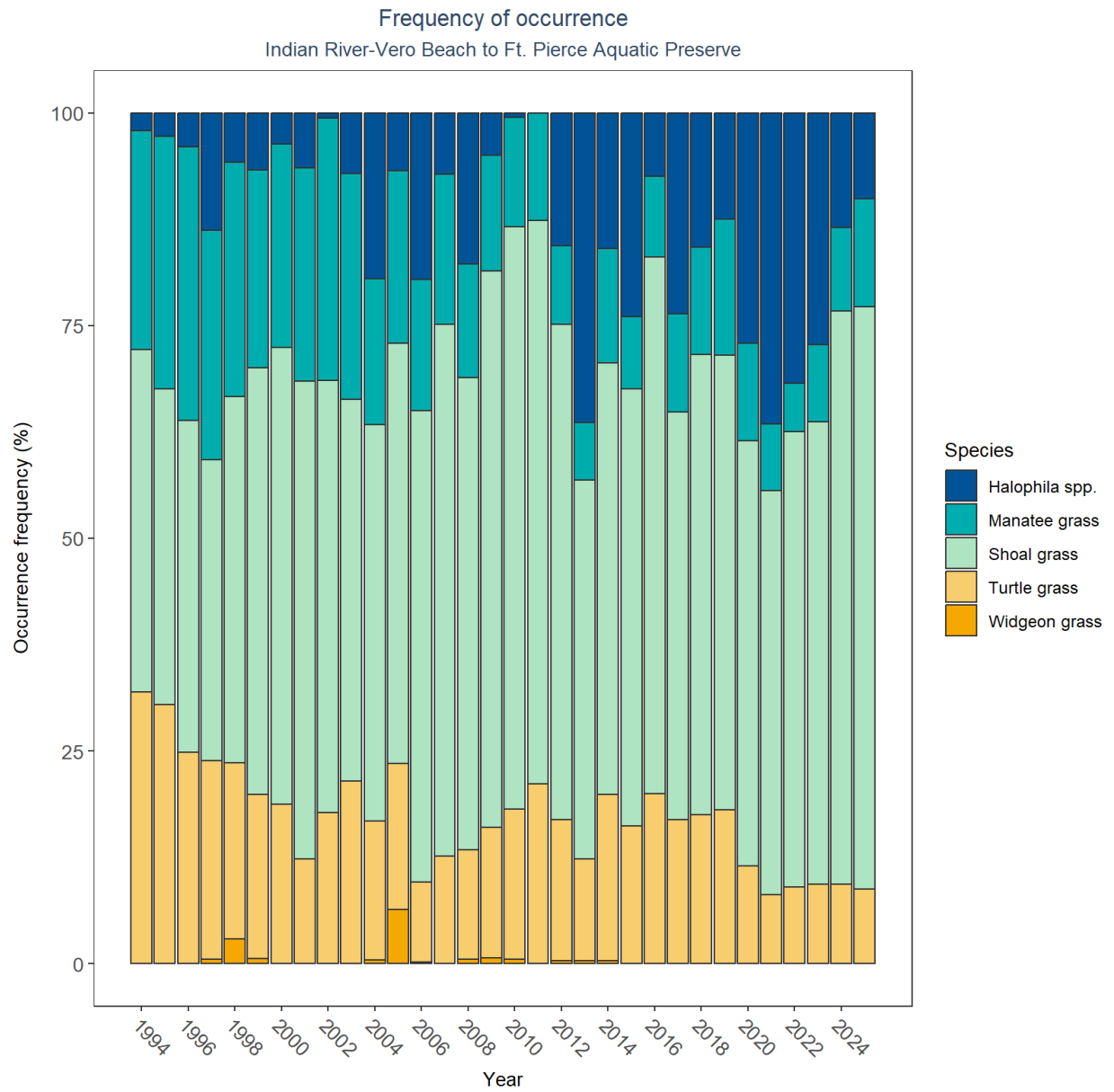


Figure 14: Map showing SAV sampling sites within the boundaries of *Indian River-Malabar to Vero Beach Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Indian River-Vero Beach to Ft. Pierce Aquatic Preserve



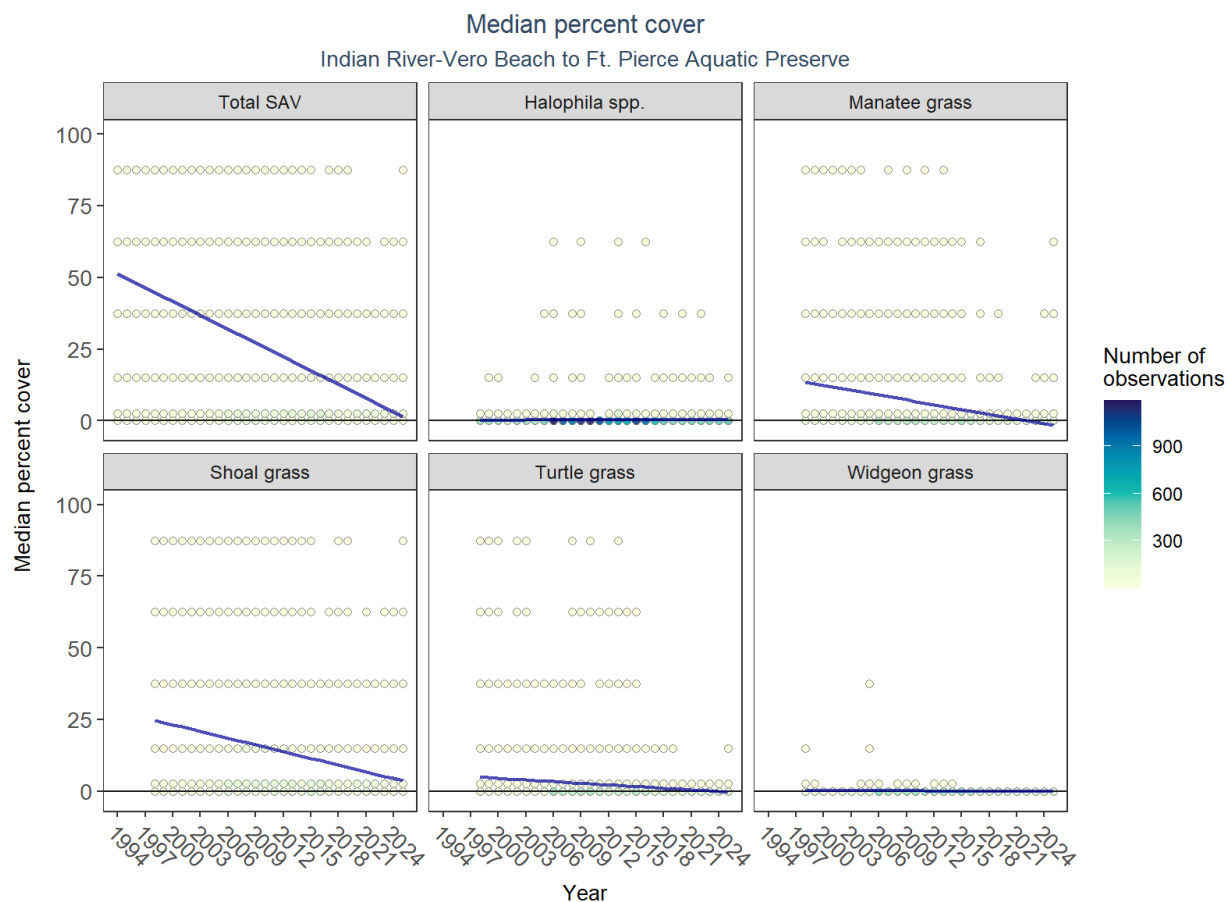
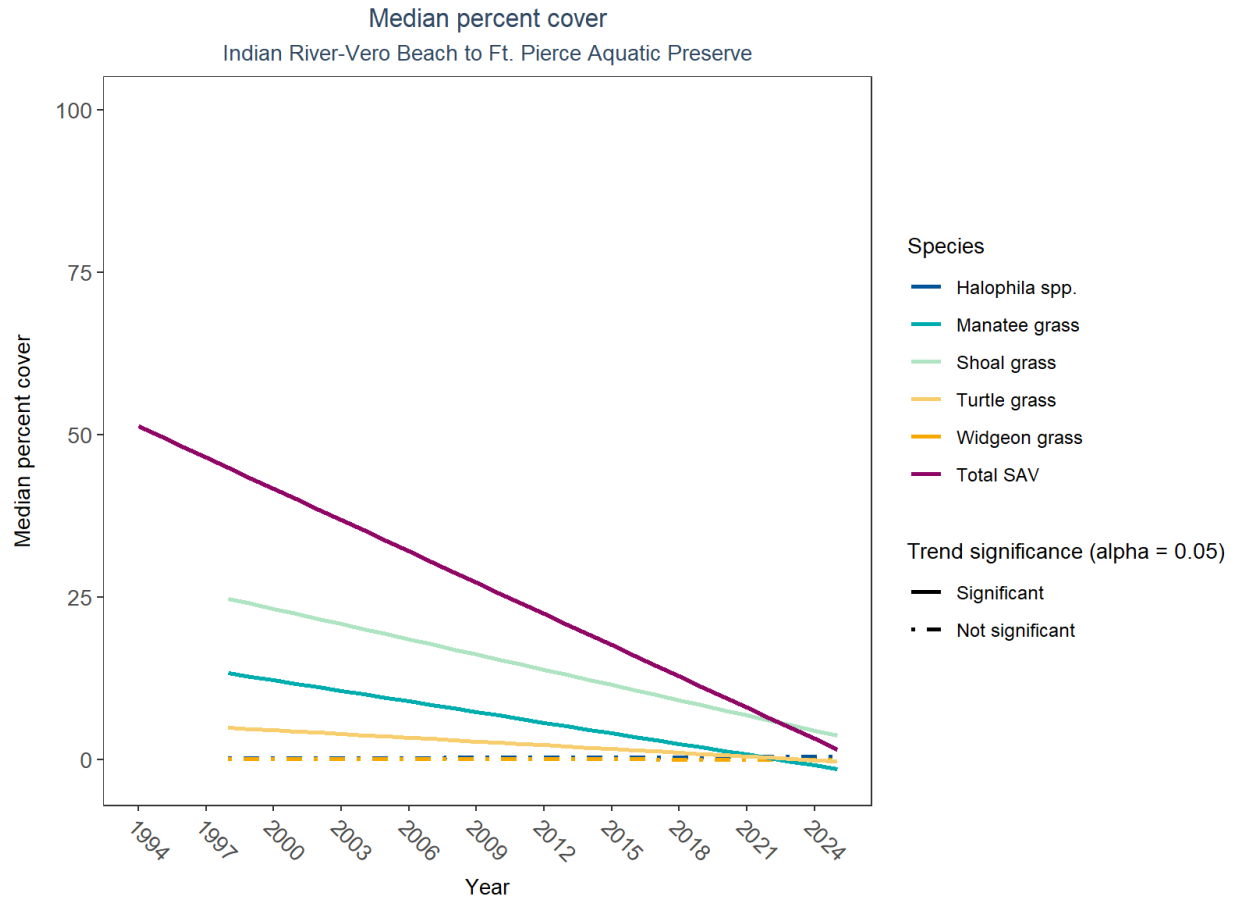


Table 15: SAV LME Results for Indian River-Vero Beach to Ft. Pierce Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Shoal grass	Decreasing trend	1998 - 2025	27.88	-0.78	0.00
Halophila spp.	No detectable trend	1998 - 2025	0.12	0.01	0.09
Widgeon grass	No detectable trend	1998 - 2025	0.13	0.00	0.12
Manatee grass	Decreasing trend	1998 - 2025	15.48	-0.55	0.04
Turtle grass	Decreasing trend	1998 - 2025	5.65	-0.19	0.01
Total SAV	Decreasing trend	1994 - 2025	51.26	-1.60	0.00

Annual decreases in percent cover were observed for total SAV (-1.60%), manatee grass (-0.55%), shoal grass (-0.78%), and turtle grass (-0.19%). No detectable change in percent cover was observed for *Halophila* spp. and widgeon grass.



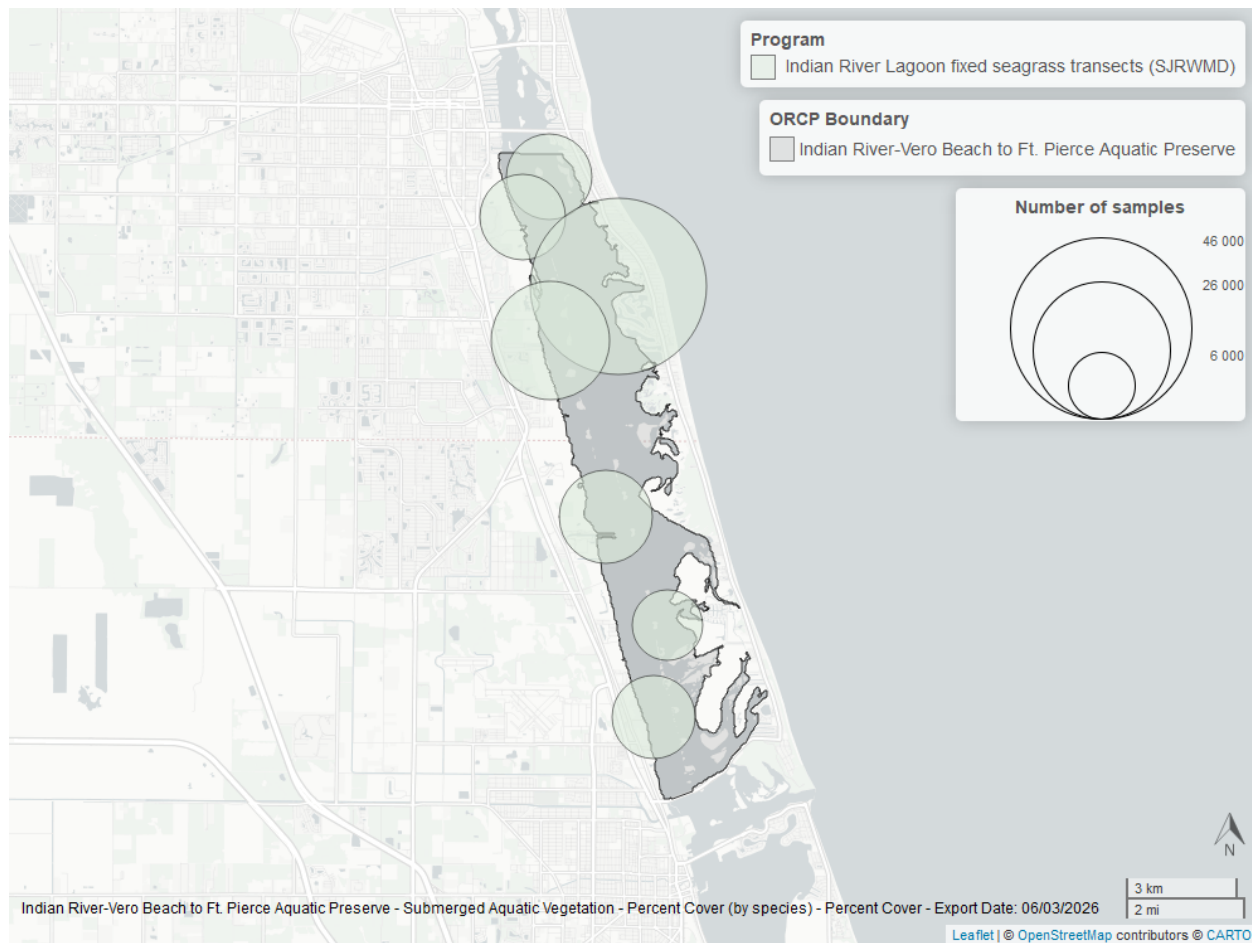
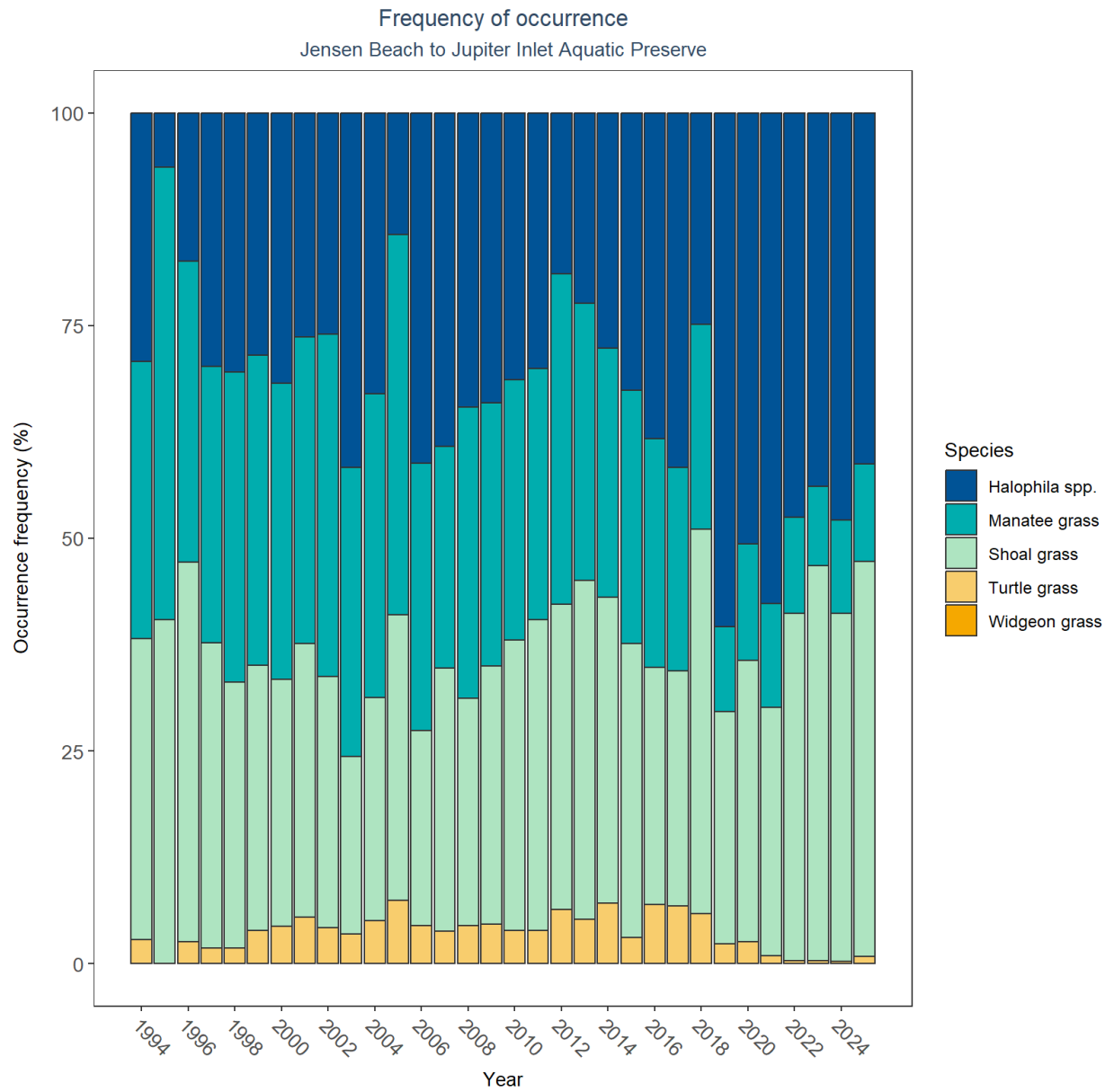


Figure 15: Map showing SAV sampling sites within the boundaries of *Indian River-Vero Beach to Ft. Pierce Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Jensen Beach to Jupiter Inlet Aquatic Preserve



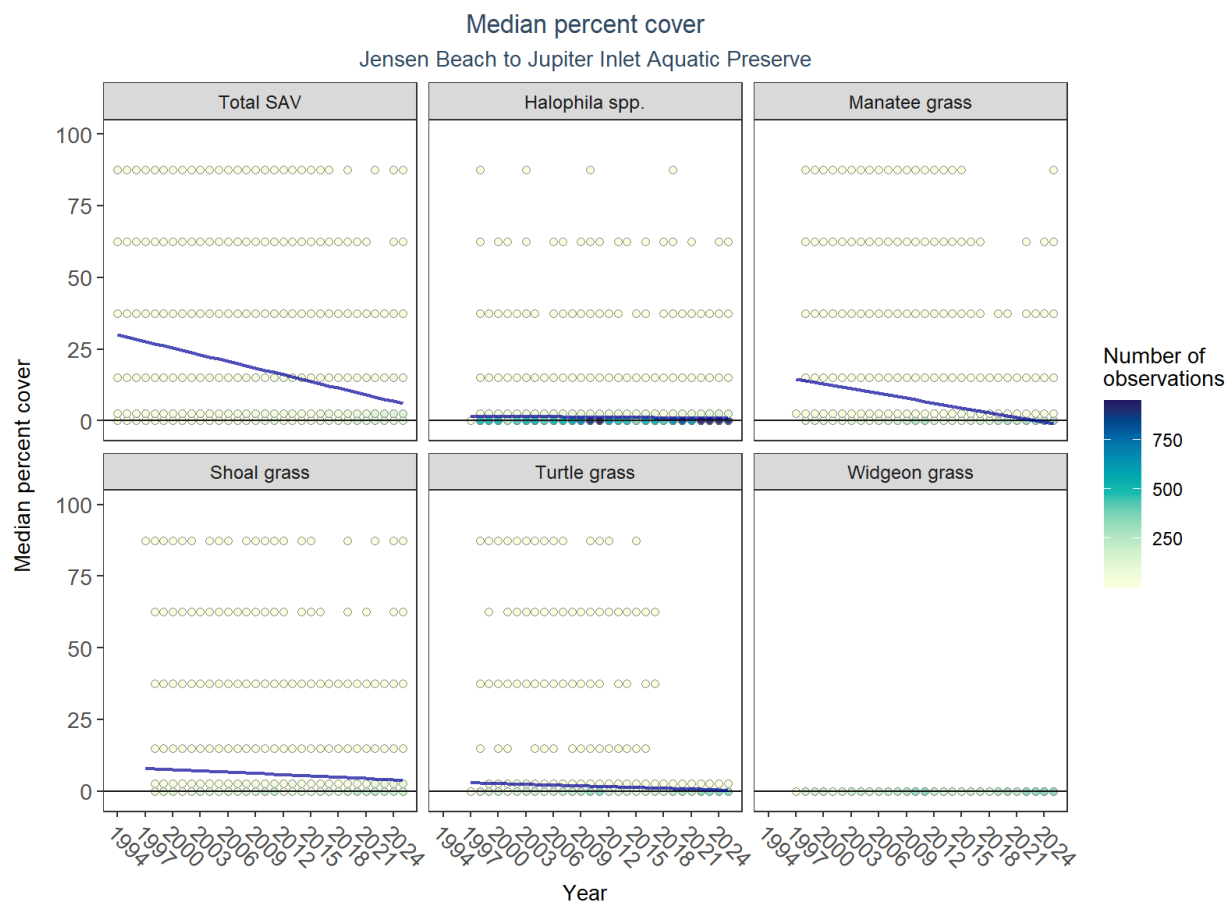
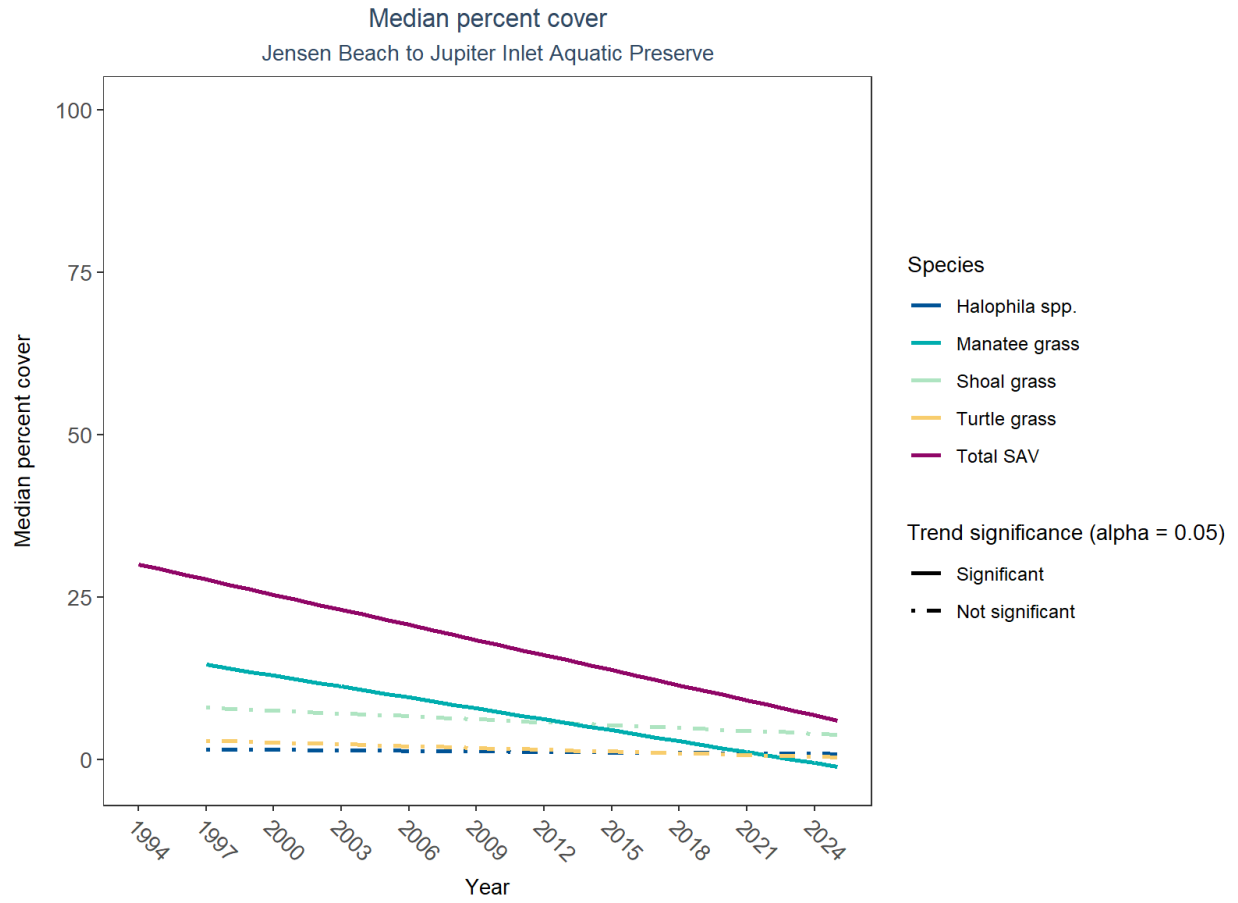


Table 16: SAV LME Results for Jensen Beach to Jupiter Inlet Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Shoal grass	No detectable trend	1997 - 2025	8.41	-0.15	0.10
Halophila spp.	No detectable trend	1997 - 2025	1.66	-0.03	0.06
Widgeon grass	Model did not fit the available data	1997 - 2025	-	-	-
Manatee grass	Decreasing trend	1997 - 2025	16.27	-0.56	0.00
Turtle grass	No detectable trend	1997 - 2025	3.16	-0.09	0.07
Total SAV	Decreasing trend	1994 - 2025	30.00	-0.77	0.00

Annual decreases in percent cover were observed for total SAV (-0.77%) and manatee grass (-0.56%). *Halophila* spp., shoal grass, and turtle grass showed no detectable change in percent cover. The model could not be fitted for widgeon grass.



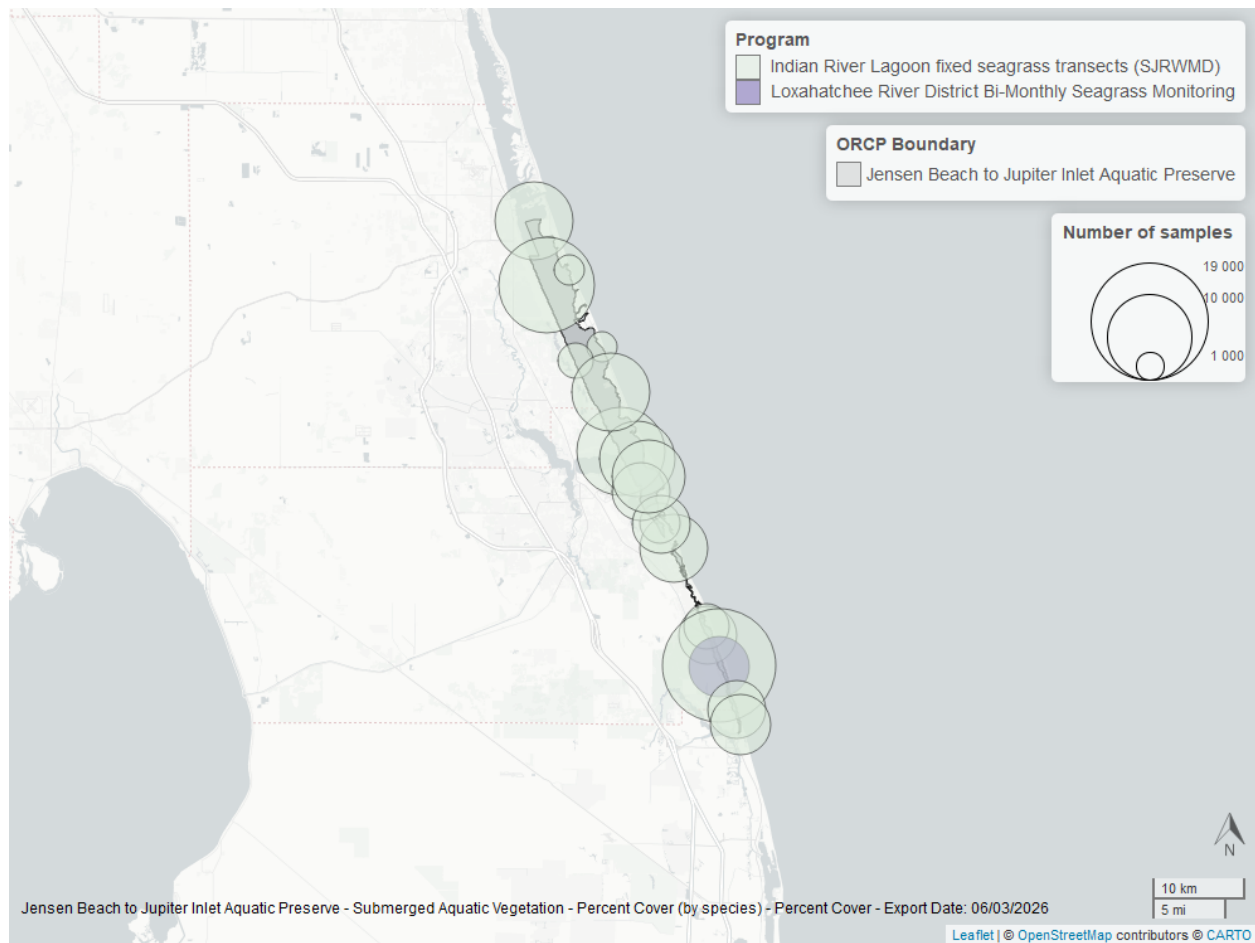
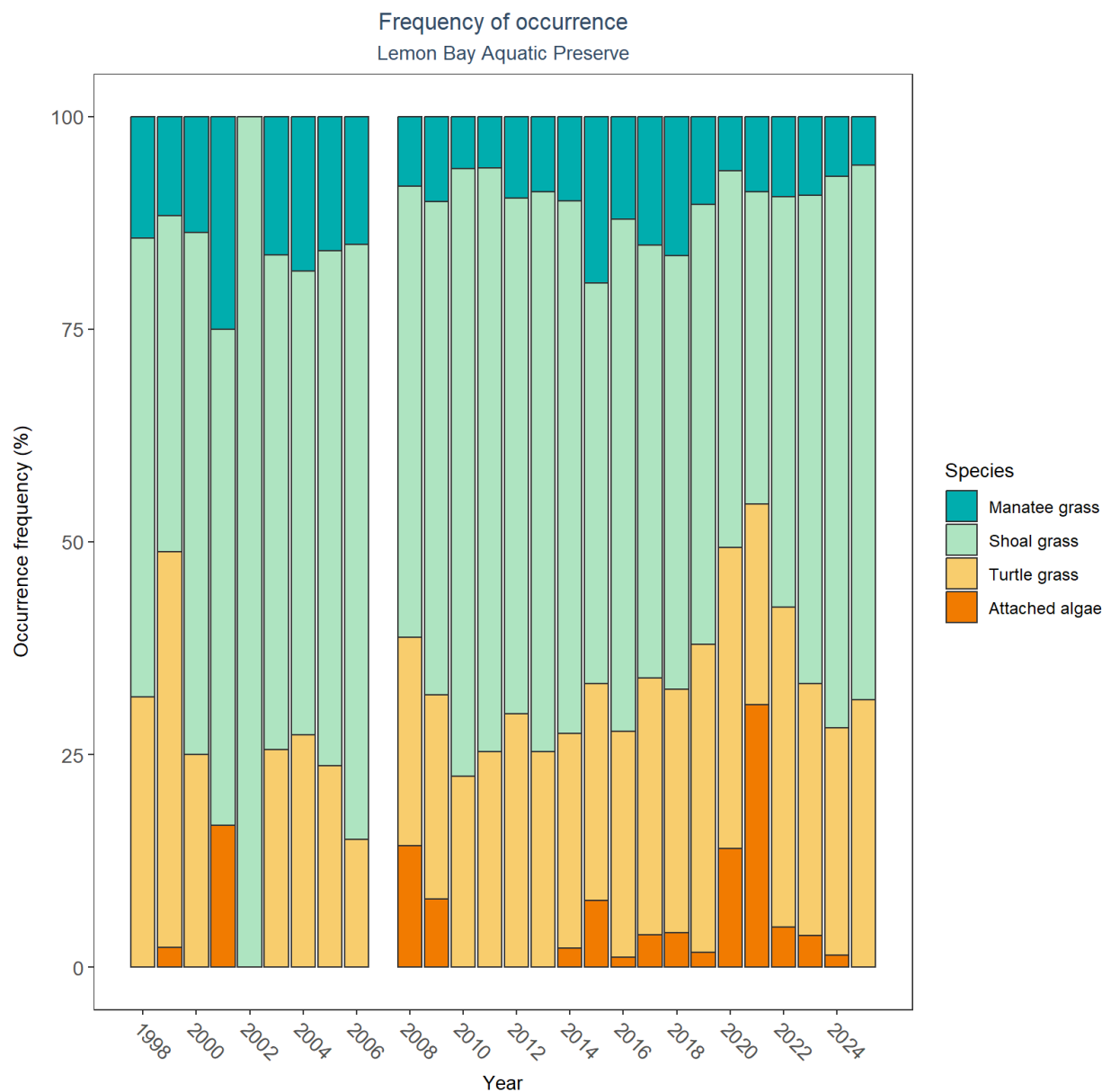


Figure 16: Map showing SAV sampling sites within the boundaries of *Jensen Beach to Jupiter Inlet Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Lemon Bay Aquatic Preserve



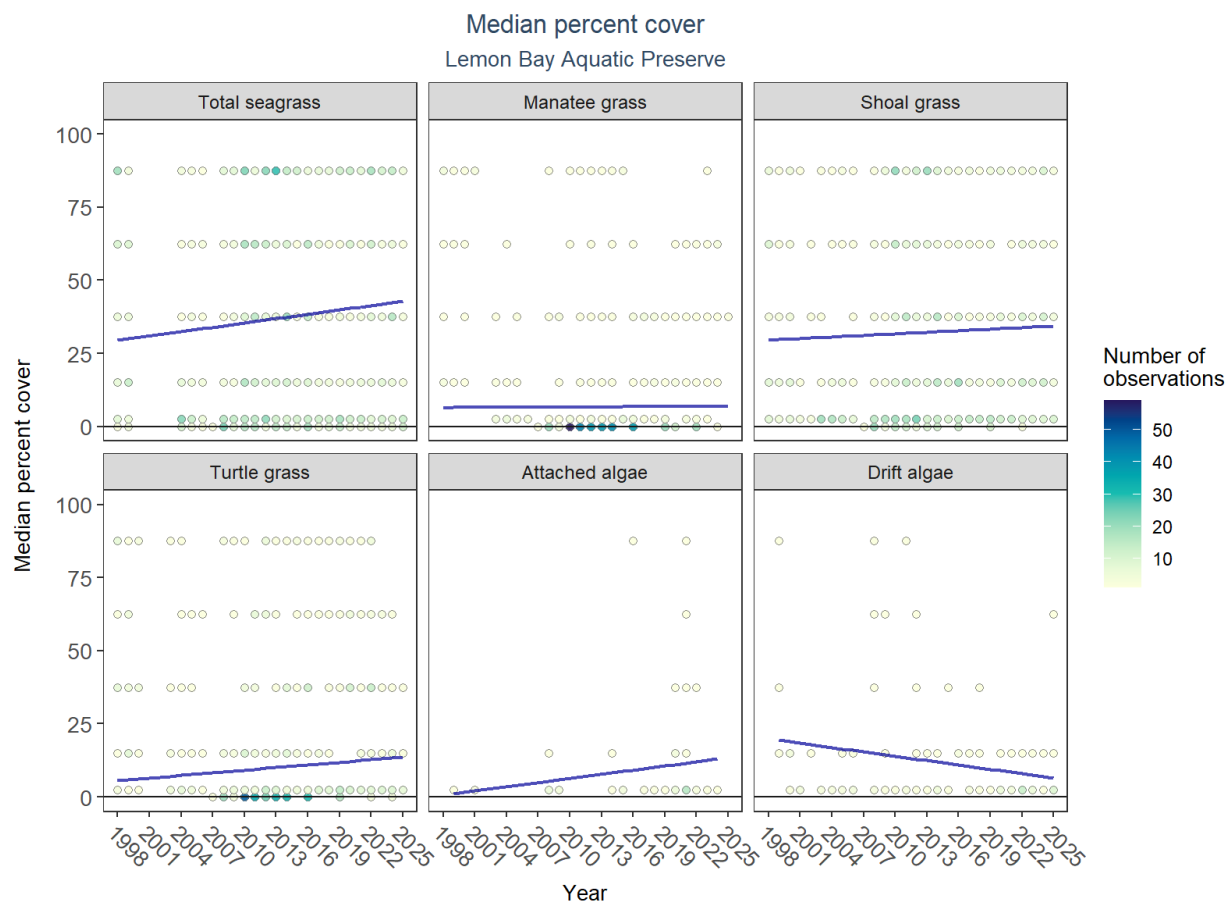
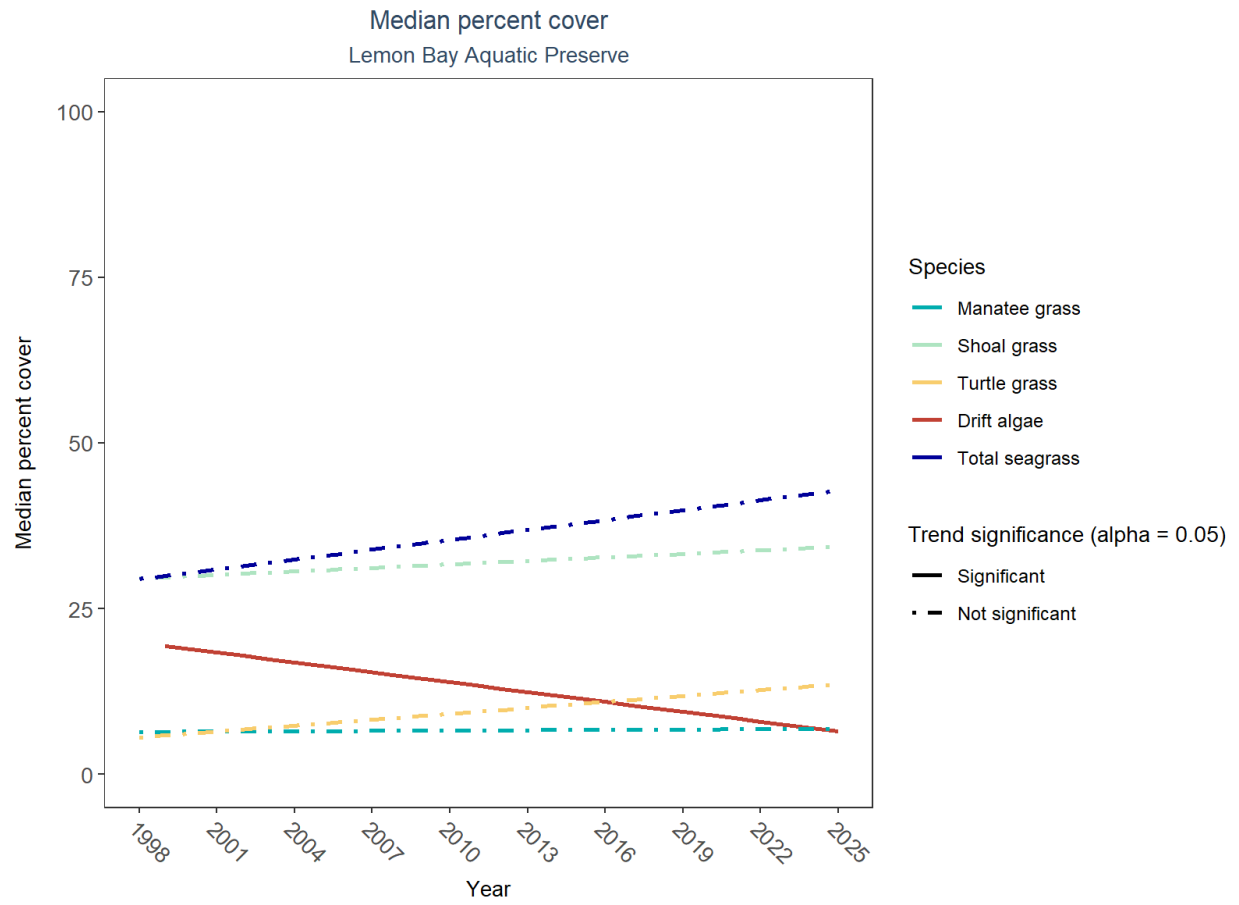


Table 17: SAV LME Results for Lemon Bay Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Attached algae	No detectable trend	1999 - 2024	-1.17	0.47	0.26
Drift algae	Decreasing trend	1999 - 2025	21.85	-0.50	0.02
Shoal grass	No detectable trend	1998 - 2025	28.87	0.18	0.62
Manatee grass	No detectable trend	1998 - 2025	6.36	0.02	0.91
Turtle grass	No detectable trend	1998 - 2025	4.40	0.30	0.46
Total seagrass	No detectable trend	1998 - 2025	27.47	0.50	0.30

An annual decrease in percent cover was observed for drift algae (-0.50%). Total seagrass, manatee grass, shoal grass, turtle grass, and attached algae showed no detectable change in percent cover.



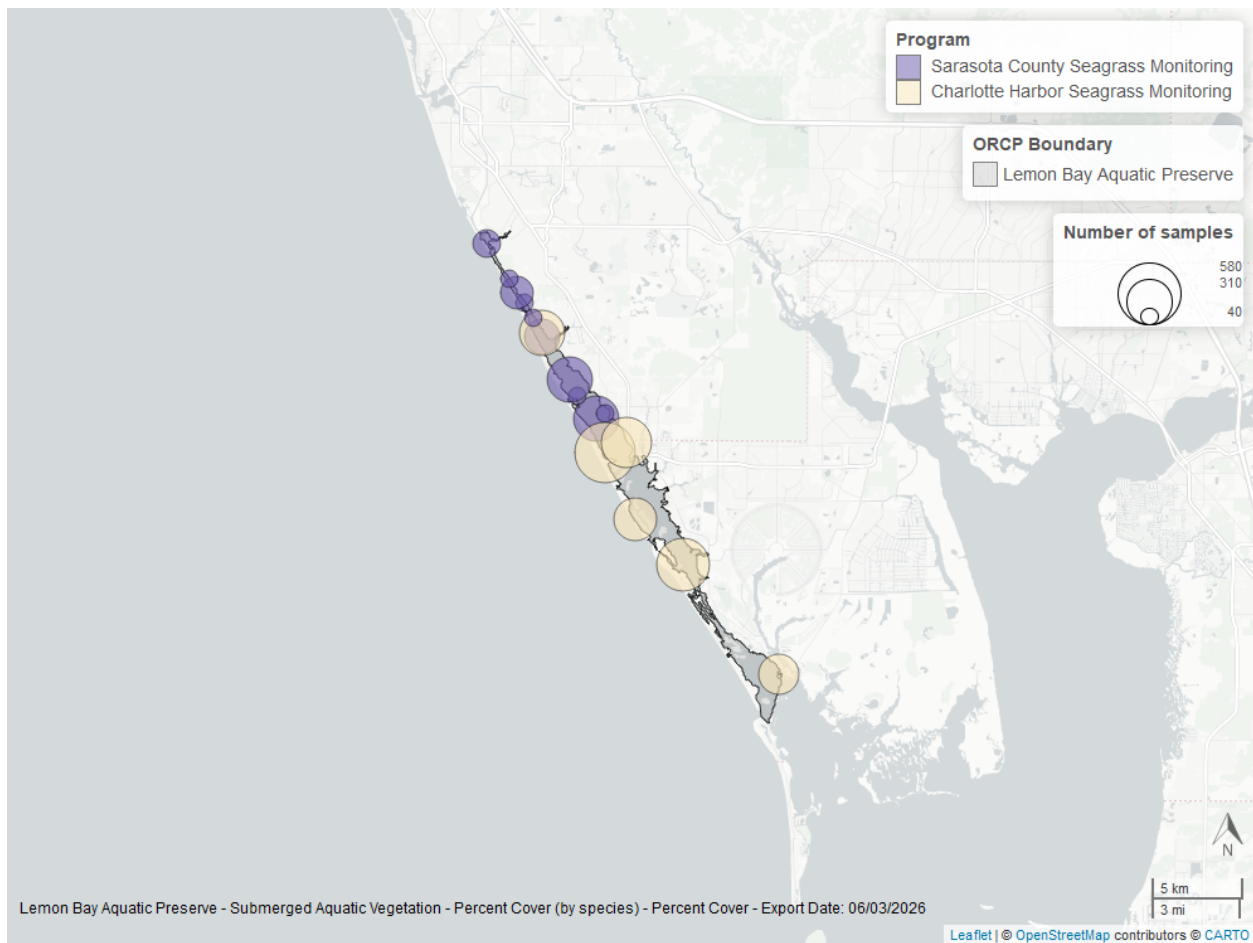


Figure 17: Map showing SAV sampling sites within the boundaries of *Lemon Bay Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Lignumvitae Key Aquatic Preserve

Not enough data to perform analyses in Lignumvitae Key Aquatic Preserve

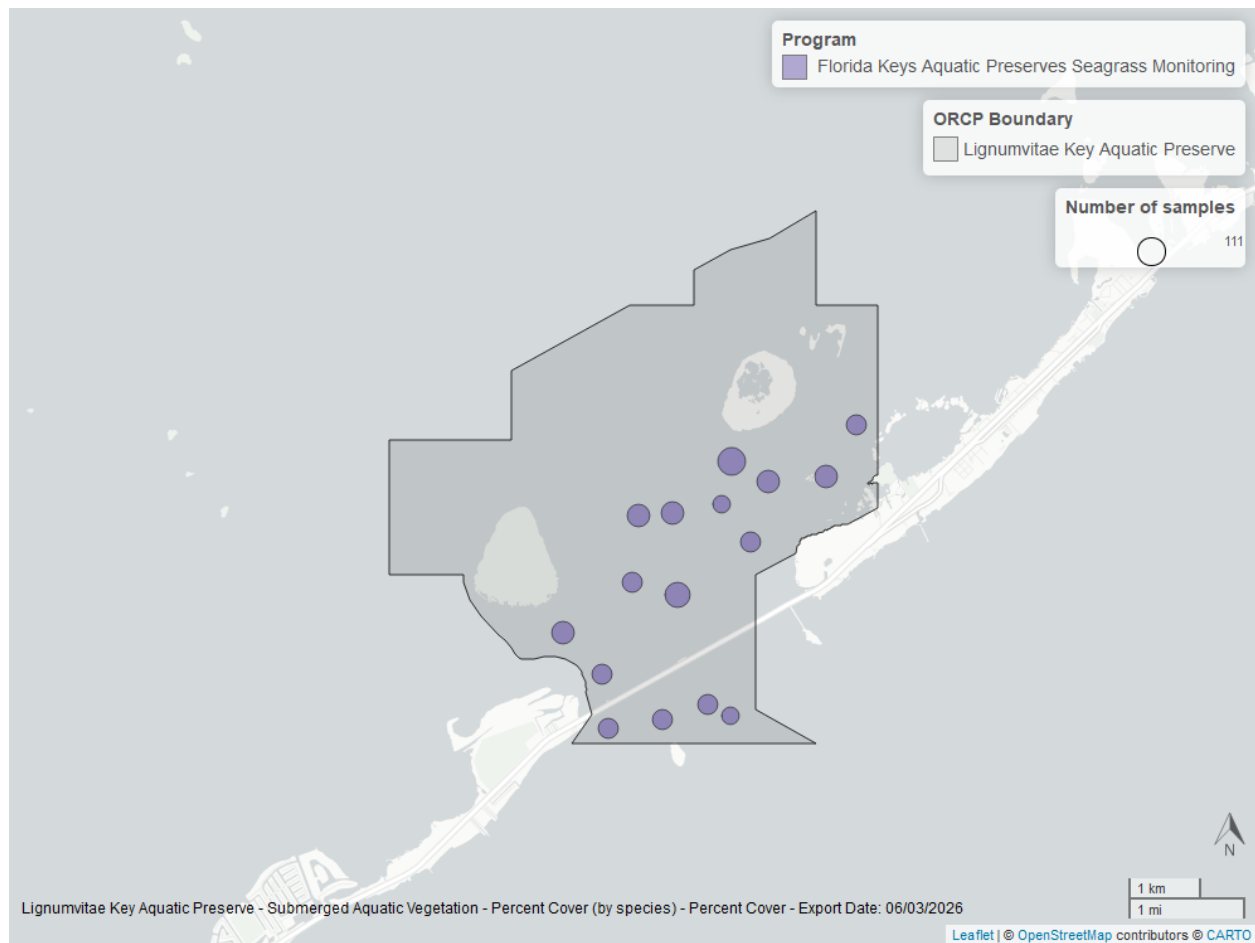
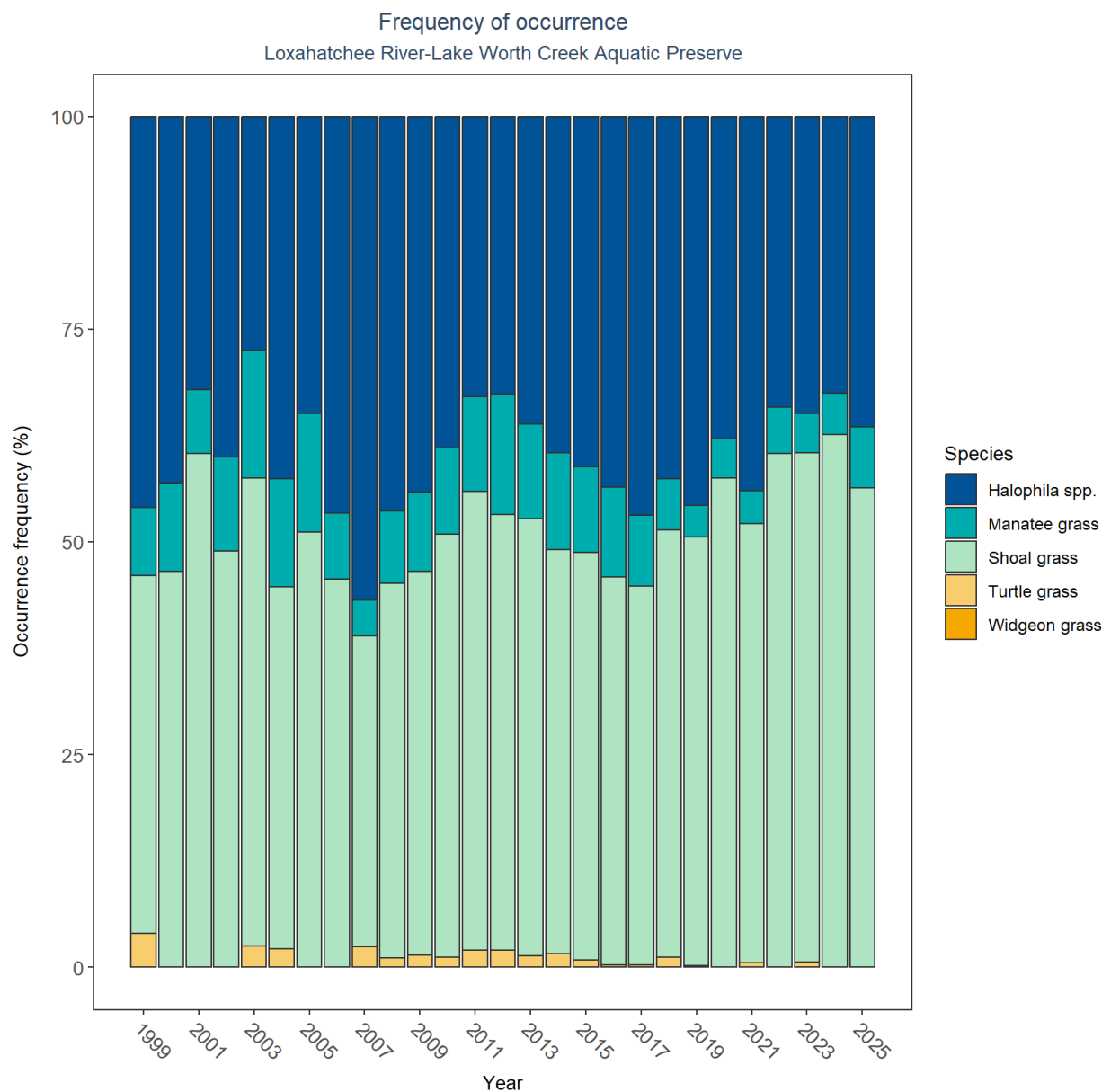


Figure 18: Map showing SAV sampling sites within the boundaries of *Lignumvitae Key Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Loxahatchee River-Lake Worth Creek Aquatic Preserve



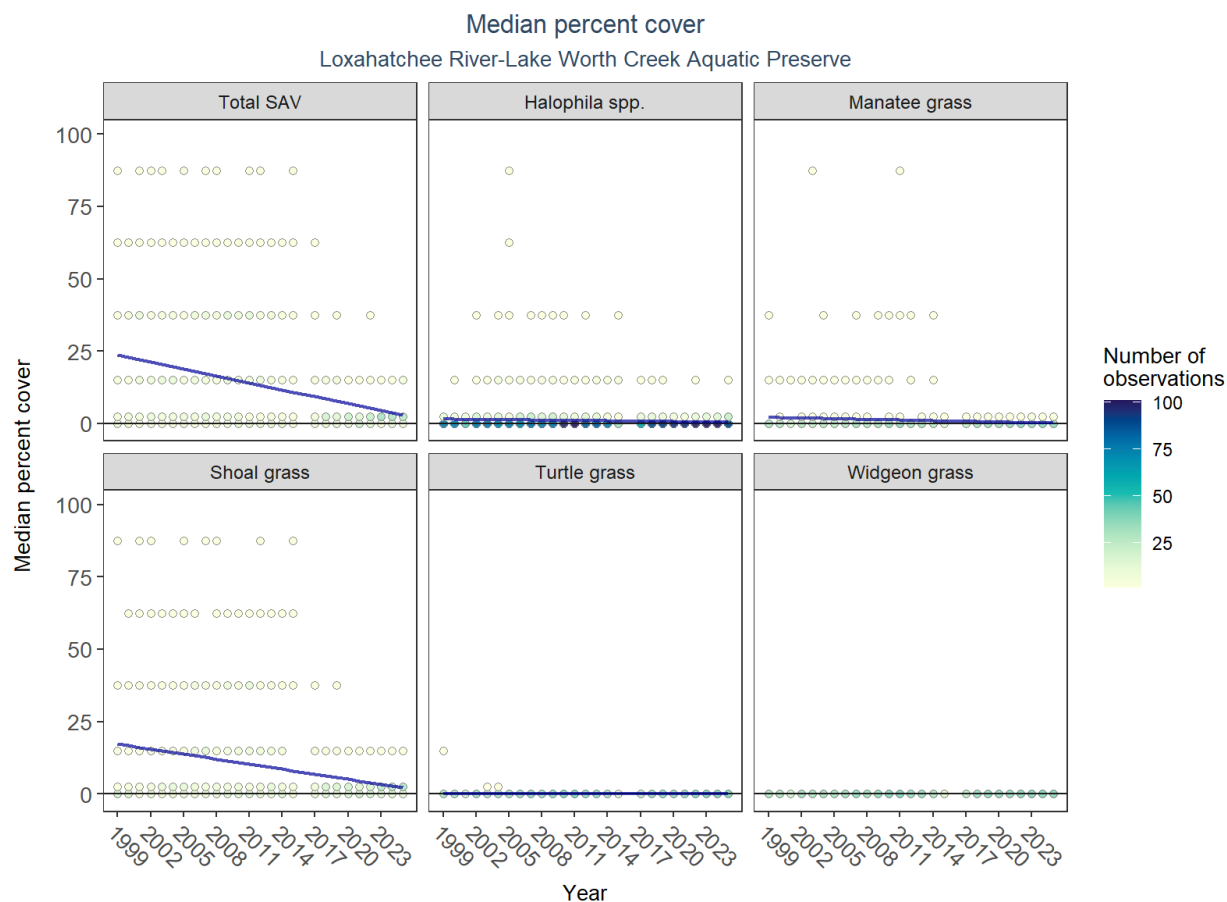
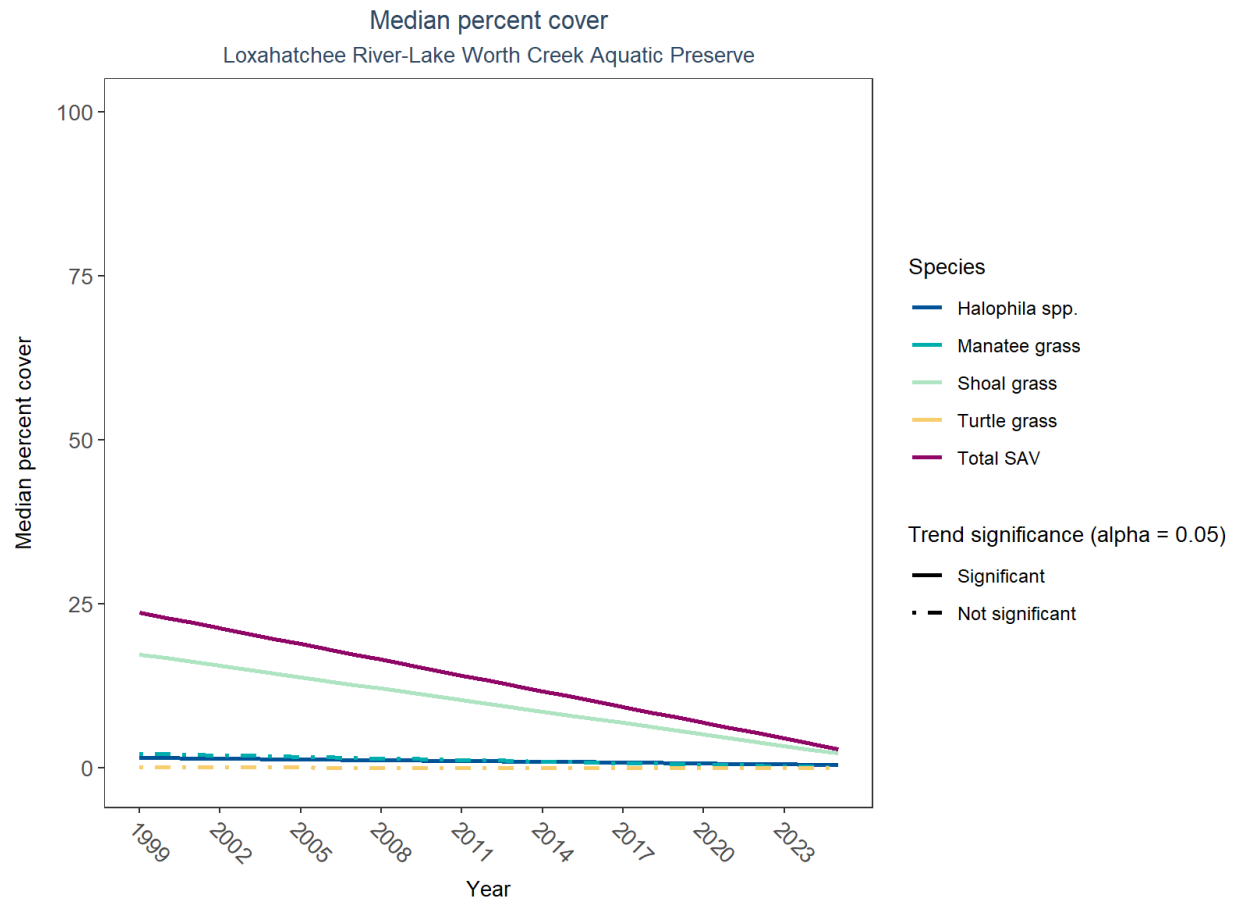


Table 18: SAV LME Results for Loxahatchee River-Lake Worth Creek Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Shoal grass	Decreasing trend	1999 - 2025	20.20	-0.58	0.00
Halophila spp.	Decreasing trend	1999 - 2025	1.71	-0.04	0.00
Widgeon grass	Model did not fit the available data	1999 - 2025	-	-	-
Manatee grass	No detectable trend	1999 - 2025	2.54	-0.08	0.34
Turtle grass	No detectable trend	1999 - 2025	0.09	0.00	0.30
Total SAV	Decreasing trend	1999 - 2025	27.66	-0.80	0.00

Annual decreases in percent cover were observed for total SAV (-0.80%), *Halophila* spp. (-0.04%), and shoal grass (-0.58%). No detectable change in percent cover was observed for manatee grass and turtle grass. The model could not be fitted for widgeon grass.



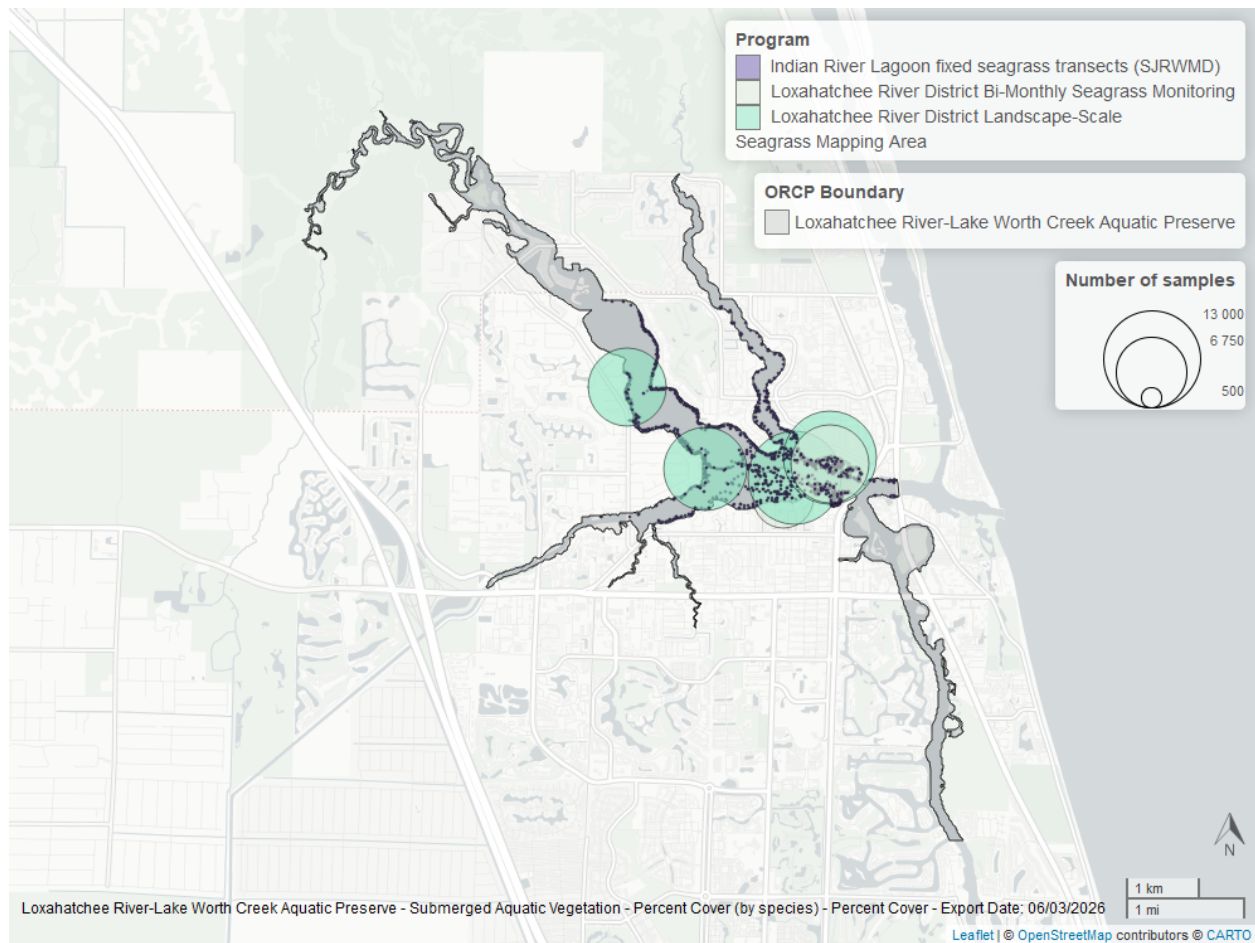
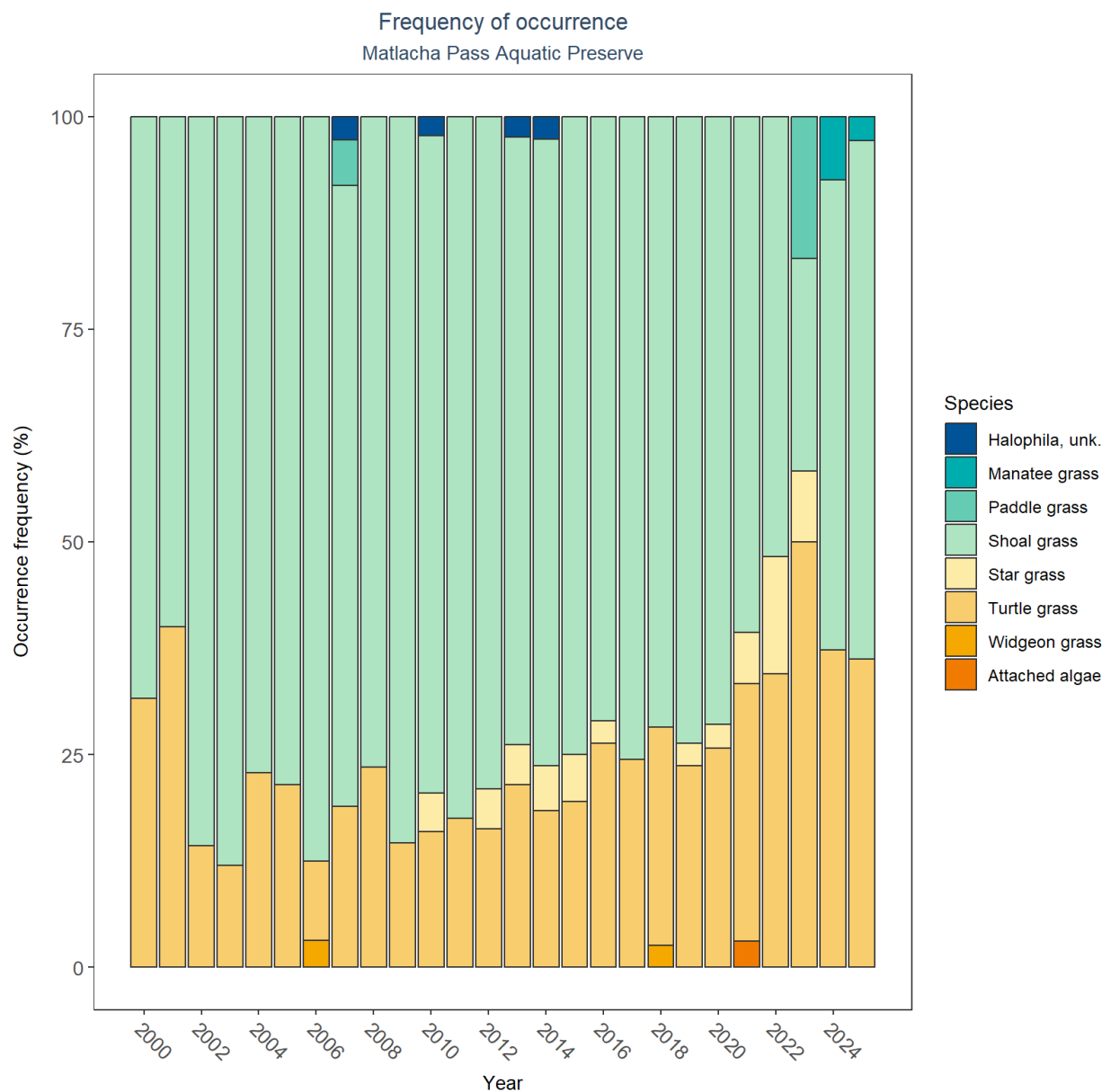


Figure 19: Map showing SAV sampling sites within the boundaries of *Loxahatchee River-Lake Worth Creek Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Matlacha Pass Aquatic Preserve



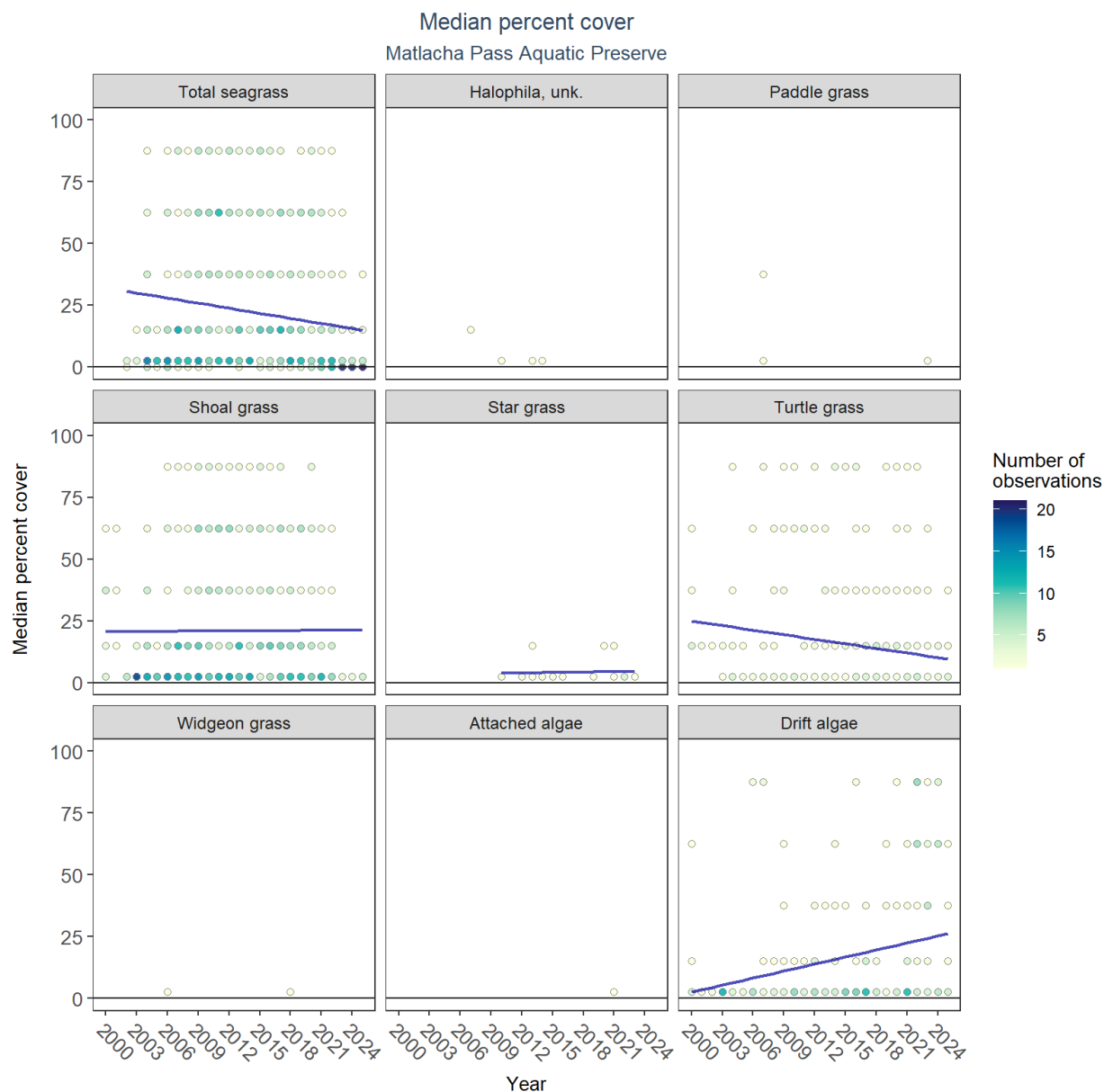
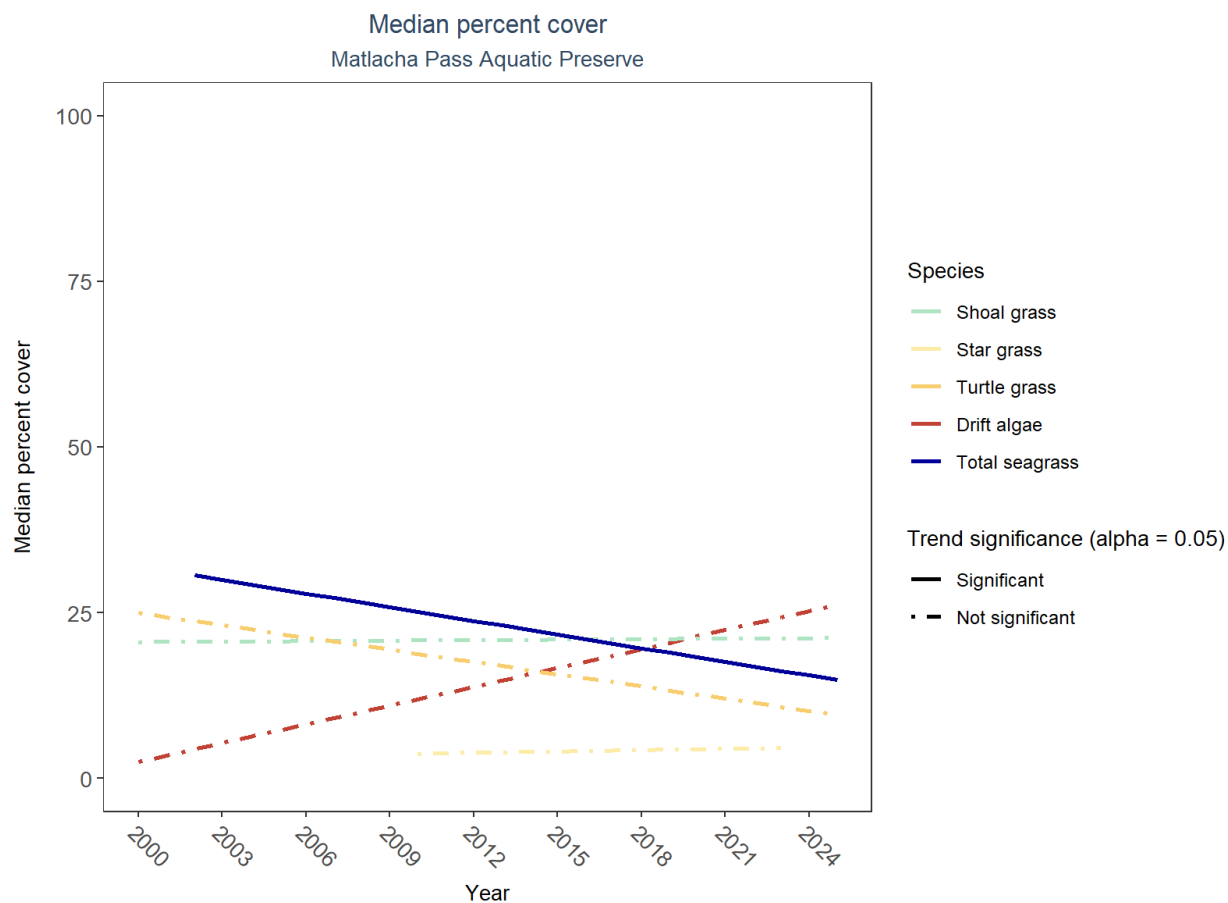


Table 19: SAV LME Results for Matlacha Pass Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Attached algae	Insufficient data	2021 - 2021	-	-	-
Drift algae	No detectable trend	2000 - 2025	-3.10	0.94	0.12
Shoal grass	No detectable trend	2000 - 2025	20.45	0.02	0.96
Paddle grass	Insufficient data	2007 - 2023	-	-	-
Star grass	No detectable trend	2010 - 2023	2.64	0.07	0.78
Widgeon grass	Insufficient data	2006 - 2018	-	-	-
Turtle grass	No detectable trend	2000 - 2025	28.66	-0.62	0.05
Total seagrass	Decreasing trend	2002 - 2025	36.12	-0.69	0.03
Halophila, unk.	Insufficient data	2007 - 2014	-	-	-

An annual decrease in percent cover was observed for total seagrass (-0.69%). Shoal grass, star grass, turtle grass, and drift algae showed no detectable change in percent cover. Trends in percent cover could not be evaluated for unknown *Halophila*, paddle grass, widgeon grass, and attached algae due to insufficient data.



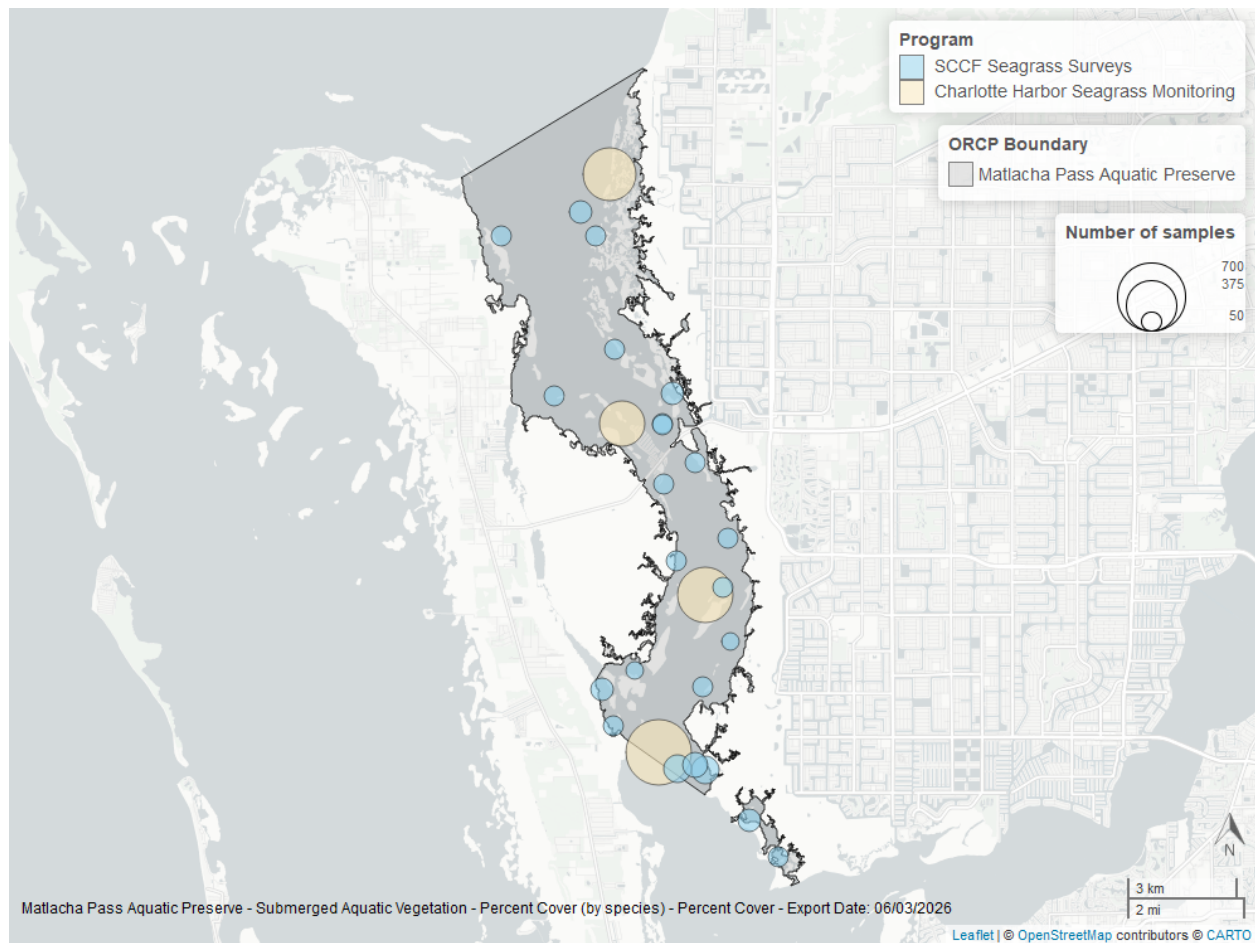
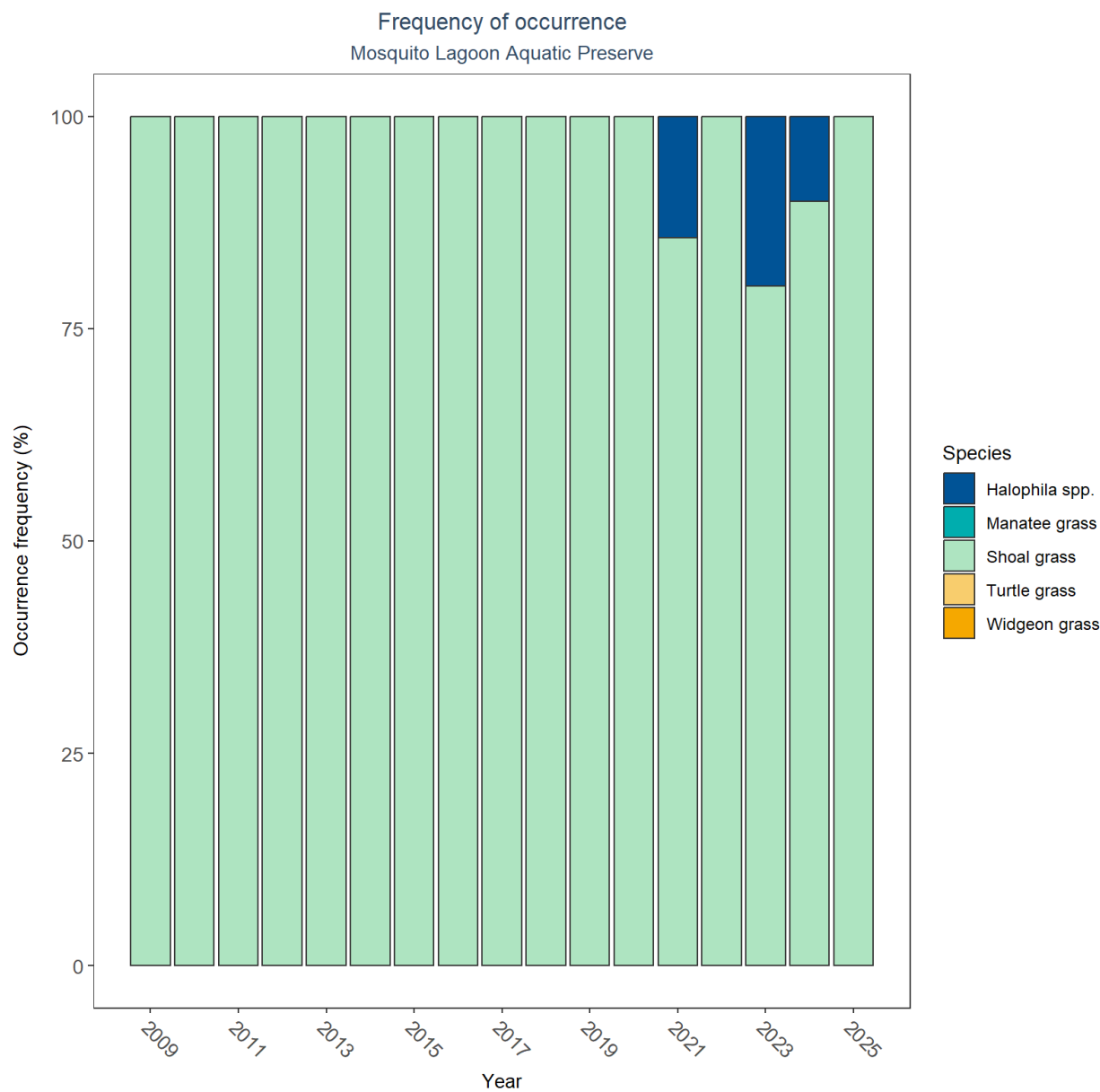


Figure 20: Map showing SAV sampling sites within the boundaries of *Matlacha Pass Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Mosquito Lagoon Aquatic Preserve



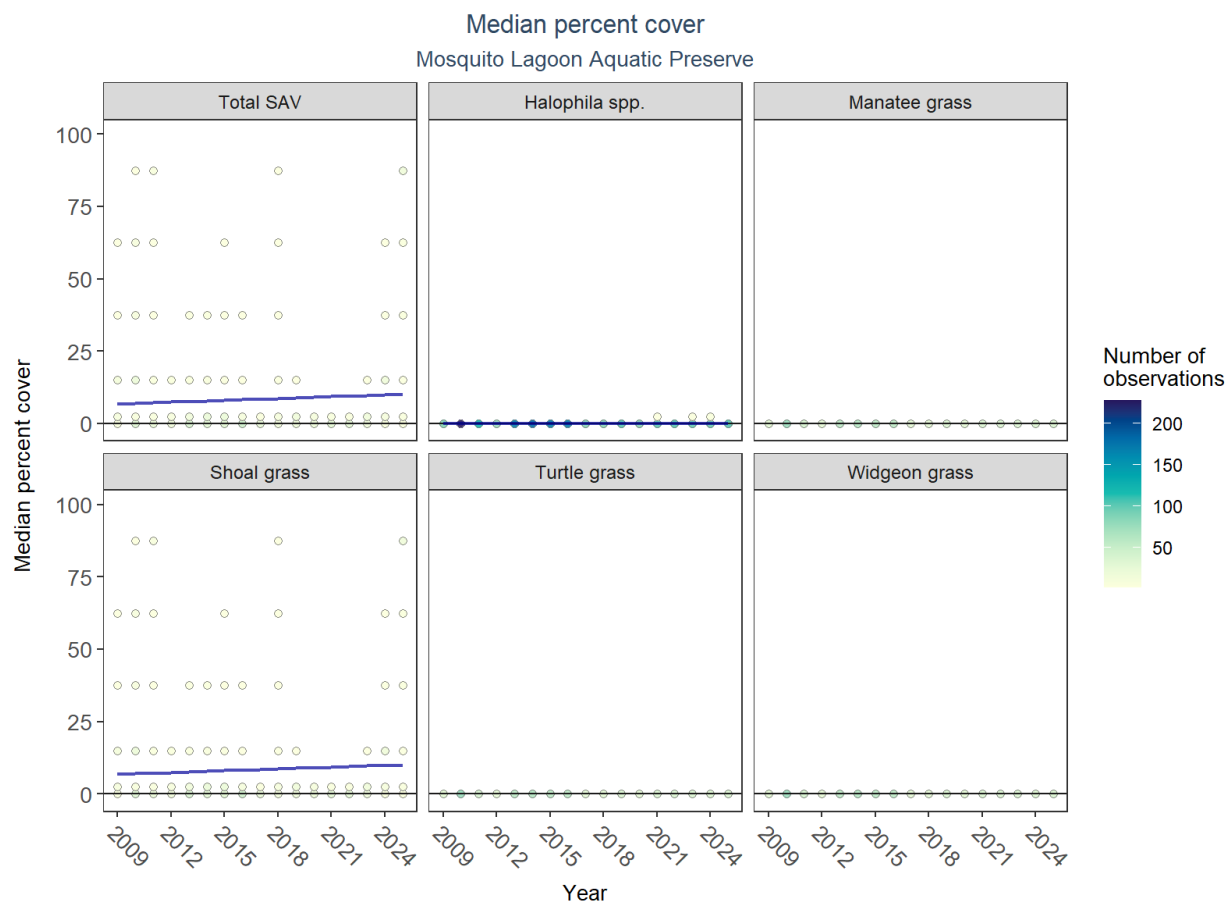
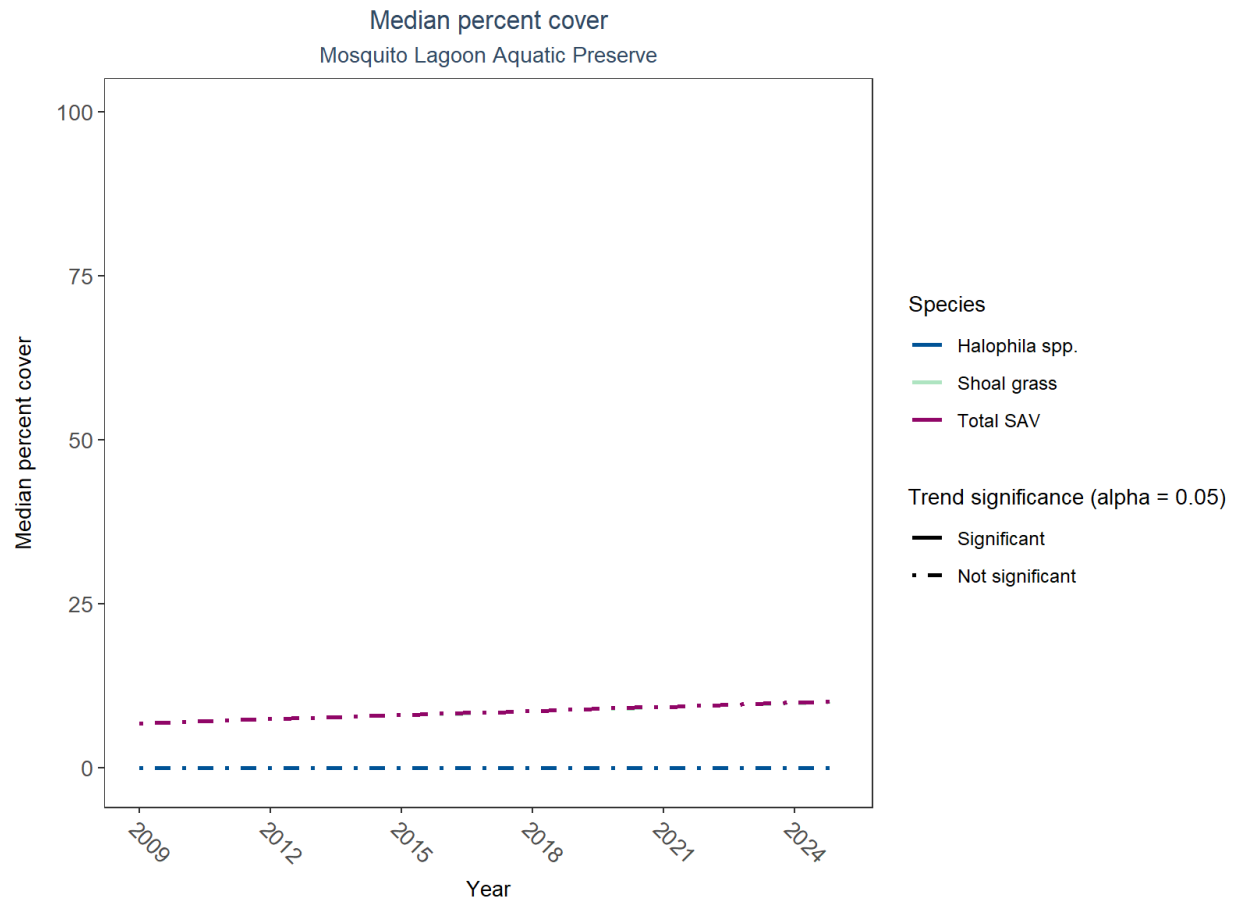


Table 20: SAV LME Results for Mosquito Lagoon Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Shoal grass	No detectable trend	2009 - 2025	3.71	0.21	0.24
Halophila spp.	No detectable trend	2009 - 2025	-0.05	0.00	0.16
Widgeon grass	Model did not fit the available data	2009 - 2025	-	-	-
Manatee grass	Model did not fit the available data	2009 - 2025	-	-	-
Turtle grass	Model did not fit the available data	2009 - 2025	-	-	-
Total SAV	No detectable trend	2009 - 2025	3.67	0.21	0.24

Total SAV, *Halophila* spp., and shoal grass showed no detectable change in percent cover. A model could not be fitted for manatee grass, turtle grass, and widgeon grass.



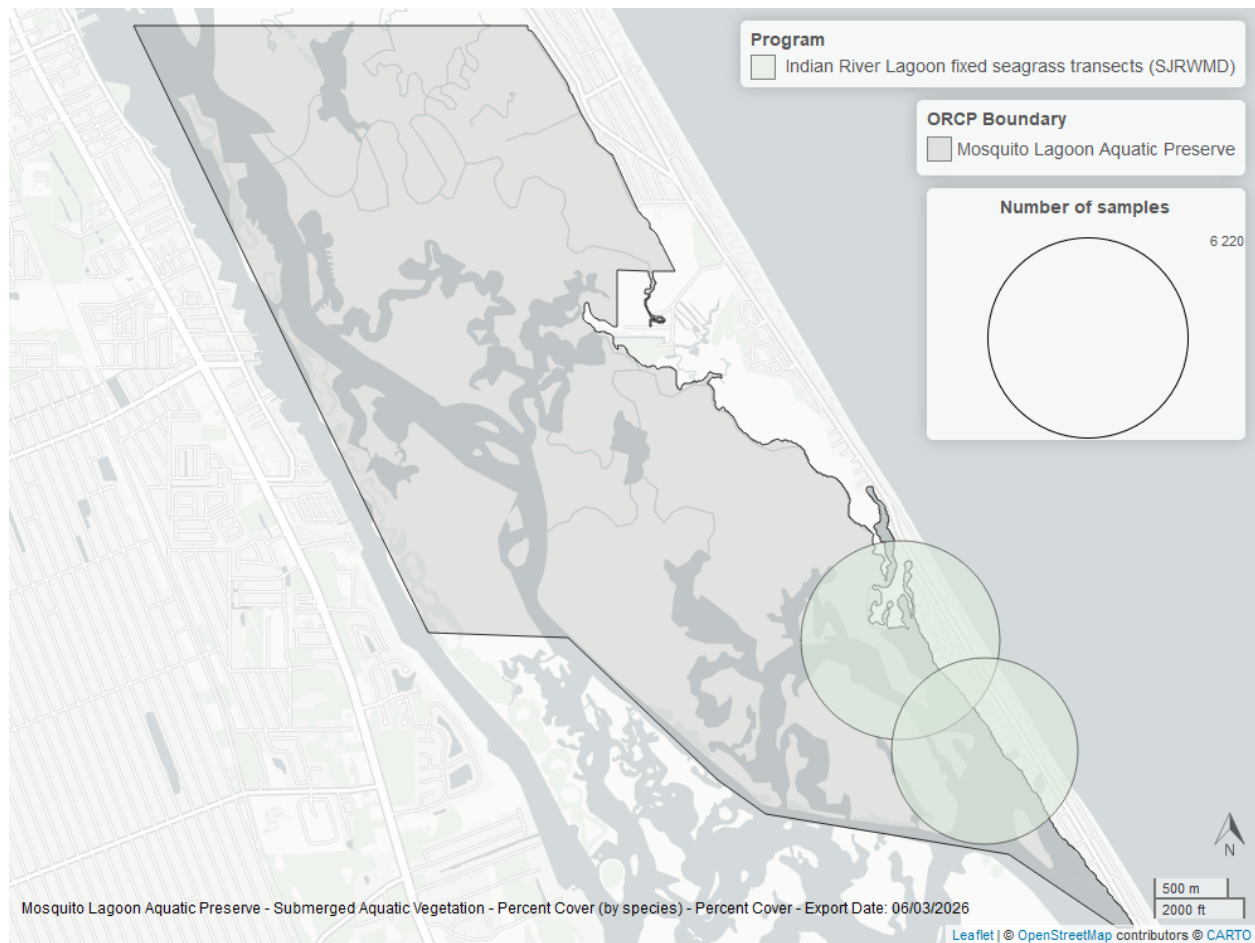
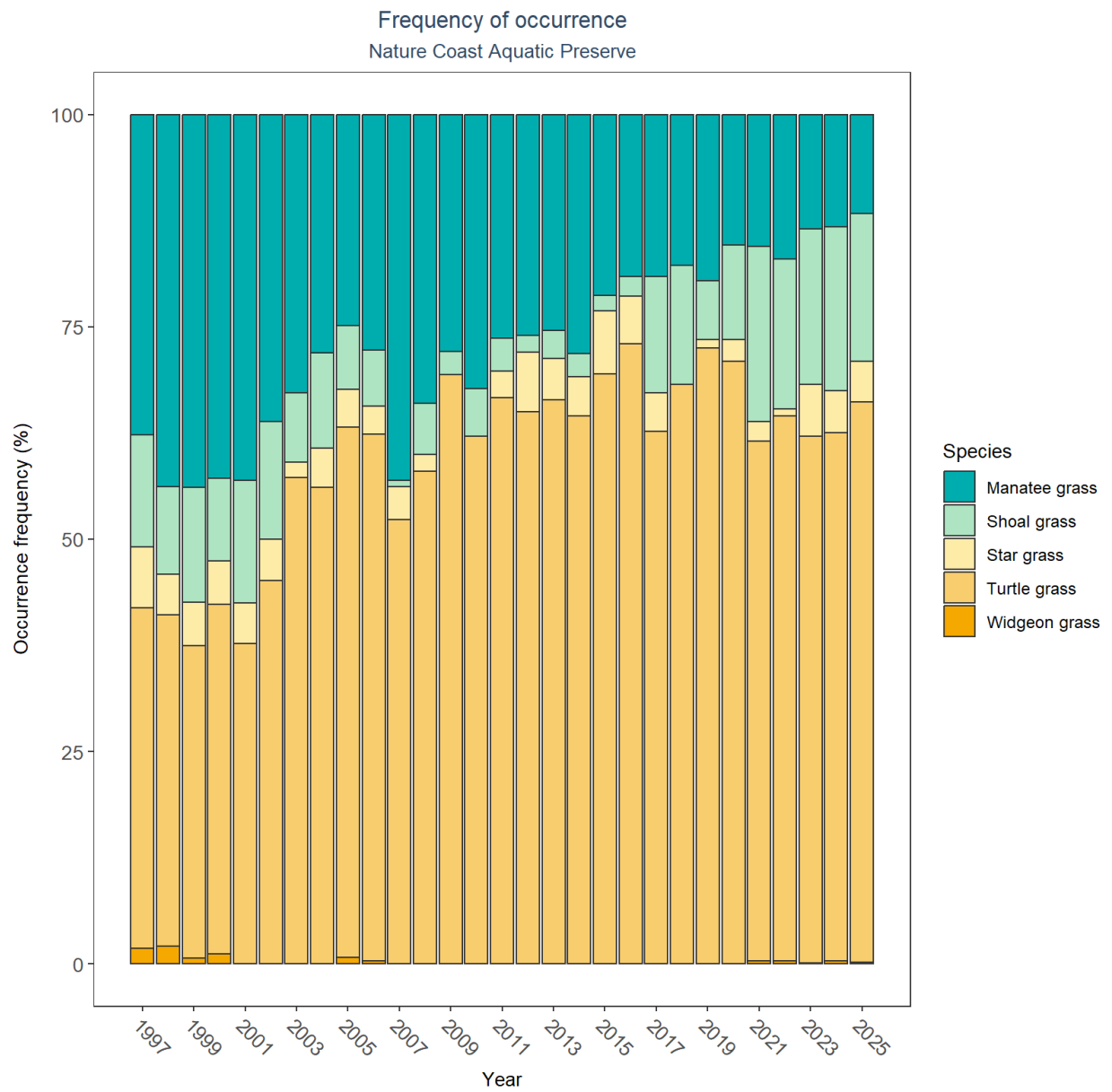


Figure 21: Map showing SAV sampling sites within the boundaries of *Mosquito Lagoon Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Nature Coast Aquatic Preserve



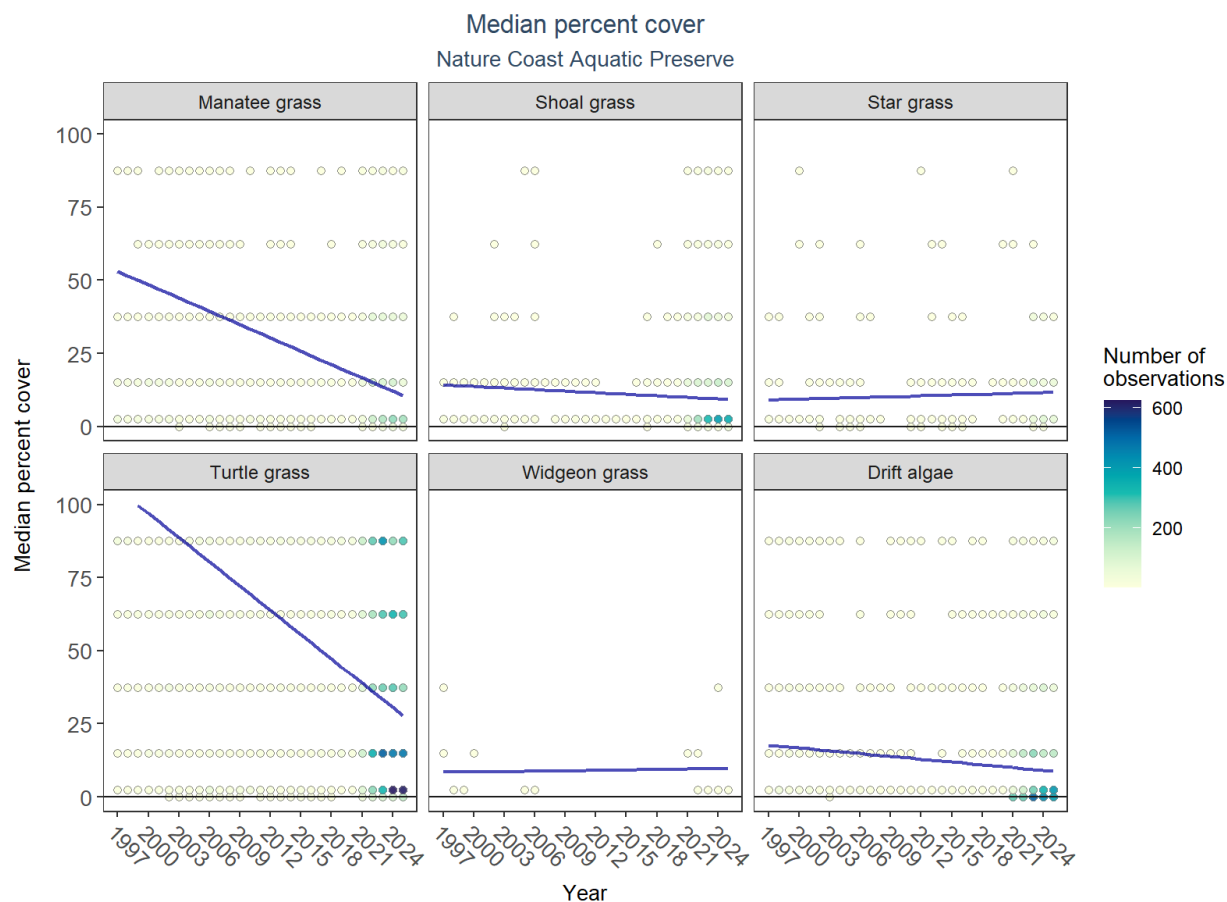
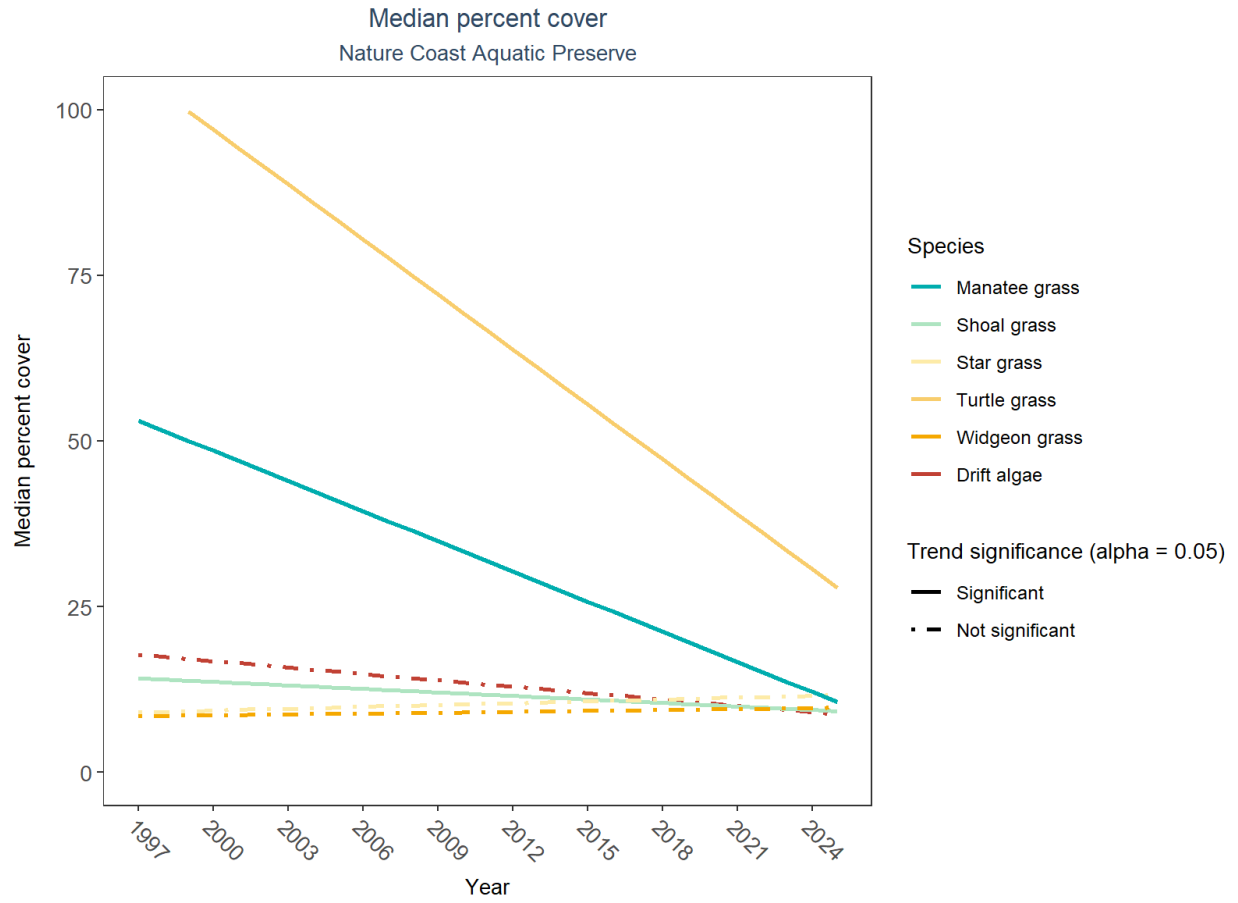


Table 21: SAV LME Results for Nature Coast Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Drift algae	No detectable trend	1997 - 2025	18.72	-0.32	0.42
Shoal grass	Decreasing trend	1997 - 2025	14.75	-0.18	0.02
Star grass	No detectable trend	1997 - 2025	8.79	0.09	0.39
Widgeon grass	No detectable trend	1997 - 2025	8.39	0.04	0.97
Manatee grass	Decreasing trend	1997 - 2025	57.63	-1.52	0.00
Turtle grass	Decreasing trend	1997 - 2025	113.58	-2.76	0.00

Annual decreases in percent cover were observed for manatee grass (-1.52%), shoal grass (-0.18%), and turtle grass (-2.76%). Star grass, widgeon grass, and drift algae showed no detectable change in percent cover.



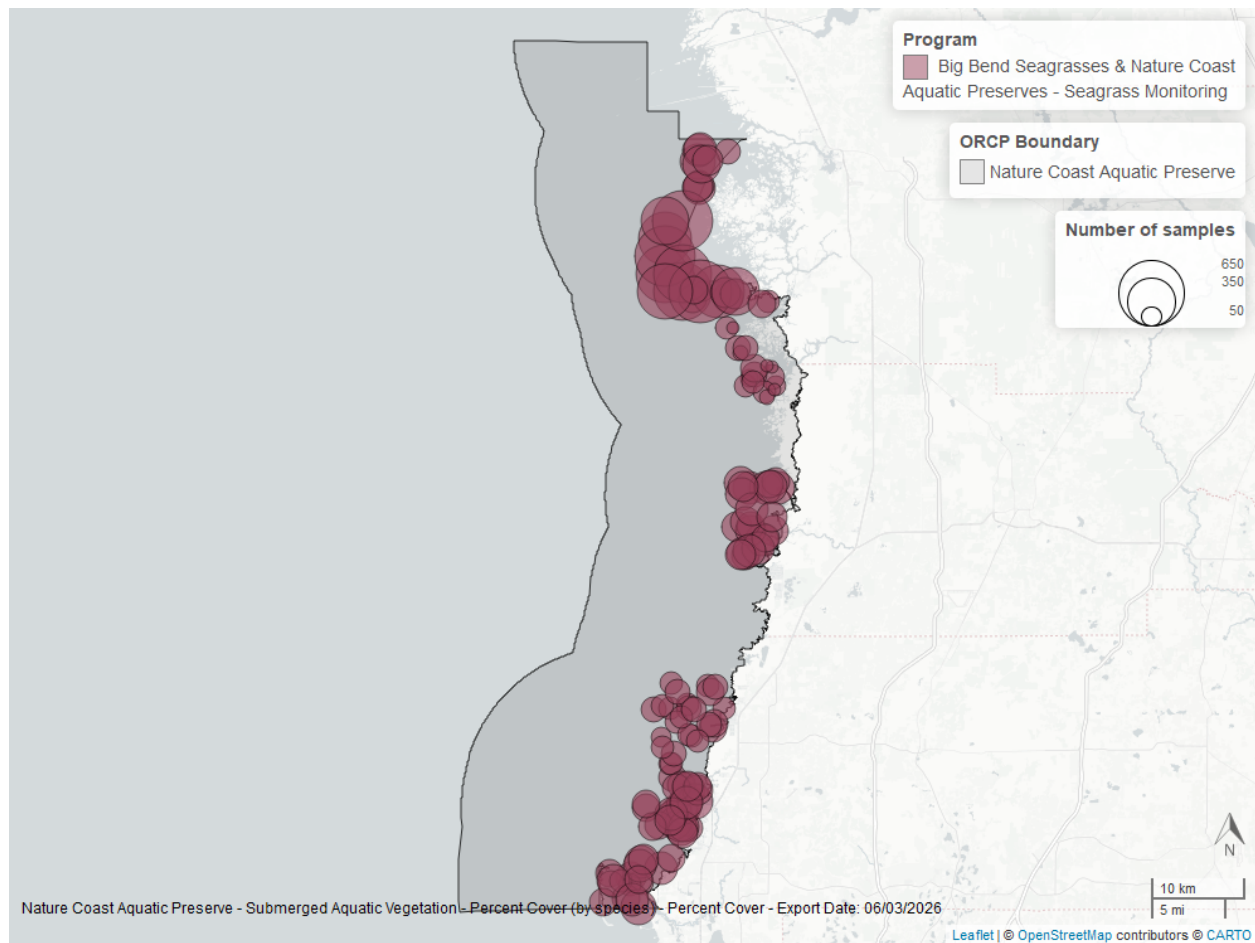
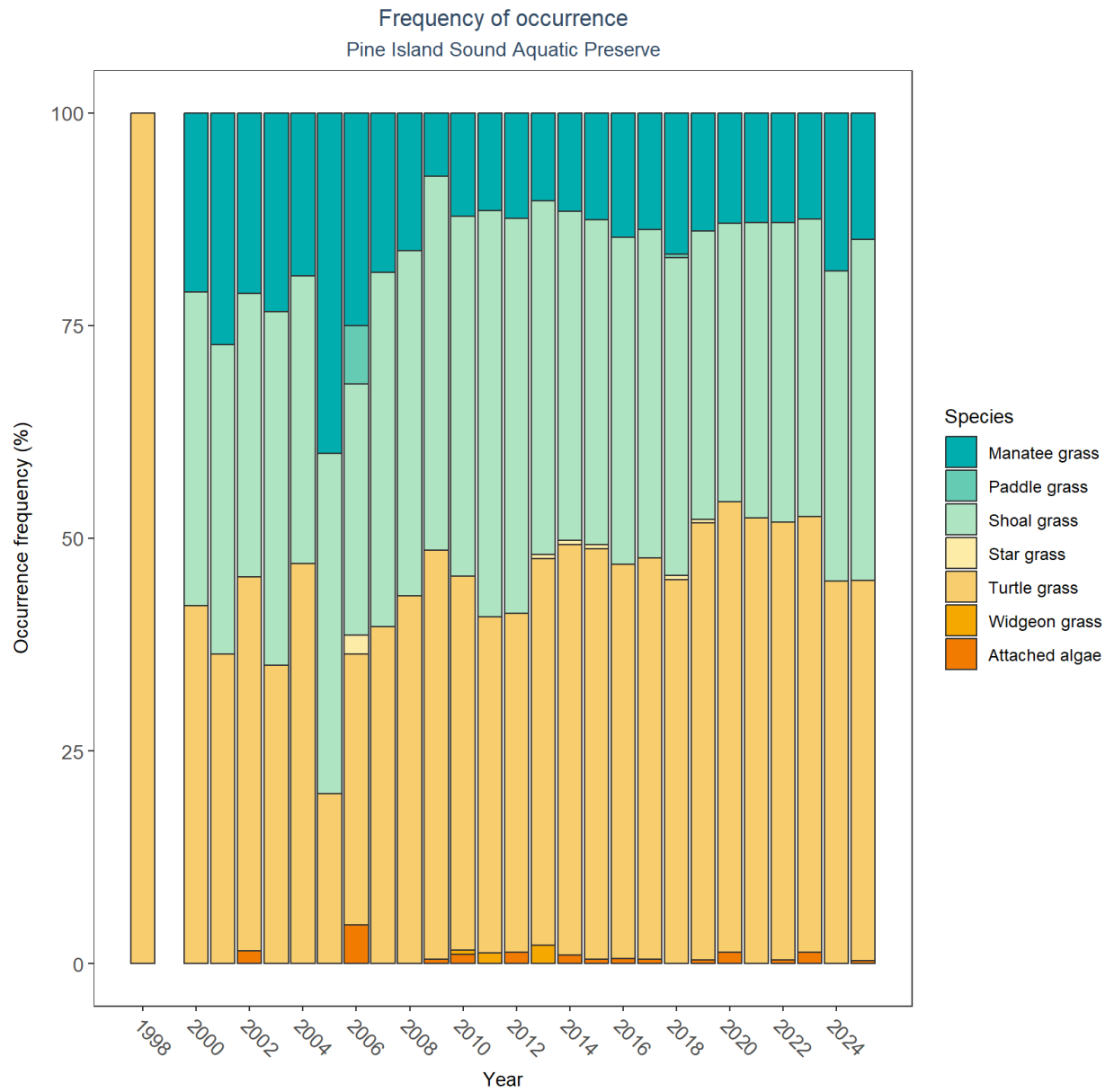


Figure 22: Map showing SAV sampling sites within the boundaries of *Nature Coast Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Pine Island Sound Aquatic Preserve



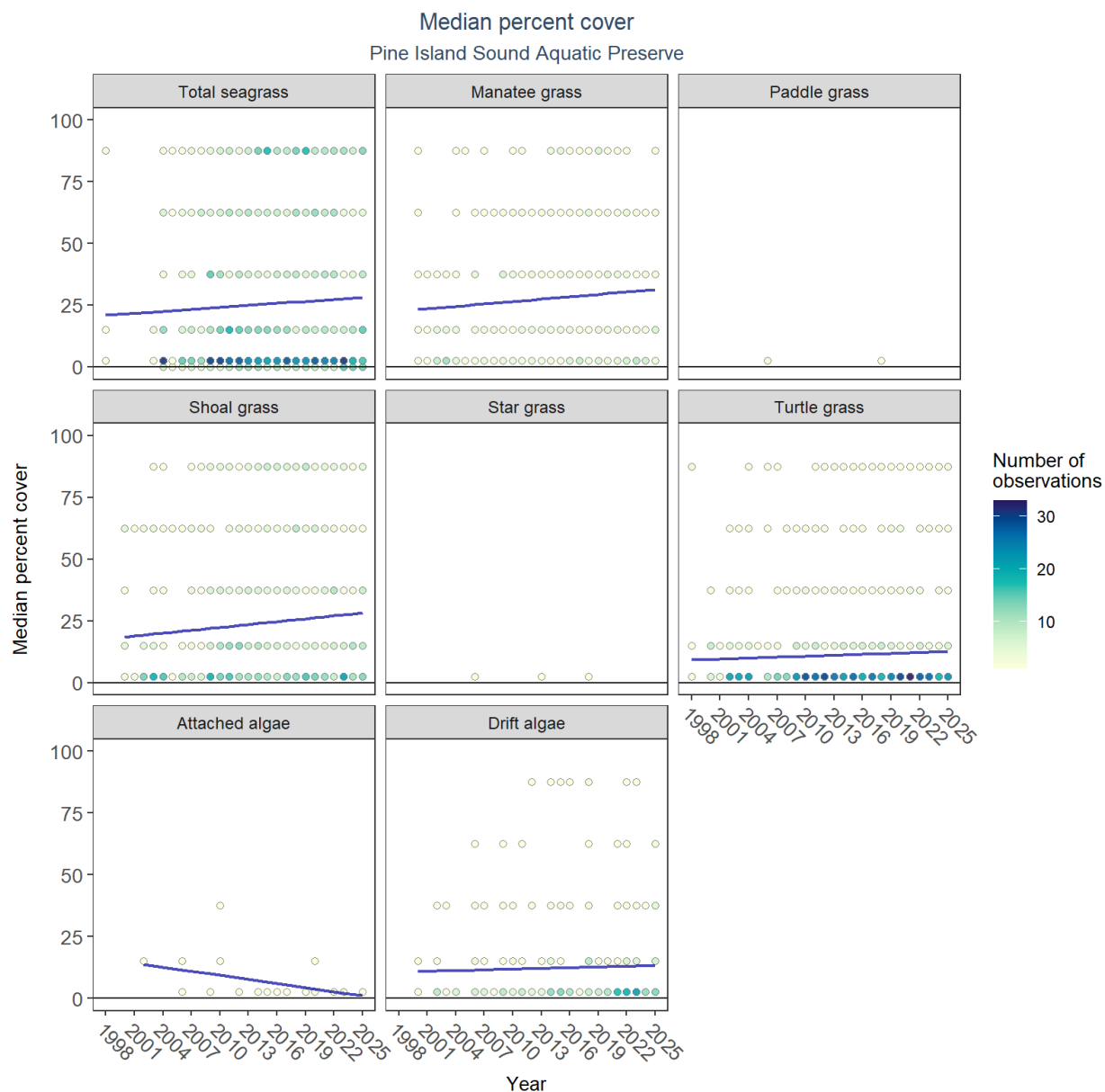
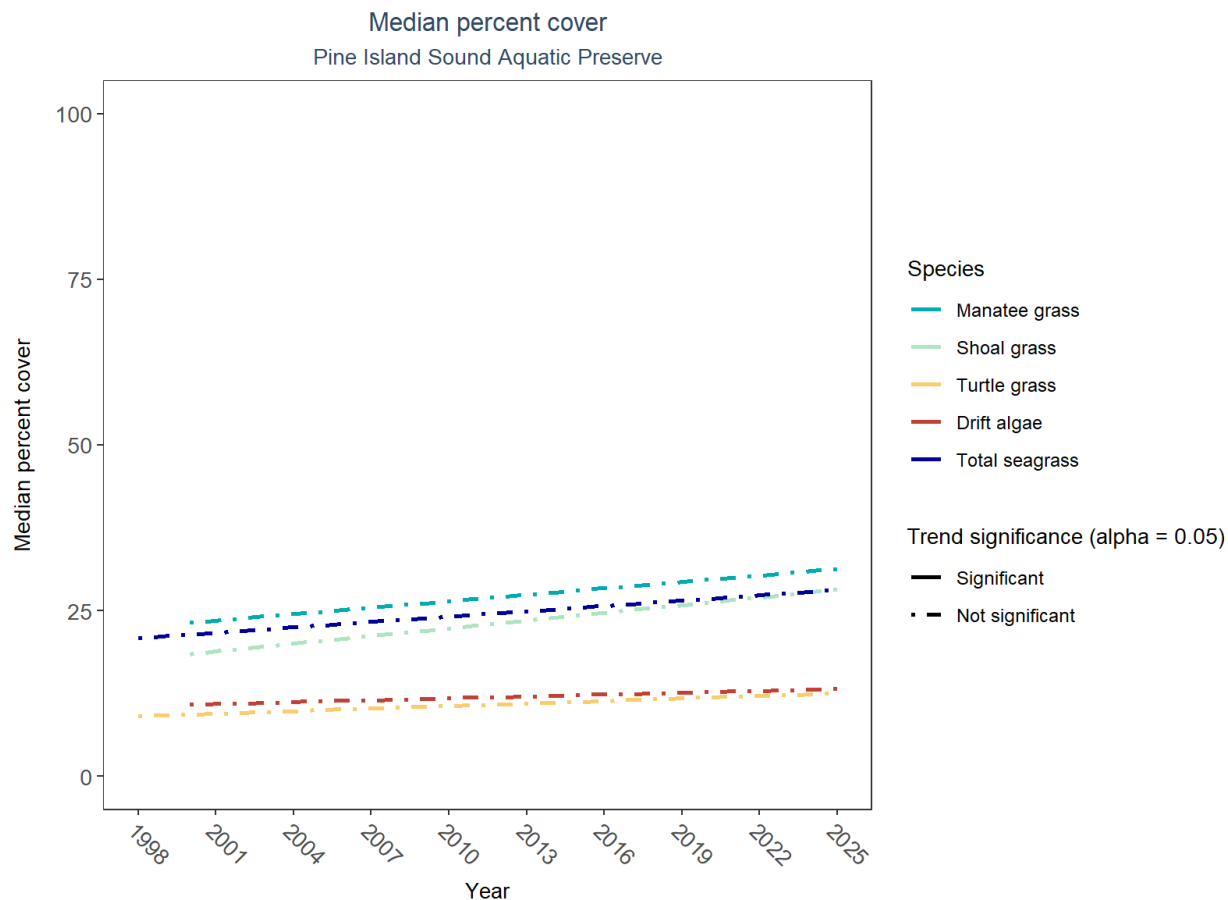


Table 22: SAV LME Results for Pine Island Sound Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Attached algae	No detectable trend	2002 - 2025	18.16	-0.55	0.05
Drift algae	No detectable trend	2000 - 2025	10.27	0.10	0.55
Shoal grass	No detectable trend	2000 - 2025	16.13	0.39	0.25
Paddle grass	Insufficient data	2006 - 2018	-	-	-
Star grass	Insufficient data	2006 - 2018	-	-	-
Manatee grass	No detectable trend	2000 - 2025	21.26	0.32	0.30
Turtle grass	No detectable trend	1998 - 2025	8.60	0.13	0.38
Total seagrass	No detectable trend	1998 - 2025	19.82	0.27	0.37

Total seagrass, manatee grass, shoal grass, turtle grass, attached algae, and drift algae showed no detectable change in percent cover. Trends in percent cover could not be evaluated for paddle grass and star grass due to insufficient data.



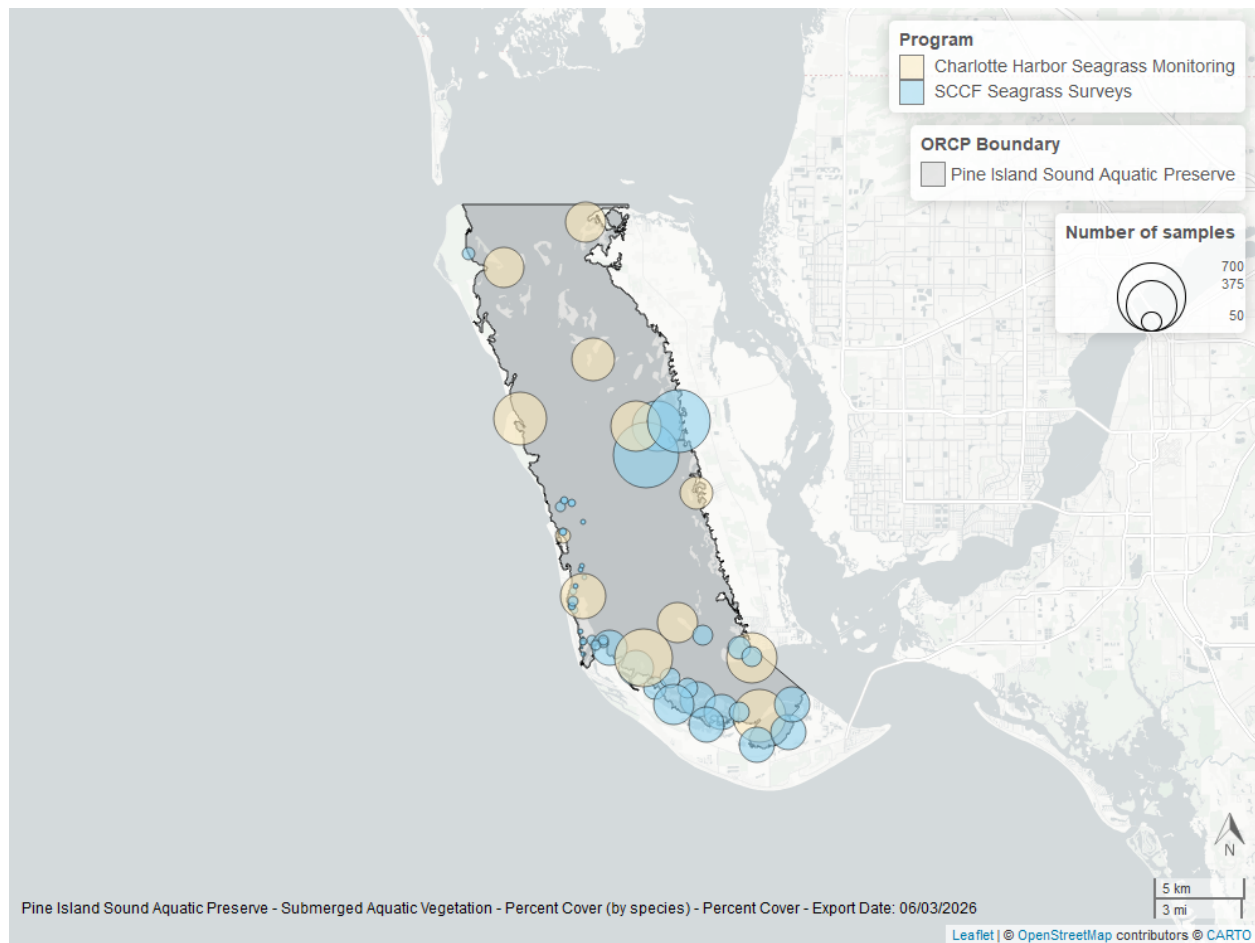
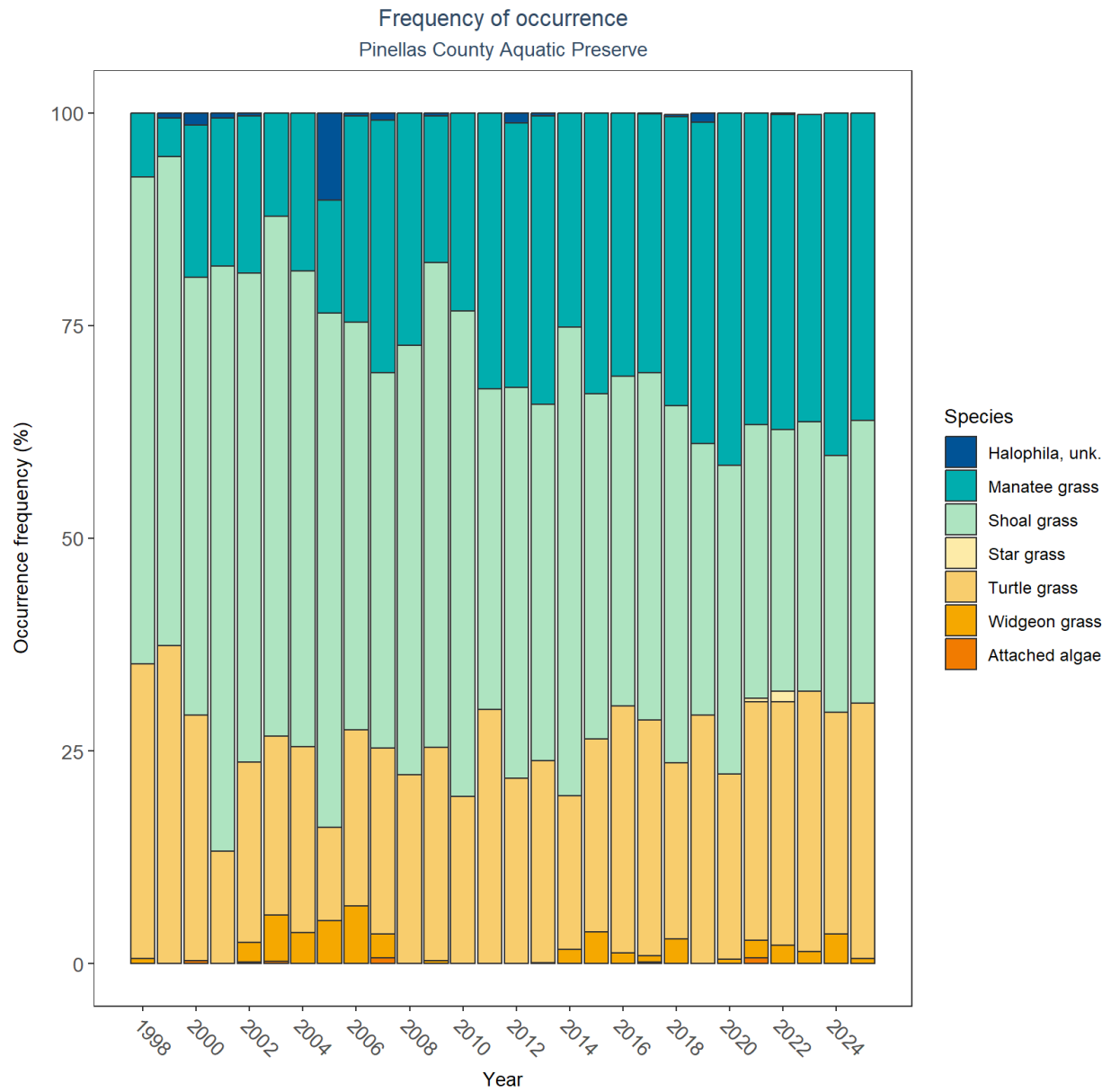


Figure 23: Map showing SAV sampling sites within the boundaries of *Pine Island Sound Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Pinellas County Aquatic Preserve



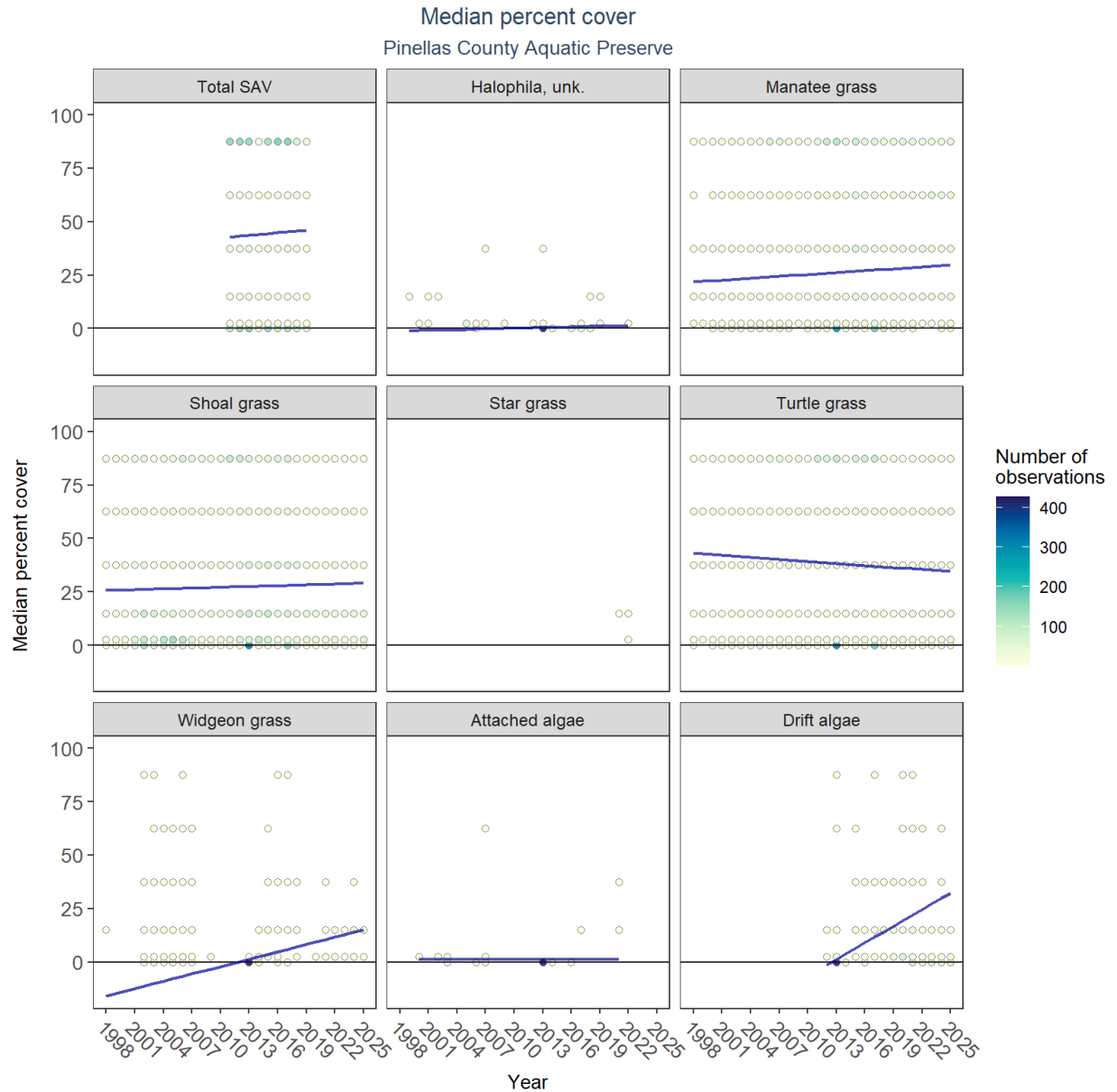
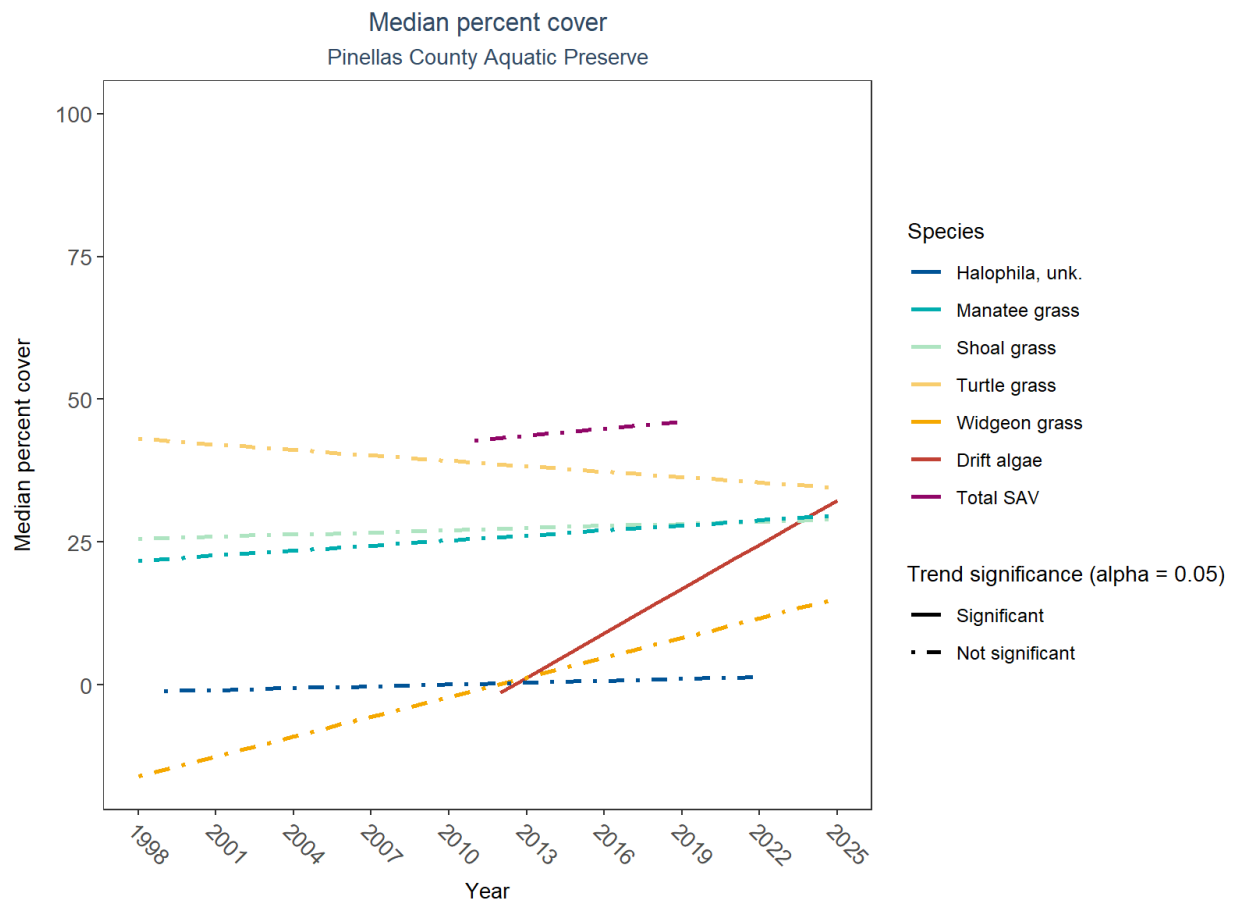


Table 23: SAV LME Results for Pinellas County Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Attached algae	No detectable trend	2000 - 2021	1.35	-0.01	0.96
Drift algae	Increasing trend	2012 - 2025	-48.03	2.59	0.00
Shoal grass	No detectable trend	1998 - 2025	25.08	0.13	0.32
Star grass	Insufficient data	2021 - 2022	-	-	-
Widgeon grass	No detectable trend	1998 - 2025	-20.51	1.15	0.37
Manatee grass	No detectable trend	1998 - 2025	20.60	0.29	0.14
Turtle grass	No detectable trend	1998 - 2025	44.35	-0.32	0.05
Total SAV	No detectable trend	2011 - 2019	35.61	0.42	0.57
Halophila, unk.	No detectable trend	1999 - 2022	-1.64	0.11	0.66

An annual increase in percent cover was observed for drift algae (2.59%). Total SAV, unknown *Halophila*, manatee grass, shoal grass, turtle grass, widgeon grass, and attached algae showed no detectable change in percent cover. Trends in percent cover could not be evaluated for star grass due to insufficient data.



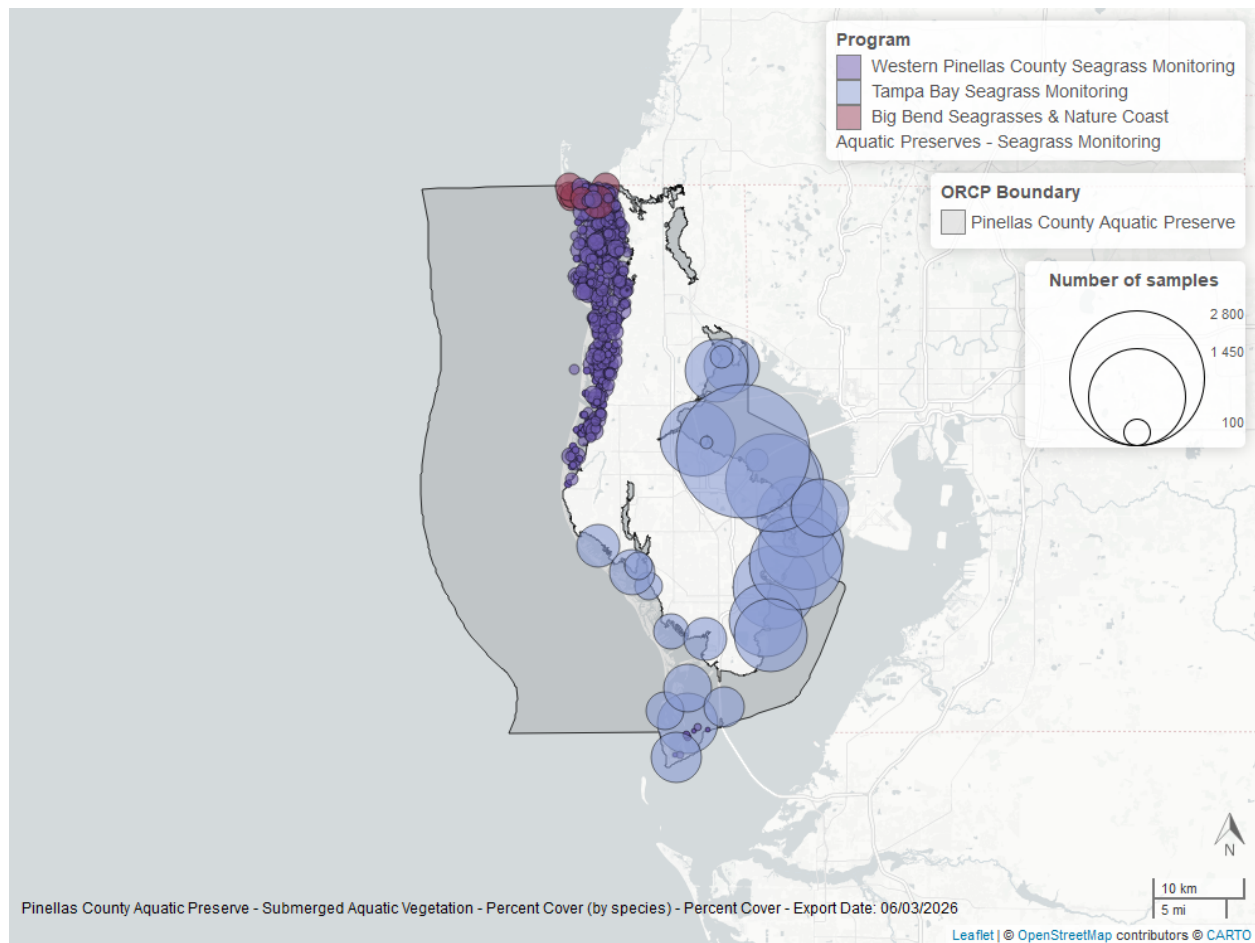
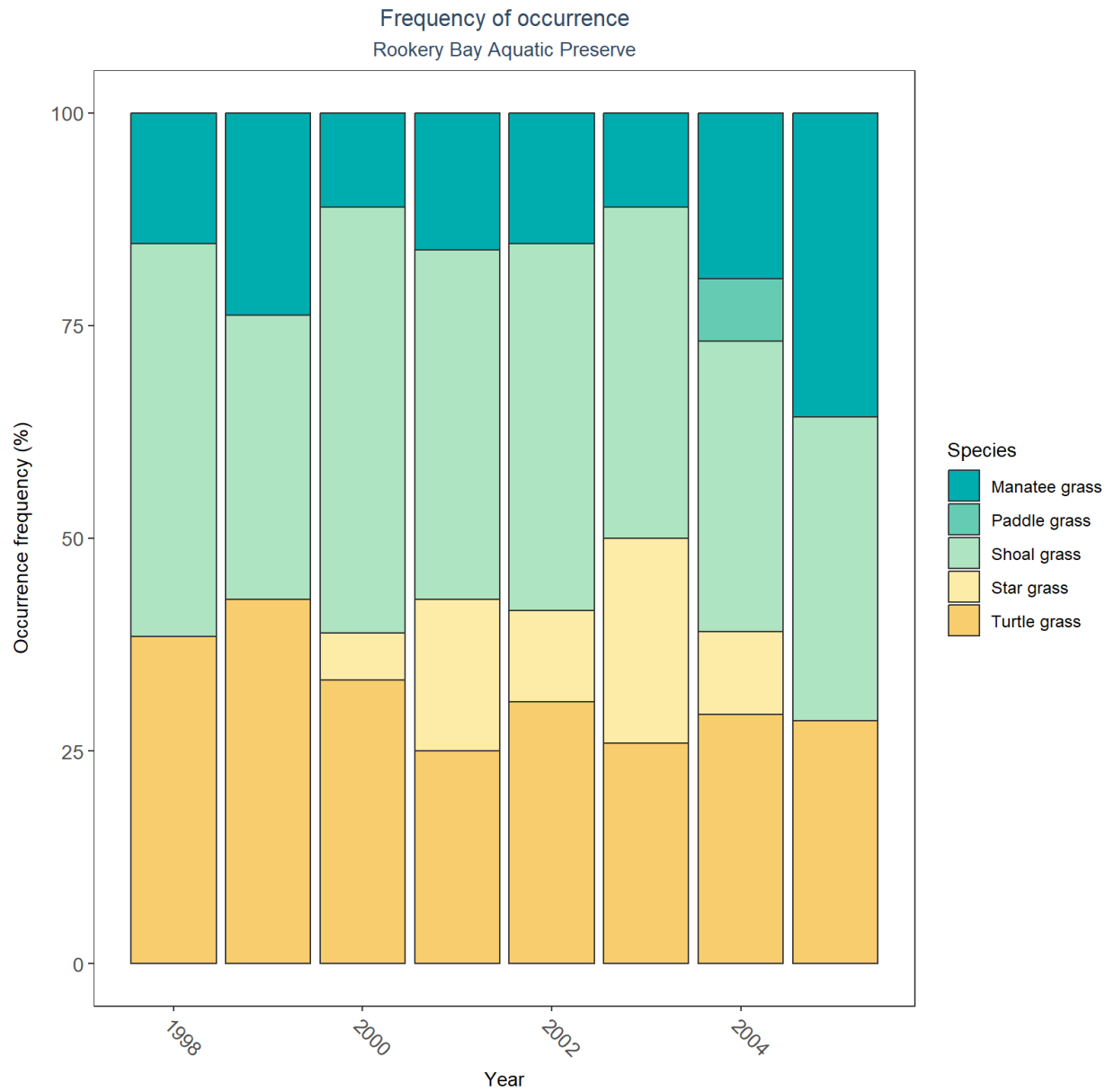


Figure 24: Map showing SAV sampling sites within the boundaries of *Pinellas County Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Rookery Bay Aquatic Preserve



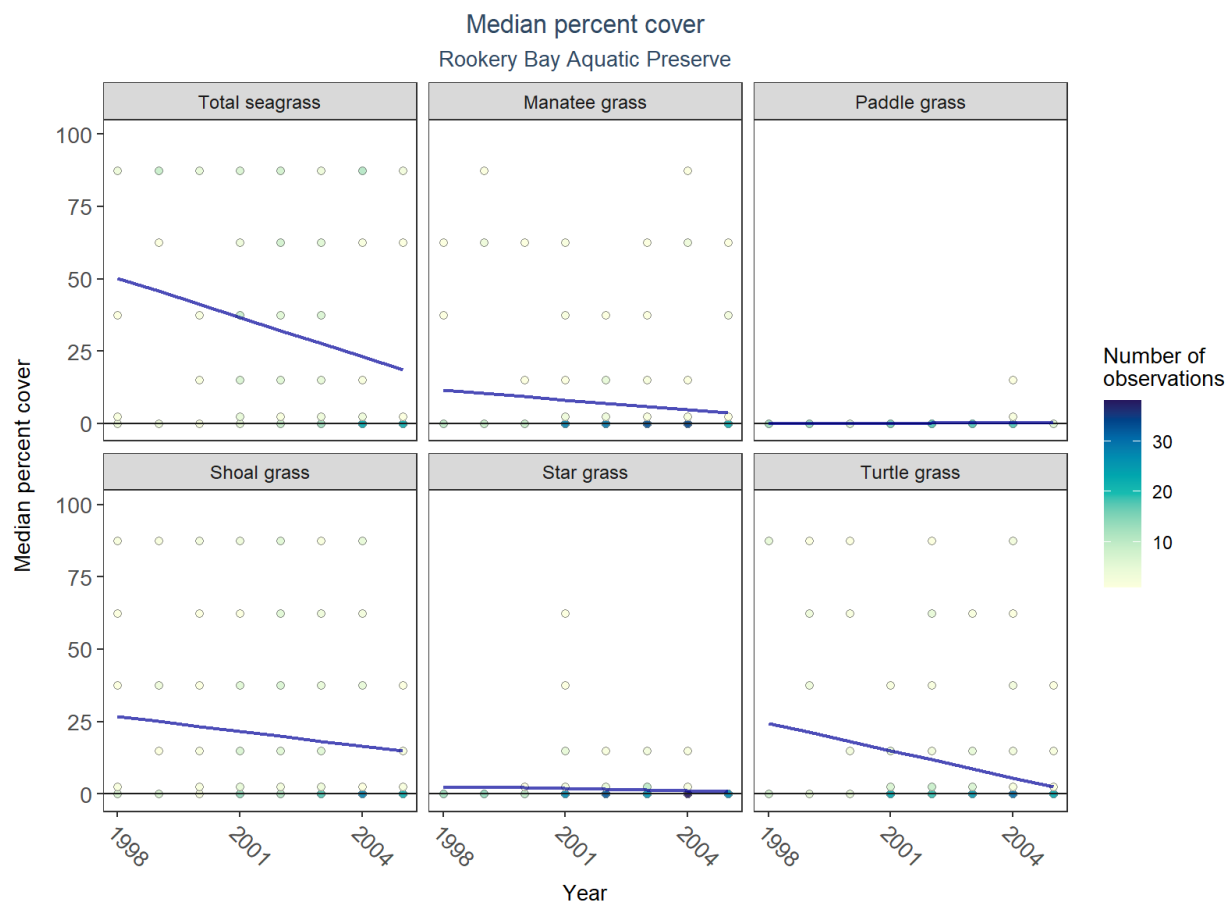
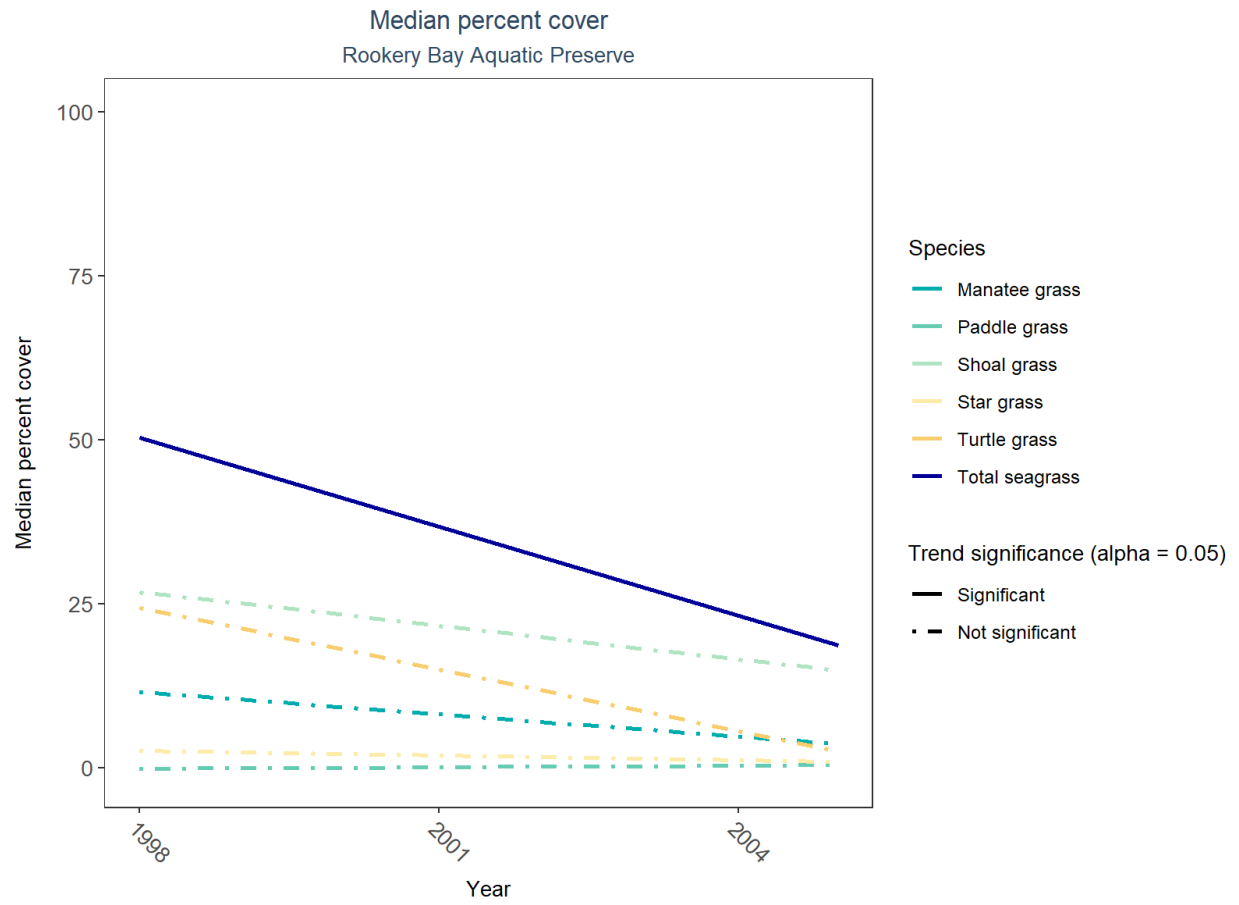


Table 24: SAV LME Results for Rookery Bay Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Shoal grass	No detectable trend	1998 - 2005	33.63	-1.71	0.25
Paddle grass	No detectable trend	1998 - 2005	-0.44	0.08	0.36
Star grass	No detectable trend	1998 - 2005	3.53	-0.23	0.44
Manatee grass	No detectable trend	1998 - 2005	16.10	-1.13	0.40
Turtle grass	No detectable trend	1998 - 2005	36.99	-3.14	0.07
Total seagrass	Decreasing trend	1998 - 2005	68.37	-4.52	0.02

An annual decrease in percent cover was observed for total seagrass (-4.52%). Manatee grass, paddle grass, shoal grass, star grass, and turtle grass showed no detectable change in percent cover.



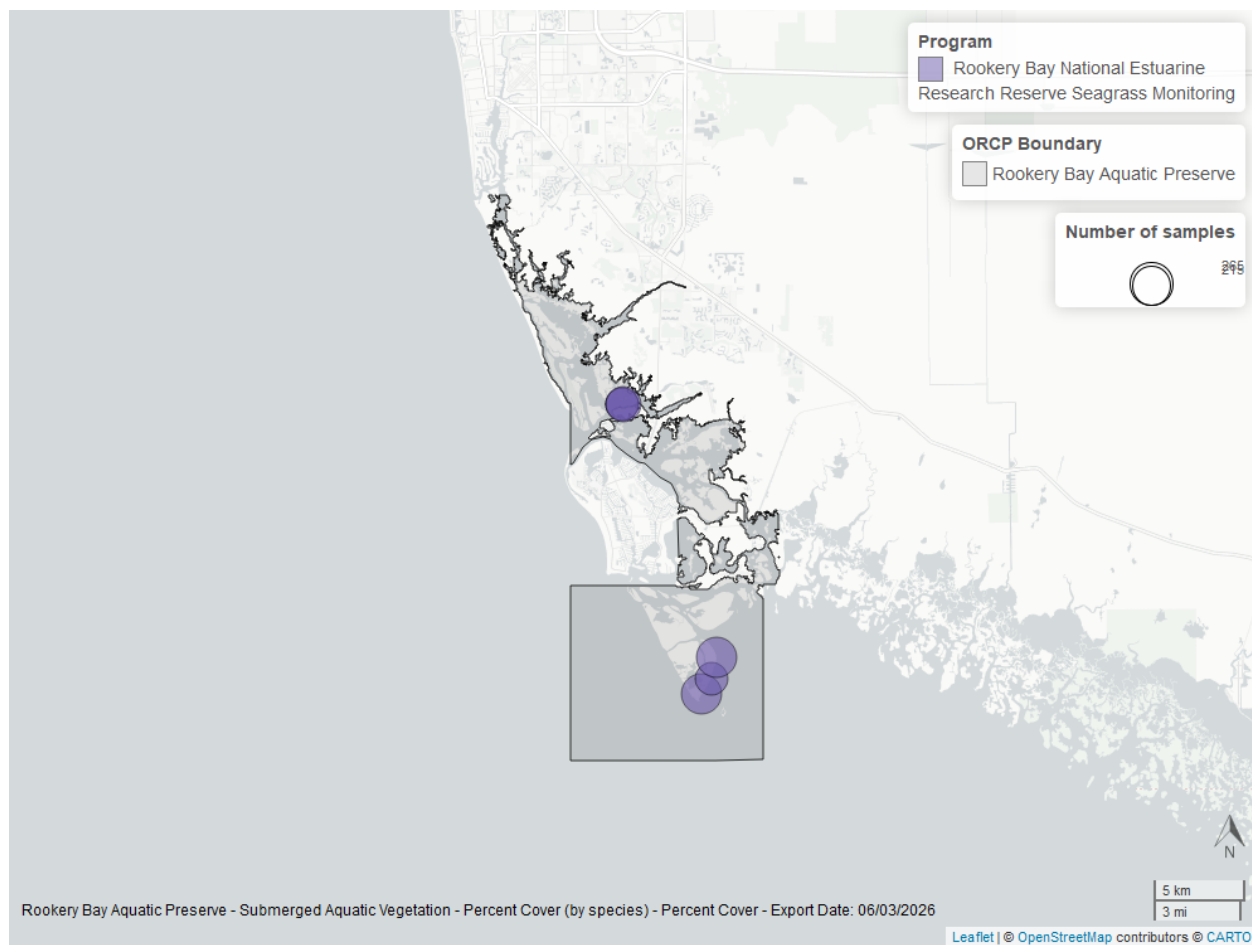
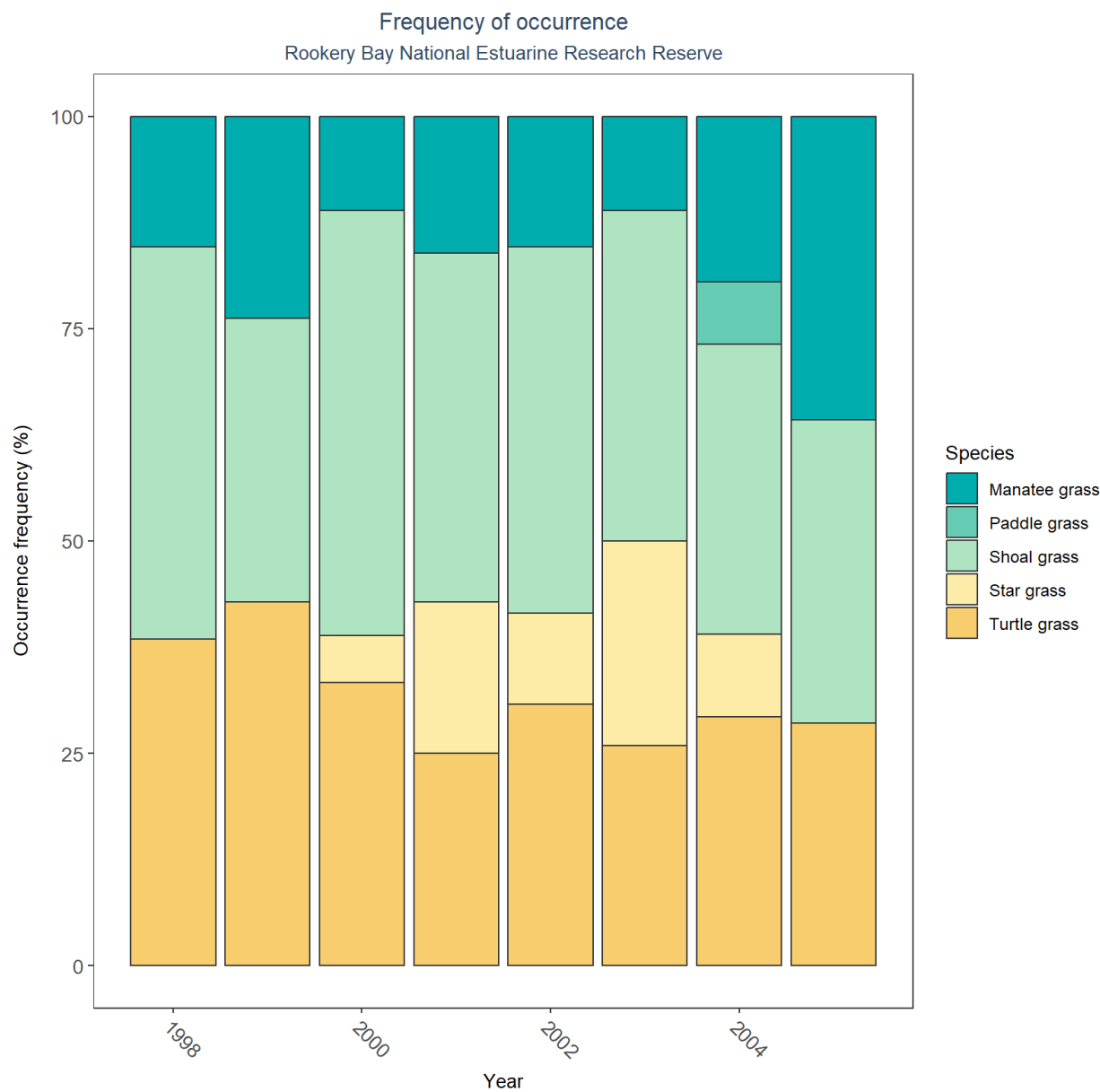


Figure 25: Map showing SAV sampling sites within the boundaries of *Rookery Bay Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Rookery Bay National Estuarine Research Reserve



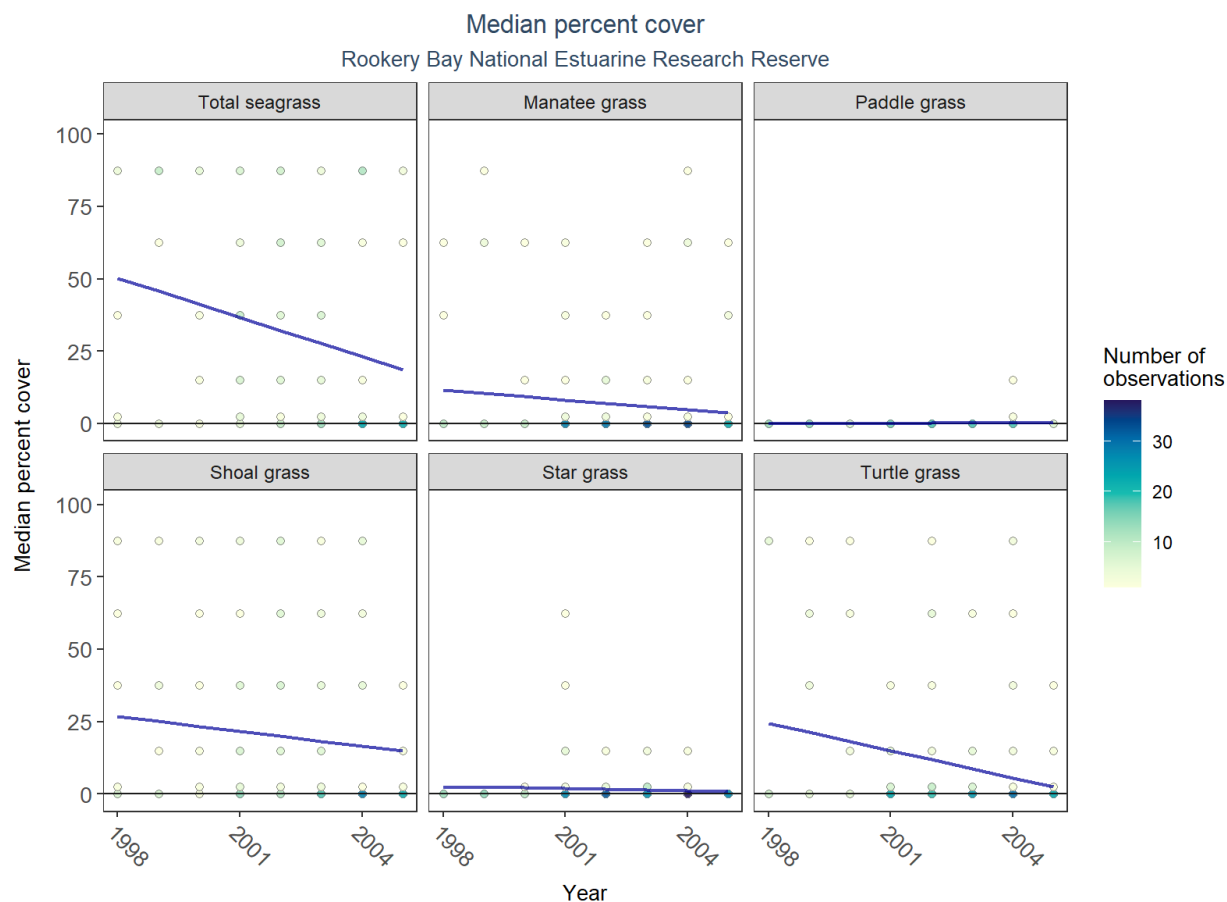
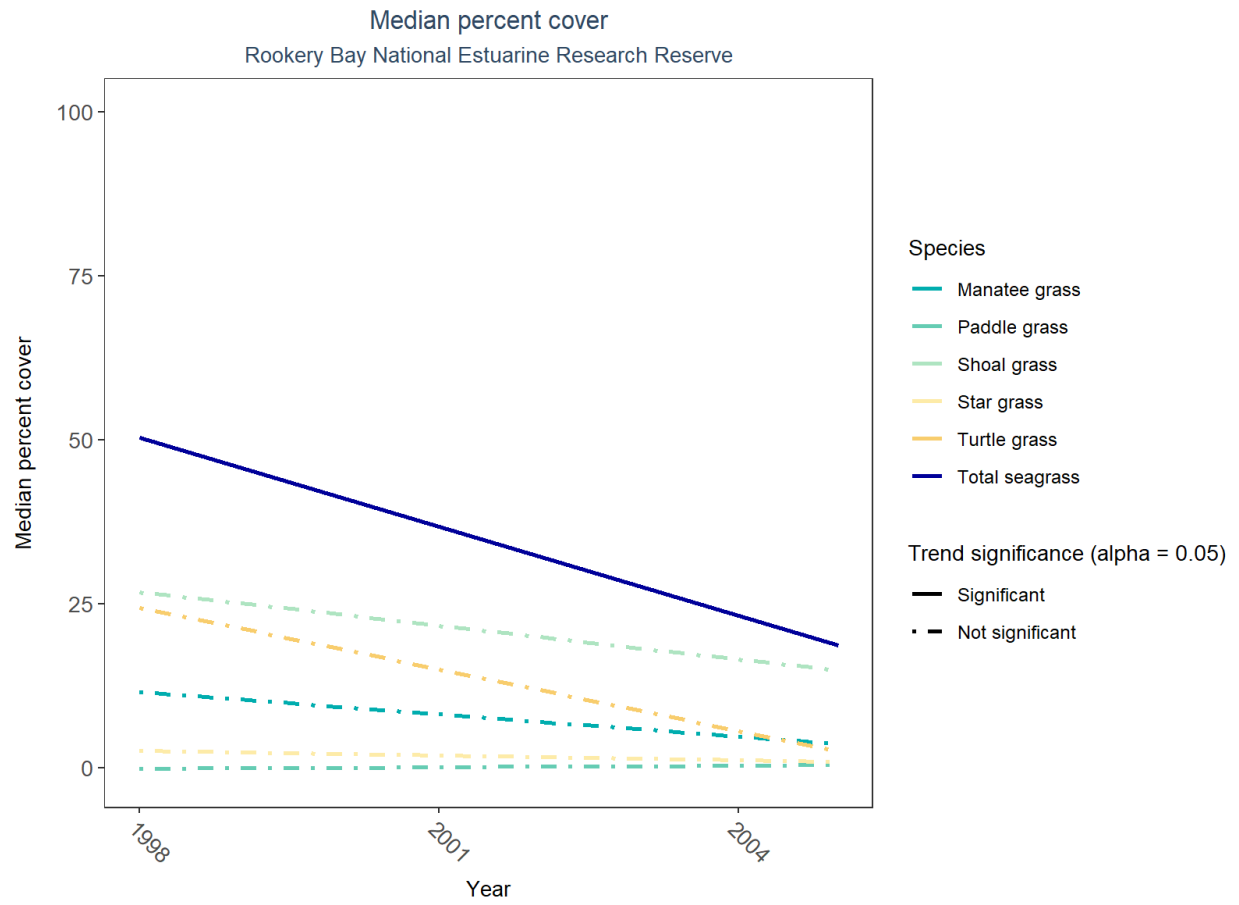


Table 25: SAV LME Results for Rookery Bay National Estuarine Research Reserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Shoal grass	No detectable trend	1998 - 2005	33.63	-1.71	0.25
Paddle grass	No detectable trend	1998 - 2005	-0.44	0.08	0.36
Star grass	No detectable trend	1998 - 2005	3.53	-0.23	0.44
Manatee grass	No detectable trend	1998 - 2005	16.10	-1.13	0.40
Turtle grass	No detectable trend	1998 - 2005	36.99	-3.14	0.07
Total seagrass	Decreasing trend	1998 - 2005	68.37	-4.52	0.02

An annual decrease in percent cover was observed for total seagrass (-4.52%). Manatee grass, paddle grass, shoal grass, star grass, and turtle grass showed no detectable change in percent cover.



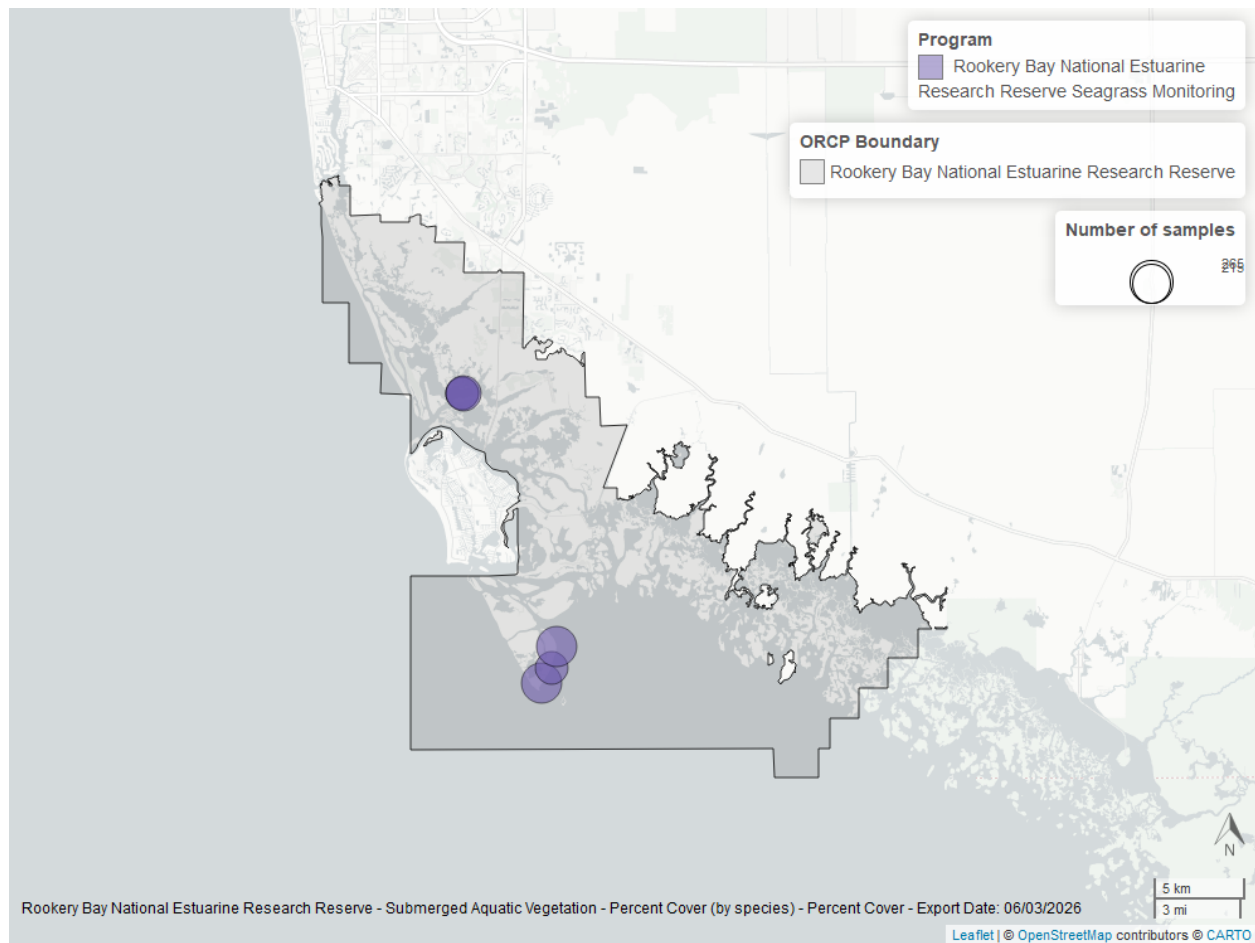
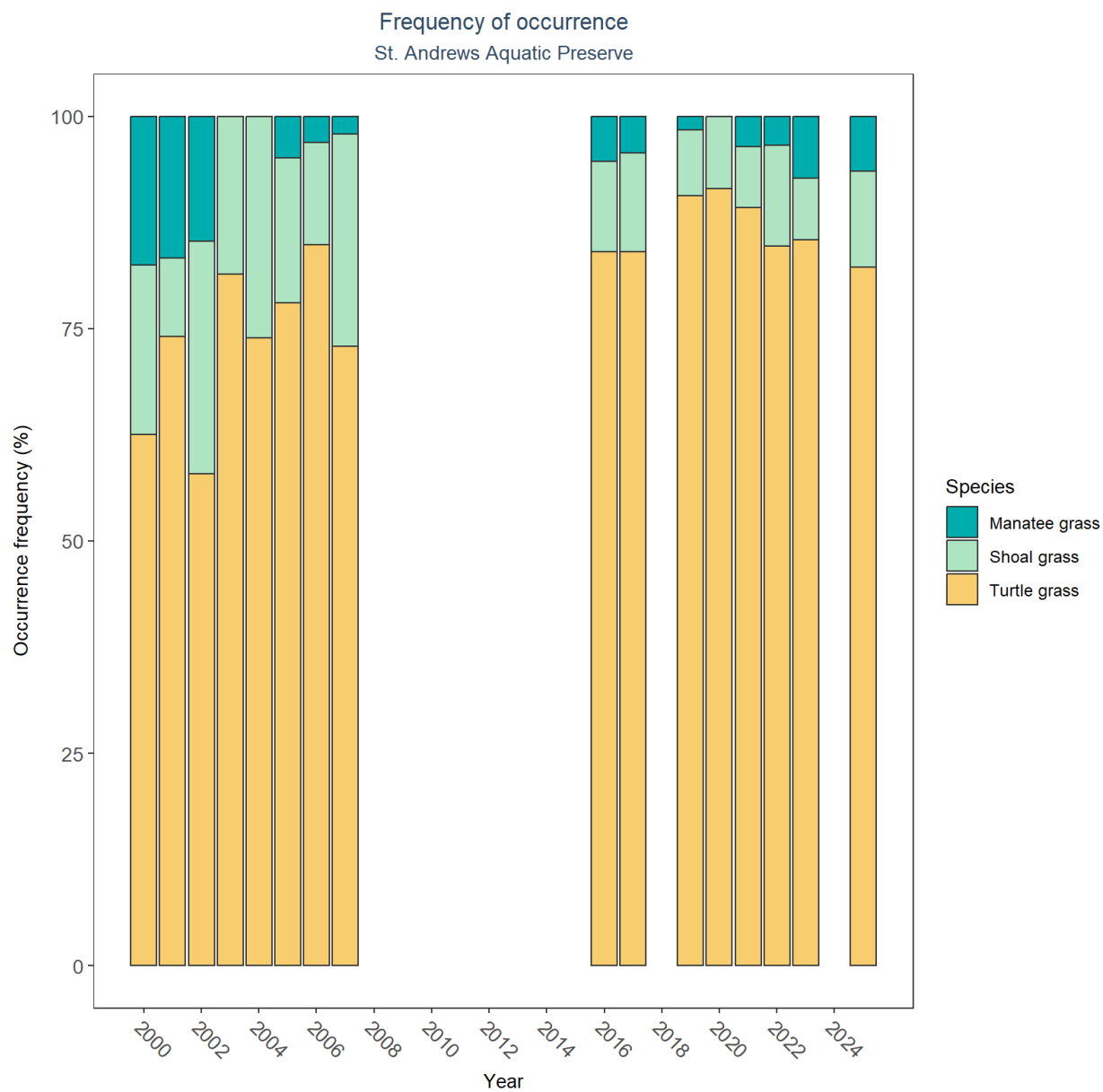


Figure 26: Map showing SAV sampling sites within the boundaries of *Rookery Bay National Estuarine Research Reserve*. The point size reflects the number of samples at a given sampling site.

St. Andrews Aquatic Preserve



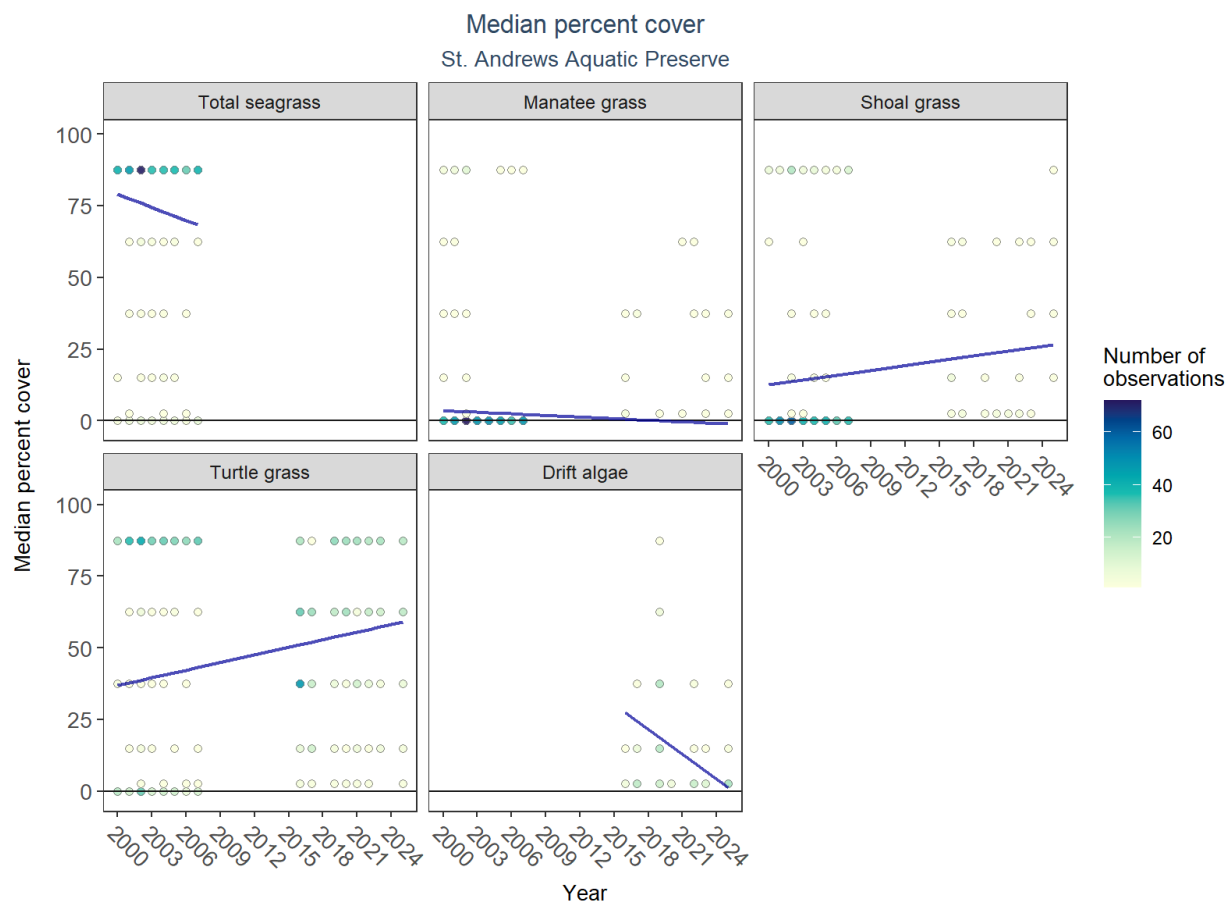
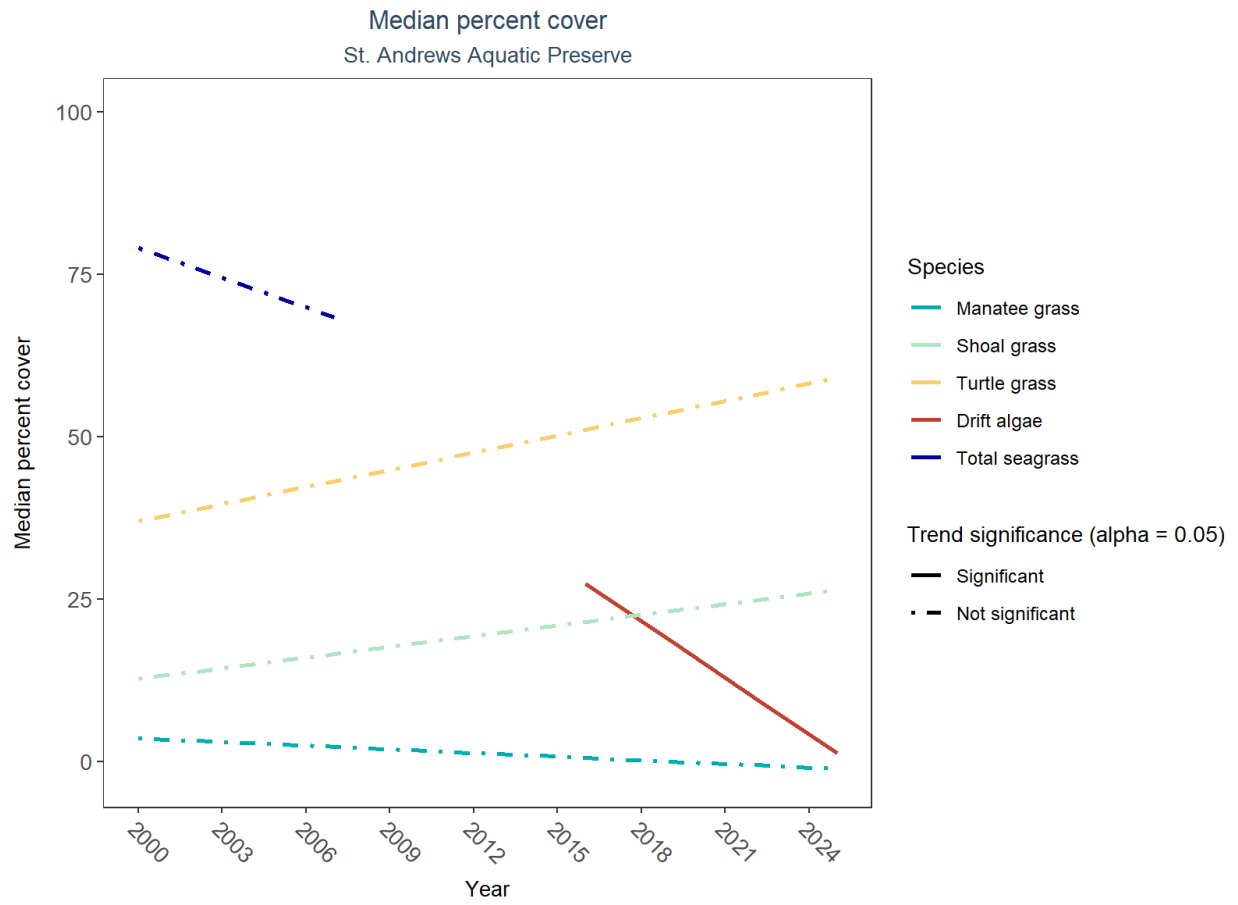


Table 26: SAV LME Results for St. Andrews Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Drift algae	Decreasing trend	2016 - 2025	91.06	-2.90	0.02
Shoal grass	No detectable trend	2000 - 2025	9.41	0.55	0.13
Manatee grass	No detectable trend	2000 - 2025	4.67	-0.19	0.77
Turtle grass	No detectable trend	2000 - 2025	31.65	0.88	0.15
Total seagrass	No detectable trend	2000 - 2007	88.14	-1.52	0.15

An annual decrease in percent cover was observed for drift algae (-2.90%). Total seagrass, manatee grass, shoal grass, and turtle grass showed no detectable change in percent cover.



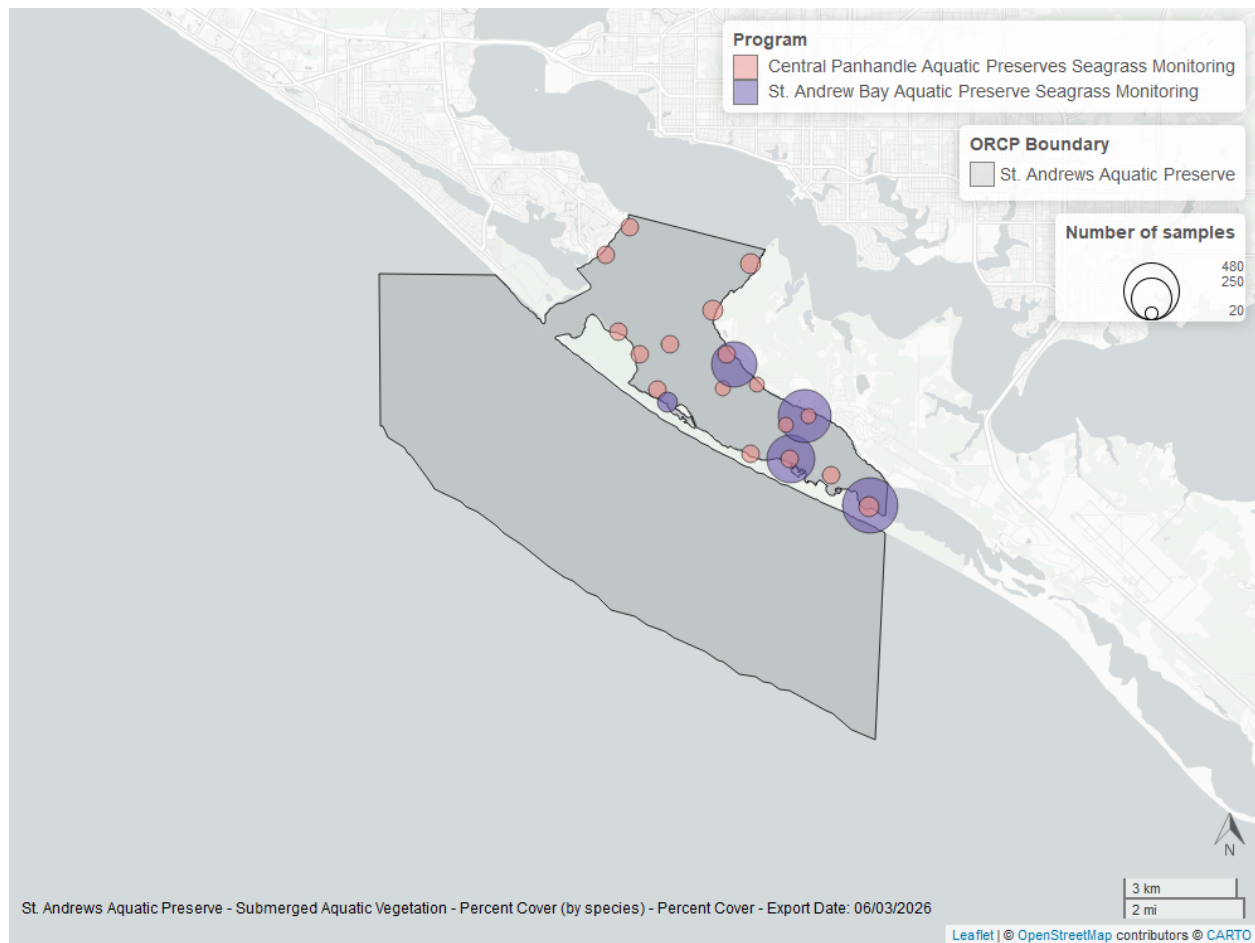
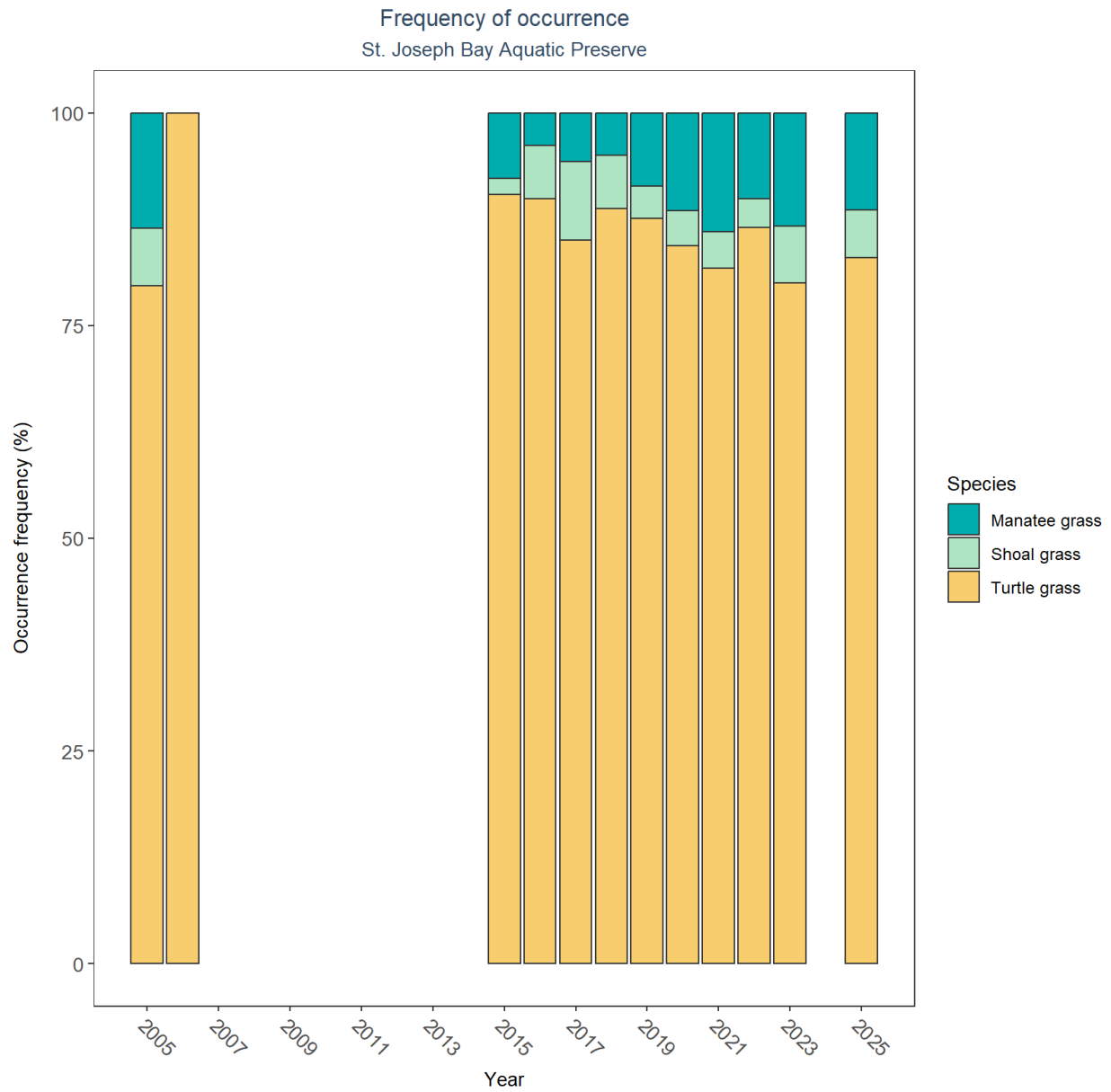


Figure 27: Map showing SAV sampling sites within the boundaries of *St. Andrews Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

St. Joseph Bay Aquatic Preserve



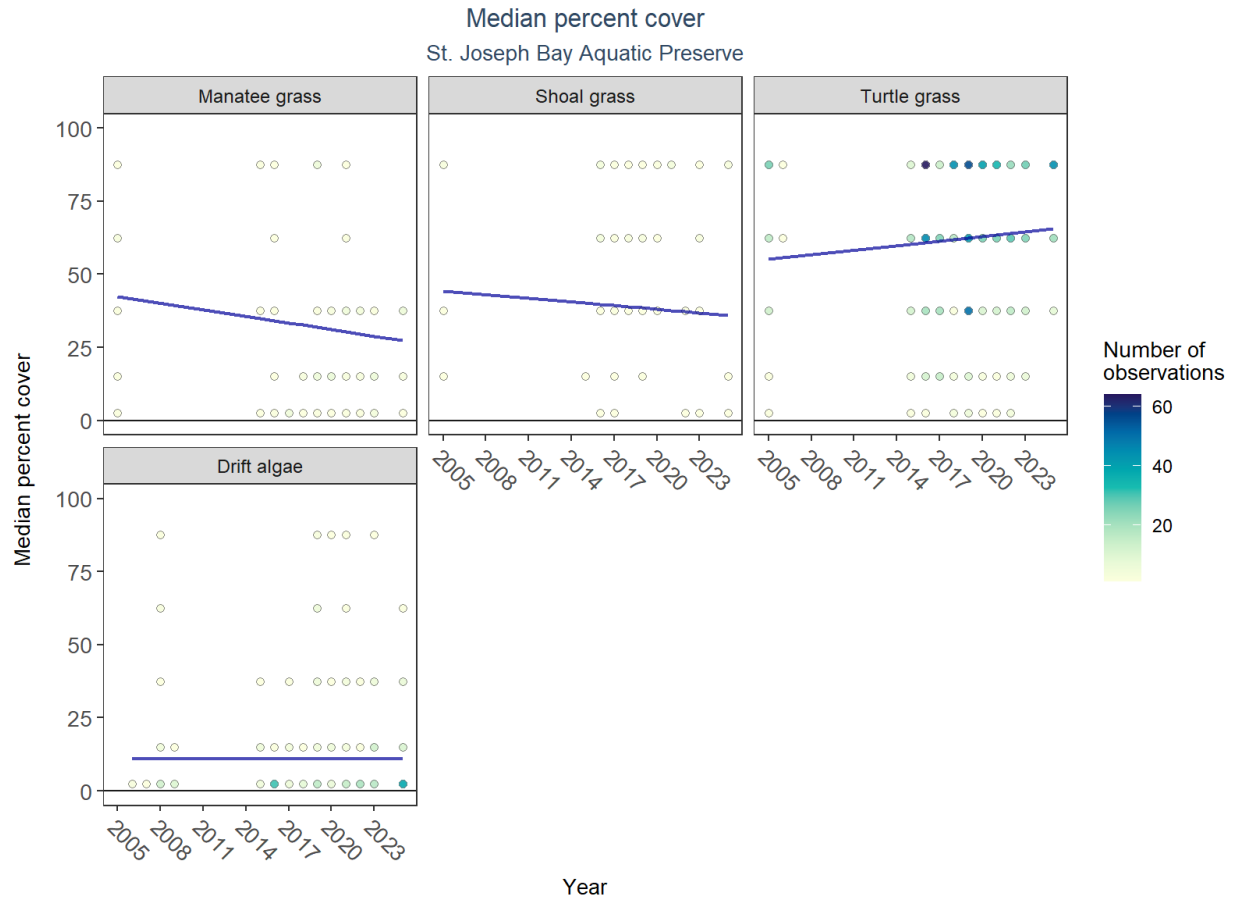
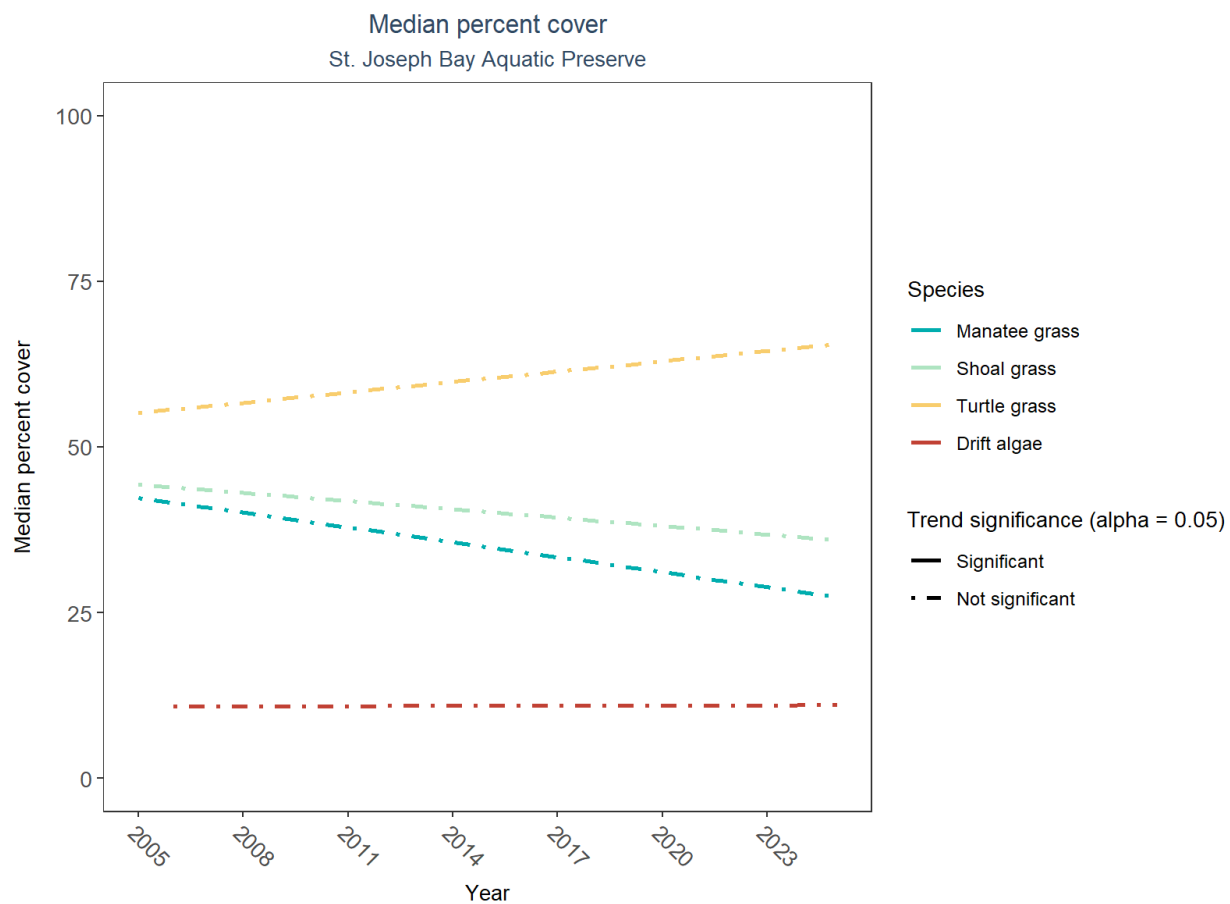


Table 27: SAV LME Results for St. Joseph Bay Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Drift algae	No detectable trend	2006 - 2025	10.71	0.01	0.97
Shoal grass	No detectable trend	2005 - 2025	48.88	-0.41	0.58
Manatee grass	No detectable trend	2005 - 2025	50.61	-0.75	0.28
Turtle grass	No detectable trend	2005 - 2025	49.47	0.52	0.11

Manatee grass, shoal grass, turtle grass, and drift algae showed no detectable change in percent cover.



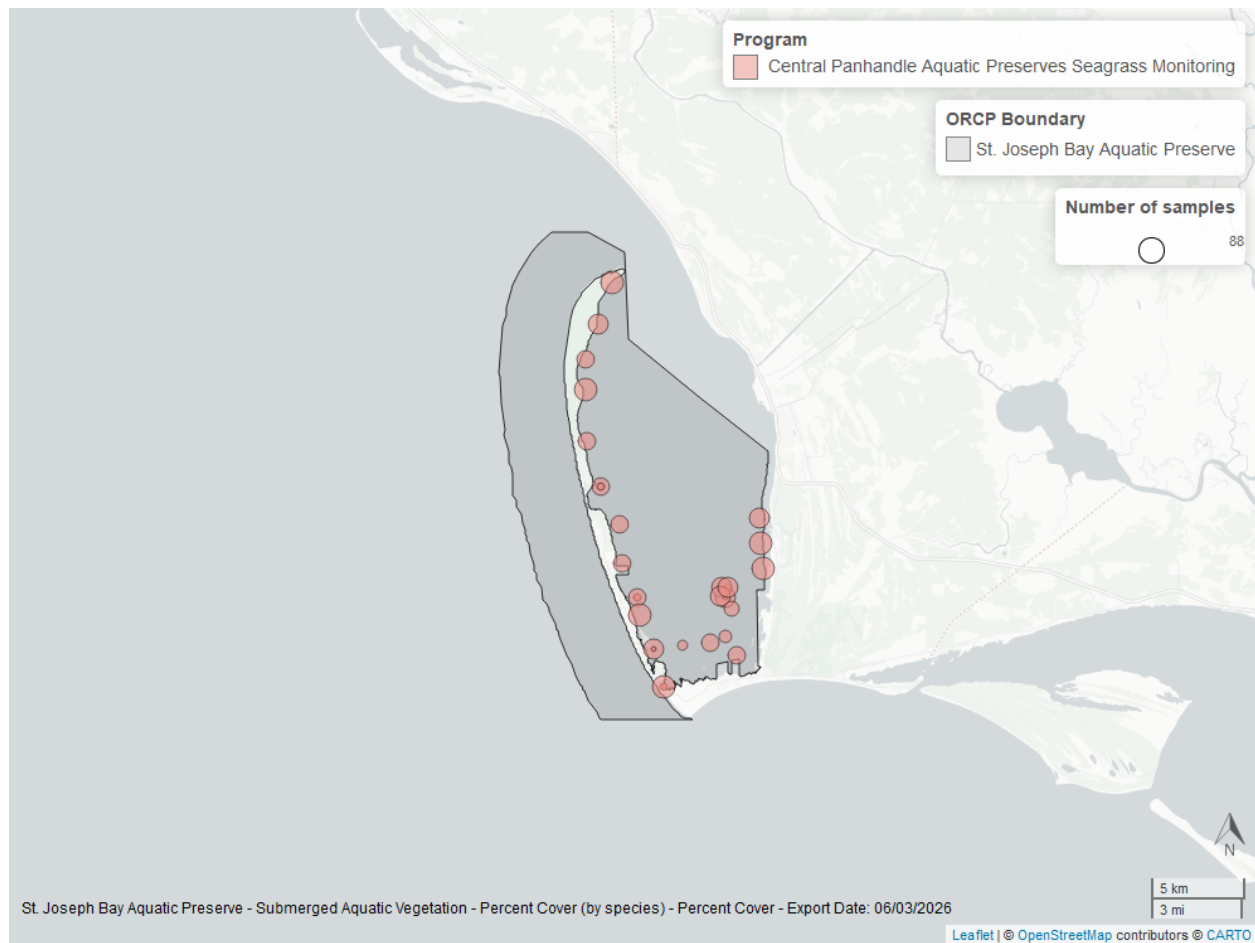
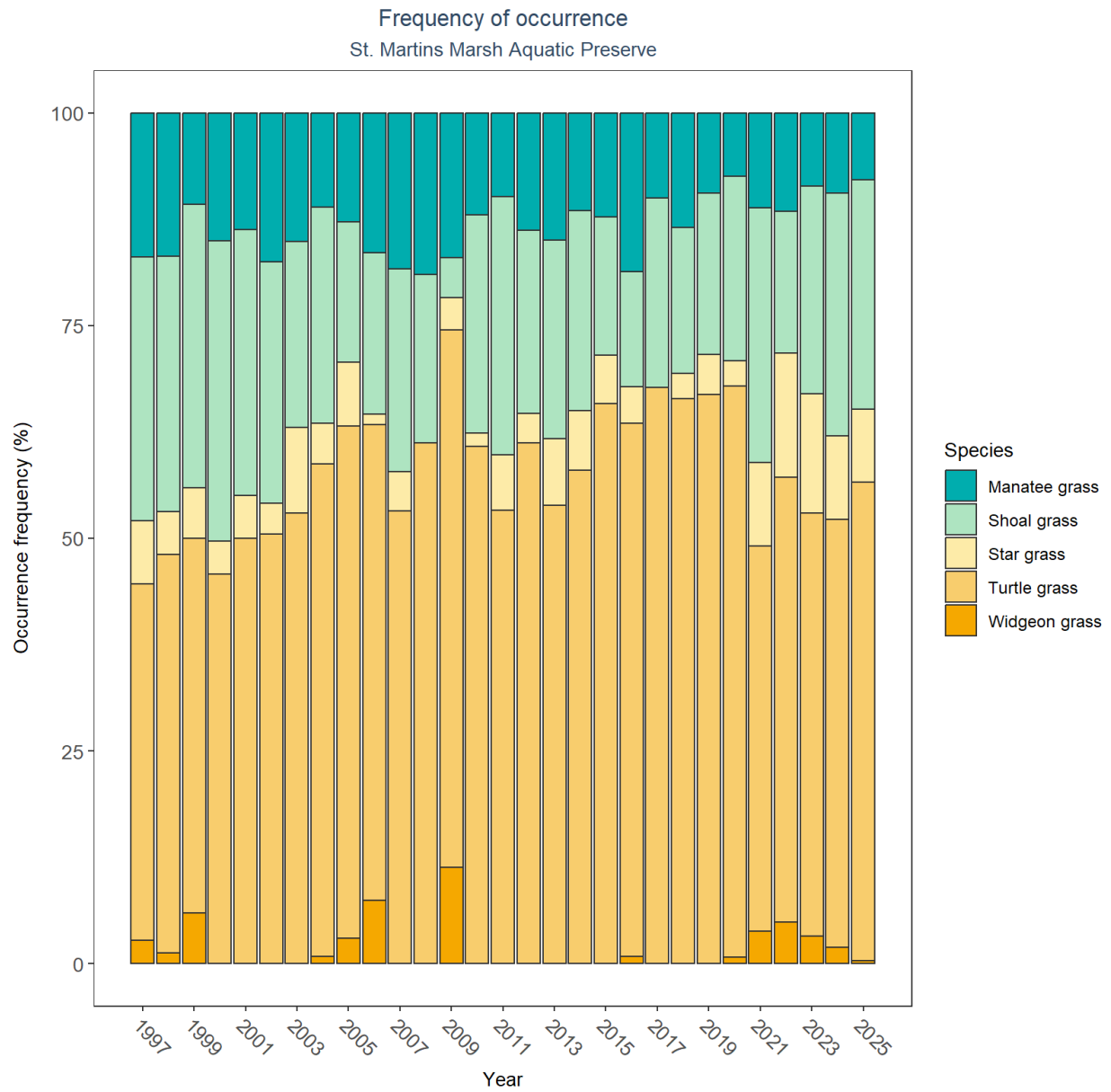


Figure 28: Map showing SAV sampling sites within the boundaries of *St. Joseph Bay Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

St. Martins Marsh Aquatic Preserve



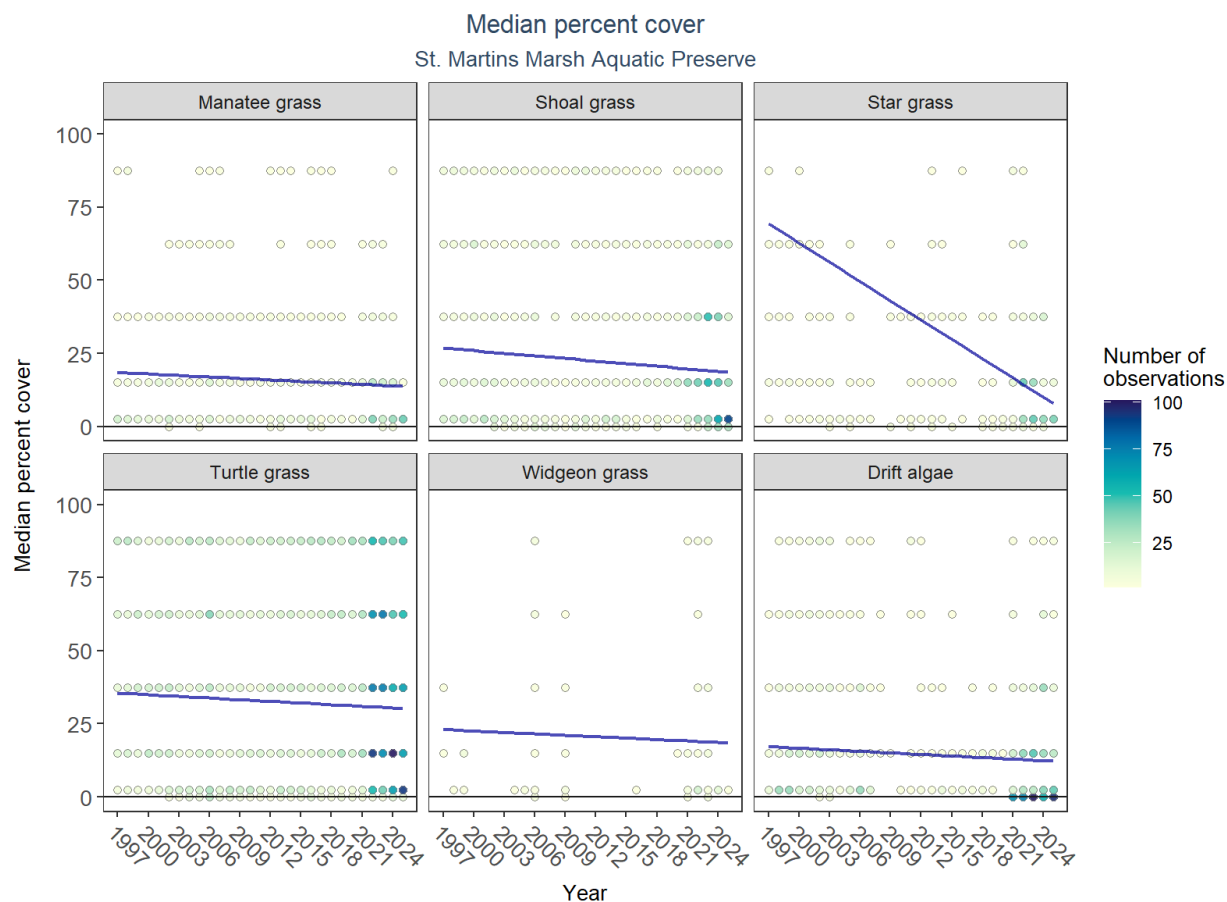
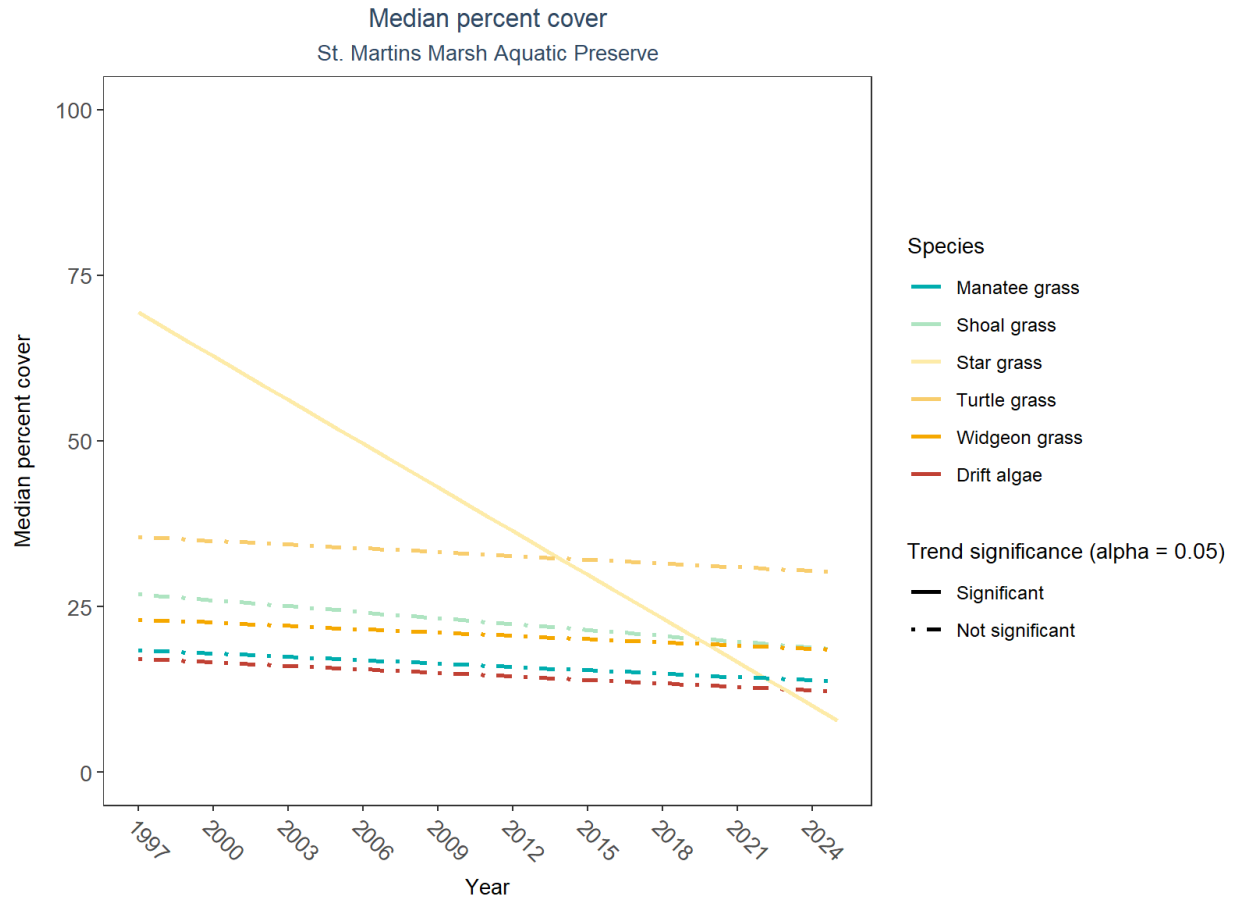


Table 28: SAV LME Results for St. Martins Marsh Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Drift algae	No detectable trend	1997 - 2025	17.68	-0.18	0.19
Shoal grass	No detectable trend	1997 - 2025	27.73	-0.29	0.10
Star grass	Decreasing trend	1997 - 2025	75.97	-2.20	0.04
Widgeon grass	No detectable trend	1997 - 2025	23.54	-0.16	0.66
Manatee grass	No detectable trend	1997 - 2025	18.96	-0.17	0.39
Turtle grass	No detectable trend	1997 - 2025	36.10	-0.19	0.20

An annual decrease in percent cover was observed for star grass (-2.20%). Manatee grass, shoal grass, turtle grass, widgeon grass, and drift algae showed no detectable change in percent cover.



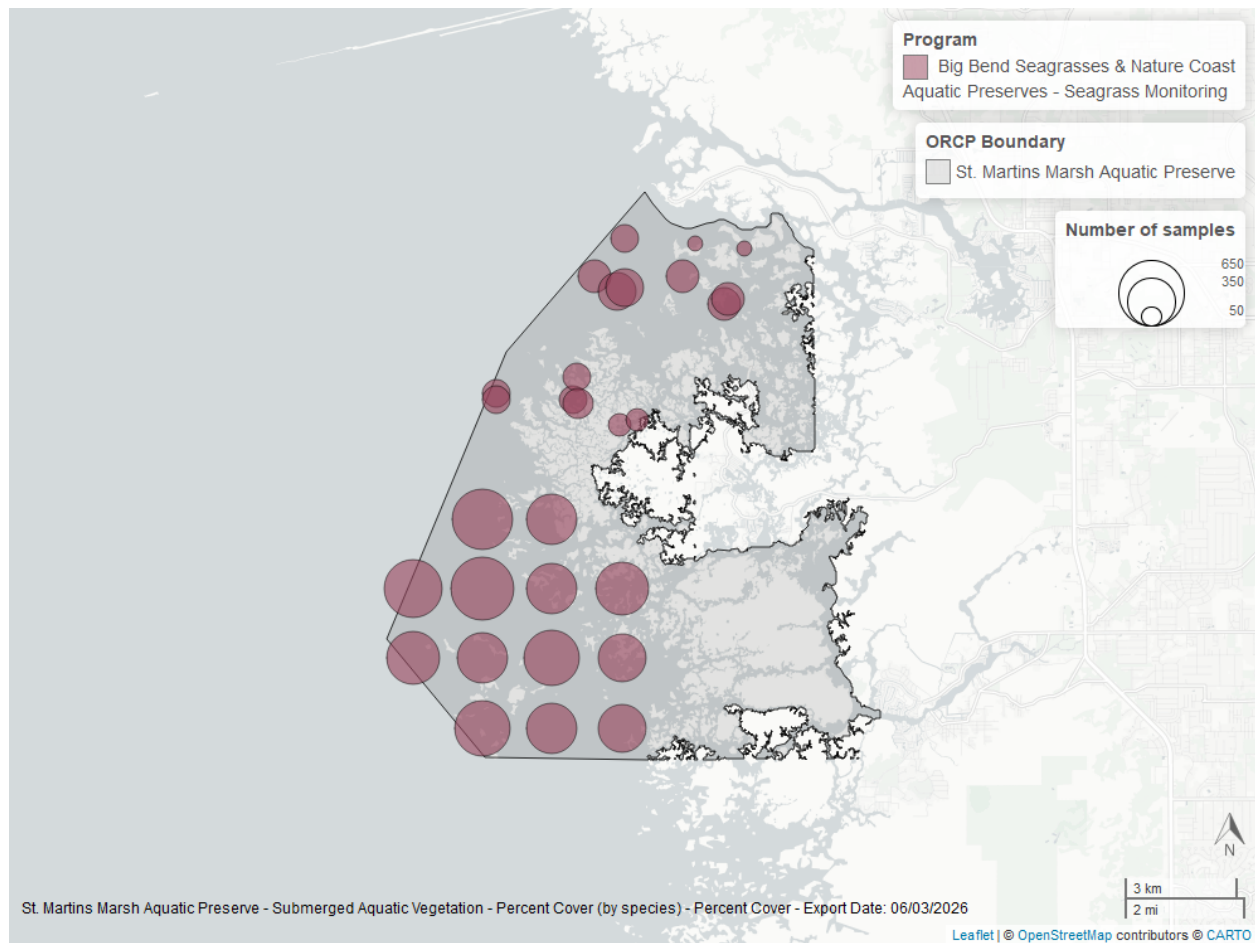
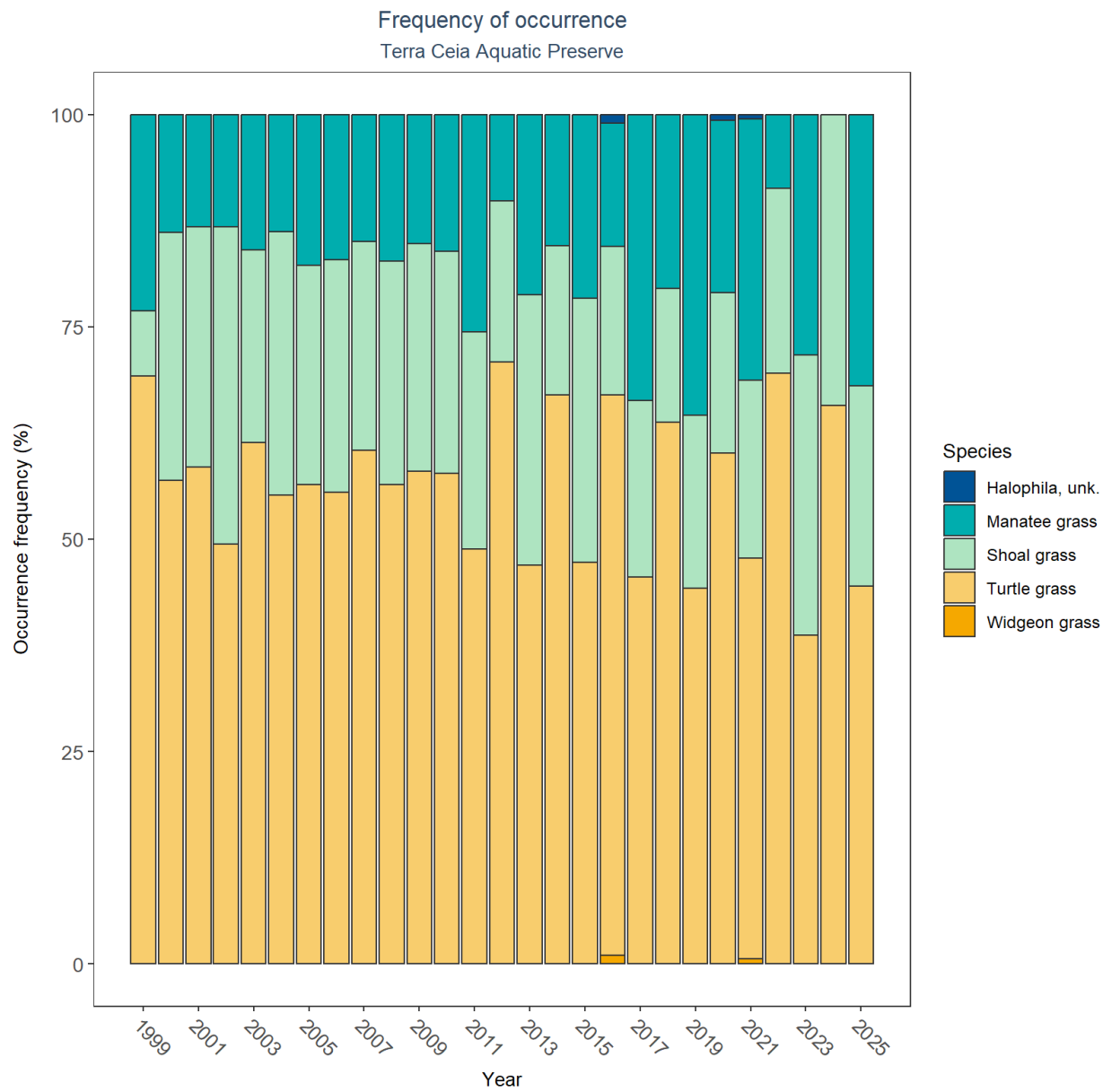


Figure 29: Map showing SAV sampling sites within the boundaries of *St. Martins Marsh Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Terra Ceia Aquatic Preserve



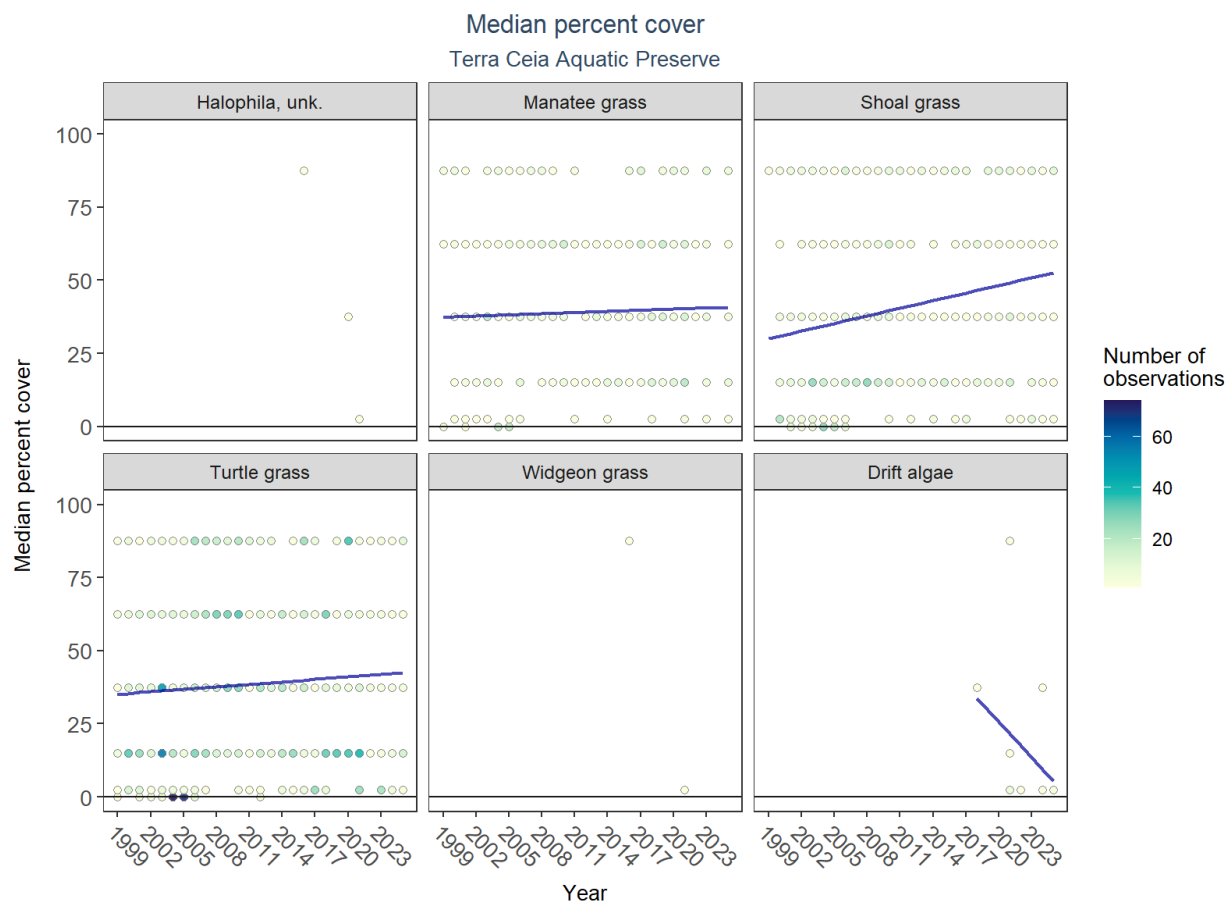
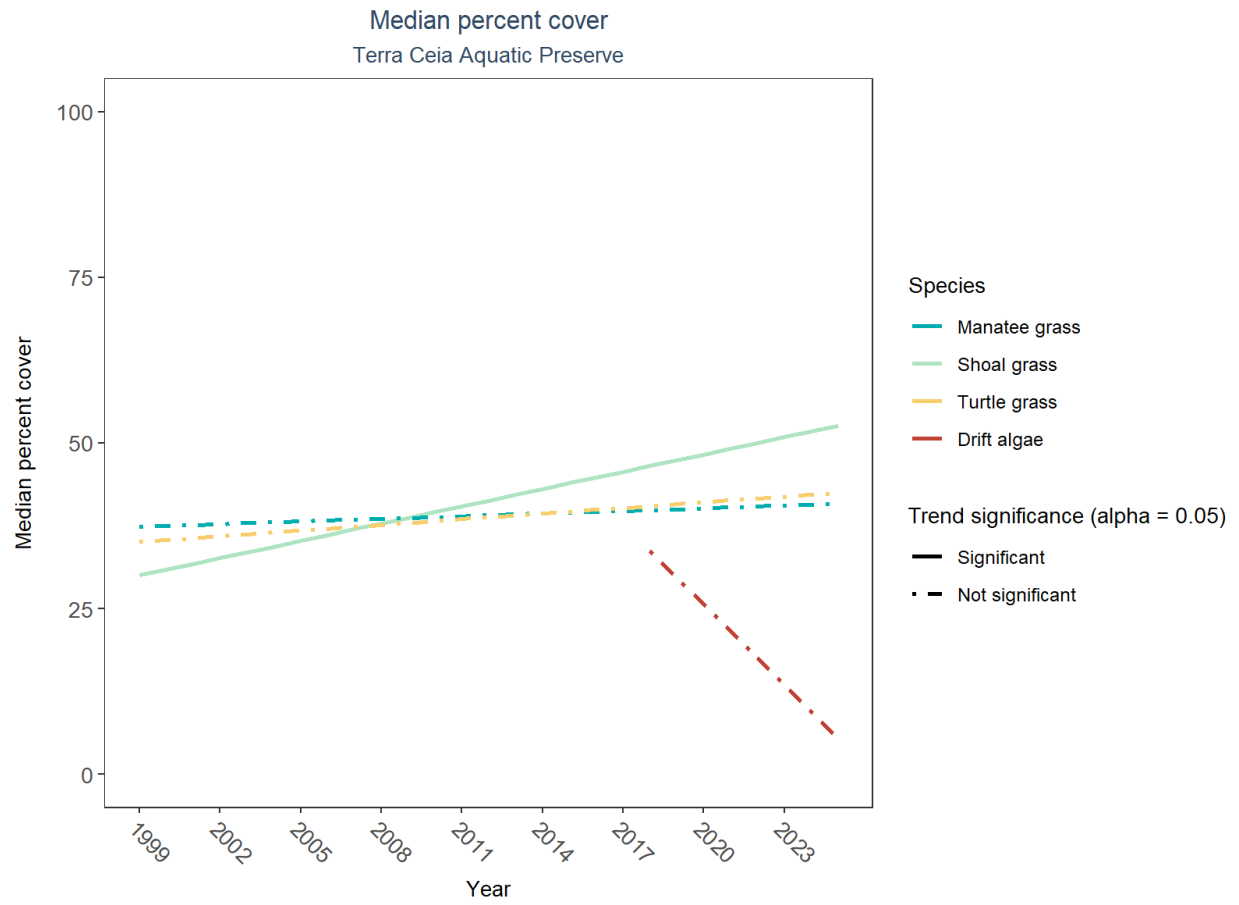


Table 29: SAV LME Results for Terra Ceia Aquatic Preserve

Species	Statistical Trend	Period of Record	LME Intercept	LME Slope	p
Drift algae	No detectable trend	2018 - 2025	131.13	-4.06	0.37
Shoal grass	Increasing trend	1999 - 2025	25.68	0.87	0.00
Widgeon grass	Insufficient data	2016 - 2021	-	-	-
Manatee grass	No detectable trend	1999 - 2025	36.73	0.13	0.72
Turtle grass	No detectable trend	1999 - 2025	33.68	0.28	0.32
Halophila, unk.	Insufficient data	2016 - 2021	-	-	-

An annual increase in percent cover was observed for shoal grass (0.87%). Manatee grass, turtle grass, and drift algae showed no detectable change in percent cover. Trends in percent cover could not be evaluated for unknown *Halophila* and widgeon grass due to insufficient data.



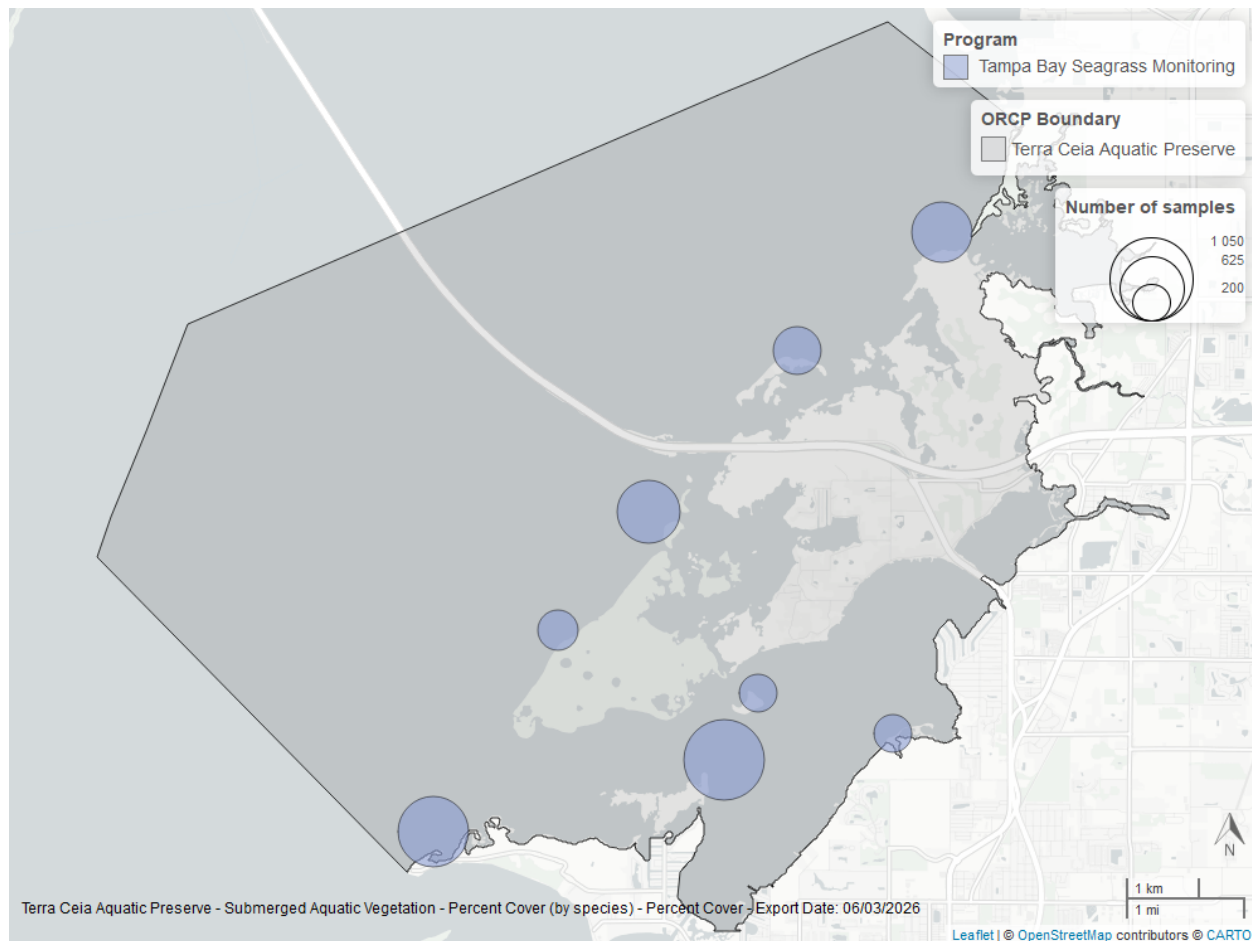


Figure 30: Map showing SAV sampling sites within the boundaries of *Terra Ceia Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Summary of SEACAR_SAV_BB_script_website.R

- Objective: Import and format SAV data, create summary plots and maps of changes over time, model changes over time using Bayesian and mixed-effects models. Results are separate for each managed area, species, and parameter (e.g., Braun Blanquet, percent cover, etc.).
- Packages: bayesplot, brms, broom.mixed, data.table, grid, gridExtra, gtable, nlme, scales, sf, tictoc, tidybayes, tidyverse
- File inputs: All_SAV_Parameters-2024-Sep-19.txt, seacar_dbo_SampleLocation_Point.shp, seacar_dbo_SampleLocation_Line.shp, ORCP_Managed_Areas.shp, Counties_-_Detailed_Shoreline.shp, MAPolygons_corners.csv
- Steps by line number:
 - 1 - 40: load libraries and import SAV file, create necessary sub-folders.
 - 43 - 132: format SAV data including renaming columns, removing NA values, and reformatting abundance/cover values. For the latter, this included removal of NA values and those out of range, and ensuring appropriate values for Braun Blanquet, modified Braun Blaunqet, percent occurrence, and percent cover.
 - 136: Import ManagedArea.csv to obtain abbreviated ManagedAreaNames.
 - 139 - 193: function for plotting model predictions (multiplots).
 - 196 - 288: function for plotting model predictions with simplified trendlines on a single plot

- (trendplots).
- 292 - 307: function used throughout script to allow for swift modification of species labels between “common” and “scientific” names. The script defaults to using scientific throughout the backend (i.e. `.rds` objects with species names default to scientific), with the `usenames` variable (line 468) converting the display of species names to the desired display type.
 - 310: EDA variable to specify what the script should produce. Setting to “plots” creates data exploration plots.
 - 313: `scope_plots` variable can be set to `TRUE` or `FALSE`. Set to `TRUE` to render spatio-temporal scope plots for each `ManagedArea`. Resulting plots are placed in the ‘output/Figures/SAV_temporal_scope_plots/’ folder.
 - 323 - 324: create `failedmods` data table to store failed model results.
 - 327 - 499: setup parameter list and objects for looping through parameters to create models and summary output, parameters include Braun Blanquet, median percent cover, visual percent cover, percent occurrence, frequency of occurrence. Write results to output `.xlsx` files.
 - * 351 - 395: setting up palettes for both species (`spcols`) and ProgramNames (`prcols`).
 - 517 - 1192: loop through parameters to create models and summary output, the following is an outline of steps in this loop.
 - * 530 - 1518: loop through managed areas using parameter from outside loop
 - * 539 - 553: create and save plot of parameter score for managed area over time by species
 - * 555 - 568: create and save plot of parameter score for managed area over time by program ID
 - * 570 - 583: create and save plot of parameter score for managed area over time grouped by species, by program ID
 - * 585 - 598: create and save plot of quadrat sizes for managed area over time by species
 - * 600 - 613: create and save plot of quadrat sizes for managed area over time by program ID
 - * 615 - 628: create and save plot of method for managed area over time by species
 - * 630 - 643: create and save plot of method for managed area over time by program ID
 - * 645 - 659: create and save plot of method for managed area by quadrat size and species
 - * 661 - 675: create and save plot of method for managed area by quadrat size and program ID
 - * 677 - 707: create and save plots of grid values over time by species and program ID if data available
 - * 709 - 739: create and save plots of depth values over time by species and program ID if data available
 - * 743 - 759: create and save a plot legend of species
 - * 762 - 785: loop through species to create and save a plot of parameter score over time
 - * 787 - 808: create and save a plot of totals for the species of parameter score over time
 - * 811 - 827: create and save a plot legend of species
 - * 830 - 853: loop through species to create and save a plot of parameter score as boxplots over time
 - * 855 - 875: create and save a plot of boxplots for the species of parameter score over time
 - * 898 - 908: setup empty objects for model results
 - * 912 - 1029: Loop through species to fit models, with separate exception statements for different parameters. The modeling workflow is similar for each parameter, with minor exceptions. The general goal of each is to assess trends in a parameter over time for a particular species and managed area. Each workflow includes error handling if models did not converge, produces summary tables of model fit, and summary plots showing model result. The models vary in the Gaussian distribution family for the response variable depending on parameter. Random effects (e.g., for `LocationID`) are used for all models.
 - * 1033 - 1124: create base plot of seagrass percent cover data over time for managed area. Add model fits if applicable, and saves object
 - 1036 - 1046: reads and sets modeled data for plots
 - * 1127 - 1185: create and save barplots of parameter results for managed area over time by species, only for Braun Blanquet and percent cover, save model results from prior loops
 - * 1197 - 1265: export `.png` of all plot types, setting height dynamically for `multiplots`.
 - * 1273 - 1277: zip all images into subfolders and create `SAV_Figures_[usenames].zip` where

`usenames` is either “scientific” or “common” based on the `usenames` variable setting.